Supplementary material for 'A Reduced-Rank Panel Data Model with Interactive Fixed Effects'

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S.1 Definitions and Notation

S.1.1 Definitions in Section 3.3

We introduce the following notation. Let

$$
\begin{aligned}
&w_{ij} = \boldsymbol{l}_{i,0}^\top \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \boldsymbol{l}_{j,0}, \quad \boldsymbol{J} = \boldsymbol{I}_{K_0} \otimes \left(\boldsymbol{0}_{(p-K_0)\times K_0}, \quad \boldsymbol{I}_{p-K_0}\right),\\
&\boldsymbol{H}_i^\perp(\boldsymbol{\Theta},\boldsymbol{F}) = \boldsymbol{I}_T - \boldsymbol{H}_i(\boldsymbol{\Theta},\boldsymbol{F}) = \boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta} \left(\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\right)^+ \boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp,\\
&\boldsymbol{\Xi}(\boldsymbol{\Theta},\boldsymbol{F}) = \frac{1}{N}\sum_{i=1}^N \boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top \otimes \left[\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{H}_i(\boldsymbol{\Theta},\boldsymbol{F}) \boldsymbol{P}_{\boldsymbol{G}}^\perp\right] \in \mathbb{R}^{TR_0\times TR_0},\\
&\boldsymbol{Q}_i(\boldsymbol{F}) = \boldsymbol{H}_i^\perp(\boldsymbol{\Theta}_0,\boldsymbol{F})\, \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \in \mathbb{R}^{T\times p},\\
&\boldsymbol{Z}_i(\boldsymbol{F}) = \boldsymbol{b}_{i,0}^\top \otimes \boldsymbol{Q}_i(\boldsymbol{F}) - \frac{1}{N}\sum_{j=1}^N \boldsymbol{l}_{i,0}^\top \otimes \boldsymbol{H}_i^\perp(\boldsymbol{\Theta}_0,\boldsymbol{F}) \cdot \boldsymbol{\Xi}^+(\boldsymbol{\Theta}_0,\boldsymbol{F}) \cdot \boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^\top \otimes \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{Q}_j(\boldsymbol{F}) \in \mathbb{R}^{T\times pK_0},\\
&\boldsymbol{D}(\boldsymbol{F}) = \frac{1}{NT}\sum_{i=1}^N \boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top \otimes \boldsymbol{Q}_i^\top(\boldsymbol{F})\boldsymbol{Q}_i(\boldsymbol{F})\\
&\qquad - \frac{1}{N^2T}\sum_{i,j=1}^N \boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{Q}_j^\top(\boldsymbol{F})\boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}^+(\boldsymbol{\Theta}_0,\boldsymbol{F}) \cdot \boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^\top \otimes \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{Q}_i(\boldsymbol{F}) \in \mathbb{R}^{pK_0\times pK_0},
\end{aligned}
\tag{S.1.1}
$$

and by replacing the general parameters with their true values, we define [1]

$$
\boldsymbol{H}_{i,0} = \boldsymbol{H}_i(\boldsymbol{\Theta}_0,\boldsymbol{F}_0), \quad \boldsymbol{H}_{i,0}^\perp = \boldsymbol{I}_T - \boldsymbol{H}_{i,0}, \quad \boldsymbol{\Xi}_0 = \boldsymbol{\Xi}(\boldsymbol{\Theta}_0,\boldsymbol{F}_0), \quad \boldsymbol{Q}_{i,0} = \boldsymbol{Q}_i(\boldsymbol{F}_0), \quad \boldsymbol{Z}_{i,0} = \boldsymbol{Z}_i(\boldsymbol{F}_0). \tag{S.1.2}
$$

The $TR_0 \times TR_0$ matrix $\boldsymbol{\Xi}$ appears in both the construction of transformed predictors $\{\boldsymbol{Z}_i(\boldsymbol{F})\}_{i=1}^N$ and definition of $\boldsymbol{D}(\boldsymbol{F})$. For the asymptotic analysis, we require $\boldsymbol{\Xi}_0$ to be positive definite, which is implied by Assumptions 2(2) and 5; see Lemma S.2.1 (3). For interpretability, consider $\boldsymbol{\Xi}_0' := N^{-1}\sum_{i=1}^N \boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top \otimes \boldsymbol{H}_{i,0}^\perp$. The positive definiteness of $\boldsymbol{\Xi}_0'$ is sufficient for that of $\boldsymbol{\Xi}_0$: for any $\boldsymbol{e} \in \mathbb{R}^{TR_0}$ with $\|\boldsymbol{e}\|_2 = 1$,

$$
\boldsymbol{e}^\top \boldsymbol{\Xi}_0 \boldsymbol{e} = \left[\left(\boldsymbol{I}_{R_0} \otimes \boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\boldsymbol{e}\right]^\top \boldsymbol{\Xi}_0' \left[\left(\boldsymbol{I}_{R_0} \otimes \boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\boldsymbol{e}\right] + \left[\left(\boldsymbol{I}_{R_0} \otimes \boldsymbol{P}_{\boldsymbol{G}}\right)\boldsymbol{e}\right]^\top \left(\boldsymbol{\Sigma}_{\boldsymbol{L}} \otimes \boldsymbol{I}_T\right)\left[\left(\boldsymbol{I}_{R_0} \otimes \boldsymbol{P}_{\boldsymbol{G}}\right)\boldsymbol{e}\right] \geq \lambda_{min}\left(\boldsymbol{\Xi}_0'\right).
$$

Intuitively, $\boldsymbol{\Xi}_0' > 0$ requires that the loadings $\boldsymbol{l}_{i,0}$ and the projected time directions $\boldsymbol{H}_{i,0}^\perp$ span a sufficiently rich space. As $T \to \infty$, $\boldsymbol{H}_{i,0}^\perp$ retains rank $T - K_0$. Coupled with $N \to \infty$, the weighted cross-sectional average

[1] The notation $\boldsymbol{D}_0$ is reserved for the probability limit of $\boldsymbol{D}(\boldsymbol{F}_0)$.

prevents $\boldsymbol{\Xi}_0'$ from becoming singular. The bias terms in (3.9) are

$$\begin{aligned}\mathbb{B}^{\boldsymbol{\xi}} = &- \boldsymbol{D}^{+}(\boldsymbol{F}_0)\cdot\frac{1}{N}\sum_{i=1}^{N}\Bigg\{\Bigg[\left(\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}-\frac{1}{N}\sum_{k=1}^{N}w_{ik}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\right)\\ &+\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^{N}\boldsymbol{b}_{l0}\boldsymbol{l}_{l0}^{\top}\otimes\boldsymbol{Q}_{l,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\right)\cdot\left(\boldsymbol{l}_{i,0}\otimes\frac{\boldsymbol{V}_{i,0}^{\top}\boldsymbol{F}_0}{\sqrt{T}}-\frac{1}{N}\sum_{k=1}^{N}w_{ik}\boldsymbol{l}_{k,0}\otimes\frac{\boldsymbol{V}_{k,0}^{\top}\boldsymbol{F}_0}{\sqrt{T}}\right)\Bigg]\\ &\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\Bigg\},\end{aligned}\tag{S.1.3}$$

and

$$\begin{aligned}\mathbb{B}^{\boldsymbol{\zeta}} = \mathbb{B}^{\boldsymbol{\zeta}_1}+\mathbb{B}^{\boldsymbol{\zeta}_2} =&\boldsymbol{D}^{+}(\boldsymbol{F}_0)\cdot\frac{1}{N}\sum_{i=1}^{N}\left[\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\otimes\boldsymbol{Q}_{i,0}^{\top}\right]\ell(\boldsymbol{\Omega}_i)\\ &-\boldsymbol{D}^{+}(\boldsymbol{F}_0)\cdot\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{Q}_{i,0}^{\top}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\ell\left(\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{U}_{j,0}\boldsymbol{\Omega}_j\boldsymbol{U}_{j,0}^{\top}\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right),\end{aligned}\tag{S.1.4}$$

where

$$\begin{aligned}&\boldsymbol{S}_{i,0}=\boldsymbol{I}_p-\boldsymbol{\Theta}_0\left(\frac{\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\frac{\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i}{T},\\ &\boldsymbol{U}_{i,0}=\boldsymbol{I}_T-\boldsymbol{V}_{i,0}=\boldsymbol{I}_T-\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp},\\ &\boldsymbol{\Omega}_i=\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^{\top}\right)=\mathrm{Diag}\left(\sigma_{i1}^2,\quad\sigma_{i2}^2,\quad\ldots,\quad\sigma_{iT}^2\right)\in\mathbb{R}^{T\times T},\quad\bar{\sigma}_i^2=\frac{1}{T}\sum_{t=1}^{T}\sigma_{it}^2.\end{aligned}$$

For simplicity, we write $\mathbb{E}_{\mathcal{C}}(\cdot)$ in place of $\mathbb{E}(\cdot\mid\mathcal{C})$ throughout. Before presenting the Nickell bias term $\mathbb{B}^{\boldsymbol{\eta}}$, we introduce the following notation:

$$\begin{aligned}&\boldsymbol{\Sigma}_{ii}=\frac{\boldsymbol{X}_i^{\top}\boldsymbol{X}_i}{T},\quad\boldsymbol{\Sigma}_{ij}=\frac{\boldsymbol{X}_i^{\top}\boldsymbol{X}_j}{T},\quad\boldsymbol{\Sigma}_{i\boldsymbol{F}}=\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T},\quad\boldsymbol{\Sigma}_{i\boldsymbol{G}}=\frac{\boldsymbol{X}_i^{\top}\boldsymbol{G}}{T},\quad\boldsymbol{\Sigma}_{\boldsymbol{G}}=\frac{\boldsymbol{G}^{\top}\boldsymbol{G}}{T},\\ &\boldsymbol{\pi}_{1i}=\left[\ell^{\top}(\boldsymbol{\Sigma}_{ii}),\ \ell^{\top}(\boldsymbol{\Sigma}_{i\boldsymbol{F}})\right]^{\top},\quad\boldsymbol{\pi}_{1i}^{e}=\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\pi}_{1i}\right),\\ &\boldsymbol{\pi}_{2ij}=\left[\ell^{\top}(\boldsymbol{\Sigma}_{ii}),\ \ell^{\top}(\boldsymbol{\Sigma}_{ij}),\ \ell^{\top}(\boldsymbol{\Sigma}_{i\boldsymbol{F}}),\ \ell^{\top}(\boldsymbol{\Sigma}_{i\boldsymbol{G}}),\ \ell^{\top}(\boldsymbol{\Sigma}_{j\boldsymbol{F}}),\ \ell^{\top}(\boldsymbol{\Sigma}_{j\boldsymbol{G}})\right]^{\top},\quad\boldsymbol{\pi}_{2ij}^{e}=\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\pi}_{2ij}\right),\\ &\boldsymbol{\Psi}_{1i}:=\boldsymbol{\Psi}_1\left(\boldsymbol{\pi}_{1i}\right)=\boldsymbol{I}_p-\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^{\top}\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^{\top},\\ &\boldsymbol{\Psi}_{2i}:=\boldsymbol{\Psi}_2\left(\boldsymbol{\pi}_{1i}\right)=\boldsymbol{\Psi}_1\left(\boldsymbol{\pi}_{1i}\right)\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1},\\ &\boldsymbol{\Psi}_{3ij}:=\boldsymbol{\Psi}_3\left(\boldsymbol{\pi}_{2ij}\right)=\left(\boldsymbol{\Sigma}_{ji}-\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}-\boldsymbol{\Sigma}_{j\boldsymbol{G}}\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{G}}^{\top}\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^{\top}\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^{\top},\\ &\boldsymbol{\Psi}_{4ij}:=\boldsymbol{\Psi}_4\left(\boldsymbol{\pi}_{2ij}\right)=\boldsymbol{\Psi}_3\left(\boldsymbol{\pi}_{2ij}\right)\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1},\\ &\boldsymbol{\Psi}_{1i}^{e}=\boldsymbol{\Psi}_1\left(\boldsymbol{\pi}_{1i}^{e}\right),\quad\boldsymbol{\Psi}_{2i}^{e}=\boldsymbol{\Psi}_2\left(\boldsymbol{\pi}_{1i}^{e}\right),\quad\boldsymbol{\Psi}_{3ij}^{e}=\boldsymbol{\Psi}_3\left(\boldsymbol{\pi}_{2ij}^{e}\right),\quad\boldsymbol{\Psi}_{4ij}^{e}=\boldsymbol{\Psi}_4\left(\boldsymbol{\pi}_{2ij}^{e}\right).\end{aligned}$$

By differentiating $\boldsymbol{\Psi}_{1i}$ w.r.t. $\boldsymbol{\pi}_{1i}$, and $\boldsymbol{\Psi}_{3ij}$ w.r.t. $\boldsymbol{\pi}_{2ij}$, we define

$$\begin{aligned}&\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}=\left.\frac{\partial\ell[\boldsymbol{\Psi}_{1i}(\boldsymbol{\pi}_1)]}{\partial\boldsymbol{\pi}_1}\right|_{\boldsymbol{\pi}_1=\boldsymbol{\pi}_{1i}},\quad\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}=\left.\frac{\partial\ell[\boldsymbol{\Psi}_{3ij}(\boldsymbol{\pi}_2)]}{\partial\boldsymbol{\pi}_2}\right|_{\boldsymbol{\pi}_2=\boldsymbol{\pi}_{2ij}},\\ &\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e}=\left.\frac{\partial\ell[\boldsymbol{\Psi}_{1i}(\boldsymbol{\pi}_1)]}{\partial\boldsymbol{\pi}_1}\right|_{\boldsymbol{\pi}_1=\boldsymbol{\pi}_{1i}^{e}},\quad\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e}=\left.\frac{\partial\ell[\boldsymbol{\Psi}_{3ij}(\boldsymbol{\pi}_2)]}{\partial\boldsymbol{\pi}_2}\right|_{\boldsymbol{\pi}_2=\boldsymbol{\pi}_{2ij}^{e}}.\end{aligned}$$

In the same way, we define $\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{2i}},\ \boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{4ij}},\ \boldsymbol{\mathcal{D}}^{e}_{\boldsymbol{\Psi}_{2i}},\ \boldsymbol{\mathcal{D}}^{e}_{\boldsymbol{\Psi}_{4ij}}\right)$. Then,

$$
\begin{aligned}
\mathbb{B}^{\boldsymbol{\eta}} =& \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right)\cdot\Bigg(\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{1i}}\boldsymbol{\pi}_{1i}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{2i}}\boldsymbol{\pi}_{1i}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\frac{1}{N^2}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}\otimes\boldsymbol{\Psi}^{e}_{1j}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{3ij}}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{4ij}}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\frac{1}{N^2}\sum_{i,j,k=1}^{N}\Bigg\{\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}^{e}_{1j}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\\
&\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_k\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^{\top}\left(\boldsymbol{\Sigma}^{e}_{kk}-\boldsymbol{\Sigma}^{e}_{k\boldsymbol{F}}\boldsymbol{\Sigma}^{-1}_{\boldsymbol{F}}\boldsymbol{\Sigma}^{e\top}_{k\boldsymbol{F}}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^{\top}\\
&\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{3ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{4ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]\Bigg\}\Bigg),
\end{aligned}
\tag{S.1.5}
$$

where D^{trun} satisfies $D^{trun}/\log N\to\infty$ and $D^{trun}/N^{1/3}\to 0$, $\tilde{\boldsymbol{l}}_{i,0}=\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\boldsymbol{l}_{i,0}$, and $\boldsymbol{\Upsilon}_0=N^{-1}\sum_{i=1}^{N}\tilde{\boldsymbol{H}}_{i,0}=N^{-1}\sum_{i=1}^{N}\tilde{\boldsymbol{l}}_{i,0}\tilde{\boldsymbol{l}}_{i,0}^{\top}\otimes\boldsymbol{P}^{\perp}_{\boldsymbol{G}}\boldsymbol{H}_{i,0}\boldsymbol{P}^{\perp}_{\boldsymbol{G}}$.

S.1.2 Definitions in Section 3.4

To estimate the parameters in Theorem 3.3, by replacing $(\boldsymbol{\Theta}_0,\ \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0)$ with $\left(\hat{\boldsymbol{\Theta}},\ \hat{\boldsymbol{B}},\ \hat{\boldsymbol{F}},\ \hat{\boldsymbol{L}}\right)$, we have

$$
\begin{aligned}
&\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}=\frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{l}}_i\hat{\boldsymbol{l}}_i^{\top},\quad \hat{w}_{ij}=\hat{\boldsymbol{l}}_i^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_j,\\
&\hat{\boldsymbol{H}}_i^{\perp}=\boldsymbol{I}_T-\hat{\boldsymbol{H}}_i=\boldsymbol{I}_T-\boldsymbol{H}_i\left(\hat{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right),\quad \hat{\boldsymbol{Q}}_i=\hat{\boldsymbol{H}}_i^{\perp}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{X}_i\\
&\hat{\boldsymbol{\Xi}}=\frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{l}}_i\hat{\boldsymbol{l}}_i^{\top}\otimes\left(\boldsymbol{I}_T-\boldsymbol{P}^{\perp}_{\boldsymbol{G}}\hat{\boldsymbol{H}}_i\boldsymbol{P}^{\perp}_{\boldsymbol{G}}\right),\\
&\hat{\boldsymbol{Z}}_i=\hat{\boldsymbol{b}}_i^{\top}\otimes\hat{\boldsymbol{Q}}_i-\hat{\boldsymbol{l}}_i^{\top}\otimes\hat{\boldsymbol{H}}_i^{\perp}\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\frac{1}{N}\sum_{j=1}^{N}\hat{\boldsymbol{l}}_j\hat{\boldsymbol{b}}_j^{\top}\otimes\boldsymbol{P}^{\perp}_{\boldsymbol{G}}\hat{\boldsymbol{Q}}_j,\quad \hat{\boldsymbol{D}}_0=\frac{1}{NT}\sum_{i=1}^{N}\hat{\boldsymbol{Z}}_i^{\top}\hat{\boldsymbol{Z}}_i,\\
&\hat{\boldsymbol{S}}_i=\boldsymbol{I}_p-\hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{X}_i,\\
&\hat{\boldsymbol{U}}_i=\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i=\boldsymbol{I}_T-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}},\\
&\hat{\boldsymbol{A}}=\left(\hat{\boldsymbol{\Theta}}^{\top}\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\otimes\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\cdot\left(\boldsymbol{I}_{p^2}+\boldsymbol{C}_{p,p}\right).
\end{aligned}
\tag{S.1.6}
$$

Let $\hat{\varepsilon}_{it}=y_{it}-\boldsymbol{x}_{it}^{\top}\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\hat{\boldsymbol{l}}_i^{\top}\hat{\boldsymbol{f}}_t$. Then, the diagonal matrix $\boldsymbol{\Omega}_i=\mathrm{Diag}\left(\sigma_{i1}^2,\ \ldots,\ \sigma_{iT}^2\right)$ is estimated by $\hat{\boldsymbol{\Omega}}_i=\mathrm{Diag}\left(\hat{\varepsilon}_{i1}^2,\ \ldots,\ \hat{\varepsilon}_{iT}^2\right)$, and $\bar{\sigma}_i^2=T^{-1}\sum_{i=1}^{N}\sigma_{it}^2$ is estimated by $\hat{\bar{\sigma}}_i^2=T^{-1}\sum_{t=1}^{T}\hat{\varepsilon}_{it}^2$. Correspondingly, the bias terms $\mathbb{B}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\zeta}}$ are respectively estimated by

$$
\begin{aligned}
\hat{\mathbb{B}}^{\boldsymbol{\xi}}=&-\hat{\boldsymbol{D}}_0^{+}\cdot\frac{1}{N}\sum_{i=1}^{N}\Bigg\{\Bigg[\left(\hat{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}}{T}-\frac{1}{N}\sum_{k=1}^{N}\hat{w}_{ik}\hat{\boldsymbol{b}}_k\otimes\hat{\boldsymbol{S}}_k^{\top}\frac{\boldsymbol{X}_k^{\top}\hat{\boldsymbol{F}}}{T}\right)\\
&+\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^{N}\hat{\boldsymbol{b}}_l\hat{\boldsymbol{l}}_l^{\top}\otimes\hat{\boldsymbol{Q}}_l^{\top}\boldsymbol{P}^{\perp}_{\boldsymbol{G}}\cdot\hat{\boldsymbol{\Xi}}^{-1}\right)\cdot\left(\hat{\boldsymbol{l}}_i\otimes\frac{\hat{\boldsymbol{V}}_i^{\top}\hat{\boldsymbol{F}}}{\sqrt{T}}-\frac{1}{N}\sum_{k=1}^{N}\hat{w}_{ik}\hat{\boldsymbol{l}}_k\otimes\frac{\hat{\boldsymbol{V}}_k^{\top}\hat{\boldsymbol{F}}}{\sqrt{T}}\right)\Bigg]\cdot\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2\Bigg\},
\end{aligned}
\tag{S.1.7}
$$

and

$$
\hat{\mathbb{B}}^{\boldsymbol{\zeta}}=\hat{\mathbb{B}}^{\boldsymbol{\zeta}_1}+\hat{\mathbb{B}}^{\boldsymbol{\zeta}_2}=\hat{\boldsymbol{D}}_0^{+}\cdot\frac{1}{N}\sum_{i=1}^{N}\left[\left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\otimes\hat{\boldsymbol{Q}}_i^{\top}\right]\ell\left(\hat{\boldsymbol{\Omega}}_i\right)
$$

$$- \hat{\boldsymbol{D}}_0^{+} \cdot \frac{1}{N^2 T} \sum_{i,j=1}^{N} \left[\hat{\boldsymbol{b}}_i \hat{\boldsymbol{l}}_i^{\top} \otimes \hat{\boldsymbol{Q}}_i^{\top} \cdot \hat{\boldsymbol{\Xi}}^{-1} \cdot \ell \left(\boldsymbol{P}_{\boldsymbol{G}}^{\perp} \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \hat{\boldsymbol{U}}_j \hat{\boldsymbol{\Omega}}_j \hat{\boldsymbol{U}}_j^{\top} \hat{\boldsymbol{F}} \right) \right]. \tag{S.1.8}$$

The estimation of $\mathbb{B}^{\boldsymbol{\eta}}$ is more complex. By replacing $(\boldsymbol{F}_0,\ \boldsymbol{\Theta}_0)$ in $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}$ and $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}$ with $(\hat{\boldsymbol{F}},\ \hat{\boldsymbol{\Theta}})$, we obtain $\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}$ and $\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}$. In the same way, we can define $\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{2i}},\ \hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{4ij}}\right)$. Let

$$\hat{\boldsymbol{\varpi}}_{1is} = \left[\ell^{\top}\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^{\top}\right) \right]^{\top},$$

$$\hat{\boldsymbol{\varpi}}_{2ijs} = \left[\ell^{\top}\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{is}\boldsymbol{x}_{js}^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{is}\boldsymbol{g}_s^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{js}\hat{\boldsymbol{f}}_s^{\top}\right),\ \ell^{\top}\left(\boldsymbol{x}_{js}\boldsymbol{g}_s^{\top}\right) \right]^{\top}.$$

Then, we estimate $\mathbb{B}^{\boldsymbol{\eta}}$ by

$$\begin{aligned}
\hat{\mathbb{B}}^{\boldsymbol{\eta}} =& \hat{\boldsymbol{D}}_0^{+} \cdot \Bigg(\frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{\boldsymbol{b}}_i \otimes \left[\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^{\top} \hat{\boldsymbol{\varpi}}_{1is}\right) \boldsymbol{x}_{it} - \mathcal{M}_{p,R_0}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{2i}}^{\top} \boldsymbol{\varpi}_{1is}\right) \hat{\boldsymbol{f}}_t \right] \hat{\varepsilon}_{it} \\
&+ \frac{1}{N^2 T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \left\{ \hat{w}_{ij} \hat{\boldsymbol{b}}_j \otimes \hat{\boldsymbol{S}}_j^{\top} \left[\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^{\top} \hat{\boldsymbol{\varpi}}_{2ijs}\right) \boldsymbol{x}_{it} - \mathcal{M}_{p,R_0}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{4ij}}^{\top} \hat{\boldsymbol{\varpi}}_{2ijs}\right) \hat{\boldsymbol{f}}_t \right] \hat{\varepsilon}_{it} \right\} \\
&+ \frac{1}{N\sqrt{T}} \sum_{j=1}^{N} \hat{\boldsymbol{b}}_j \hat{\boldsymbol{l}}_j^{\top} \otimes \hat{\boldsymbol{Q}}_j^{\top} \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \cdot \hat{\boldsymbol{\Xi}}^{-1} \cdot \frac{1}{N} \sum_{k=1}^{N} \Bigg\{ \hat{\boldsymbol{l}}_k \hat{\boldsymbol{l}}_k^{\top} \hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1} \otimes \frac{\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{X}_k \hat{\boldsymbol{\Theta}}}{\sqrt{T}} \left(\frac{\hat{\boldsymbol{\Theta}}^{\top} \boldsymbol{X}_k^{\top} \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{X}_k \hat{\boldsymbol{\Theta}}}{T} \right)^{-1} \hat{\boldsymbol{\Theta}}^{\top} \\
&\cdot \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{\boldsymbol{l}}_i \otimes \left[\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ik}}^{\top} \hat{\boldsymbol{\varpi}}_{2iks}\right) \boldsymbol{x}_{it} - \mathcal{M}_{p,R_0}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{4ik}}^{\top} \hat{\boldsymbol{\varpi}}_{2iks}\right) \hat{\boldsymbol{f}}_t \right] \hat{\varepsilon}_{it} \Bigg\} \Bigg),
\end{aligned} \tag{S.1.9}$$

where the bandwidth $m \to \infty$ and $m/T \to 0$. Note that $\left(\boldsymbol{\Theta}_{[K_0],0}^{-1} \otimes \boldsymbol{I}_p\right) \hat{\boldsymbol{A}}$ and $\boldsymbol{A}_0$ are close, and $\left(\boldsymbol{\Theta}_{[K_0],0}^{\top} \otimes \boldsymbol{I}_p\right) \hat{\mathbb{B}}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\xi}}$ are close. Instead of analyzing the differences $\hat{\boldsymbol{A}} - \boldsymbol{A}_0$ or $\hat{\mathbb{B}}^{\boldsymbol{\xi}} - \mathbb{B}^{\boldsymbol{\xi}}$ individually, we focus on $\hat{\boldsymbol{A}}^{\top}\hat{\mathbb{B}}^{\boldsymbol{\xi}} - \boldsymbol{A}_0^{\top}\mathbb{B}^{\boldsymbol{\xi}}$. The same approach applies to the terms $\hat{\boldsymbol{A}}^{\top}\hat{\mathbb{B}}^{\boldsymbol{\zeta}} - \boldsymbol{A}_0^{\top}\mathbb{B}^{\boldsymbol{\zeta}}$ and $\hat{\boldsymbol{A}}^{\top}\hat{\mathbb{B}}^{\boldsymbol{\eta}} - \boldsymbol{A}_0^{\top}\mathbb{B}^{\boldsymbol{\eta}}$.

S.1.3 List of Other Notation

We list the notation before proof.

- $(N,\ T)$: the number of sections and time periods
- $\boldsymbol{\Theta}_0 \in \mathbb{R}^{p \times K_0}$: $\boldsymbol{\Theta}_0^{\top}\boldsymbol{x}_{it}$ is the common components, or constructed factors
- $\boldsymbol{B}_0 \in \mathbb{R}^{N \times K_0}$: common component loadings, the i-th row is $\boldsymbol{b}_{i,0}^{\top}$
- $\boldsymbol{\Sigma}_{\boldsymbol{B}} = N^{-1}\boldsymbol{B}_0^{\top}\boldsymbol{B}_0$
- $(p_1,\ p_2)$: p_1 and p_2 are the number of high-rank and low-rank predictors respectively, and $p_1 + p_2 = p$
- $(\boldsymbol{\gamma}_{\cdot j},\ \boldsymbol{g}_{\cdot j})$: $\boldsymbol{X}_{\cdot j} = \boldsymbol{g}_{\cdot j}\boldsymbol{\gamma}_{\cdot j}^{\top}$ with $\boldsymbol{g}_{\cdot j} \in \mathbb{R}^T$ and $\boldsymbol{\gamma}_{\cdot j} \in \mathbb{R}^N$ for $j = p_1 + 1, \ldots, p$
- $(\boldsymbol{\Gamma},\ \boldsymbol{G})$: $\boldsymbol{\Gamma} = \left(\boldsymbol{\gamma}_{\cdot p_1+1},\ \ldots,\ \boldsymbol{\gamma}_{\cdot p}\right) = \left(\boldsymbol{\gamma}_1,\ \ldots,\ \boldsymbol{\gamma}_N\right)^{\top}$, $\boldsymbol{G} = \left(\boldsymbol{g}_{\cdot p_1+1},\ \ldots,\ \boldsymbol{g}_{\cdot p}\right) = \left(\boldsymbol{g}_1,\ \ldots,\ \boldsymbol{g}_T\right)^{\top}$
- $(\boldsymbol{F}_0,\ \boldsymbol{L}_0)$: latent factors and latent factor loadings

- $\boldsymbol{\Sigma_F} = T^{-1}\boldsymbol{F}_0^\top \boldsymbol{F}_0, \quad \boldsymbol{\Sigma_L} = N^{-1}\boldsymbol{L}_0^\top \boldsymbol{L}_0$
- $K_0 = \dim(\boldsymbol{b}_{i,0})$: the number of common components
- $R_0 = \dim(\boldsymbol{f}_{t0}) = \dim(\boldsymbol{l}_{i,0})$: the number of latent factors
- C or c: different constants
- $\boldsymbol{I}_n$: $n \times n$ identity matrix
- $\boldsymbol{M} \in \mathbb{R}^{m_1 \times m_2}$: a general matrix
- $m_{NT} = \min\left(\sqrt{N}, \sqrt{T}\right)$, $M_{NT} = \max\left(\sqrt{N}, \sqrt{T}\right)$
- $\boldsymbol{P_M}$ and $\boldsymbol{P_M^\perp}$: $\boldsymbol{P_M} = \boldsymbol{M}(\boldsymbol{M}^\top \boldsymbol{M})^+ \boldsymbol{M}^\top$, $\boldsymbol{P_M^\perp} = \boldsymbol{I}_{m_1} - \boldsymbol{P_M}$
- $\mathbb{B}$: bias
- $\mathbb{E}$: expectation
- $\mathbb{E}_{\mathcal{S}}$ or $\mathbb{E}_{\mathcal{C}}$: conditional expectation given the event $\mathcal{S}$ or the sigma field $\mathcal{C}$
- $\mathcal{C}$: Sigma field generated by $(\boldsymbol{F}_0, \boldsymbol{L}_0, \boldsymbol{B}_0)$
- $\mathcal{F}_t = \sigma\{(\boldsymbol{x}_{is}, \varepsilon_{i,s-1}), s \le t\}$
- $\boldsymbol{\mathcal{E}} = \left(\boldsymbol{\varepsilon}_1, \quad \boldsymbol{\varepsilon}_2, \quad \ldots, \quad \boldsymbol{\varepsilon}_N\right) \in \mathbb{R}^{T \times N}$

Let

- $\boldsymbol{W}_i(\boldsymbol{\Theta}, \boldsymbol{F}) = \boldsymbol{X}_i \boldsymbol{\Theta} \left(\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P_F^\perp} \boldsymbol{X}_i \boldsymbol{\Theta}\right)^{-1} \boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top$
- $\boldsymbol{V}_i(\boldsymbol{\Theta}, \boldsymbol{F}) = \boldsymbol{W}_i(\boldsymbol{\Theta}, \boldsymbol{F}) \boldsymbol{P_F^\perp}$
- $\boldsymbol{U}_i(\boldsymbol{\Theta}, \boldsymbol{F}) = \boldsymbol{I}_T - \boldsymbol{V}_i(\boldsymbol{\Theta}, \boldsymbol{F})$
- $\boldsymbol{\Phi}_{1i}(\boldsymbol{F}) = \boldsymbol{\Theta}_0 \left(\dfrac{\boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{P_F^\perp} \boldsymbol{X}_i \boldsymbol{\Theta}_0}{T}\right)^{-1}, \quad \boldsymbol{\Phi}_{2i}(\boldsymbol{F}) = \dfrac{\boldsymbol{X}_i^\top \boldsymbol{P_F^\perp} \boldsymbol{F}_0}{T}, \quad \boldsymbol{\Phi}_{3i}(\boldsymbol{F}) = \dfrac{\boldsymbol{X}_i^\top \boldsymbol{P_F^\perp} \boldsymbol{\varepsilon}_i}{T}$
- $\boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}, \boldsymbol{F}) = \dfrac{\boldsymbol{X}_i \boldsymbol{\Theta}}{\sqrt{T}} \left(\dfrac{\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P_F^\perp} \boldsymbol{X}_i \boldsymbol{\Theta}}{T}\right)^{-1} \boldsymbol{\Theta}^\top$
- $\boldsymbol{\Upsilon}(\boldsymbol{\Theta}, \boldsymbol{F}) = \dfrac{1}{N} \sum_{i=1}^{N} \tilde{\boldsymbol{l}}_{i0} \tilde{\boldsymbol{l}}_{i0}^\top \otimes \boldsymbol{P_G^\perp} \boldsymbol{H}_i(\boldsymbol{\Theta}, \boldsymbol{F}) \boldsymbol{P_G^\perp}$

For notational simplicity, we suppress the explicit dependence on model parameters for the following quantities:

$$
\begin{aligned}
&\boldsymbol{W}_i := \boldsymbol{W}_i(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{W}_{i,0} := \boldsymbol{W}_i(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \hat{\boldsymbol{W}}_i := \boldsymbol{W}_i\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right),\\
&\boldsymbol{V}_i := \boldsymbol{V}_i(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{V}_{i,0} := \boldsymbol{V}_i(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \hat{\boldsymbol{V}}_i := \boldsymbol{V}_i\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right),\\
&\boldsymbol{U}_i := \boldsymbol{U}_i(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{U}_{i,0} := \boldsymbol{U}_i(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \hat{\boldsymbol{U}}_i := \boldsymbol{U}_i\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right),\\
&\boldsymbol{H}_i := \boldsymbol{H}_i(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{H}_{i,0} := \boldsymbol{H}_i(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \hat{\boldsymbol{H}}_i := \boldsymbol{H}_i\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right),\\
&\boldsymbol{\Phi}_{4i} := \boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{\Phi}_{4i,0} := \boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \hat{\boldsymbol{\Phi}}_{4i} := \boldsymbol{\Phi}_{4i}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right),\\
&\boldsymbol{\Xi} := \boldsymbol{\Xi}(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{\Xi}_0 := \boldsymbol{\Xi}(\boldsymbol{\Theta}_0, \boldsymbol{F}_0), \quad \boldsymbol{\Upsilon} := \boldsymbol{\Upsilon}(\boldsymbol{\Theta}, \boldsymbol{F}), \quad \boldsymbol{\Upsilon}_0 := \boldsymbol{\Upsilon}(\boldsymbol{\Theta}_0, \boldsymbol{F}_0),\\
&\boldsymbol{S}_{i,0} := \boldsymbol{S}_i(\boldsymbol{F}_0), \quad \boldsymbol{Q}_{i,0} := \boldsymbol{Q}_i(\boldsymbol{F}_0).
\end{aligned}
$$

For $\boldsymbol{V}_i\left(\boldsymbol{\Theta}_0, \hat{\boldsymbol{F}}\right)$ or $\boldsymbol{V}_i\left(\hat{\boldsymbol{\Theta}}, \boldsymbol{F}_0\right)$, we keep the model parameters. Note that we continue to use the notation $\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)$, $\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)$, $\boldsymbol{\Xi}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right)$, $\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)$ etc. when proving the consistency rate, while reserving $\hat{\boldsymbol{S}}_i$, $\hat{\boldsymbol{Q}}_i$, $\hat{\boldsymbol{\Xi}}$, $\hat{\boldsymbol{D}}_0$ for the bias correction stage to maintain the notational consistency; see (S.1.6). To clarify the difference, consider $\boldsymbol{\Xi}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right)$ versus $\hat{\boldsymbol{\Xi}}$. The true values $\{\boldsymbol{l}_{i,0}\}_{i=1}^N$ are used in $\boldsymbol{\Xi}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right)$, while the estimated values $\left\{\hat{\boldsymbol{l}}_i\right\}_{i=1}^N$ are used in $\hat{\boldsymbol{\Xi}}$. In addition, let

$$
\begin{aligned}
&\bar{\boldsymbol{H}}(\boldsymbol{\Theta}, \boldsymbol{F}) = \frac{1}{N}\sum_{i=1}^N \boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top \otimes \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{H}_i(\boldsymbol{\Theta}, \boldsymbol{F})\boldsymbol{P}_{\boldsymbol{G}}^\perp,\\
&\boldsymbol{\Phi}_{4i,0}^e = \frac{\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\boldsymbol{\Theta}_0}{\sqrt{T}}\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top,\\
&\boldsymbol{S}_{i,0}^e = \boldsymbol{I}_p - \boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right).
\end{aligned}
$$

S.2 Proofs of Theorems

S.2.1 Proof of Theorem 2.1 in Section 2

To simplify the notation, we replace the superscript *high* or *low* with h or l hereafter, such as $\boldsymbol{X}_i^h$ and $\boldsymbol{X}_i^l$, instead of $\boldsymbol{X}_i^{high}$ and $\boldsymbol{X}_i^{low}$.

Proof of Theorem 2.1. Define $g(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}) = \sum_{i=1}^N \mathbb{E}\left(\|\boldsymbol{y}_i - \boldsymbol{X}_i\boldsymbol{\Theta}\boldsymbol{b}_i - \boldsymbol{F}\boldsymbol{l}_i\|_2^2 \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right)$. Then,

$$
\begin{aligned}
&g(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L})\\
&= \sum_{i=1}^N \mathbb{E}\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right)\\
&\quad + \sum_{i=1}^N \mathbb{E}\left\{\left[(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i)\right]^\top\left[(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i)\right] \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right\}\\
&=: \sum_{i=1}^N \mathbb{E}\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) + g^*(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}).
\end{aligned}
$$

Since minimizing g is equivalent to minimizing g^*, we work with $g^*(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L})$. We have

$$
\begin{aligned}
& g^*(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}) \\
\geq & \sum_{i=1}^{N} \mathbb{E}\left\{ \left[(\boldsymbol{F}_0 \boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i) \right]^\top \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0, \boldsymbol{F}, \boldsymbol{G})} \left[(\boldsymbol{F}_0 \boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i) \right] \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right\} \\
= & \sum_{i=1}^{N} \mathbb{E}\left[(\boldsymbol{\Theta}_0^h \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^h \boldsymbol{b}_i)^\top \boldsymbol{X}_i^{h\top} \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0, \boldsymbol{F}, \boldsymbol{G})} \boldsymbol{X}_i^h (\boldsymbol{\Theta}_0^h \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^h \boldsymbol{b}_i) \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right] =: g_1\left(\boldsymbol{\Theta}^h, \boldsymbol{B}, \boldsymbol{F}\right).
\end{aligned}
$$

Let $\left(\bar{\boldsymbol{\Theta}},\ \bar{\boldsymbol{B}},\ \bar{\boldsymbol{F}},\ \bar{\boldsymbol{L}}\right)$ be the minimizer of $g^*(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L})$. Since $g_1\left(\boldsymbol{\Theta}^h, \boldsymbol{B}, \boldsymbol{F}\right) \geq 0$, $g^*\left(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}\right) \geq 0$, and $g^*\left(\boldsymbol{\Theta}_0, \boldsymbol{B}_0, \boldsymbol{F}_0, \boldsymbol{L}_0\right) = 0$, we have $g_1\left(\bar{\boldsymbol{\Theta}}^h, \bar{\boldsymbol{B}}, \bar{\boldsymbol{F}}\right) = 0$ and $g^*\left(\bar{\boldsymbol{\Theta}}, \bar{\boldsymbol{B}}, \bar{\boldsymbol{F}}, \bar{\boldsymbol{L}}\right) = 0$. By Assumption 1(3), $\bar{\boldsymbol{\Theta}}^h \bar{\boldsymbol{b}}_i = \boldsymbol{\Theta}_0^h \boldsymbol{b}_{i,0}$ for all $i = 1, \ldots, N$, or

$$
\bar{\boldsymbol{\Theta}}^h \bar{\boldsymbol{B}}^\top = \boldsymbol{\Theta}_0^h \bar{\boldsymbol{B}}_0^\top. \tag{S.2.1}
$$

Similarly, we have

$$
\begin{aligned}
& g^*(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}) \\
\geq & \sum_{i=1}^{N} \mathbb{E}\left\{ \left[(\boldsymbol{F}_0 \boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i) \right]^\top \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0, \boldsymbol{F})} \left[(\boldsymbol{F}_0 \boldsymbol{l}_{i,0} - \boldsymbol{F}\boldsymbol{l}_i) + \boldsymbol{X}_i(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i) \right] \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right\} \\
= & \sum_{i=1}^{N} \mathbb{E}\left\{ (\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i)^\top \boldsymbol{X}_i^\top \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0, \boldsymbol{F})} \boldsymbol{X}_i (\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}\boldsymbol{b}_i) \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right\} =: g_2\left(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}\right).
\end{aligned}
$$

At the minimizer, $g_2\left(\bar{\boldsymbol{\Theta}}, \bar{\boldsymbol{B}}, \bar{\boldsymbol{F}}\right) = 0$. Using (S.2.1) and the constraint $(\boldsymbol{F}_0,\ \boldsymbol{F}) \perp \boldsymbol{G}$, we obtain

$$
\begin{aligned}
0 &= \sum_{i=1}^{N} \mathbb{E}\left[(\boldsymbol{\Theta}_0^l \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^l \boldsymbol{b}_i)^\top \boldsymbol{X}_i^{l\top} \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0, \boldsymbol{F})} \boldsymbol{X}_i^l (\boldsymbol{\Theta}_0^l \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^l \boldsymbol{b}_i) \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right] \\
&= \sum_{i=1}^{N} \mathbb{E}\left[(\boldsymbol{\Theta}_0^l \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^l \boldsymbol{b}_i)^\top \boldsymbol{X}_i^{l\top} \boldsymbol{X}_i^l (\boldsymbol{\Theta}_0^l \boldsymbol{b}_{i,0} - \boldsymbol{\Theta}^l \boldsymbol{b}_i) \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right].
\end{aligned}
$$

By Assumption 1(4), $\bar{\boldsymbol{\Theta}}^l \bar{\boldsymbol{b}}_i = \boldsymbol{\Theta}_0^l \boldsymbol{b}_{i,0}$ for all $i = 1, \ldots, N$, or

$$
\bar{\boldsymbol{\Theta}}^l \bar{\boldsymbol{B}}^\top = \boldsymbol{\Theta}_0^l \bar{\boldsymbol{B}}_0^\top. \tag{S.2.2}
$$

From (S.2.1) and (S.2.2),

$$
\bar{\boldsymbol{\Theta}} \bar{\boldsymbol{B}}^\top = \begin{pmatrix} \bar{\boldsymbol{\Theta}}^h \bar{\boldsymbol{B}}^\top \\ \bar{\boldsymbol{\Theta}}^l \bar{\boldsymbol{B}}^\top \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Theta}_0^h \boldsymbol{B}_0^\top \\ \boldsymbol{\Theta}_0^l \boldsymbol{B}_0^\top \end{pmatrix} = \boldsymbol{\Theta}_0 \boldsymbol{B}_0^\top.
$$

Finally, since $g^*\left(\bar{\boldsymbol{\Theta}}, \bar{\boldsymbol{B}}, \bar{\boldsymbol{F}}, \bar{\boldsymbol{L}}\right) = \sum_{i=1}^{N} \|\boldsymbol{F}_0 \boldsymbol{l}_{i,0} - \bar{\boldsymbol{F}} \bar{\boldsymbol{l}}_{i,0}\|_2^2 = 0$, we conclude that $\bar{\boldsymbol{F}} \bar{\boldsymbol{L}}^\top = \boldsymbol{F}_0 \boldsymbol{L}_0^\top$. This establishes the identification result. □

S.2.2 Proof of Theorem 3.1 in Section 3.2

Proof of Theorem 3.1. We prove the results in four steps: (I) the consistency of $\hat{\boldsymbol{\Theta}}$; (II) the consistency of $\left\{\hat{\boldsymbol{b}}_i\right\}_{i \in \mathcal{I}_{\hat{\boldsymbol{F}}}}$ in the mean sense; (III) the consistency of $\hat{\boldsymbol{F}}$; (IV) the consistency of $\left\{\hat{\boldsymbol{b}}_i\right\}_{i=1}^N$ in the mean sense.

(I) Let

$$
f_{1,NT}(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}) = \frac{1}{NT} \sum_{i=1}^{N} \|\boldsymbol{y}_i - \boldsymbol{X}_i \boldsymbol{\Theta}\boldsymbol{b}_i - \boldsymbol{F}\boldsymbol{l}_i\|_2^2.
$$

From the optimization problem (3.3), we have

$$
\begin{aligned}
0 \geq & f_{1,NT}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{B}}, \hat{\boldsymbol{F}}, \hat{\boldsymbol{L}}\right) - f_{1,NT}\left(\boldsymbol{\Theta}_0, \boldsymbol{B}_0, \boldsymbol{F}_0, \boldsymbol{L}_0\right) \\
\geq & \frac{1}{NT}\sum_{i=1}^{N}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right) + \boldsymbol{\varepsilon}_i\right]^\top \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right) + \boldsymbol{\varepsilon}_i\right] - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i \\
\geq & \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right) + \boldsymbol{\varepsilon}_i\right]^\top \boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right) + \boldsymbol{\varepsilon}_i\right] - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i \\
= & \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right)^\top \frac{\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i}{T}\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right) \\
& + \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right) + \left(\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\right) \\
=: & \Delta_1 + \Delta_2 + \Delta_3.
\end{aligned}
$$

Here in the third step, $\left\{\tilde{\boldsymbol{b}}_i\right\}_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}$ is the minimizer of the expression in the previous step, that is,

$$
\tilde{\boldsymbol{b}}_i = \left(\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{+}\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\left(\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} + \boldsymbol{\varepsilon}_i\right).
$$

Decompose Δ_2:

$$
\begin{aligned}
\Delta_2 = & \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} \\
& - \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{+}\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} \\
& - \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{+}\hat{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i \\
=: & \Delta_{2.1} + \Delta_{2.2} + \Delta_{2.3}.
\end{aligned}
$$

For $\Delta_{2.1}$,

$$
\Delta_{2.1} = \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} =: \Delta_{2.1.1} + \Delta_{2.1.2},
$$

where

$$
\begin{aligned}
|\Delta_{2.1.1}| \leq & \frac{2}{NT}\sum_{i=1}^{N}\left(\left\|\boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\right\|_2 \cdot \left\|\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right\|_2\right) \lesssim \frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\right\|_2 = o_p(1), \\
|\Delta_{2.1.2}| \lesssim & \left(\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i\right)^{1/2} \cdot \left(\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}^\top\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right)^{1/2} \lesssim \left(\frac{2R_0\|\boldsymbol{\mathcal{E}}\|_2^2}{NT}\right)^{1/2} = o_p(1).
\end{aligned}
$$

Let $\boldsymbol{M}_i^{hh} = T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i^h$, $\boldsymbol{M}_i^{hl} = T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i^l$, $\boldsymbol{M}_i^{lh} = \boldsymbol{M}_i^{hl\top}$, $\boldsymbol{M}_i^{ll} = T^{-1}\boldsymbol{X}_i^{l\top}\boldsymbol{X}_i^l$, and

$$
\boldsymbol{M}_i = \begin{pmatrix}\boldsymbol{M}_i^{hh} & \boldsymbol{M}_i^{hl} \\ \boldsymbol{M}_i^{lh} & \boldsymbol{M}_i^{ll}\end{pmatrix} = \frac{\boldsymbol{X}_i^\top\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i}{T}.
$$

Note that $\boldsymbol{M}_i^{hh} - \boldsymbol{M}_i^{hl}\left(\boldsymbol{M}_i^{ll}\right)^{-1}\boldsymbol{M}_i^{lh} = T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}},\boldsymbol{G})}\boldsymbol{X}_i^h$. Then, for $i \in \mathcal{I}_{\hat{\boldsymbol{F}}}$,

$$
\lambda_{min}\left(\boldsymbol{M}_i\right) \geq \frac{\det(\boldsymbol{M}_i)}{\lambda_{max}^{p-1}(\boldsymbol{M}_i)} = \frac{\det(\boldsymbol{M}_i^{ll})\det(\boldsymbol{M}_i^{hh} - \boldsymbol{M}_i^{hl}(\boldsymbol{M}_i^{ll})^{-1}\boldsymbol{M}_i^{lh})}{\lambda_{max}^{p-1}(\boldsymbol{M}_i)} \geq \frac{C_6^{p_2}C_7^{p_1}}{C_5^{p-1}},
$$

where the first step follows from $\det(\boldsymbol{M}_i) = \prod_{j=1}^{p} \lambda_j(\boldsymbol{M}_i) \le \lambda_{max}^{p-1}(\boldsymbol{M}_i)\lambda_{min}(\boldsymbol{M}_i)$, the second step follows from Schur's Formula in Theorem 1.1 of Zhang (2006), and the last step follows from $\lambda_{max}(\boldsymbol{M}_i) \le \lambda_{max}\left(T^{-1}\boldsymbol{X}_i^\top \boldsymbol{X}_i\right) \le C_5$. Therefore,

$$\lambda_{max}\left[\hat{\boldsymbol{\Theta}}^\top \left(\frac{\hat{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}}{T}\right)^{-1}\hat{\boldsymbol{\Theta}}\right] \le \frac{1}{\lambda_{min}(\boldsymbol{M}_i)}$$

is bounded for $i \in \mathcal{I}_{\hat{\boldsymbol{F}}}$. For $\Delta_{2.2}$,

$$|\Delta_{2.2}| \lesssim \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \left\|\boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\right\|_2 = o_p(1),$$

where the analysis of the second step parallels that of $\Delta_{2.1}$. For $\Delta_{2.3}$,

$$|\Delta_{2.3}| \lesssim \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i + \frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{X}_i^\top \boldsymbol{X}_i\boldsymbol{\varepsilon}_i = o_p(1).$$

For Δ_3,

$$\Delta_3 = -\frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i - \frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i = o_p(1),$$

where

$$\frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}},\boldsymbol{G})}\boldsymbol{\varepsilon}_i \le \frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_i \le \frac{|\mathcal{I}^c_{\hat{\boldsymbol{F}}}|\cdot\|\boldsymbol{\mathcal{E}}\|_2^2}{NT} = o_p(1).$$

Since $\Delta_1 \ge 0$, $\Delta_2 = o_p(1)$, $\Delta_3 = o_p(1)$, we have

$$\begin{aligned}
o_p(1) =& \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right)^\top \frac{\boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i}{T}\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right)\\
\gtrsim& \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right)^\top \left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i\right)\\
\ge& \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \boldsymbol{b}_{i,0}^\top \boldsymbol{\Theta}_0^\top \left[\boldsymbol{I}_p - \hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{\Theta}}^\top\hat{\boldsymbol{\Theta}}\right)^+ \hat{\boldsymbol{\Theta}}^\top\right]\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\\
=& \operatorname{tr}\left(\boldsymbol{\Theta}_0^\top \boldsymbol{P}^\perp_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0 \cdot \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\right).
\end{aligned}$$

In the third step, $\boldsymbol{b}_i = \left(\hat{\boldsymbol{\Theta}}^\top\hat{\boldsymbol{\Theta}}\right)^+ \hat{\boldsymbol{\Theta}}^\top\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}$ minimizes $\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\boldsymbol{b}_i\right)^\top\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}\boldsymbol{b}_i\right)$. Since $N^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}} \boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top$ is positive definite, $\boldsymbol{P}^\perp_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0 = o_p(1)$, or $\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}} - \boldsymbol{\Theta}_0 = o_p(1)$, where $\boldsymbol{O}_{\boldsymbol{\Theta}} = \left(\hat{\boldsymbol{\Theta}}^\top\hat{\boldsymbol{\Theta}}\right)^+ \hat{\boldsymbol{\Theta}}^\top\boldsymbol{\Theta}_0$. By examining the first $K_0 \times K_0$ blocks of $\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}} - \boldsymbol{\Theta}_0$, we obtain $\boldsymbol{O}_{\boldsymbol{\Theta}} - \boldsymbol{\Theta}_{[K_0],0} = o_p(1)$. Therefore,

$$\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0} - \boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}\left(\boldsymbol{\Theta}_{[K_0],0} - \boldsymbol{O}_{\boldsymbol{\Theta}}\right) + \left(\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}} - \boldsymbol{\Theta}_0\right) = o_p(1).$$

Here we use the bound $\left\|\hat{\boldsymbol{\Theta}}\right\|_2 \le c_1$ and this fact is implicitly applied throughout the consistency proof.

(II) From the optimization problem (3.3), we have

$$0 \ge f_{1,NT}\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{B}}, \hat{\boldsymbol{F}}, \hat{\boldsymbol{L}}\right) - f_{1,NT}\left(\boldsymbol{\Theta}_0, \boldsymbol{B}_0, \boldsymbol{F}_0, \boldsymbol{L}_0\right)$$

$$
\begin{aligned}
&\geq \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\boldsymbol{X}_i\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}+\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right)+\boldsymbol{\varepsilon}_i\right]^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\left[\boldsymbol{X}_i\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}+\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right)+\boldsymbol{\varepsilon}_i\right]-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i\\
&=\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right)^{\top}\frac{\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}}{T}\left(\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right)+\frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\boldsymbol{X}_i\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}+\boldsymbol{\varepsilon}_i\right)^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\left(\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right)\\
&\quad+\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{b}_{i,0}^{\top}\boldsymbol{\Theta}_0^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i}{T}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}+\left(\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{\varepsilon}_i-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i\right)\\
&\quad+\frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\\
&=:\Delta_1+\ldots+\Delta_5.
\end{aligned}
$$

Let $\boldsymbol{B}_{\mathcal{I}}$ be an $|\mathcal{I}_{\hat{\boldsymbol{F}}}|\times K_0$ matrix with each row $\boldsymbol{b}_i^{\top}$, where $i\in\mathcal{I}_{\hat{\boldsymbol{F}}}$. For Δ_1,

$$
\Delta_1\gtrsim\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right\|_2^2=\frac{\|\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}-\hat{\boldsymbol{B}}_{\mathcal{I}}\|_f^2}{N}.
$$

For Δ_2,

$$
\begin{aligned}
|\Delta_2|&\lesssim\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\left(\|\boldsymbol{b}_{i,0}\|_2\cdot\left\|\boldsymbol{X}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right\|_2\cdot\left\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\right\|_2+\left\|\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right\|_2\right)\cdot\left\|\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right\|_2\right]\\
&\lesssim\left\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{\Theta}}}\boldsymbol{\Theta}_0\right\|_2\cdot\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right\|_2+\left(\frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right\|_2^2\right)^{1/2}\\
&=o_p(1)\cdot\left(\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}-\hat{\boldsymbol{b}}_i\right\|_2^2\right)^{1/2}=o_p\left(\frac{\|\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}-\hat{\boldsymbol{B}}_{\mathcal{I}}\|_f}{\sqrt{N}}\right).
\end{aligned}
$$

It is easy to show that

$$
\Delta_3=o_p(1),\quad\Delta_4=o_p(1),\quad\Delta_5=o_p(1).
$$

Therefore,

$$
\frac{\|\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}-\hat{\boldsymbol{B}}_{\mathcal{I}}\|_f^2}{N}+o_p\left(\frac{\|\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}-\hat{\boldsymbol{B}}_{\mathcal{I}}\|_f}{\sqrt{N}}\right)+o_p(1)\leq 0,
$$

leading to

$$
\frac{\|\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}-\hat{\boldsymbol{B}}_{\mathcal{I}}\|_f^2}{N}=o_p(1).
$$

(III) From the optimization problem (3.3), we have

$$
\begin{aligned}
0&\geq f_{1,NT}\left(\hat{\boldsymbol{\Theta}},\hat{\boldsymbol{B}},\hat{\boldsymbol{F}},\hat{\boldsymbol{L}}\right)-f_{1,NT}\left(\boldsymbol{\Theta}_0,\boldsymbol{B}_0,\boldsymbol{F}_0,\boldsymbol{L}_0\right)\\
&\geq\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{y}_i-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\hat{\boldsymbol{F}}\hat{\boldsymbol{l}}_i\right)^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\left(\boldsymbol{y}_i-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\hat{\boldsymbol{F}}\hat{\boldsymbol{l}}_i\right)-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i\\
&\geq\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)+\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right]^{\top}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)+\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right]-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{l}_{i,0}^{\top}\boldsymbol{F}_0^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0} + \left\{\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)+\boldsymbol{\varepsilon}_i\right]^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)+\boldsymbol{\varepsilon}_i\right] - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i\right\} \\
&\quad + \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)+\boldsymbol{\varepsilon}_i\right]^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0} \\
&=: \Delta_1 + \Delta_2 + \Delta_3.
\end{aligned}
$$

Decompose Δ_2:

$$
\begin{aligned}
\Delta_2 =& \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right) \\
&+ \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}-\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right) + \left(\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{\varepsilon}_i\right) \\
=:& \Delta_{2.1} + \Delta_{2.2} + \Delta_{2.3},
\end{aligned}
$$

where

$$
\begin{aligned}
|\Delta_{2.1}| \lesssim& \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right)^{\top}\left(\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right) \\
\le& \frac{2}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\hat{\boldsymbol{b}}_i-\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}\right)^{\top}\hat{\boldsymbol{\Theta}}^{\top}\hat{\boldsymbol{\Theta}}\left(\hat{\boldsymbol{b}}_i-\boldsymbol{O}_{\boldsymbol{\Theta}}\boldsymbol{b}_{i,0}\right) + \frac{2}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{b}_{i,0}^{\top}\left(\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0\right)^{\top}\left(\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0\right)\boldsymbol{b}_{i,0} \\
\lesssim& \frac{\left\|\hat{\boldsymbol{B}}_{\mathcal{I}}-\boldsymbol{B}_{\mathcal{I},0}\boldsymbol{O}_{\boldsymbol{\Theta}}^{\top}\right\|_f^2}{N} + \frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\|\boldsymbol{b}_{i,0}\|_2^2\cdot\left\|\hat{\boldsymbol{\Theta}}\boldsymbol{O}_{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0\right\|_2^2 = o_p(1), \\
|\Delta_{2.2}| \lesssim& \left(\frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left\|\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right\|_2^2\right)^{1/2} = o_p(1), \\
|\Delta_{2.3}| \le& \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}\boldsymbol{\varepsilon}_i + \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}^c}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i = o_p(1).
\end{aligned}
$$

It is straightforward to show that $\Delta_3 = o_p(1)$ by the following decomposition:

$$
\Delta_3 = \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\left(\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i-\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right)^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0} + \frac{2}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \frac{1}{NT^2}\sum_{i\in\mathcal{I}_F}\boldsymbol{\varepsilon}_i^{\top}\hat{\boldsymbol{F}}\hat{\boldsymbol{F}}^{\top}\boldsymbol{F}_0\boldsymbol{l}_{i,0}.
$$

Therefore,

$$
\Delta_1 = \operatorname{tr}\left[\frac{\boldsymbol{F}_0^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0}{T}\cdot\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}\right] = o_p(1).
$$

Since $N^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}$ is positive definite, we have $T^{-1}\boldsymbol{F}_0^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0 = o_p(1)$, or $\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f = o_p(1)$.

(IV) In (3.4), since

$$
\begin{aligned}
\hat{\boldsymbol{b}}_i =& \left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{y}_i \\
=& \boldsymbol{\Theta}_{[K_0],0}\boldsymbol{b}_{i,0} + \left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0}\right)\boldsymbol{b}_{i,0} \\
&+ \left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0} + \left(\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\right)^{-1}\hat{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i
\end{aligned}
$$

$$=\boldsymbol{\Theta}_{[K_0],0}\boldsymbol{b}_{i,0}+\Delta_{\boldsymbol{B}.i.1}+\Delta_{\boldsymbol{B}.i.2}+\Delta_{\boldsymbol{B}.i.3}, \tag{S.2.3}$$

we have

$$\frac{\|\hat{\boldsymbol{B}}-\boldsymbol{B}_0\boldsymbol{\Theta}_{[K_0],0}^\top\|_f^2}{N}\lesssim\frac{1}{N}\sum_{i=1}^N\left(\|\Delta_{\boldsymbol{B}.i.1}\|_2^2+\|\Delta_{\boldsymbol{B}.i.2}\|_2^2+\|\Delta_{\boldsymbol{B}.i.3}\|_2^2\right)=:\Delta_{\boldsymbol{B}.1}+\Delta_{\boldsymbol{B}.2}+\Delta_{\boldsymbol{B}.3}.$$

It is straightforward to show that

$$\Delta_{\boldsymbol{B}.1}=O_p\left(\left\|\boldsymbol{\Theta}_0-\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0}\right\|_2^2\right)=o_p(1),\quad \Delta_{\boldsymbol{B}.2}=O_p\left(\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f^2\right)=o_p(1),$$
$$\Delta_{\boldsymbol{B}.3}\lesssim\frac{1}{NT^2}\sum_{i=1}^N\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_i\right\|_2^2\lesssim\frac{1}{NT^2}\sum_{i=1}^N\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^2+\frac{1}{NT^2}\sum_{i=1}^N\left\|\hat{\boldsymbol{F}}^\top\boldsymbol{\varepsilon}_i\right\|_2^2=o_p(1).$$

This completes the proof. □

S.2.3 Proofs of Theorems in Section 3.3

To further study the consistency rate of $\hat{\boldsymbol{\Theta}}$, define

$$\boldsymbol{\vartheta}_0=\ell\left(\boldsymbol{\Theta}_{-[K_0],0}\right),\quad \boldsymbol{\vartheta}=\ell\left(\boldsymbol{\Theta}_{-[K_0]}\boldsymbol{\Theta}_{[K_0],0}\right),\quad \tilde{\boldsymbol{\vartheta}}=\ell\left(\hat{\boldsymbol{\Theta}}_{-[K_0]}\boldsymbol{\Theta}_{[K_0],0}\right),$$

and let

$$\tilde{\boldsymbol{\Theta}}=\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0}=\begin{pmatrix}\boldsymbol{\Theta}_{[K_0],0}\\ \hat{\boldsymbol{\Theta}}_{-[K_0]}\boldsymbol{\Theta}_{[K_0],0}\end{pmatrix}=\begin{pmatrix}\boldsymbol{\Theta}_{[K_0],0}\\ \mathcal{M}_{p-K_0,K_0}\left(\tilde{\boldsymbol{\vartheta}}\right)\end{pmatrix}.$$

Then, $\tilde{\boldsymbol{\Theta}}$ is also a solution to the minimization problem (3.8). Although $\tilde{\boldsymbol{\Theta}}$ incorporates the known block $\boldsymbol{\Theta}_{[K_0],0}$, it spans the same column space as $\hat{\boldsymbol{\Theta}}$, i.e., $\mathrm{Span}\left(\tilde{\boldsymbol{\Theta}}\right)=\mathrm{Span}\left(\hat{\boldsymbol{\Theta}}\right)$. Consequently, $\tilde{\boldsymbol{\Theta}}$ and $\hat{\boldsymbol{\Theta}}$ share the same consistency rate, and it suffices to study the behavior of $\tilde{\boldsymbol{\vartheta}}$. For notational simplicity and ease of differentiation in (S.2.6) and Lemma S.2.8, we work with $\tilde{\boldsymbol{\Theta}}$ and $\tilde{\boldsymbol{\vartheta}}$ throughout the proof. Since $\mathrm{Span}\left(\tilde{\boldsymbol{\Theta}}\right)=\mathrm{Span}\left(\hat{\boldsymbol{\Theta}}\right)$, we have $\boldsymbol{U}_i\left(\hat{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)=\boldsymbol{U}_i\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)$ and $\boldsymbol{V}_i\left(\hat{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)=\boldsymbol{V}_i\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)$. Note that many results below can find analogues in Bai (2009). However, extra terms require additional conditions and new approaches in proof, and may lead to different results.

By replacing $\left\{\hat{\boldsymbol{b}}_i\right\}_{i=1}^N$ in (3.5) with the expansion in (S.2.3), it follows that

$$\begin{aligned}
\hat{\boldsymbol{F}}\boldsymbol{\Lambda}_{NT}=&\left(\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{y}_i\boldsymbol{y}_i^\top\hat{\boldsymbol{U}}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\hat{\boldsymbol{F}}\\
=&\left\{\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}+\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right]\left[\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}+\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right]^\top\hat{\boldsymbol{U}}_i^\top\right\}\hat{\boldsymbol{F}}\\
=&\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{U}}_i^\top\hat{\boldsymbol{F}}\\
&+\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{U}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{U}}_i^\top\hat{\boldsymbol{F}}\\
&+\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{U}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^N\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{U}}_i\boldsymbol{\varepsilon}_i\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{U}}_i^\top\hat{\boldsymbol{F}}
\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{F}}\\
&-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{F}}-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}\\
&-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}\\
&-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{F}}-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}\\
&-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}}\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}\\
=:&\boldsymbol{\delta}_1+\ldots+\boldsymbol{\delta}_{21}.
\end{aligned}
\tag{S.2.4}
$$

In the second equality, we use $\hat{\boldsymbol{U}}_i\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}=\boldsymbol{0}_{T\times K_0}$.

Lemma S.2.1 (Property of $\boldsymbol{\Xi}$ and $\boldsymbol{\Upsilon}$)**.** *The following results hold for $\boldsymbol{\Xi}$ and $\boldsymbol{\Upsilon}$.*

(1) Expansion of $\boldsymbol{\Xi}^{-1}$:

$$
\begin{aligned}
\boldsymbol{\Xi}^{-1}=&\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T+\boldsymbol{\Xi}^{-1}\cdot\bar{\boldsymbol{H}}\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\\
=&\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T\cdot\left[\boldsymbol{I}_{TR_0}+\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}\right)^{-1}\boldsymbol{\Upsilon}\right]\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T,
\end{aligned}
$$

(2) Projection property:

$$
\begin{aligned}
\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}\cdot\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}^{\perp}=\boldsymbol{0}_{TR_0\times TR_0},\\
\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{G}}\cdot\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}=\boldsymbol{0}_{TR_0\times TR_0}.
\end{aligned}
$$

(3) Under Assumption 2(2) and 5, $\boldsymbol{\Xi}_0$ is strictly positive definite, and $\|\boldsymbol{\Upsilon}_0\|_2<a_1<1$ w.p.a. 1.

(4) Under Assumption 2(2) and 5, if $\log T=o\left(N^{1/3}\right)$, then $\left\|\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^D\right)\right\|_2\le ca_2^D$ for any $D=o\left(N^{1/3}\right)$, where $a_2<1$.

Lemma S.2.2. *Suppose Assumptions 2, 5, and 6 hold. The following rates hold for $\boldsymbol{\delta}_k$:*

(1) $\dfrac{\|\boldsymbol{\delta}_k\|_f^2}{T}=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right),\quad 1\le k\le 5.$

(2) $\dfrac{\|\boldsymbol{\delta}_k\|_f^2}{T}=O_p\left(\dfrac{1}{m_{NT}^2}\right),\quad k=6,7,10,12,16.$

(3) $\dfrac{\|\boldsymbol{\delta}_k\|_f^2}{T}=O_p\left(\dfrac{1}{m_{NT}^4}\right),\quad k=8,15,17.$

(4) $\dfrac{\|\boldsymbol{\delta}_k\|_f^2}{T}=O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^2}{Tm_{NT}^2}\right),\quad k=9,11,14,20.$

(5) $\dfrac{\|\boldsymbol{\delta}_{13}\|_f^2}{T} = O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^2}{Tm_{NT}}\right)$.

(6) $\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{19}\|_f^2}{T} = O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^4}{T^2}\right)$.

Proof of Theorem 3.2.

(1) We first establish the consistency rate of $T^{-1/2}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_f$ because a direct analysis of $T^{-1/2}\left\|\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0\right\|_f$ would involve non-negligible terms $\boldsymbol{\delta}_{18}$ and $\boldsymbol{\delta}_{19}$. [2]

Left multiply by $\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}$ and right multiply by $\left(T^{-1}\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}\right)^{-1}$ on both sides of (S.2.4), we have $\boldsymbol{0}_{T\times R_0} = \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\sum_{k=1}^{21}\boldsymbol{\delta}_k\left(T^{-1}\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}\right)^{-1}$. Since

$$\begin{aligned}
\ell\left[\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\delta}_{18}+\boldsymbol{\delta}_{21}\right)\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\right] =& \ell\left[\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{H}}_i\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right)\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}\right] \\
=& \left[\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}\otimes\left(\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{H}}_i\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right)\right]\cdot\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right) \\
=& \boldsymbol{\Xi}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right),
\end{aligned} \tag{S.2.5}$$

it follows that

$$\begin{aligned}
\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}} =& \frac{\|\ell(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0)\|_2}{\sqrt{T}} \\
=& \frac{1}{\sqrt{T}}\left\|\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\sum_{k=1}^{17}\boldsymbol{\delta}_k+\boldsymbol{\delta}_{19}+\boldsymbol{\delta}_{20}\right)\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\|_2 \\
\le& \left\|\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\right\|_2\cdot\frac{1}{\sqrt{T}}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\sum_{k=1}^{17}\boldsymbol{\delta}_k+\boldsymbol{\delta}_{19}+\boldsymbol{\delta}_{20}\right)\right\|_f\cdot\left\|\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\right\|_2.
\end{aligned}$$

Since $\boldsymbol{\Xi}_0$ is strictly positive definite, and $\left\|\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)-\boldsymbol{\Xi}_0^{-1}\right\|_2 = o_p(1)$ by Lemma S.4.1 (2g), it follows that $\left\|\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\right\|_2 \le c_1$ for some constant $c_1 > 0$. By Lemma S.2.2,

$$\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{1}{m_{NT}}\right) + o_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right),$$

where the last term is asymptotically negligible. Moving the term $\boldsymbol{\delta}_{21}$ to the LHS, we obtain

$$\begin{aligned}
\frac{\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0}{\sqrt{T}} =& \frac{1}{\sqrt{T}}\left(\hat{\boldsymbol{F}}\boldsymbol{\Lambda}_{NT}-\boldsymbol{\delta}_{21}\right)\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \\
=& \frac{1}{\sqrt{T}}\sum_{k=1}^{20}\boldsymbol{\delta}_k\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}.
\end{aligned}$$

The terms $\boldsymbol{\delta}_k$, $k \in \{1,2,\ldots,17\}\cup\{20\}$ are analyzed in Lemma S.2.2, and they have the same rate as above. For $\boldsymbol{\delta}_{18}$ and $\boldsymbol{\delta}_{19}$,

$$\frac{\|\boldsymbol{\delta}_{18}\|_f}{\sqrt{T}} \le \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\right\|_2\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_2}{\sqrt{T}}\cdot\frac{\bar{l}^2\|\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}\|_f}{T}$$

[2] In the analysis of $\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0$, the term $\boldsymbol{\delta}_{18}$ is grouped with $\boldsymbol{\delta}_{21}$ via the matrix $\boldsymbol{\Xi}$. By studying $\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{19}$ instead of $\boldsymbol{\delta}_{19}$ directly, we obtain an improved consistency rate.

$$
\begin{aligned}
&= O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{1}{m_{NT}}\right), \\
\frac{\|\boldsymbol{\delta}_{19}\|_f}{\sqrt{T}} &\leq \frac{\|\boldsymbol{F}_0\|_2 \bar{\boldsymbol{l}}^2}{\sqrt{T}} \cdot \frac{\|\boldsymbol{F}_0^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\|_2}{\sqrt{T}} \cdot \frac{1}{N}\sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i^\top\right\|_2 \cdot \frac{\|\hat{\boldsymbol{F}}\|_f}{\sqrt{T}} \\
&= O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

This completes the proof of our result.

(2) By left multiplying $T^{-1}\hat{\boldsymbol{F}}^\top$ and $T^{-1}\boldsymbol{F}_0^\top$ on both sides of

$$
\hat{\boldsymbol{F}}\boldsymbol{\Lambda}_{NT} - \boldsymbol{F}_0 \cdot \frac{\boldsymbol{L}_0^\top \boldsymbol{L}_0}{N} \cdot \frac{\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}}{T} = \sum_{k=1}^{20} \boldsymbol{\delta}_k,
$$

it follows that

$$
\begin{aligned}
\boldsymbol{\Lambda}_{NT} &= \frac{\hat{\boldsymbol{F}}^\top \boldsymbol{F}_0}{T} \cdot \frac{\boldsymbol{L}_0^\top \boldsymbol{L}_0}{N} \cdot \frac{\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}}{T} + o_p(1), \\
\frac{\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}}{T} \cdot \boldsymbol{\Lambda}_{NT} &= \left(\frac{\boldsymbol{F}_0^\top \boldsymbol{F}_0}{T} \cdot \frac{\boldsymbol{L}_0^\top \boldsymbol{L}_0}{N}\right) \frac{\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}}{T} + o_p(1).
\end{aligned}
$$

The invertibility and the probability limit of $\boldsymbol{\Lambda}_{NT}$ are shown following a similar argument in Bai (2009). □

Lemma S.2.3. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$
\frac{\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_2^2}{T^2} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(\frac{1}{m_{NT}^4}\right), \quad \textit{for each } i = 1, \ldots, N.
$$

Lemma S.2.4. *Suppose Assumptions 2, 5, and 6 hold. Then:*

(1) $\dfrac{\|\boldsymbol{X}_i^\top(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0)\|_2^2}{T^2} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(\dfrac{1}{m_{NT}^4}\right), \quad$ *for each* $i = 1, \ldots, N$.

(2) $\dfrac{\boldsymbol{F}_0^\top(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0)}{T} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{1}{m_{NT}^2}\right)$.

(3) $\dfrac{\hat{\boldsymbol{F}}^\top(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0)}{T} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{1}{m_{NT}^2}\right)$.

(4) $\boldsymbol{O}_{\boldsymbol{F}}^\top \boldsymbol{O}_{\boldsymbol{F}} = \boldsymbol{\Sigma}_{\boldsymbol{F}} + O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{1}{m_{NT}^2}\right)$.

Remark S.2.1. *The result in Lemma S.2.3 is crucial for establishing the consistency rate in Lemma S.2.4. A direct analysis of* $T^{-1}\boldsymbol{X}_i^\top\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0\right)$ *does not yield a satisfactory consistency rate for the terms* $T^{-1}\boldsymbol{X}_i^\top \boldsymbol{\delta}_{18}$ *and* $T^{-1}\boldsymbol{X}_i^\top \boldsymbol{\delta}_{19}$.

In the following proof, we first derive an initial representation (S.2.9) of $\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right)$ in Proposition S.2.1 from the first-order condition (S.2.10). We then gradually eliminate estimated quantities and analyze the asymptotic bias in the representation step by step, leading to the final form (S.2.17) in Corollary S.2.3.

Let $f_{2,NT}(\boldsymbol{\vartheta})$ be the objective function in (3.8), that is,

$$f_{2,NT}(\boldsymbol{\vartheta}) = \frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}\left(\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{y}_i\right\|_2^2.$$

The first order derivative of $f_{2,NT}$ evaluated at $\boldsymbol{\vartheta} = \boldsymbol{\vartheta}_0$ is given by

$$\begin{aligned}\frac{\partial f_{2,NT}(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}} =& \boldsymbol{J}\cdot\frac{2}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{y}_i\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{y}_i\\ =& \boldsymbol{J}\cdot\frac{2}{NT}\sum_{i=1}^{N}\left[\boldsymbol{b}_{i,0}+\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right)\right]\\ &\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i\right).\end{aligned}\tag{S.2.6}$$

Lemmas S.2.5–S.2.7 analyze the terms in (S.2.6).

Lemma S.2.5 (Dominated terms in (S.2.6))**.** *Suppose Assumptions 2, 5, and 6 hold. Then:*

(1) $\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{F}_0\boldsymbol{l}_{i,0} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$

(2) $\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_i = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).$

(3) $\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{F}_0\boldsymbol{l}_{i,0} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).$

Lemma S.2.6 (Leading term 1 in (S.2.6))**.** *Suppose Assumptions 2, 5, and 6 hold. Then:*

$$\begin{aligned}&\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_i\\ =&\frac{\zeta_{1,NT}^{\dagger}}{T}+O_p\left(\frac{\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),\end{aligned}$$

where

$$\zeta_{1,NT}^{\dagger} = \frac{1}{N}\sum_{i=1}^{N}\left[(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\otimes\boldsymbol{Q}_{i,0}^{\top}\right]\ell(\boldsymbol{\Omega}_i) = O_p(1).\tag{S.2.7}$$

Lemma S.2.7 (Leading term 2 in (S.2.6))**.** *Suppose Assumptions 2, 5, and 6 hold. Then:*

$$\begin{aligned}&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\left[\boldsymbol{I}_p-\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0)^{-1}\boldsymbol{\Theta}_0^{\top}\right]\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\\ =&\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{Q}_i^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_j\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^{\top}\cdot\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\\ &-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{Q}_i^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{j,0}\otimes\left[\boldsymbol{I}_T-\boldsymbol{H}_j\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_j\\ &+\frac{\zeta_{2,NT}^{\dagger}}{T}+O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),\end{aligned}$$

where

$$\zeta_{2,NT}^{\dagger} = -\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{Q}_{i,0}^{\top}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\ell\left(\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{U}_{j,0}\boldsymbol{\Omega}_j\boldsymbol{U}_{j,0}^{\top}\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right) = O_p(1).\tag{S.2.8}$$

Lemma S.2.8 (The second-order and third-order derivatives of $f_{2,NT}$)**.** *Suppose Assumptions 2, 5, and 6 hold. Then:*

(1) $\dfrac{\partial^2 f_{2,NT}(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^\top} = -\boldsymbol{J}\cdot\dfrac{2}{NT}\sum_{i=1}^N \boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top \otimes \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\left[\boldsymbol{I}_T - \boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i\cdot\boldsymbol{J}^\top + O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{1}{m_{NT}}\right).$

(2) For $1\le l_1,l_2,l_3\le (p-K_0)K_0$, $\left|\dfrac{\partial^3 f_{2,NT}(\boldsymbol{\vartheta})}{\partial\vartheta_{l_1}\partial\vartheta_{l_2}\partial\vartheta_{l_3}}\right| = O_p(1)$, *for* $\boldsymbol{\vartheta}$ *satisfying* $\left\|\boldsymbol{\vartheta}-\boldsymbol{\vartheta}_0\right\|_2 = o(1)$.

Proposition S.2.1. *Suppose Assumptions 2 and 5–7 hold. Then:*

$$\begin{aligned}
&\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\\
=&\boldsymbol{J}\cdot\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\frac{1}{\sqrt{NT}}\sum_{i=1}^N\left\{\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)-\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{i,0}\otimes\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\right\}\boldsymbol{\varepsilon}_i\\
&+\sqrt{\frac{N}{T}}\boldsymbol{J}\cdot\boldsymbol{\zeta}_{NT}+\sqrt{NT}\cdot\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right)\right],
\end{aligned}\tag{S.2.9}$$

where

$$\boldsymbol{\zeta}_{NT}=\boldsymbol{\zeta}_{1,NT}+\boldsymbol{\zeta}_{2,NT},\quad \boldsymbol{\zeta}_{1,NT}=\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\zeta}_{1,NT}^\dagger,\quad \boldsymbol{\zeta}_{2,NT}=\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\zeta}_{2,NT}^\dagger$$

with $\boldsymbol{\zeta}_{1,NT}^\dagger$ *and* $\boldsymbol{\zeta}_{2,NT}^\dagger$ *defined in* (S.2.7) *and* (S.2.8) *respectively.*

Proof of Proposition S.2.1. We have the first order condition

$$\begin{aligned}
\boldsymbol{0}_{(p-K_0)K_0} =& \frac{\partial f_{2,NT}(\tilde{\boldsymbol{\vartheta}})}{\partial\boldsymbol{\vartheta}}\\
=&\frac{\partial f_{2,NT}(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}}+\frac{\partial^2 f_{2,NT}(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^\top}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)+\boldsymbol{r}_{\boldsymbol{\vartheta}},
\end{aligned}\tag{S.2.10}$$

where the l-th element of $\boldsymbol{r}_{\boldsymbol{\vartheta}}$ is

$$\frac{1}{2}\sum_{1\le l_1,l_2\le(p-K_0)K_0}\left[\frac{\partial^3 f_{2,NT}(\boldsymbol{\vartheta}_0+\alpha_l(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0))}{\partial\vartheta_l\partial\vartheta_{l_1}\partial\vartheta_{l_2}}\left(\tilde{\vartheta}_{l_1}-\vartheta_{l_1,0}\right)\left(\tilde{\vartheta}_{l_2}-\vartheta_{l_2,0}\right)\right],$$

with $0\le\alpha_l\le 1$ for $1\le l\le(p-K_0)K_0$. By (S.2.6), (S.2.10), and Lemmas S.2.5–S.2.8, we have

$$\begin{aligned}
\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)=\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}\cdot\Bigg\{&\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{Z}_i^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{\varepsilon}_i+\sqrt{\frac{N}{T}}\left(\boldsymbol{\zeta}_{1,NT}^\dagger+\boldsymbol{\zeta}_{2,NT}^\dagger\right)\\
&+\sqrt{NT}\cdot\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right)\right]\Bigg\},
\end{aligned}$$

where the positive definiteness of $\boldsymbol{D}_{22}:=\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top$ follows from Assumption 7 and $\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}\left(\boldsymbol{F}_0\right)=o_p(1)$; see Lemma S.2.15 (1). Since $\boldsymbol{J}\boldsymbol{J}^\top=\boldsymbol{I}_{(p-K_0)K_0}$, it remains to show that $\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}=\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)$. Let

$$\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)=\begin{pmatrix}(\boldsymbol{D}_{11})_{K_0^2\times K_0^2} & (\boldsymbol{D}_{12})_{K_0^2\times(p-K_0)K_0}\\ (\boldsymbol{D}_{12}^\top)_{(p-K_0)K_0\times K_0^2} & (\boldsymbol{D}_{22})_{(p-K_0)K_0\times(p-K_0)K_0}\end{pmatrix}.$$

Then,

$$\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}=\begin{pmatrix}\boldsymbol{0}_{K_0^2\times K_0^2} & \boldsymbol{0}_{K_0^2\times(p-K_0)K_0}\\ \boldsymbol{0}_{(p-K_0)K_0\times K_0^2} & \boldsymbol{D}_{22}^{-1}\end{pmatrix}.$$

On the other hand,

$$\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}=\begin{pmatrix}\boldsymbol{D}_{11}\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}_{[K_0]}+\boldsymbol{D}_{12}\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}_{-[K_0]}\\ \boldsymbol{D}_{12}^\top\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}_{[K_0]}+\boldsymbol{D}_{22}\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}_{-[K_0]}\end{pmatrix}=\begin{pmatrix}\boldsymbol{0}_{K_0^2\times K_0^2}\\ \boldsymbol{0}_{(p-K_0)K_0\times K_0^2}\end{pmatrix},$$

which implies that

$$\boldsymbol{D}_{11}=-\boldsymbol{D}_{12}\cdot\boldsymbol{I}_{K_0}\otimes\hat{\boldsymbol{\Theta}}_{-[K_0]}\hat{\boldsymbol{\Theta}}_{[K_0]}^{-1}=\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top.$$

It follows that

$$\begin{aligned}\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)&=\begin{pmatrix}\left(\boldsymbol{D}_{11}-\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\right)^+ & -\left(\boldsymbol{D}_{11}-\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\right)^+\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\\ -\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\left(\boldsymbol{D}_{11}-\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\right)^+ & \boldsymbol{D}_{22}^{-1}+\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\left(\boldsymbol{D}_{11}-\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\boldsymbol{D}_{12}^\top\right)^+\boldsymbol{D}_{12}\boldsymbol{D}_{22}^{-1}\end{pmatrix}\\ &=\begin{pmatrix}\boldsymbol{0}_{K_0^2\times K_0^2} & \boldsymbol{0}_{K_0^2\times(p-K_0)K_0}\\ \boldsymbol{0}_{(p-K_0)K_0\times K_0^2} & \boldsymbol{D}_{22}^{-1}\end{pmatrix}.\end{aligned}$$

This completes the proof. □

Lemma S.2.9 prepares for Lemma S.2.10 and the representation (S.2.12) in Corollary S.2.1, which separates $\sum_{i=1}^N \boldsymbol{Z}_i^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{\varepsilon}_i$ into $\sum_{i=1}^N \boldsymbol{Z}_{i,0}^\top\boldsymbol{\varepsilon}_i$ and an additional asymptotic bias term.

Lemma S.2.9. *Let $\boldsymbol{M}_i\in\mathbb{R}^{n\times R_0}$ for $i=1,\ldots,N$, be deterministic matrices given $\mathcal{C}$. In addition, $\max_i\|\boldsymbol{M}_i\|_f\le c_1$. Suppose Assumptions 2, 5, and 6 hold. Then:*

$$\frac{1}{NT}\sum_{i=1}^N\boldsymbol{M}_i\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\right)^\top\boldsymbol{\varepsilon}_i=\frac{1}{N^2}\sum_{i,j=1}^N\boldsymbol{M}_i\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).$$

Lemma S.2.10. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$\begin{aligned}&\frac{1}{\sqrt{NT}}\sum_{i=1}^N\left\{\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)-\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{i,0}\otimes\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\right\}\boldsymbol{\varepsilon}_i\\ =&\frac{1}{\sqrt{NT}}\sum_{i=1}^N\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_{i,0}^\top-\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\otimes\left(\boldsymbol{I}_T-\boldsymbol{H}_{i,0}\right)\right]\boldsymbol{\varepsilon}_i+\sqrt{\frac{T}{N}}\boldsymbol{\xi}_{NT}^\dagger\\ &+\sqrt{NT}\cdot\left[O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right)\right],\end{aligned}$$

where

$$\begin{aligned}\boldsymbol{\xi}_{NT}^\dagger=&-\frac{1}{N}\sum_{i,j=1}^N\left\{\left(\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e-\frac{1}{N}\sum_{k=1}^N w_{ik}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^\top\boldsymbol{\Sigma}_{k\boldsymbol{F}}\right)\right.\\ &\left.+\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^N\boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^\top\otimes\boldsymbol{Q}_{l0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\right)\cdot\left[\boldsymbol{l}_{i,0}\otimes\frac{\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{T}-\frac{1}{N}\sum_{k=1}^N w_{ik}\boldsymbol{l}_{k,0}\otimes\frac{\boldsymbol{X}_k\boldsymbol{\Phi}_{4k,0}^\top\boldsymbol{F}_0}{T}\right]\right\}\\ &\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)=O_p(1).\end{aligned}\tag{S.2.11}$$

Corollary S.2.1. *Suppose Assumptions 2 and 5–7 hold. Then:*

$$
\begin{aligned}
&\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\\
=&\boldsymbol{J}\cdot\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right)\cdot\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_{i,0}^{\top}-\frac{1}{N}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_{j,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\otimes\left(\boldsymbol{I}_T-\boldsymbol{H}_{i,0}\right)\right]\boldsymbol{\varepsilon}_i\\
&+\sqrt{\frac{T}{N}}\boldsymbol{J}\cdot\boldsymbol{\xi}_{NT}+\sqrt{\frac{N}{T}}\boldsymbol{J}\cdot\boldsymbol{\zeta}_{NT}+\sqrt{NT}\cdot\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right)\right],
\end{aligned}
\tag{S.2.12}
$$

where $\boldsymbol{\xi}_{NT}=\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\xi}_{NT}^{\dagger}$ *with* $\boldsymbol{\xi}_{NT}^{\dagger}$ *defined in* (S.2.11).

Next, we will show the asymptotic normality of the following term with additional Nickell bias separated:

$$
\begin{aligned}
\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{Z}_{i,0}^{\top}\boldsymbol{\varepsilon}_i=&\frac{1}{NT}\sum_{i=1}^{N}\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_{i,0}^{\top}-\frac{1}{N}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_{j,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\left(\boldsymbol{I}_T-\boldsymbol{H}_{i,0}\right)\right]\boldsymbol{\varepsilon}_i\\
=&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\varepsilon}_i+\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i\\
&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\varepsilon}_i\\
=:&\Delta_1+\Delta_2+\Delta_3.
\end{aligned}
\tag{S.2.13}
$$

By definition,

$$
\begin{aligned}
\boldsymbol{Z}_{i,0}=&\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\cdot\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{\Psi}_{1i}+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}\right]^{\top}\\
&-\tilde{\boldsymbol{l}}_{i,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\cdot\frac{1}{N}\sum_{j=1}^{N}\tilde{\boldsymbol{l}}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{X}_j\boldsymbol{\Psi}_{1j}^{\top}.
\end{aligned}
$$

Let

$$
\tilde{\boldsymbol{X}}_i=\boldsymbol{X}_i-\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i),\quad \boldsymbol{\mathfrak{X}}_i=\tilde{\boldsymbol{X}}_i+\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right).
$$

Lemma S.2.11. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$
\begin{aligned}
\Delta_1=&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{\Psi}_{1i}^{e}\boldsymbol{\mathfrak{X}}_i^{\top}\boldsymbol{\varepsilon}_i\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\pi}_{1i}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{2i}}^{e\top}\boldsymbol{\pi}_{1i}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]+O_p\left(\frac{1}{Tm_{NT}}\right).
\end{aligned}
$$

Lemma S.2.12. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$
\begin{aligned}
\Delta_2=&\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{e}\boldsymbol{\mathfrak{X}}_i^{\top}\boldsymbol{\varepsilon}_i+\frac{1}{NT}\sum_{i=1}^{N}\left[\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\boldsymbol{\mathfrak{X}}_i^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{4ij}}^{e\top}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\right.
\end{aligned}
$$

$$\cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{3ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i - \mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{4ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right]\Bigg\} + O_p\left(\frac{1}{m_{NT}^3}\right),$$

where

$$\boldsymbol{\mathcal{V}} = \frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\tilde{\boldsymbol{l}}_{j,0}\boldsymbol{b}_{j,0}^\top \otimes \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j\right)\boldsymbol{\Psi}^{e\top}_{1j}.$$

Lemma S.2.13. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$\Delta_3 = -\boldsymbol{\mathcal{V}}^\top\cdot\left[\boldsymbol{I}_{TR_0} + \sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\frac{1}{N\sqrt{T}}\sum_{k=1}^{N}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top\otimes\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\boldsymbol{\Phi}^e_{4k,0}\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_k^\top)\right]$$
$$\cdot\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\boldsymbol{\varepsilon}_i + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).$$

Let

$$\begin{aligned}\boldsymbol{\mathfrak{Z}}_{i,0} =& \boldsymbol{\mathfrak{X}}_i\cdot\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{\Psi}^e_{1i} + \frac{1}{N}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}^e_{1j}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}^e_{3ij} + \boldsymbol{\mathcal{V}}^\top\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}^e_{4i,0}\right]^\top\\ &- \sqrt{T}\tilde{\boldsymbol{l}}_{i,0}^\top\otimes\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\cdot\left[\boldsymbol{I}_{TR_0} + \frac{1}{N\sqrt{T}}\sum_{k=1}^{N}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top\otimes\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_k)\boldsymbol{\Phi}^{e\top}_{4k,0}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\right]\cdot\boldsymbol{\mathcal{V}}.\end{aligned}\tag{S.2.14}$$

Rearranging the terms in Lemmas S.2.11–S.2.13 leads to Corollary S.2.2.

Corollary S.2.2. *Suppose Assumptions 2, 5, and 6 hold. Then,*

$$\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{Z}_{i,0}^\top\boldsymbol{\varepsilon}_i = \boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i + \frac{\boldsymbol{\eta}_{NT}}{T} + o_p\left(\frac{1}{\sqrt{NT}}\right),$$

where $\boldsymbol{\eta}_{NT} = \boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\eta}^{\dagger}_{NT}$, *and*

$$\begin{aligned}\boldsymbol{\eta}^{\dagger}_{NT} =& \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{1i}}\boldsymbol{\pi}_{1i}\right)\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i - \mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{2i}}\boldsymbol{\pi}_{1i}\right)\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right]\\ &+ \frac{1}{N^2}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}^e_{1j}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{3ij}}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i - \mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{4ij}}\boldsymbol{\pi}_{2ij}\right)\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right]\\ &+ \frac{1}{N^2}\sum_{i,k=1}^{N}\Bigg\{\boldsymbol{\mathcal{V}}^\top\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top\otimes\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{G})}\boldsymbol{\Phi}^e_{4k,0}\\ &\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{3ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i - \mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}^{e\top}_{\boldsymbol{\Psi}_{4ik}}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right]\Bigg\}.\end{aligned}\tag{S.2.15}$$

We establish the asymptotic normality for the first term in the next lemma.

Lemma S.2.14. *Suppose Assumptions 2 and 5–7 hold. Then,*

$$\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i \xrightarrow{\mathbb{D}} \mathcal{N}\left(\boldsymbol{0}_{(p-K_0)K_0},\ \boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\right).$$

Lemma S.2.15. *Suppose Assumptions 2 and 5–7 hold. Then*

(1) $\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{D}\left(\boldsymbol{F}_0\right) = o_p(1)$.

If we further assume that $\sqrt{N}/T \to 0$ *and* $\sqrt{T}/N \to 0$*, then*

(2) $\sqrt{T/N}\left[\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right)\right] = o_p(1)$.

(3) $\sqrt{N/T}\left[\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right)\right] = o_p(1)$.

(4) $\sqrt{T/N}\left(\boldsymbol{\xi}_{NT} - \mathbb{B}^{\boldsymbol{\xi}}\right) = o_p(1)$*, where* $\mathbb{B}^{\boldsymbol{\xi}}$ *is defined in* (S.1.3).

(5) $\sqrt{N/T}\left(\boldsymbol{\zeta}_{NT} - \mathbb{B}^{\boldsymbol{\zeta}}\right) = o_p(1)$*, where* $\mathbb{B}^{\boldsymbol{\zeta}}$ *is defined in* (S.1.4).

(6) $\sqrt{N/T}\left(\boldsymbol{\eta}_{NT} - \mathbb{B}^{\boldsymbol{\eta}}\right) = o_p(1)$*, where* $\mathbb{B}^{\boldsymbol{\eta}}$ *is defined in* (S.1.5).

In the proof of Lemma S.2.15 (2), we show that $\tilde{\boldsymbol{\vartheta}}$ admits the following expansion:

$$\begin{aligned}\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0 =& \boldsymbol{J} \cdot \left[\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right) \cdot \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top}\boldsymbol{\varepsilon}_i + \frac{\boldsymbol{\xi}_{NT}}{N} + \frac{\boldsymbol{\zeta}_{NT} + \boldsymbol{\eta}_{NT}}{T}\right] \\ &+ o_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{1}{m_{NT}^3}\right) + o_p\left(\frac{1}{\sqrt{NT}}\right). \end{aligned} \tag{S.2.16}$$

This representation establishes the consistency rate of $\tilde{\boldsymbol{\vartheta}}$. Note that this rate does not require $\sqrt{N}/T \to 0$ or $\sqrt{T}/N \to 0$.

Proposition S.2.2 (Consistency rate of $\hat{\boldsymbol{\Theta}}$)**.** *Suppose Assumptions 2 and 5–7 hold. Then*

$$m_{NT}^2 \cdot \left(\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K],0} - \boldsymbol{\Theta}_0\right) = O_p(1).$$

Equation (S.2.16) and Lemma S.2.15 together lead to Corollary S.2.3. The proof closely follows Proposition A.3 in Bai (2009) and is therefore omitted.

Corollary S.2.3. *Suppose Assumptions 2 and 5–8 hold. If* $\sqrt{N}/T \to 0$ *and* $\sqrt{T}/N \to 0$*, then*

$$\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right) = \boldsymbol{J} \cdot \left[\boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right) \cdot \frac{1}{\sqrt{NT}}\sum_{i=1}^{N} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top}\boldsymbol{\varepsilon}_i + \sqrt{\frac{T}{N}}\mathbb{B}^{\boldsymbol{\xi}} + \sqrt{\frac{N}{T}}\left(\mathbb{B}^{\boldsymbol{\zeta}} + \mathbb{B}^{\boldsymbol{\eta}}\right)\right] + o_p(1), \tag{S.2.17}$$

where $\mathbb{B}^{\boldsymbol{\xi}}$*,* $\mathbb{B}^{\boldsymbol{\zeta}}$*, and* $\mathbb{B}^{\boldsymbol{\eta}}$ *are defined in* (S.1.3)*,* (S.1.4)*, and* (S.1.5) *respectively.*

Proof of Theorem 3.3. Since $\operatorname{plim}_{(N,T)\to\infty} \boldsymbol{D}(\boldsymbol{F}_0) = \boldsymbol{D}_0$ by Assumption 7, combined with Lemma S.2.14, we obtain

$$\begin{aligned}\boldsymbol{J} \cdot \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right) \cdot \frac{1}{\sqrt{NT}}\sum_{i=1}^{N} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top}\boldsymbol{\varepsilon}_i =& \left[\boldsymbol{J} \cdot \boldsymbol{D}\left(\boldsymbol{F}_0\right) \cdot \boldsymbol{J}^{\top}\right]^{-1} \cdot \frac{1}{\sqrt{NT}}\sum_{i=1}^{N} \boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^{\top}\boldsymbol{\varepsilon}_i \\ &\xrightarrow{\mathbb{D}} \mathcal{N}\left[\boldsymbol{0}_{(p-K_0)K_0}, \left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^{\top}\right)^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^{\top}\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^{\top}\right)^{-1}\right]. \end{aligned}$$

Under $T/N \to \rho$, by Assumption 8, it follows that

$$\operatorname{plim}_{(N,T)\to\infty}\sqrt{\frac{T}{N}}\mathbb{B}^{\boldsymbol{\xi}} = \rho^{1/2}\mathbb{B}_0^{\boldsymbol{\xi}}, \quad \operatorname{plim}_{(N,T)\to\infty}\sqrt{\frac{N}{T}}\mathbb{B}^{\boldsymbol{\zeta}} = \rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\zeta}}, \quad \operatorname{plim}_{(N,T)\to\infty}\sqrt{\frac{N}{T}}\mathbb{B}^{\boldsymbol{\eta}} = \rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\eta}}.$$

By the representation (S.2.17), we obtain

$$\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right) \xrightarrow{\mathbb{D}} \mathcal{N}\left[\boldsymbol{J}\left(\rho^{1/2}\mathbb{B}_0^{\boldsymbol{\xi}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\zeta}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\eta}}\right),\ \left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)\right]. \tag{S.2.18}$$

We apply the delta method to (S.2.18) with function $g(\boldsymbol{\vartheta}) = \ell(\boldsymbol{P}_{\boldsymbol{\Theta}})$, where $\partial g(\boldsymbol{\vartheta}_0)/\partial\boldsymbol{\vartheta} = \boldsymbol{J}\cdot\boldsymbol{A}_0$, and $\boldsymbol{A}_0$ is defined in Theorem 3.3. The bias term is

$$\frac{\partial^\top g(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}}\cdot\boldsymbol{J}\left(\rho^{1/2}\mathbb{B}_0^{\boldsymbol{\xi}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\zeta}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\eta}}\right) = \boldsymbol{A}_0^\top\left(\rho^{1/2}\mathbb{B}_0^{\boldsymbol{\xi}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\zeta}}+\rho^{-1/2}\mathbb{B}_0^{\boldsymbol{\eta}}\right),$$

since $\boldsymbol{J}^\top\boldsymbol{J}\boldsymbol{D}_0^+ = \boldsymbol{J}^\top\boldsymbol{J}\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}^\top = \boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}^\top = \boldsymbol{D}_0^+$. The covariance term is

$$\frac{\partial^\top g(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}}\cdot\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0^{-1}\boldsymbol{J}^\top\right)\cdot\frac{\partial g(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}} = \boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0.$$

This completes the proof. □

Proof of Theorem 3.4.

(1) When $T/N \to 0$, since $\mathbb{B}^{\boldsymbol{\xi}} = O_p(1)$, we have $\sqrt{T/N}\mathbb{B}^{\boldsymbol{\xi}} = o_p(1)$. When the serial heteroskedasticity is absent, $\boldsymbol{\Omega}_i = \sigma_i^2\boldsymbol{I}_T$. For $\mathbb{B}^{\boldsymbol{\zeta}_1}$, we have

$$\begin{aligned}\mathbb{B}^{\boldsymbol{\zeta}_1} =& \boldsymbol{D}^+(\boldsymbol{F}_0)\cdot\frac{1}{N}\sum_{i=1}^N \ell\left[\boldsymbol{Q}_{i,0}^\top\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\right]\\ =& \boldsymbol{D}^+(\boldsymbol{F}_0)\cdot\frac{1}{N}\sum_{i=1}^N \sigma_i^2\ell\left[\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{H}_{i,0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\right]\\ =& \boldsymbol{0}_{pK_0},\end{aligned}$$

where the last step is due to $\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{H}_{i,0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0 = \boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0 - \boldsymbol{H}_{i,0}\boldsymbol{X}_i\boldsymbol{\Theta}_0 = \boldsymbol{0}_{T\times K_0}$. For $\mathbb{B}^{\boldsymbol{\zeta}_2}$, we have

$$\mathbb{B}^{\boldsymbol{\zeta}_2} = -\boldsymbol{D}^+(\boldsymbol{F}_0)\cdot\frac{1}{N^2T}\sum_{i,j=1}^N \boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^\top\otimes\boldsymbol{Q}_{i,0}^\top\cdot\boldsymbol{\Xi}_0^{-1}\cdot\sigma_j^2\ell\left(\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{U}_{j,0}\boldsymbol{U}_{j,0}^\top\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right).$$

Note that

$$\begin{aligned}\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{U}_{j,0}\boldsymbol{U}_{j,0}^\top\boldsymbol{F}_0 =& \boldsymbol{P}_{\boldsymbol{F}_0}^\perp\left(\boldsymbol{I}_T-\boldsymbol{V}_{j,0}\right)\left(\boldsymbol{I}_T-\boldsymbol{V}_{j,0}^\top\right)\boldsymbol{F}_0\\ =& \boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{F}_0-\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}\boldsymbol{F}_0-\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}^\top\boldsymbol{F}_0+\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}\boldsymbol{V}_{j,0}^\top\boldsymbol{F}_0\\ =& \boldsymbol{0}_{T\times R_0},\end{aligned}$$

where the last step is due to $\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{F}_0 = \boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}\boldsymbol{F}_0 = \boldsymbol{0}_{T\times R_0}$, and $\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}^\top\boldsymbol{F}_0 = \boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{V}_{j,0}\boldsymbol{V}_{j,0}^\top\boldsymbol{F}_0 = \boldsymbol{V}_{j,0}^\top\boldsymbol{F}_0$. Therefore, $\mathbb{B}^{\boldsymbol{\zeta}_2} = \boldsymbol{0}_{pK_0}$. To conclude, $\mathbb{B}^{\boldsymbol{\zeta}} = \mathbb{B}^{\boldsymbol{\zeta}_1}+\mathbb{B}^{\boldsymbol{\zeta}_2} = \boldsymbol{0}_{pK_0}$. It is straightforward to show $\mathbb{B}^{\boldsymbol{\eta}} = \boldsymbol{0}_{(p-K_0)K_0}$ when $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$ are conditionally independent of $\{\boldsymbol{X}_i\}_{i=1}^N$ given $\mathcal{C}$. By the representation (S.2.16) and the asymptotic distribution in (3.12), it follows that

$$\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right) \xrightarrow{\mathbb{D}} \mathcal{N}\left[\boldsymbol{0}_{(p-K_0)K_0},\ \left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}\boldsymbol{D}_1\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\right].$$

Applying the delta method with function $g(\boldsymbol{\vartheta}) = \ell(\boldsymbol{P}_{\boldsymbol{\Theta}})$ yields the desired result.

(2) When $N/T \to 0$, since $\mathbb{B}^{\boldsymbol{\zeta}}$ and $\mathbb{B}^{\boldsymbol{\eta}}$ are both $O_p(1)$, we have $\sqrt{N/T}\mathbb{B}^{\boldsymbol{\zeta}} = o_p(1)$ and $\sqrt{N/T}\mathbb{B}^{\boldsymbol{\eta}} = o_p(1)$. When the cross-sectional heteroskedasticity is absent, $T^{-1}\sum_{t=1}^{T}\sigma_{it}^2 = T^{-1}\sum_{t=1}^{T}\tau_t^2 =: \bar{\sigma}^2$. Then, $\mathbb{B}^{\boldsymbol{\xi}}$ simplifies to

$$\begin{aligned}\mathbb{B}^{\boldsymbol{\xi}} = &-\boldsymbol{D}^{+}(\boldsymbol{F}_0)\cdot\frac{1}{N}\sum_{i=1}^{N}\Bigg[\left(\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}-\frac{1}{N}\sum_{k=1}^{N}w_{ik}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\right)\\ &+\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^{N}\boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^{\top}\otimes\boldsymbol{Q}_{l0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\right)\cdot\left(\boldsymbol{l}_{i,0}\otimes\frac{\boldsymbol{V}_{i,0}^{\top}\boldsymbol{F}_0}{\sqrt{T}}-\frac{1}{N}\sum_{k=1}^{N}w_{ik}\boldsymbol{l}_{k,0}\otimes\frac{\boldsymbol{V}_{k,0}^{\top}\boldsymbol{F}_0}{\sqrt{T}}\right)\Bigg]\\ &\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}^2 =: \Delta_1+\Delta_2.\end{aligned}$$

Since

$$\frac{1}{N}\sum_{i=1}^{N}w_{ik}\boldsymbol{l}_{i,0} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{k,0} = \boldsymbol{l}_{k,0},$$

we have

$$\begin{aligned}\frac{1}{N^2}\sum_{i,k=1}^{N}w_{ik}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}^2 &= \frac{1}{N}\sum_{k=1}^{N}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\left(\frac{1}{N}\sum_{i=1}^{N}w_{ik}\boldsymbol{l}_{i,0}\right)\bar{\sigma}^2\\ &= \frac{1}{N}\sum_{k=1}^{N}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{k,0}\bar{\sigma}^2\\ &= \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}^2,\end{aligned}$$

where the last step simply relabels $k \mapsto i$, indicating that $\Delta_1 = \mathbf{0}_{pK_0}$. Using the same approach, we can also show $\Delta_2 = \mathbf{0}_{pK_0}$. To conclude, $\mathbb{B}^{\boldsymbol{\xi}} = \Delta_1 + \Delta_2 = \mathbf{0}_{pK_0}$. By the representation (S.2.16) and the asymptotic distribution in (3.13), it follows that

$$\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\xrightarrow{\mathbb{D}}\mathcal{N}\left[\mathbf{0}_{(p-K_0)K_0},\ \left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^{\top}\right)^{-1}\boldsymbol{J}\boldsymbol{D}_2\boldsymbol{J}^{\top}\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^{\top}\right)^{-1}\right].$$

Again, applying the delta method with function $g(\boldsymbol{\vartheta}) = \ell(\boldsymbol{P}_{\boldsymbol{\Theta}})$ yields the desired result.

(3) Using the results in (1) and (2), when $\{\varepsilon_{it}\}$ are i.i.d. with mean zero and variance σ^2, we have $\mathbb{B}^{\boldsymbol{\zeta}} = \mathbb{B}^{\boldsymbol{\xi}} = \mathbf{0}_{pK_0}$. Also, $\mathbb{B}^{\boldsymbol{\eta}} = \mathbf{0}_{pK_0}$ due to the conditional independence of $\{\boldsymbol{\varepsilon}_i\}_{i=1}^{N}$ and $\{\boldsymbol{X}_i\}_{i=1}^{N}$. The matrix $\boldsymbol{D}_1$ or $\boldsymbol{D}_2$ simplifies to $\sigma^2\boldsymbol{D}_0$. It follows that

$$\sqrt{NT}\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\xrightarrow{\mathbb{D}}\mathcal{N}\left[\mathbf{0}_{(p-K_0)K_0},\ \sigma^2\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^{\top}\right)^{-1}\right].$$

This completes the proof. □

S.2.4 Proofs of Theorems in Section 3.4

For convenience of discussion, we replace $\hat{\boldsymbol{\Theta}}$ and $\hat{\boldsymbol{b}}_i$ with $\tilde{\boldsymbol{\Theta}}$ and $\tilde{\boldsymbol{b}}_i = \left(\tilde{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\tilde{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}\right)^{-1}\tilde{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\tilde{\boldsymbol{F}}}^{\perp}\boldsymbol{y}_i$ respectively, and study $\left(\tilde{\boldsymbol{Z}}_i,\ \tilde{\boldsymbol{D}}_0,\ \tilde{\boldsymbol{A}},\ \tilde{\mathbb{B}}^{\boldsymbol{\xi}},\ \tilde{\mathbb{B}}^{\boldsymbol{\zeta}},\ \tilde{\mathbb{B}}^{\boldsymbol{\eta}}\right)$ instead of $\left(\hat{\boldsymbol{Z}}_i,\ \hat{\boldsymbol{D}}_0,\ \hat{\boldsymbol{A}},\ \hat{\mathbb{B}}^{\boldsymbol{\xi}},\ \hat{\mathbb{B}}^{\boldsymbol{\zeta}},\ \hat{\mathbb{B}}^{\boldsymbol{\eta}}\right)$. Since $\tilde{\boldsymbol{\Theta}}$ is the product of

$\hat{\boldsymbol{\Theta}}$ and some invertible matrix, some quantities remain unchanged, such as $\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i = \tilde{\boldsymbol{\Theta}}\tilde{\boldsymbol{b}}_i$, $\tilde{\boldsymbol{S}}_i = \hat{\boldsymbol{S}}_i$, $\tilde{\boldsymbol{Q}}_i = \hat{\boldsymbol{Q}}_i$, $\boldsymbol{U}_i(\tilde{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}) = \boldsymbol{U}_i(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}})$, $\hat{\varepsilon}_{it} = \tilde{\varepsilon}_{it}$, and $\tilde{\boldsymbol{A}}^\top\left(\tilde{\mathbb{B}}^{\boldsymbol{\xi}}, \tilde{\mathbb{B}}^{\boldsymbol{\zeta}}, \tilde{\mathbb{B}}^{\boldsymbol{\eta}}\right) = \hat{\boldsymbol{A}}^\top\left(\hat{\mathbb{B}}^{\boldsymbol{\xi}}, \hat{\mathbb{B}}^{\boldsymbol{\zeta}}, \hat{\mathbb{B}}^{\boldsymbol{\eta}}\right)$ especially.

Lemma S.2.16. *Suppose Assumptions 2, 5-7, and 9. If $T = O(N)$, then:*

(1) $\tilde{\mathbb{B}}^{\boldsymbol{\xi}} - \mathbb{B}^{\boldsymbol{\xi}} = o_p(1)$.

(2) $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} - \mathbb{B}^{\boldsymbol{\zeta}} = o_p(1)$.

(3) $\tilde{\mathbb{B}}^{\boldsymbol{\eta}} - \mathbb{B}^{\boldsymbol{\eta}} = o_p(1)$.

Proof of Theorem 3.5. By the representation (S.2.17) and Lemma S.2.16,

$$\begin{aligned}
&\sqrt{NT}\left[\tilde{\boldsymbol{\vartheta}} - \boldsymbol{J}\cdot\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right) - \boldsymbol{\vartheta}_0\right] \\
=&\boldsymbol{J}\cdot\left[\boldsymbol{D}^+(\boldsymbol{F}_0)\cdot\frac{1}{\sqrt{NT}}\sum_{i=1}^N \boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i + \sqrt{\frac{T}{N}}\left(\mathbb{B}^{\boldsymbol{\xi}} - \tilde{\mathbb{B}}^{\boldsymbol{\xi}}\right) + \sqrt{\frac{N}{T}}\left(\mathbb{B}^{\boldsymbol{\zeta}} - \tilde{\mathbb{B}}^{\boldsymbol{\zeta}}\right) + \sqrt{\frac{N}{T}}\left(\mathbb{B}^{\boldsymbol{\eta}} - \tilde{\mathbb{B}}^{\boldsymbol{\eta}}\right)\right] + o_p(1) \\
\xrightarrow{\mathbb{D}}&\mathcal{N}\left[\boldsymbol{0}_{(p-K_0)K_0},\ \left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\right].
\end{aligned}$$

Applying the delta method with function $g(\boldsymbol{\vartheta}) = \ell(\boldsymbol{P}_{\boldsymbol{\Theta}})$ to the above expression, we have

$$\sqrt{NT}\left[g\left(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{J}\cdot\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right)\right) - g(\boldsymbol{\vartheta}_0)\right] \xrightarrow{\mathbb{D}} \mathcal{N}\left(\boldsymbol{0}_{p^2},\ \boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0\right). \tag{S.2.19}$$

By Taylor expansion,

$$\begin{aligned}
g\left(\tilde{\boldsymbol{\vartheta}} - \boldsymbol{J}\cdot\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right)\right) =& g\left(\tilde{\boldsymbol{\vartheta}}\right) - \frac{\partial g^\top(\tilde{\boldsymbol{\vartheta}})}{\partial\boldsymbol{\vartheta}}\cdot\boldsymbol{J}\cdot\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right) + \boldsymbol{r}_g \\
=& \ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}}\right) - \tilde{\boldsymbol{A}}^\top\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right) + O_p\left(\frac{1}{m_{NT}^4}\right)
\end{aligned} \tag{S.2.20}$$

In the second step, $g\left(\tilde{\boldsymbol{\vartheta}}\right) = \ell\left(\boldsymbol{P}_{\tilde{\boldsymbol{\Theta}}}\right) = \ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}}\right)$, and the l-th element of $\boldsymbol{r}_g$ is

$$\begin{aligned}
r_{g,l} =& \frac{1}{2}\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right)^\top\cdot\boldsymbol{J}^\top\cdot\frac{\partial^2 g_l(\tilde{\boldsymbol{\vartheta}} - \alpha_l N^{-1}\tilde{\mathbb{B}}^{\boldsymbol{\xi}} - \alpha_l T^{-1}\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} - \alpha_l T^{-1}\tilde{\mathbb{B}}^{\boldsymbol{\eta}})}{\partial\boldsymbol{\vartheta}\partial^\top\boldsymbol{\vartheta}}\cdot\boldsymbol{J}\cdot\left(\frac{\tilde{\mathbb{B}}^{\boldsymbol{\xi}}}{N} + \frac{\tilde{\mathbb{B}}^{\boldsymbol{\zeta}} + \tilde{\mathbb{B}}^{\boldsymbol{\eta}}}{T}\right) \\
=& O_p\left(\frac{1}{m_{NT}^2}\right)\cdot O_p(1)\cdot O_p\left(\frac{1}{m_{NT}^2}\right) = O_p\left(\frac{1}{m_{NT}^4}\right),
\end{aligned}$$

with $0 \le \alpha_l \le 1$ for $l = 1, \ldots, p^2$. The result is showed by (S.2.19) and (S.2.20). □

We postpone the proof of Proposition 3.1 after that of Lemma S.2.16 in Section S.4 as they are closely related.

S.2.5 Proof of Theorem 3.6 in Section 3.5

Lemma S.2.17. *For a pair (K, R) where $K_0 \le K \le K_{max}$ and $R_0 \le R \le R_{max}$,*

$$f_{K,R}^* = \frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i + O_p\left(\frac{1}{m_{NT}^2}\right).$$

Lemma S.2.18. *For a pair (K, R) where $0 \le K < K_0$, there exists a $\tau_K > 0$ such that*

$$\operatorname{plim}\inf_{(N,T)\to\infty}\left(f^*_{K,R} - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\right) \ge \tau_K.$$

Lemma S.2.19. *For a pair (K, R) where $0 \le R < R_0$, there exists a $\tau_R > 0$ such that*

$$\operatorname{plim}\inf_{(N,T)\to\infty}\left(f^*_{K,R} - \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\right) \ge \tau_R.$$

Proof of Theorem 3.6. Since

$$PC_p(K,R) - PC_p(K_0,R_0) = f^*_{K,R} - f^*_{K_0,R_0} - (K_0 - K)g_{1,NT} - (R_0 - R)g_{2,NT},$$

it suffices to show that

$$\lim_{(N,T)\to\infty}\mathbb{P}\left[f^*_{K,R} - f^*_{K_0,R_0} \le (K_0 - K)g_{1,NT} + (R_0 - R)g_{2,NT}\right] = 0.$$

For any pair (K, R) such that $0 \le K < K_0$, Lemmas S.2.17 and S.2.18 imply that $f^*_{K,R} - f^*_{K_0,R_0}$ converges to a positive limit bounded below by some constant $\tau_K > 0$, while $g_{1,NT}, g_{2,NT} \to 0$ as $(N, T) \to \infty$. Therefore,

$$\lim_{(N,T)\to\infty}\mathbb{P}\left[PC_p(K,R) \le PC_p(K_0,R_0)\right] = 0.$$

The same argument applies to any pair (K, R) with $0 \le R < R_0$, using Lemmas S.2.17 and S.2.19.

Now consider the case where $K_0 \le K \le K_{\max}$, $R_0 \le R \le R_{\max}$, and $(K - K_0) + (R - R_0) \ge 1$. By Lemma S.2.17, $f^*_{K,R} - f^*_{K_0,R_0} = O_p\left(m_{NT}^{-2}\right)$, while $(K_0 - K)g_{1,NT} + (R_0 - R)g_{2,NT} \le -\min\left(g_{1,NT}, g_{2,NT}\right)$, which vanishes at a slower rate than $O_p\left(m_{NT}^{-2}\right)$. Hence,

$$\lim_{(N,T)\to\infty}\mathbb{P}\left[PC_p(K,R) \le PC_p(K_0,R_0)\right] = 0$$

also holds in this case. □

S.3 Examples for Assumption 5 and Proofs

We first introduce the following two types of random variables. A random variable ξ_1 is called sub-exponential if there exists a constant $c > 0$, such that

$$\mathbb{P}\left(|\xi_1| \ge t\right) \le 2\exp\left(-ct\right), \quad t \ge 0.$$

The sub-exponential norm of ξ_1 is defined as

$$\|\xi_1\|_{\psi_1} := \inf\left\{t > 0 : \mathbb{E}\left[\exp\left(\frac{|\xi_1|}{t}\right)\right] \le 2\right\}.$$

A random variable ξ_2 is sub-Gaussian if there exists some $c > 0$, such that

$$\mathbb{P}\left(|\xi_2| \ge t\right) \le 2\exp\left(-ct^2\right), \quad t \ge 0.$$

Its sub-Gaussian norm is given by

$$\|\xi_2\|_{\psi_2} := \inf\left\{t > 0 : \mathbb{E}\left[\exp\left(\frac{\xi_2^2}{t^2}\right)\right] \le 2\right\}.$$

Gaussian, Bernoulli and any bounded random variables are classical examples of sub-Gaussian distributions. Importantly, if ξ_2 is sub-Gaussian, then ξ_2^2 is sub-exponential and $\|\xi_2\|_{\psi_2}^2 = \left\|\xi_2^2\right\|_{\psi_1}$. See Vershynin (2020) for further properties of the sub-exponential and sub-Gaussian random variables.

We show that $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_1) = 1$ in Examples S.3.1 and S.3.2. We establish that $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_1') = 1$ specifically, where

$$\mathcal{S}_1' = \left\{C_8 \le \frac{\boldsymbol{X}_i^\top \boldsymbol{X}_i}{T} \le C_5, \quad i = 1, \ldots, N\right\}.$$

We omit the lower bound condition in $\mathcal{S}_1$ from Assumption 5(1), as it is implied by $\mathcal{S}_2$. However, this lower bound in $\mathcal{S}_1'$ will be used in Example S.3.3. The two examples considered below assume temporal independence of the data. Extending these results to dependent data structures, such as strongly mixing processes, martingale difference sequences, Markov chains, or m-dependent sequences, would be desirable; however, we do not pursue this direction further.

Example S.3.1. *We assume that:*

(c1) For each $i = 1, \ldots, N$, the sequence $\{\boldsymbol{x}_{it}\}_{t=1}^T$ is i.i.d.

(c2) There exist constants $0 < c_1 < c_2 < \infty$ such that

$$c_1 \le \min_{1\le i\le N} \lambda_{min}(\boldsymbol{\Sigma}_i) \le \max_{1\le i\le N} \lambda_{max}(\boldsymbol{\Sigma}_i) \le c_2,$$

where $\boldsymbol{\Sigma}_i := \mathbb{E}\left(\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\right)$.

(c3) There exists $\tau_1 \ge 1$ such that for any $\boldsymbol{u} \in \mathbb{R}^p$,

$$\|\boldsymbol{u}^\top \boldsymbol{x}_{it}\|_{\psi_2} \le \tau_1 \sqrt{\boldsymbol{u}^\top \boldsymbol{\Sigma}_i \boldsymbol{u}}.$$

Then, $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_1') = 1$.

Example S.3.2. *We retain conditions (c1) and (c2) from the previous example, and as an alternative to the sub-Gaussian assumption (c3), we impose the following condition:*

(c3') There exists $\tau_1 \ge 1$ such that

$$\|\boldsymbol{x}_{it}\|_2^2 \le \tau_1^2 \mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^2\right).$$

Here $\mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^2\right) = \mathrm{tr}(\boldsymbol{\Sigma}_i)$. Then, $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_1') = 1$.

Proof of Examples S.3.1 and S.3.2. Let $\hat{\boldsymbol{\Sigma}}_i = T^{-1}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^{\top}$. Applying the tail bound in Exercise 4.7.3 of Vershynin (2020), we obtain

$$\mathbb{P}\left[\left\|\hat{\boldsymbol{\Sigma}}_i - \boldsymbol{\Sigma}_i\right\|_2 \geq c_3\tau_1^2\left\|\boldsymbol{\Sigma}_i\right\|_2\left(\frac{p+d_1}{T}+\sqrt{\frac{p+d_1}{T}}\right)\right] \leq 2e^{-d_1}, \quad d_1 \geq 0,$$

or equivalently,

$$\mathbb{P}\left(\left\|\hat{\boldsymbol{\Sigma}}_i - \boldsymbol{\Sigma}_i\right\|_2 \geq 2c_3\tau_1^2\left\|\boldsymbol{\Sigma}_i\right\|_2 d_2\right) \leq 2\exp\left[-T\min\left(d_2, d_2^2\right)+p\right], \quad d_2 \geq 0.$$

Let

$$\begin{aligned}
&d_2 = \frac{\min_{1\leq j\leq N}\lambda_{min}(\boldsymbol{\Sigma}_j)}{4c_3\tau^2\max_{1\leq j\leq N}\lambda_{max}(\boldsymbol{\Sigma}_j)}, \quad d_3 = \min(d_2, d_2^2), \quad C_8 = \frac{c_1}{2}, \quad C_5 = \frac{3c_2}{2},\\
&\mathcal{S}_{1i(1)} = \left\{C_8 \leq \hat{\boldsymbol{\Sigma}}_i \leq C_5\right\}, \quad \mathcal{S}_{1i(2)} = \left\{\frac{\lambda_{min}(\boldsymbol{\Sigma}_i)}{2} \leq \hat{\boldsymbol{\Sigma}}_i \leq \lambda_{max}(\boldsymbol{\Sigma}_i) + \frac{\lambda_{min}(\boldsymbol{\Sigma}_i)}{2}\right\},\\
&\mathcal{S}_{1i(3)} = \left\{\left\|\hat{\boldsymbol{\Sigma}}_i - \boldsymbol{\Sigma}_i\right\|_2 \leq \frac{\lambda_{min}(\boldsymbol{\Sigma}_i)}{2}\right\}, \quad \mathcal{S}_{1i(4)} = \left\{\left\|\hat{\boldsymbol{\Sigma}}_i - \boldsymbol{\Sigma}_i\right\|_2 \leq \frac{\lambda_{max}(\boldsymbol{\Sigma}_i)\min_{1\leq j\leq N}\lambda_{min}(\boldsymbol{\Sigma}_j)}{2\max_{1\leq j\leq N}\lambda_{max}(\boldsymbol{\Sigma}_j)}\right\}.
\end{aligned}$$

If $T^{-1}\log N \to 0$, then

$$\mathbb{P}\left[(\mathcal{S}_1')^c\right] \leq \sum_{i=1}^{N}\mathbb{P}\left(\mathcal{S}_{1i(1)}^c\right) \leq \sum_{i=1}^{N}\mathbb{P}\left(\mathcal{S}_{1i(2)}^c\right) \leq \sum_{i=1}^{N}\mathbb{P}\left(\mathcal{S}_{1i(3)}^c\right) \leq \sum_{i=1}^{N}\mathbb{P}\left(\mathcal{S}_{1i(4)}^c\right) \leq 2N\exp\left(-Td_3+p\right) \to 0,$$

where we use the relationship $\mathcal{S}_1' = \cap_{1\leq i\leq N}\mathcal{S}_{1i(1)}$ and $\mathcal{S}_{1i(1)} \supseteq \mathcal{S}_{1i(2)} \supseteq \mathcal{S}_{1i(3)} \supseteq \mathcal{S}_{1i(4)}$. In Example S.3.2, the result can still be established by applying the tail bound in Exercise 5.6.4 of Vershynin (2020). □

Assumption 5(2) holds in the following example. To simplify notation, we consider $\boldsymbol{P}_{\boldsymbol{F}}$ with $\boldsymbol{F} \in \mathbb{R}^{T\times R_0}$ instead of $\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}$; otherwise, one can similarly analyze $\boldsymbol{P}_{\tilde{\boldsymbol{F}}}$ with $\tilde{\boldsymbol{F}} \in \mathbb{R}^{T\times(2R_0+p_2)}$.

Example S.3.3. *We assume that:*

(c1) $\{\boldsymbol{X}_i\}_{i=1}^N$ *are cross-sectionally independent.*

(c2) For each $i = 1,\ldots,N$, $C_8 \leq T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{X}_i^h \leq C_5$.

(c3) For each $i = 1,\ldots,N$, *for any* $\boldsymbol{u}_1 \in \mathbb{R}^T$ *and* $\boldsymbol{u}_2 \in \mathbb{R}^{p_1}$,

$$\left\|\boldsymbol{u}_1^{\top}\boldsymbol{X}_i^h\boldsymbol{u}_2\right\|_{\psi_2} \leq \tau_2\left\|\boldsymbol{u}_1\right\|_2\left\|\boldsymbol{u}_2\right\|_2.$$

Then for any matrix $\boldsymbol{F} \in \mathbb{R}^{T\times R_0}$, *there exists an index set* $\mathcal{I}_{\boldsymbol{F}} \subset [N]$ *satisfying* $|\mathcal{I}_{\boldsymbol{F}}^c| \leq \phi$, *where* ϕ *is specified in* (S.3.1), *such that for* $i \in \mathcal{I}_{\boldsymbol{F}}$,

$$\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}}^{\perp}\boldsymbol{X}_i^h}{T} \geq \frac{C_8}{4}.$$

Condition (c2) is the event $\mathcal{S}_1'$ from previous two examples.

Proof of Example S.3.3. Let $\mathcal{K} = \left\{\boldsymbol{P}_{\boldsymbol{F}} : \boldsymbol{F} \in \mathbb{R}^{T\times R_0}\right\}$. Define the metric on $\mathcal{K}$ by $\rho\left(\boldsymbol{P}_{\boldsymbol{F}^{(1)}}, \boldsymbol{P}_{\boldsymbol{F}^{(2)}}\right) = \left\|\boldsymbol{P}_{\boldsymbol{F}^{(1)}} - \boldsymbol{P}_{\boldsymbol{F}^{(2)}}\right\|_f$. Denote by $D = D\left(\mathcal{K}, \epsilon\right)$ the ϵ-covering number of $\mathcal{K}$, that is, the minimal number of balls $\{B_d\}_{d=1}^{D}$ of radius $\epsilon = \frac{C_8}{4C_5}$ with center $\boldsymbol{P}_{\boldsymbol{F}^{(d)}}$ whose union contains $\mathcal{K}$. Applying Lemma 1 in Cai et al. (2021), we obtain

$$D\left(\mathcal{K}, \epsilon\right) \le \left(\frac{c_1}{\epsilon}\right)^{R_0(T-R_0)}.$$

Let

$$\begin{aligned}
\mathcal{S}_d^{\phi} &= \left\{\lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}^{\perp}\boldsymbol{X}_i^h}{T}\right) \le \frac{C_8}{2} \text{ for } i \in \mathcal{J}_{\boldsymbol{F}^{(d)}}^c, \quad \left|\mathcal{J}_{\boldsymbol{F}^{(d)}}^c\right| = \phi\right\},\\
\tilde{\mathcal{S}}_d^{\phi} &= \left\{\lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}^{\perp}\boldsymbol{X}_i^h}{T}\right) \le \frac{C_8}{2} \text{ for } i \in \mathcal{J}_{\boldsymbol{F}^{(d)}}^c, \quad \left|\mathcal{J}_{\boldsymbol{F}^{(d)}}^c\right| \ge \phi\right\},\\
\tilde{\mathcal{S}}^{\phi} &= \bigcup_{1\le d\le D} \tilde{\mathcal{S}}_d^{\phi}.
\end{aligned}$$

Since for any matrix $\boldsymbol{F} \in \mathbb{R}^{T\times R_0}$, there exists one $\boldsymbol{F}^{(d)}$ such that $\left\|\boldsymbol{P}_{\boldsymbol{F}} - \boldsymbol{P}_{\boldsymbol{F}^{(d)}}\right\|_f \le \epsilon$, and for $i \in \mathcal{J}_{\boldsymbol{F}^{(d)}}$,

$$\lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}}^{\perp}\boldsymbol{X}_i^h}{T}\right) \ge \lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}^{\perp}\boldsymbol{X}_i^h}{T}\right) - \left\|\frac{\boldsymbol{X}_i^{h\top}(\boldsymbol{P}_{\boldsymbol{F}^{(d)}} - \boldsymbol{P}_{\boldsymbol{F}})\boldsymbol{X}_i^h}{T}\right\|_2 \ge \frac{C_8}{2} - C_5\epsilon = \frac{C_8}{4},$$

the complement of $\tilde{\mathcal{S}}^{\phi}$ is subset of the event in Assumption 5(2) by setting $C_7 = C_8/4$ and $\left|\mathcal{I}_{\boldsymbol{F}}^c\right| \le \left|\mathcal{J}_{\boldsymbol{F}^{(d)}}^c\right| < \phi$. It remains to show $\lim_{(N,T)\to\infty} \mathbb{P}\left(\tilde{\mathcal{S}}^{\phi}\right) = 0$. For $i \in \mathcal{J}_{\boldsymbol{F}^{(d)}}^c$,

$$\lambda_{max}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}\boldsymbol{X}_i^h}{T}\right) \ge \lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{X}_i^h}{T}\right) - \lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}^{\perp}\boldsymbol{X}_i^h}{T}\right) \ge \frac{C_8}{2}.$$

Let

$$\tilde{\boldsymbol{F}}^{(d)} = \frac{\boldsymbol{F}^{(d)}}{\sqrt{T}}\left(\frac{\boldsymbol{F}^{(d)\top}\boldsymbol{F}^{(d)}}{T}\right)^{-1/2} = \left(\tilde{\boldsymbol{f}}_{\cdot 1}^{(d)}, \quad \ldots, \quad \tilde{\boldsymbol{f}}_{\cdot R_0}^{(d)}\right).$$

For any $\boldsymbol{u} \in \mathbb{R}^{p_1}$,

$$\begin{aligned}
\left\|\boldsymbol{u}^{\top}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}\boldsymbol{X}_i^h\boldsymbol{u}\right\|_{\psi_1} &= \left\|\sum_{r=1}^{R_0}\left(\boldsymbol{u}^{\top}\boldsymbol{X}_i^{h\top}\tilde{\boldsymbol{f}}_{\cdot r}^{(d)}\right)^2\right\|_{\psi_1}\\
&\le \sum_{r=1}^{R_0}\left\|\tilde{\boldsymbol{f}}_{\cdot r}^{(d)\top}\boldsymbol{X}_i^h\boldsymbol{u}\right\|_{\psi_2}^2\\
&\le \tau_2^2\|\boldsymbol{u}\|_2^2 \cdot \sum_{r=1}^{R_0}\left\|\tilde{\boldsymbol{f}}_{\cdot r}^{(d)}\right\|_2^2\\
&= \tau_2^2 R_0\|\boldsymbol{u}\|_2^2.
\end{aligned}$$

Let $\boldsymbol{e}_j \in \mathbb{R}^d$ denote the unit vector with 1 in j–th position and 0 elsewhere. Since

$$\begin{aligned}
\left\|\lambda_{max}\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}\boldsymbol{X}_i^h\right)\right\|_{\psi_1} &\le \left\|\mathrm{tr}\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}\boldsymbol{X}_i^h\right)\right\|_{\psi_1}\\
&= \left\|\sum_{j=1}^{p_1}\boldsymbol{e}_j^{\top}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{F}^{(d)}}\boldsymbol{X}_i^h\boldsymbol{e}_j\right\|_{\psi_1}
\end{aligned}$$

$$
\begin{aligned}
&\leq \sum_{j=1}^{p_1} \left\| \boldsymbol{e}_j^\top \boldsymbol{X}_i^{h\top} \boldsymbol{P}_{\boldsymbol{F}^{(d)}} \boldsymbol{X}_i^h \boldsymbol{e}_j \right\|_{\psi_1} \\
&= \tau_2^2 p_1 R_0,
\end{aligned}
$$

it follows that

$$
\begin{aligned}
\mathbb{P}\left[\lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top} \boldsymbol{P}_{\boldsymbol{F}^{(d)}}^{\perp} \boldsymbol{X}_i^h}{T}\right) \leq \frac{C_8}{2}\right] &\leq \mathbb{P}\left[\lambda_{max}\left(\frac{\boldsymbol{X}_i^{h\top} \boldsymbol{P}_{\boldsymbol{F}^{(d)}} \boldsymbol{X}_i^h}{T}\right) \geq \frac{C_8}{2}\right] \\
&\leq \exp\left(-\frac{c_2 C_8 T}{\tau_2^2 p_1 R_0}\right) =: \kappa.
\end{aligned}
$$

Let

$$
\phi = \left\lceil \frac{2\tau_2^2 p_1 R_0^2 \log(4c_1 C_5 / C_8)}{c_2 C_8} \right\rceil. \tag{S.3.1}
$$

If $\log N/T \to 0$, then

$$
\begin{aligned}
\mathbb{P}\left(\tilde{\mathcal{S}}^{\phi}\right) &\leq \sum_{d=1}^{D} \mathbb{P}\left(\tilde{\mathcal{S}}_d^{\phi}\right) \\
&= \sum_{d=1}^{D} \sum_{n=\phi}^{N} \mathbb{P}\left(\mathcal{S}_d^n\right) \\
&\leq D \sum_{n=\phi}^{N} C_N^n \kappa^n \\
&\leq D C_N^{\phi} \kappa^{\phi} \sum_{n=0}^{N-\phi} \left(\frac{N\kappa}{\phi}\right)^n \\
&\leq \frac{2}{\phi!} \exp\left[R_0(T - R_0) \log\left(\frac{4c_1 C_5}{C_8}\right) - \frac{c_2 C_8 T \phi}{\tau_2^2 p_1 R_0} + \phi \log N\right] \\
&\leq \frac{2}{\phi!} \exp\left[-R_0 \log\left(\frac{4c_1 C_5}{C_8}\right) T + \phi \log N - R_0^2 \log\left(\frac{4c_1 C_5}{C_8}\right)\right] \to 0,
\end{aligned}
$$

where the fourth step follows from $\frac{C_N^{n+1}}{C_N^n} = \frac{N-n}{n+1} \leq \frac{N}{\phi}$ for $n \geq \phi$, and the fifth step follows from $\sum_{n=0}^{N-\phi} \left(\frac{N\kappa}{\phi}\right)^n \leq \frac{1}{1 - N\kappa/\phi} \leq 2$ and $C_N^{\phi} \leq \frac{N^{\phi}}{\phi!}$. □

The following example provides a uniform lower bound for $\left\{T^{-1} \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}_0}^{\perp} \boldsymbol{X}_i\right\}_{i=1}^N$ in Assumption 5(1).

Example S.3.4. *We assume that:*

(c1) For each $i = 1, \ldots, N$, $T^{-1} \boldsymbol{X}_i^\top \boldsymbol{X}_i \leq C_5$.

(c2) For each $i = 1, \ldots, N$, $T^{-1} \boldsymbol{X}_i^{h\top} \boldsymbol{P}_{(\boldsymbol{F}_0, \boldsymbol{G})}^{\perp} \boldsymbol{X}_i^h \geq C_8$.

(c3) For each $i = 1, \ldots, N$, $T^{-1} \boldsymbol{X}_i^{l\top} \boldsymbol{X}_i^l \geq C_9$.

Then, $\lim_{(N,T)\to\infty} \mathbb{P}\left(\mathcal{S}_2\right) = 1$.

Conditions (c1) and (c2) are from Assumptions 5(1) and 5(2), respectively. Condition (c2) holds with probability approaching one under the same setting in Example S.3.3. If there are no low-rank predictors, that is, $p_2 = 0$, the event $\mathcal{S}_2$ is directly implied by (c2). Condition (c3) holds if $\boldsymbol{G}^\top \boldsymbol{G}$ is positive definite and all elements of $\{\boldsymbol{\gamma}_j\}_{j=p_1+1}^{p}$ are uniformly bounded away from 0. The proof is omitted here; see Part (I) of the proof of Theorem 3.1 for a similar argument establishing the positive definiteness of $T^{-1}\boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}})}\boldsymbol{X}_i$.

S.4 Proofs of Lemmas

S.4.1 Proofs of Lemmas in Section S.2.3

Proof of Lemma S.2.1.

(1) The equality follows from

$$\begin{aligned}\boldsymbol{\Xi}^{-1} - \boldsymbol{\Xi}^{-1}\cdot\bar{\boldsymbol{H}}\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T =& \boldsymbol{\Xi}^{-1}\cdot\left(\boldsymbol{I}_{TR_0} - \bar{\boldsymbol{H}}\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\right)\\ =&\boldsymbol{\Xi}^{-1}\cdot\left(\boldsymbol{\Sigma}_{\boldsymbol{L}}\otimes\boldsymbol{I}_T - \bar{\boldsymbol{H}}\right)\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\\ =&\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T,\end{aligned}$$

and it is straightforward to verify that $\boldsymbol{\Xi}^{-1}\cdot\bar{\boldsymbol{H}}\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T = \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T\cdot\left(\boldsymbol{I}_{TR_0} - \boldsymbol{\Upsilon}\right)^{-1}\boldsymbol{\Upsilon}\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T$.

(2) Suppose that the matrix $\boldsymbol{\Sigma}_{\boldsymbol{L}}$ has the spectral decomposition $\boldsymbol{V}_{\boldsymbol{L}}\boldsymbol{\Lambda}_{\boldsymbol{L}}\boldsymbol{V}_{\boldsymbol{L}}^\top$, where $\boldsymbol{V}_{\boldsymbol{L}} = \left(\boldsymbol{v}_{\boldsymbol{L}_1},\quad \boldsymbol{v}_{\boldsymbol{L}_2},\quad \ldots,\quad \boldsymbol{v}_{\boldsymbol{L}_{R_0}}\right)$ and $\boldsymbol{\Lambda}_{\boldsymbol{L}} = \mathrm{Diag}\left(\lambda_{\boldsymbol{L}_1},\quad \lambda_{\boldsymbol{L}_2},\quad \ldots,\quad \lambda_{\boldsymbol{L}_{R_0}}\right)$ with $\lambda_{\boldsymbol{L}_1}\geq\lambda_{\boldsymbol{L}_2}\geq\cdots\geq\lambda_{\boldsymbol{L}_{R_0}}>0$. Then, $\boldsymbol{\Xi}\cdot\boldsymbol{V}_{\boldsymbol{L}}\otimes\boldsymbol{P}_{\boldsymbol{F}} = \boldsymbol{V}_{\boldsymbol{L}}\boldsymbol{\Lambda}_{\boldsymbol{L}}\otimes\boldsymbol{P}_{\boldsymbol{F}}$, or equivalently, $\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{V}_{\boldsymbol{L}}\otimes\boldsymbol{P}_{\boldsymbol{F}} = \boldsymbol{V}_{\boldsymbol{L}}\boldsymbol{\Lambda}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{P}_{\boldsymbol{F}}$. It follows that

$$\begin{aligned}\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}\cdot\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}^\perp =& \boldsymbol{V}_{\boldsymbol{L}}\otimes\boldsymbol{I}_T\cdot\boldsymbol{V}_{\boldsymbol{L}}^\top\otimes\boldsymbol{P}_{\boldsymbol{F}}\cdot\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}^\perp\\ =&\boldsymbol{V}_{\boldsymbol{L}}\otimes\boldsymbol{I}_T\cdot\boldsymbol{\Lambda}_{\boldsymbol{L}}^{-1}\boldsymbol{V}_{\boldsymbol{L}}^\top\otimes\boldsymbol{P}_{\boldsymbol{F}}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{F}}^\perp\\ =&\boldsymbol{0}_{TR_0\times TR_0}.\end{aligned}$$

In the same way, we can show that $\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{G}}\cdot\boldsymbol{\Xi}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\boldsymbol{P}_{\boldsymbol{G}}^\perp = \boldsymbol{0}_{TR_0\times TR_0}$.

(3) We first show that, for a matrix $\boldsymbol{M}\in\mathbb{R}^{T\times R_0}$ with $\mathrm{Rank}(\boldsymbol{M}) = R_0$, there exists some $c\geq 0$,

$$\boldsymbol{M}^\top\left(\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{H}_{i,0}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\boldsymbol{M}\geq c_1\boldsymbol{M}^\top\boldsymbol{M},\quad \text{for } i\in\mathcal{I}_{\boldsymbol{M}}, \tag{S.4.1}$$

which is equivalent to

$$\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1/2}\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp(\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{M}}\boldsymbol{P}_{\boldsymbol{G}}^\perp)\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1/2}\geq c_1\boldsymbol{I}_{K_0},\quad \text{for } i\in\mathcal{I}_{\boldsymbol{M}}.$$

Note that, for $i\in\mathcal{I}_{\boldsymbol{M}}$,

$$\lambda_{min}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp(\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{M}}\boldsymbol{P}_{\boldsymbol{G}}^\perp)\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)$$

$$
\begin{aligned}
&\geq \lambda_{min}\left(\frac{\boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{M})}\boldsymbol{X}_i}{T}\right)\cdot \lambda_{min}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0\right)\\
&\geq \frac{\det(T^{-1}\boldsymbol{X}_i^{l\top}\boldsymbol{X}_i^l)\cdot \det(T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{M},\boldsymbol{X}_i^l)}\boldsymbol{X}_i^h)}{\lambda_{max}^{p-1}(T^{-1}\boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{M})}\boldsymbol{X}_i)}\cdot \lambda_{min}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0\right)\\
&\geq \frac{C_6^{p_2}C_7^{p_1}}{C_5^{p-1}}\cdot \lambda_{min}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0\right).
\end{aligned}
$$

Therefore, for $i\in\mathcal{I}_{\boldsymbol{M}}$,

$$
\begin{aligned}
&\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{\boldsymbol{F}_0}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1/2}\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{\boldsymbol{F}_0}(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{M}}\boldsymbol{P}_{\boldsymbol{G}}^\perp)\boldsymbol{P}^\perp_{\boldsymbol{F}_0}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{\boldsymbol{F}_0}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1/2}\\
&\geq \lambda_{max}^{-1}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{\boldsymbol{F}_0}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)\cdot\lambda_{min}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{\boldsymbol{F}_0}(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{M}}\boldsymbol{P}_{\boldsymbol{G}}^\perp)\boldsymbol{P}^\perp_{\boldsymbol{F}_0}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)\cdot\boldsymbol{I}_{K_0}\\
&\geq \frac{C_6^{p_2}C_7^{p_1}}{C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}\cdot\boldsymbol{I}_{K_0}.
\end{aligned}
$$

If $\mathrm{Rank}(\boldsymbol{M})=R_1<R_0$, let $\boldsymbol{M}=\boldsymbol{M}'\times\boldsymbol{O}_{\boldsymbol{M}}$, where $\boldsymbol{M}'\in\mathbb{R}^{T\times R_1}$ and $\boldsymbol{O}_{\boldsymbol{M}}\in\mathbb{R}^{R_1\times R_0}$. Then, it is straightforward to show that

$$
\boldsymbol{M}'^\top\left(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{H}_{i,0}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\boldsymbol{M}'-\frac{C_6^{p_2}C_7^{p_1}}{C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}\cdot\boldsymbol{M}'^\top\boldsymbol{M}'\geq 0,\quad \text{for } i\in\mathcal{I}_{\boldsymbol{M}}.
$$

By left multiplying $\boldsymbol{O}_{\boldsymbol{M}}^\top$ and right multiplying $\boldsymbol{O}_{\boldsymbol{M}}$ on both sides of the inequality, (S.4.1) still holds with $c_1=(C_6^{p_2}C_7^{p_1}/C_5^p)\cdot[\lambda_{min}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0\right)/\lambda_{max}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0\right)]$. Then, for any $\boldsymbol{e}\in\mathbb{R}^{TR_0}$ with $\|\boldsymbol{e}\|_2=1$ and corresponding $\boldsymbol{M}_{\boldsymbol{e}}=\mathcal{M}_{T,R_0}(\boldsymbol{e})$, we have

$$
\begin{aligned}
\boldsymbol{e}^\top\boldsymbol{\Xi}_0\boldsymbol{e}=&\,\mathrm{tr}\left[\frac{1}{N}\sum_{i=1}^N\boldsymbol{M}_{\boldsymbol{e}}^\top\left(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{H}_{i,0}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right)\boldsymbol{M}_{\boldsymbol{e}}\cdot\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\right]\\
\geq&\,\frac{C_6^{p_2}C_7^{p_1}}{C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}\cdot\mathrm{tr}\left[\boldsymbol{M}_{\boldsymbol{e}}^\top\boldsymbol{M}_{\boldsymbol{e}}\cdot\frac{1}{N}\sum_{i\in\mathcal{I}_{\boldsymbol{M}_{\boldsymbol{e}}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\right]\\
\geq&\,\frac{C_6^{p_2}C_7^{p_1}}{C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}\cdot\lambda_{min}\left(\frac{1}{N}\sum_{i\in\mathcal{I}_{\boldsymbol{M}_{\boldsymbol{e}}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\right)\cdot\mathrm{tr}\left(\boldsymbol{M}_{\boldsymbol{e}}^\top\boldsymbol{M}_{\boldsymbol{e}}\right)\\
\geq&\,\frac{C_3C_6^{p_2}C_7^{p_1}}{2C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)},
\end{aligned}
$$

which shows that $\boldsymbol{\Xi}_0$ is strictly positive definite. On the other hand, we have

$$
\begin{aligned}
\boldsymbol{e}^\top\boldsymbol{\Upsilon}_0\boldsymbol{e}=&\,1-\boldsymbol{e}^\top\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)\boldsymbol{e}\\
=&\,1-\boldsymbol{e}^\top\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T\cdot\boldsymbol{\Xi}_0\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{I}_T\cdot\boldsymbol{e}\\
\leq&\,1-\frac{\lambda_{min}(\boldsymbol{\Xi}_0)}{\lambda_{max}(\boldsymbol{\Sigma}_{\boldsymbol{L}})}=1-\frac{C_3C_6^{p_2}C_7^{p_1}}{2C_4C_5^{p}}\cdot\frac{\lambda_{min}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}{\lambda_{max}(\boldsymbol{\Theta}_0^\top\boldsymbol{\Theta}_0)}.
\end{aligned}
$$

This completes the proof.

(4) Since $\|\boldsymbol{\Upsilon}_0\|_2\leq a_1$ w.p.a. 1, and $\|\boldsymbol{\Upsilon}_0\|_2\leq 1$, Jensen's inequality implies that $\|\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\|_2\leq\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{\Upsilon}_0\|_2\right)\leq$

$a_2 < 1$. For notational convenience, denote $\bar{\boldsymbol{\Upsilon}}_0 := \mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_0)$ and define $\Delta := \boldsymbol{\Upsilon}_0 - \bar{\boldsymbol{\Upsilon}}_0$. Consider the following decomposition:

$$\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^D\right) = \mathbb{E}_{\mathcal{C}}\left[\left(\bar{\boldsymbol{\Upsilon}}_0 + \Delta\right)^D\right] = \bar{\boldsymbol{\Upsilon}}_0^D + \sum_{d=1}^{D} \sum_{i_1+\cdots+i_{d+1}=D-d} \bar{\boldsymbol{\Upsilon}}_0^{i_1} \Delta \bar{\boldsymbol{\Upsilon}}_0^{i_2} \Delta \cdots \Delta \bar{\boldsymbol{\Upsilon}}_0^{i_{d+1}}.$$

For the leading term, we have $\left\|\bar{\boldsymbol{\Upsilon}}_0^D\right\|_2 \le \left\|\bar{\boldsymbol{\Upsilon}}_0\right\|_2^D \le a_2^D$. By the matrix Bernstein inequality together with standard tail-to-moment arguments (see Tropp et al., 2015; Vershynin, 2020), for any fixed integer $d \ge 1$,

$$\left[\mathbb{E}_{\mathcal{C}}\left(\|\Delta\|_2^d\right)\right]^{1/d} \lesssim \sqrt{\frac{c_1\left(d + \log T\right)}{N}} + \frac{c_2(d + \log T)}{N},$$

where

$$c_1 = \frac{1}{N}\left\|\sum_{i=1}^{N} \mathbb{E}\left\{\left[\tilde{\boldsymbol{l}}_{i,0}\tilde{\boldsymbol{l}}_{i,0}^\top \otimes \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{H}_{i,0} - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{H}_{i,0}\right)\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right]^2\right\}\right\|_2, \quad c_2 = \max_i \left\|\tilde{\boldsymbol{l}}_{i,0}\tilde{\boldsymbol{l}}_{i,0}^\top \otimes \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{H}_{i,0} - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{H}_{i,0}\right)\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right\|_2,$$

and both c_1 and c_2 are bounded by finite constants. Under the condition $D = o\left(N^{1/3}\right)$ and $\log T = o(N^{1/3})$, we have $\left[\mathbb{E}_{\mathcal{C}}\left(\|\Delta\|_2^d\right)\right]^{1/d} \lesssim \sqrt{c_3\left(D + \log T\right)/N}$. Therefore,

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{d=1}^{D} \sum_{i_1+\cdots+i_{d+1}=D-d} \bar{\boldsymbol{\Upsilon}}_0^{i_1} \Delta \bar{\boldsymbol{\Upsilon}}_0^{i_2} \Delta \cdots \Delta \bar{\boldsymbol{\Upsilon}}_0^{i_{d+1}}\right\|_2\right) \\
\le& \sum_{d=1}^{D} \sum_{i_1+\cdots+i_{d+1}=D-d} \left[\mathbb{E}_{\mathcal{C}}\left(\|\Delta\|_2^d\right) \cdot \left\|\bar{\boldsymbol{\Upsilon}}_0\right\|_2^{D-d}\right] \\
\le& \sum_{d=1}^{D} \left\{D^d \cdot \left[\frac{c_3\left(D + \log T\right)}{N}\right]^{d/2} \cdot a_2^{D-d}\right\} \\
\le& a_2^D \cdot \sum_{d=1}^{D} \left[\frac{c_3(D^3 + D^2 \log T)}{a_2^2 N}\right]^{d/2} \le c_4 \cdot a_2^D,
\end{aligned}$$

where $c_3(D^3 + D^2 \log T)/(a_2^2 N) = o(1)$. It follows that $\left\|\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^D\right)\right\|_2 \le (1 + c_4)a_2^D$. □

The following technical lemma is crucial. For the sake of the narrative flow, we postpone proofs of all auxiliary lemmas in Section S.4 at the end of this section.

Lemma S.4.1. *Suppose Assumptions 2, 5, and 6 hold. Then:*

(1) $\dfrac{c_1\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}} \le \left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}} - \boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f \le \dfrac{c_2\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}.$

(2a) $\max_i \left\|\hat{\boldsymbol{W}}_i - \boldsymbol{W}_{i,0}\right\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right).$

(2b) $\max_i \left\|\hat{\boldsymbol{V}}_i - \boldsymbol{V}_{i,0}\right\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right).$

(2c) $\max_i \left\|\hat{\boldsymbol{\Phi}}_{1i} - \boldsymbol{\Phi}_{1i,0}\right\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\dfrac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right).$

(2d) $\max_i \left\|\hat{\boldsymbol{\Phi}}_{4i} - \boldsymbol{\Phi}_{4i,0}\right\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right)$.

(2e) $\max_i \left\|\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{S}_{i,0}\right\|_f = O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right)$.

(2f) $\frac{1}{\sqrt{T}}\max_i \left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{Q}_{i,0}\right\|_f = O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right)$.

(2g) $\left\|\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right) - \boldsymbol{\Xi}_0^{-1}\right\|_2 = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}}\right)$.

(3a) $\mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\right\|_f^4\right) \le c_3T^2$.

(3b) $\mathbb{E}\left(\left\|\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\right)^\top\boldsymbol{F}_0\right\|_f^4\right) \le c_3T^2$.

(3c) $\mathbb{E}\left(\left\|\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\right)^\top\boldsymbol{G}\right\|_f^4\right) \le c_3T^2$.

(3d) $\mathbb{E}\left(\left\|\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\right)^\top\boldsymbol{X}_j\right\|_f^4\right) \le c_3T^2$ *for any* $i \ne j$.

(3e) $\mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^4\right) \le c_3T^2$.

(3f) $\mathbb{E}\left(\left\|\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right\|_2^4\right) \le c_3T^2$.

In Lemma S.4.1, statement (1) shows the equivalence between $\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}} - \boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f$ and $T^{-1/2}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_f$. Statements (2a)–(2g) control the deviations of estimated matrices from their true counterparts. Statements (3a)–(3f) establish bounds on the fourth moments of different terms, using moment inequalities for sum of strong mixing processes (Rio et al., 2017) or the Burkholder-Davis-Gundy (BDG) inequality for martingale difference sequences (Davis and Song, 2011).

Proof of Lemma S.2.2. For simplicity, we ignore the sign of $\boldsymbol{\delta}_k$ whenever only the consistency rate is of interest.

(1) Take $\boldsymbol{\delta}_1$ for example. Decompose $\boldsymbol{\delta}_1$:

$$\begin{aligned}
\boldsymbol{\delta}_1 =& \frac{1}{NT}\sum_{i=1}^N \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{F}} \\
&- \frac{1}{NT}\sum_{i=1}^N \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{F}} \\
&- \frac{1}{NT}\sum_{i=1}^N \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}} \\
&+ \frac{1}{NT}\sum_{i=1}^N \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{V}}_i\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top\left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_i^\top\hat{\boldsymbol{V}}_i^\top\hat{\boldsymbol{F}} \\
=:& \boldsymbol{\delta}_{1.1} + \boldsymbol{\delta}_{1.2} + \boldsymbol{\delta}_{1.3} + \boldsymbol{\delta}_{1.4}.
\end{aligned}$$

For $\boldsymbol{\delta}_{1.1}$,

$$\frac{\|\boldsymbol{\delta}_{1.1}\|_f^2}{T} \le \left\|\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right\|_2^4 \cdot \frac{\|\hat{\boldsymbol{F}}\|_f^2}{T} \cdot \frac{1}{NT^2} \sum_{i=1}^N \left(\|\boldsymbol{X}_i\|_2^4 \cdot \|\boldsymbol{b}_{i,0}\|_2^4\right) = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^4\right).$$

For $\boldsymbol{\delta}_{1.2}$,

$$\begin{aligned}\frac{\|\boldsymbol{\delta}_{1.2}\|_f^2}{T} \le& \frac{1}{NT^3} \sum_{i=1}^N \left\|\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\hat{\boldsymbol{V}}_i - \boldsymbol{V}_{i,0} + \boldsymbol{V}_{i,0}\right) \boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right) \boldsymbol{b}_{i,0} \boldsymbol{b}_{i,0}^\top (\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}})^\top \boldsymbol{X}_i^\top \hat{\boldsymbol{F}}\right\|_f^2 \\ \lesssim& \left\|\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right\|_2^4 \cdot \left(\max_i \left\|\hat{\boldsymbol{V}}_i - \boldsymbol{V}_{i,0}\right\|_2^2 + \max_i \|\boldsymbol{V}_{i,0}\|_2^2\right) \cdot \frac{1}{NT^2} \sum_{i=1}^N \left(\|\boldsymbol{X}_i\|_2^4 \cdot \|\boldsymbol{b}_{i,0}\|_2^4\right) \\ =& O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^4\right),\end{aligned}$$

where the last step uses Lemma S.4.1 (2b). By a similar argument for $\boldsymbol{\delta}_{1.2}$, both $T^{-1}\|\boldsymbol{\delta}_{1.3}\|_f^2$ and $T^{-1}\|\boldsymbol{\delta}_{1.4}\|_f^2$ are $O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^4\right)$. Similarly, we can show that $T^{-1}\|\boldsymbol{\delta}_k\|_f^2 = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right)$ for $2 \le k \le 5$.

(2) For $k = 6, 7$, we have

$$\begin{aligned}\frac{\|\boldsymbol{\delta}_6\|_f^2}{T} \le& \frac{1}{N} \cdot \frac{\|\boldsymbol{F}_0\|_f^2}{T} \cdot \left\|\frac{1}{\sqrt{NT}} \sum_{i=1}^N \boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top\right\|_2^2 \cdot \frac{\|\hat{\boldsymbol{F}}\|_2^2}{T} = O_p\left(\frac{1}{N}\right), \\ \frac{\|\boldsymbol{\delta}_7\|_f^2}{T} \le& \frac{1}{N} \cdot \left\|\frac{1}{\sqrt{NT}} \sum_{i=1}^N \boldsymbol{\varepsilon}_i \boldsymbol{l}_{i,0}^\top\right\|_2^2 \cdot \frac{\|\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}\|_2^2}{T^2} = O_p\left(\frac{1}{N}\right).\end{aligned}$$

Decompose $\boldsymbol{\delta}_{10}$:

$$\boldsymbol{\delta}_{10} = \frac{1}{NT} \sum_{i=1}^N \boldsymbol{F}_0 \boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{W}}_i \hat{\boldsymbol{F}} - \frac{1}{NT^2} \sum_{i=1}^N \boldsymbol{F}_0 \boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{F}} \hat{\boldsymbol{F}}^\top \hat{\boldsymbol{W}}_i \hat{\boldsymbol{F}} =: \boldsymbol{\delta}_{10.1} + \boldsymbol{\delta}_{10.2}.$$

For $\boldsymbol{\delta}_{10.1}$, we have

$$\frac{\|\boldsymbol{\delta}_{10.1}\|_f^2}{T} \le \frac{1}{T} \cdot \frac{\|\boldsymbol{F}_0\|_f^2}{T} \cdot \frac{1}{NT} \sum_{i=1}^N \left\|\boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top \boldsymbol{X}_i\right\|_2^2 \cdot \frac{1}{N} \sum_{i=1}^N \left\|\boldsymbol{\Phi}_{4i}^\top \left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right)\right\|_2^2 \cdot \frac{\|\hat{\boldsymbol{F}}\|_f^2}{T} = O_p\left(\frac{1}{T}\right).$$

For $\boldsymbol{\delta}_{10.2}$, we have

$$\frac{\|\boldsymbol{\delta}_{10.2}\|_f^2}{T} \lesssim \frac{1}{NT^2} \sum_{i=1}^N \left\|\boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{F}}\right\|_2^2 \cdot \frac{1}{NT^2} \sum_{i=1}^N \left\|\hat{\boldsymbol{F}}^\top \hat{\boldsymbol{W}}_i \hat{\boldsymbol{F}}\right\|_2^2 = O_p\left(\frac{1}{m_{NT}^2}\right),$$

where

$$\frac{1}{NT^2} \sum_{i=1}^N \left\|\boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{F}}\right\|_2^2 = \operatorname{tr}\left(\frac{\hat{\boldsymbol{F}}^\top}{\sqrt{T}} \cdot \frac{\sum_{i=1}^N \boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_i^\top}{NT} \cdot \frac{\hat{\boldsymbol{F}}}{\sqrt{T}}\right) \le \frac{R_0 \|\boldsymbol{\mathcal{E}}\|_2^2}{NT} = O_p\left(\frac{1}{m_{NT}^2}\right).$$

It follows that $T^{-1}\|\boldsymbol{\delta}_{10}\|_f^2 = O_p\left(m_{NT}^{-2}\right)$. For $\boldsymbol{\delta}_{12}$, we have

$$\frac{\|\boldsymbol{\delta}_{12}\|_f^2}{T} \lesssim \frac{1}{NT} \sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i \boldsymbol{\varepsilon}_i \boldsymbol{l}_{i,0}^\top\right\|_2^2 \cdot \frac{\|\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}\|_f^2}{T^2} = O_p\left(\frac{1}{m_{NT}^2}\right).$$

For $\boldsymbol{\delta}_{16}$, we have

$$\frac{\|\boldsymbol{\delta}_{16}\|_f^2}{T} \lesssim \frac{1}{NT} \sum_{i=1}^N \|\boldsymbol{\varepsilon}_i\|_2^2 \cdot \frac{1}{NT} \sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i \boldsymbol{\varepsilon}_i\right\|_2^2 = O_p\left(\frac{1}{m_{NT}^2}\right).$$

(3) For $\boldsymbol{\delta}_8$, we have

$$\frac{\|\boldsymbol{\delta}_8\|_f^2}{T} \leq \frac{\|\boldsymbol{\mathcal{E}}\|_2^4}{N^2T^2} \cdot \frac{\|\hat{\boldsymbol{F}}\|_f^2}{T} = O_p\left(\frac{1}{m_{NT}^4}\right).$$

For $\boldsymbol{\delta}_{15}$, we have

$$\frac{\|\boldsymbol{\delta}_{15}\|_f^2}{T} \leq \frac{1}{NT}\sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\right\|_2^2 \cdot \frac{1}{NT^2}\sum_{i=1}^N \left\|\boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{F}}\right\|_2^2 = O_p\left(\frac{1}{m_{NT}^2}\right) \cdot O_p\left(\frac{1}{m_{NT}^2}\right) = O_p\left(\frac{1}{m_{NT}^4}\right).$$

For $\boldsymbol{\delta}_{17}$, we have

$$\frac{\|\boldsymbol{\delta}_{17}\|_f^2}{T} \lesssim \left(\frac{1}{NT}\sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\right\|_2^2\right)^2 = O_p\left(\frac{1}{m_{NT}^4}\right).$$

(4)–(6) For $\boldsymbol{\delta}_9$, we have

$$\frac{\|\boldsymbol{\delta}_9\|_f^2}{T} \leq \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{T} \cdot \frac{1}{N}\sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i\right\|_2^2 \cdot \frac{1}{NT^2}\sum_{i=1}^N \left\|\boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{F}}\right\|_2^2 = O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{Tm_{NT}^2}\right).$$

For $\boldsymbol{\delta}_{11}$, we have

$$\begin{aligned}
\frac{\|\boldsymbol{\delta}_{11}\|_f^2}{T} \lesssim& \left\|\frac{1}{NT}\sum_{i=1}^N \hat{\boldsymbol{W}}_i \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0 \boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top \hat{\boldsymbol{V}}_i^\top\right\|_2^2 \\
\leq& \frac{1}{NT}\sum_{i=1}^N \left\|\hat{\boldsymbol{W}}_i \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0 \boldsymbol{l}_{i,0}\right\|_2^2 \cdot \frac{1}{NT}\sum_{i=1}^N \left\|\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\right\|_2^2 \\
=& O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{T}\right) \cdot O_p\left(\frac{1}{m_{NT}^2}\right) = O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{Tm_{NT}^2}\right).
\end{aligned}$$

For $\boldsymbol{\delta}_{13}$, we have

$$\begin{aligned}
\frac{\|\boldsymbol{\delta}_{13}\|_f^2}{T} \lesssim& \left\|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i \boldsymbol{l}_{i,0}^\top \boldsymbol{F}_0^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \hat{\boldsymbol{W}}_i\right\|_f^2 \\
\leq& \left\|\frac{1}{N\sqrt{T}}\sum_{i=1}^N \hat{\boldsymbol{W}}_i \otimes \boldsymbol{\varepsilon}_i \boldsymbol{l}_{i,0}^\top\right\|_f^2 \cdot \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{T} \\
=& O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{F}_0\|_f^2}{Tm_{NT}}\right).
\end{aligned}$$

where

$$\begin{aligned}
\left\|\frac{1}{N\sqrt{T}}\sum_{i=1}^N \hat{\boldsymbol{W}}_i \otimes \boldsymbol{\varepsilon}_i \boldsymbol{l}_{i,0}^\top\right\|_f^2 =& \frac{1}{N^2T}\sum_{i,j=1}^N \left[\operatorname{tr}\left(\hat{\boldsymbol{W}}_i^{1/2}\hat{\boldsymbol{W}}_j\hat{\boldsymbol{W}}_i^{1/2}\right) \cdot \boldsymbol{l}_{i,0}^\top \boldsymbol{l}_{j,0} \cdot \boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j\right] \\
\lesssim& \frac{1}{N^2T}\sum_{i,j=1}^N \left|\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j\right| \\
\leq& \left[\frac{1}{N^2T^2}\sum_{i,j=1}^N \left(\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j\right)^2\right]^{1/2} \cdot \left(\frac{1}{N^2}\sum_{i,j=1}^N 1^2\right)^{1/2} = O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}$$

Note that

$$\mathbb{E}_{\mathcal{C}}\left[\frac{1}{N^2T^2}\sum_{i,j=1}^N \left(\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j\right)^2\right]$$

$$
\begin{aligned}
&= \frac{1}{N^2T^2}\sum_{i,j=1}^{N}\sum_{s,t=1}^{T}\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}\varepsilon_{it}\varepsilon_{js}\varepsilon_{jt}\right)\\
&= \frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^4\right)+\frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{j\neq i}\sum_{t=1}^{T}\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^2\right)\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{jt}^2\right)+\frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{s=1}^{T}\sum_{t\neq s}\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}^2\varepsilon_{it}^2\right)\\
&= O\left(\frac{1}{NT}\right)+O\left(\frac{1}{T}\right)+O\left(\frac{1}{N}\right)=O\left(\frac{1}{m_{NT}^2}\right),
\end{aligned}
$$

where $\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}\varepsilon_{it}\varepsilon_{js}\varepsilon_{jt}\right)=0$ when $i\neq j$ and $s\neq t$. By a similar argument for $\boldsymbol{\delta}_{11}$, we have

$$
\frac{\|\boldsymbol{\delta}_{14}\|_f^2}{T}\lesssim\frac{1}{NT}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\right\|_2^2\cdot\frac{1}{NT}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{W}}_i\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\right\|_2^2=O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^2}{Tm_{NT}^2}\right).
$$

For $k=19,20$,

$$
\begin{aligned}
\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{19}\|_f^2}{T}&\leq\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^4}{T^2}\cdot\frac{\|\hat{\boldsymbol{F}}\|_f^2\bar{\boldsymbol{l}}^4}{T}\cdot\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{V}}_i\right\|_2^2=O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^4}{T^2}\right),\\
\frac{\|\boldsymbol{\delta}_{20}\|_f^2}{T}&\leq\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^4}{T}\cdot\frac{\|\hat{\boldsymbol{F}}\|_f^2\bar{\boldsymbol{l}}^4}{T}\cdot\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{V}}_i\right\|_2^4=O_p\left(\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^4}{T^2}\right).
\end{aligned}
$$

□

Proof of Lemma S.2.3. From Equation (S.2.5), we obtain

$$
\begin{aligned}
\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0}{T}&=\frac{1}{T}\boldsymbol{X}_i^{\top}\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\delta}_{18}+\boldsymbol{\delta}_{21}\right)\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\\
&=-\frac{1}{T}\boldsymbol{X}_i^{\top}\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\sum_{k=1}^{17}\boldsymbol{\delta}_k+\boldsymbol{\delta}_{19}+\boldsymbol{\delta}_{20}\right)\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\\
&=:\sum_{k=1}^{17}\Delta_k+\Delta_{19}+\Delta_{20}.
\end{aligned}
\tag{S.4.2}
$$

It is straightforward to verify that the consistency rate of $\|\Delta_k\|_2^2$ is at least as fast as that of $T^{-1}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_k\right\|_f^2$ or $T^{-1}\|\boldsymbol{\delta}_k\|_f^2$. Therefore, we only need to study $T^{-1}\|\boldsymbol{\delta}_k\|_f^2$, $T^{-1}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_k\right\|_f^2$, or $\|\Delta_k\|_2^2$ for $k=6,7,10,12,13,16$, as the remaining terms are already showed in Lemma S.2.2 (1), (3), (4), and (6). For $k=6$, decompose $\boldsymbol{\delta}_6$:

$$
\boldsymbol{\delta}_6=\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^{\top}\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}=:\boldsymbol{\delta}_{6.1}+\boldsymbol{\delta}_{6.2}.
$$

For $\boldsymbol{\delta}_{6.1}$, we have

$$
\frac{\|\boldsymbol{\delta}_{6.1}\|_f^2}{T}\leq\frac{\|\boldsymbol{F}_0\|_f^2}{T}\cdot\bar{\boldsymbol{l}}^2\cdot\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{\varepsilon}_i^{\top}\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)\right\|_2^2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right),
$$

where

$$
\begin{aligned}
\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{\varepsilon}_i^{\top}\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)\right\|_2^2&=\operatorname{tr}\left(\frac{(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1})^{\top}}{\sqrt{T}}\cdot\frac{\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^{\top}}{NT}\cdot\frac{\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}}{\sqrt{T}}\right)\\
&\leq\frac{\|\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\|_f^2}{T}\cdot\frac{\|\boldsymbol{\mathcal{E}}\|_2^2}{NT}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
\end{aligned}
$$

For $\boldsymbol{\delta}_{6.2}$,

$$\frac{\|\boldsymbol{\delta}_{6.2}\|_f^2}{T} \lesssim \frac{\|\boldsymbol{F}_0\|_f^2}{T} \cdot \left\| \frac{1}{NT} \sum_{i=1}^N \boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top \boldsymbol{F}_0 \right\|_f^2 = O_p\left(\frac{1}{NT}\right).$$

The result follows from

$$\begin{aligned}
\mathbb{E}\left(\left\| \frac{1}{NT} \sum_{i=1}^N \boldsymbol{l}_{i,0} \boldsymbol{\varepsilon}_i^\top \boldsymbol{F}_0 \right\|_f^2\right) =& \mathbb{E}\left\{ \frac{1}{N^2T^2} \sum_{i=1}^N \left[\|\boldsymbol{l}_{i,0}\|_2^2 \cdot \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\varepsilon}_i^\top \boldsymbol{F}_0 \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right)\right]\right\} \\
\lesssim& \frac{1}{N^2T^2} \sum_{i=1}^N \sum_{t=1}^T \mathbb{E}\left[\|\boldsymbol{f}_{t,0}\|_2^2 \cdot \mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^2\right)\right] \\
\lesssim& \frac{1}{N^2T^2} \sum_{i=1}^N \sum_{t=1}^T \mathbb{E}\left(\|\boldsymbol{f}_{t,0}\|_2^2\right) = O\left(\frac{1}{NT}\right),
\end{aligned}$$

where the first step uses the conditional cross-sectional independence of $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$, and the second uses the conditional serial uncorrelatedness of $\{\varepsilon_{it}\}_{t=1}^T$. Decompose Δ_7:

$$\begin{aligned}
\Delta_7 =& \frac{1}{T} \boldsymbol{X}_i^\top \mathcal{M}_{T,R_0}\left[\boldsymbol{\Xi}_0^{-1} \cdot \ell\left(\frac{1}{N} \sum_{j=1}^N \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right)\right] + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}}\right) + O_p\left(\frac{1}{\sqrt{N} m_{NT}}\right) \\
=& \frac{1}{T} \boldsymbol{X}_i^\top \mathcal{M}_{T,R_0}\left[\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \otimes \boldsymbol{I}_T \cdot \ell\left(\frac{1}{N} \sum_{j=1}^N \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right)\right] \\
&+ \frac{1}{T} \boldsymbol{X}_i^\top \mathcal{M}_{T,R_0}\left[\boldsymbol{\Xi}_0^{-1} \cdot \bar{\boldsymbol{H}}_0 \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \otimes \boldsymbol{I}_T \cdot \ell\left(\frac{1}{N} \sum_{j=1}^N \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right)\right] + \textit{low-order terms} \\
=:& \Delta_{7.1} + \Delta_{7.2} + \textit{low-order terms},
\end{aligned}$$

where the first step follows from Lemma S.4.1 (2g) and $(NT)^{-1/2} \left\|\sum_{j=1}^N \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top\right\|_f = O_p(1)$, and the second step follows from the expansion in Lemma S.2.1 (1). For $\Delta_{7.1}$,

$$\begin{aligned}
\|\Delta_{7.1}\|_2^2 =& \left\| \frac{1}{NT} \sum_{j=1}^N \boldsymbol{X}_i^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \right\|_2^2 \\
\lesssim& \left\| \frac{1}{NT} \sum_{j=1}^N \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right\|_2^2 + \left\| \frac{1}{NT} \sum_{j=1}^N \boldsymbol{G}^\top \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right\|_2^2 + \left\| \frac{1}{NT} \sum_{j=1}^N \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right\|_2^2 = O_p\left(\frac{1}{NT}\right),
\end{aligned}$$

where

$$\begin{aligned}
\mathbb{E}_{\mathcal{C}}\left(\left\| \frac{1}{NT} \sum_{j=1}^N \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top \right\|_f^2\right) =& \frac{1}{N^2T^2} \sum_{j=1}^N \left[\|\boldsymbol{l}_{j,0}\|_2^2 \cdot \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\varepsilon}_j^\top \boldsymbol{X}_i \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_j\right)\right] \\
\lesssim& \frac{1}{N^2T^2} \sum_{j=1}^N \sum_{s,t=1}^T \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{x}_{is}^\top \boldsymbol{x}_{it} \varepsilon_{js} \varepsilon_{jt}\right) \\
=& \frac{1}{N^2T^2} \sum_{j=1}^N \sum_{t=1}^T \mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\|_2^2 \cdot \varepsilon_{jt}^2\right) = O\left(\frac{1}{NT}\right),
\end{aligned}$$

and other terms are similarly analyzed before. For $\Delta_{7.2}$,

$$\|\Delta_{7.2}\|_2^2 \leq \frac{\|\boldsymbol{X}_i\|_2^2}{T} \cdot \|\boldsymbol{\Xi}_0^{-1}\|_2^2 \cdot \left\| \bar{\boldsymbol{H}}_0 \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \otimes \boldsymbol{I}_T \cdot \frac{1}{N\sqrt{T}} \sum_{j=1}^N \boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \boldsymbol{\varepsilon}_j \right\|_2^2$$

$$
\begin{aligned}
&\lesssim \frac{1}{N}\sum_{i=1}^{N}\left(\left\|\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\frac{\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{\sqrt{T}}\left(\frac{\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\boldsymbol{\Theta}_0^{\top}\right\|_2^2\cdot\left\|\frac{1}{NT}\sum_{j=1}^{N}\boldsymbol{l}_{j,0}\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\varepsilon}_j\right\|_2^2\right)\\
&=O_p\left(\frac{1}{NT}\right).
\end{aligned}
$$

To conclude, $\|\Delta_7\|_f^2=O_p\left(N^{-1}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(N^{-1}m_{NT}^{-2}\right)$. For $k=10$, we have

$$
\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{10}\|_f^2}{T}\lesssim\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^2}{T}\cdot\frac{1}{NT}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{V}}_i\boldsymbol{\varepsilon}_i\right\|_2^2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
$$

Using the same type of decomposition for $k=7$, it suffices to study $T^{-2}\left\|\boldsymbol{X}_j^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{12}\right\|_f^2$. Decompose $\boldsymbol{\delta}_{12}$:

$$
\begin{aligned}
\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{12}=&\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{\Phi}}_{4i}\cdot\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right)\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\\
=&\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\left[\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{\Phi}}_{4i}-\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\Phi}_{4i,0}\right)\cdot\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right]\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\\
&+\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\left[\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4i,0}\cdot\left(\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i-\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\varepsilon}_i\right)\boldsymbol{l}_{i,0}^{\top}\right]\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\\
&+\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\left(\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4i,0}\cdot\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right)\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\\
=:&\boldsymbol{\delta}_{12.1}'+\boldsymbol{\delta}_{12.2}'+\boldsymbol{\delta}_{12.3}'.
\end{aligned}
$$

For $\boldsymbol{\delta}_{12.1}'$, we have

$$
\frac{\|\boldsymbol{\delta}_{12.1}'\|_f^2}{T}\lesssim\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{\Phi}}_{4i}-\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\Phi}_{4i,0}\right\|_2^2\cdot\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\right\|_2^2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
$$

For $\boldsymbol{\delta}_{12.2}'$, we have

$$
\begin{aligned}
\frac{\|\boldsymbol{\delta}_{12.2}'\|_f^2}{T}\lesssim&\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\left(\boldsymbol{P}_{\boldsymbol{F}_0}-\boldsymbol{P}_{\hat{\boldsymbol{F}}}\right)\boldsymbol{\varepsilon}_i\right\|_2^2\\
\lesssim&\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\left[\boldsymbol{X}_i^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}-\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\boldsymbol{\varepsilon}_i\right\|_2^2\\
&+\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\cdot\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\right)^{\top}\boldsymbol{\varepsilon}_i\right\|_2^2\\
=&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
\end{aligned}
$$

Let

$$
\begin{aligned}
&\boldsymbol{\mathcal{U}}_{12.1.ij}=\boldsymbol{\mathcal{U}}_{12.1}\left(\boldsymbol{\pi}_{2ij}\right)=\left(\boldsymbol{\Sigma}_{ji}-\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}-\boldsymbol{\Sigma}_{j\boldsymbol{G}}\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{G}}^{\top}\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^{\top}\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^{\top},\\
&\boldsymbol{\mathcal{U}}_{12.2.ij}=\boldsymbol{\mathcal{U}}_{12.2}\left(\boldsymbol{\pi}_{2ij}\right)=\boldsymbol{\mathcal{U}}_{12.1}(\boldsymbol{\pi}_{2ij})\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1},\quad\boldsymbol{\mathcal{U}}_{12.1.ij}^{e}=\boldsymbol{\mathcal{U}}_{12.1}\left(\boldsymbol{\pi}_{2ij}^{e}\right),\quad\boldsymbol{\mathcal{U}}_{12.2.ij}^{e}=\boldsymbol{\mathcal{U}}_{12.2}\left(\boldsymbol{\pi}_{2ij}^{e}\right).
\end{aligned}
\tag{S.4.3}
$$

Decompose $\boldsymbol{X}_j^{\top}\boldsymbol{\delta}_{12.3}'$:

$$
\boldsymbol{X}_j^{\top}\boldsymbol{\delta}_{12.3}'=\left\{\frac{1}{N}\sum_{i=1}^{N}\left[\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.1.ij}}^{\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\cdot\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right]+\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{\mathcal{U}}_{12.1.ij}^{e}\cdot\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right)\right.
$$

$$
\begin{aligned}
&-\frac{1}{N}\sum_{i=1}^{N}\left[\mathcal{M}_{p,R_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.2.ij}}^{\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\cdot\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right]-\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{\mathcal{U}}_{12.2.ij}^{e}\cdot\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right)\Bigg\}\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\\
=:&\left(\boldsymbol{\delta}_{12.3.1}''+\cdots+\boldsymbol{\delta}_{12.3.4}''\right)\cdot\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T},
\end{aligned}
$$

where $\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.k.ij}}$ denotes the Jacobian matrix of $\boldsymbol{\mathcal{U}}_{12.k.ij}$ for $k=1,2$. In particular, the $((q_1-1)p+q_2)$-th column of $\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.1.ij}}$ is given by

$$
\left.\frac{\partial u_{12.1,q_1q_2}(\boldsymbol{\pi}_2)}{\partial\boldsymbol{\pi}_2}\right|_{\boldsymbol{\pi}_2=\alpha_{q_1q_2}\boldsymbol{\pi}_{2ij}^{e}+(1-\alpha_{q_1q_2})\boldsymbol{\pi}_{2ij}},
$$

where $u_{12.1,q_1q_2}$ denotes the (q_1,q_2)-th element of $\boldsymbol{\mathcal{U}}_{12.1}$ and $0\le\alpha_{q_1q_2}\le 1$ for $q_1,q_2=1,\ldots,p$, and $\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.2.ij}}$ can be similarly defined. Then,

$$
\begin{aligned}
\frac{\|\boldsymbol{\delta}_{12.3.1}''\|_2^2}{T^2}\le&\max_{i,j}\left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.1.ij}}\right\|_2^2\cdot\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right\|_2^2\right)\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\boldsymbol{l}_{i,0}^{\top}\right\|_f^2\right)\\
=&O_p(1)\cdot O_p\left(\frac{1}{T}\right)\cdot O_p\left(\frac{1}{T}\right)=O_p\left(\frac{1}{T^2}\right),
\end{aligned}
$$

and

$$
\frac{\mathbb{E}_{\mathcal{C}}(\|\boldsymbol{\delta}_{12.3.2}''\|_f^2)}{T^2}\lesssim\frac{1}{NT}\cdot\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\boldsymbol{\mathcal{U}}_{12.1.ij}^{e}\right\|_2^2\cdot\frac{\mathbb{E}_{\mathcal{C}}(\|\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\|_2^2)}{T}\right)=O_p\left(\frac{1}{NT}\right).
$$

Similarly, we can show that $T^{-2}\left\|\boldsymbol{\delta}_{12.3.3}''\right\|_2^2=O_p\left(T^{-2}\right)$, and $T^{-2}\left\|\boldsymbol{\delta}_{12.3.4}''\right\|_2^2=O_p\left((NT)^{-1}\right)$. It follows that $T^{-2}\left\|\boldsymbol{X}_j^{\top}\boldsymbol{\delta}_{12.3}'\right\|_2^2=O_p\left(T^{-1}m_{NT}^{-2}\right)$. To conclude,

$$
\frac{\|\boldsymbol{X}_j^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{12}\|_f^2}{T^2}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
$$

By the same type of decomposition for $k=7$, it is straightforward to show that

$$
\|\Delta_{13}\|_f^2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right),\quad\|\Delta_{16}\|_f^2=O_p\left(\frac{1}{m_{NT}^4}\right).
$$

□

Proof of Lemma S.2.4. The proof of (2) is similar to that of (1). Statement (3) follows from

$$
\frac{\hat{\boldsymbol{F}}^{\top}(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)}{T}=\frac{\boldsymbol{O}_{\boldsymbol{F}}^{-\top}(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)^{\top}(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)}{T}+\frac{\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^{\top}(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)}{T},
$$

the consistency rate of $\hat{\boldsymbol{F}}$ established in Theorem 3.2, and (2). For a similar argument regarding (4), see Lemma A.7 (1) in Bai (2009). We only provide a detailed proof for (1) below. Let

$$
\frac{\boldsymbol{X}_i^{\top}(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)}{T}=\frac{1}{T}\sum_{k=1}^{20}\boldsymbol{X}_i^{\top}\boldsymbol{\delta}_k\cdot\left(\frac{\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}}{T}\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}=:\sum_{k=1}^{20}\Delta_k.
$$

By a similar argument in the proof of Lemma S.2.3, it suffices to study $\|\Delta_k\|_f^2$ or $T^{-1}\|\boldsymbol{\delta}_k\|_f^2$ for $k \in \{6,7,10,12,13,16,18\}$. For $k=6$, it is shown in the proof of Lemma S.2.3 that $T^{-1}\|\boldsymbol{\delta}_6\|_f^2$ is $O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(m_{NT}^{-4}\right)$. For $k=7$, we have

$$\|\Delta_7\|_f^2 \lesssim \left\|\frac{1}{NT}\sum_{j=1}^N \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\varepsilon}_j \boldsymbol{l}_{j,0}^\top\right\|_f^2 = O_p\left(\frac{1}{NT}\right).$$

Let

$$\begin{aligned}
&\boldsymbol{\mathcal{U}}_{10.1.j}=\boldsymbol{\mathcal{U}}_{10.1}(\boldsymbol{\pi}_{1j})=\frac{\boldsymbol{\Phi}_{4j,0}^\top \boldsymbol{F}_0}{\sqrt{T}}, \quad \boldsymbol{\mathcal{U}}_{10.2.j}=\boldsymbol{\mathcal{U}}_{10.2}(\boldsymbol{\pi}_{1j})=\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{j\boldsymbol{F}}^\top \boldsymbol{\mathcal{U}}_{10.1}(\boldsymbol{\pi}_{1j}),\\
&\boldsymbol{\mathcal{U}}_{10.1.j}^e=\boldsymbol{\mathcal{U}}_{10.1}\left(\boldsymbol{\pi}_{1j}^e\right), \quad \boldsymbol{\mathcal{U}}_{10.2.j}^e=\boldsymbol{\mathcal{U}}_{10.2}\left(\boldsymbol{\pi}_{1j}^e\right).
\end{aligned} \tag{S.4.4}$$

Decompose $\boldsymbol{\delta}_{10}$:

$$\begin{aligned}
\boldsymbol{\delta}_{10}=&\frac{1}{NT}\sum_{j=1}^N\left(\boldsymbol{F}_0\boldsymbol{l}_{j,0}\boldsymbol{\varepsilon}_j^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_j\cdot\frac{\hat{\boldsymbol{\Phi}}_{4j}^\top\hat{\boldsymbol{F}}-\boldsymbol{\Phi}_{4j,0}^\top\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}}{\sqrt{T}}\right)+\frac{1}{NT^{3/2}}\sum_{j=1}^N\boldsymbol{F}_0\boldsymbol{l}_{j,0}\left(\boldsymbol{\varepsilon}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_j-\boldsymbol{\varepsilon}_j^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_j\right)\boldsymbol{\Phi}_{4j,0}^\top\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\\
&+\frac{1}{NT}\sum_{j=1}^N\left[\boldsymbol{F}_0\boldsymbol{l}_{j,0}\boldsymbol{\varepsilon}_j^\top\boldsymbol{X}_j\cdot\mathcal{M}_{p,R_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{10.1.j}}^\top\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^e\right)\right)\right]\boldsymbol{O}_{\boldsymbol{F}}^{-1}+\frac{1}{NT}\sum_{j=1}^N\left(\boldsymbol{F}_0\boldsymbol{l}_{j,0}\boldsymbol{\varepsilon}_j^\top\boldsymbol{X}_j\cdot\boldsymbol{\mathcal{U}}_{10.1.j}^e\right)\boldsymbol{O}_{\boldsymbol{F}}^{-1}\\
&-\frac{1}{NT}\sum_{j=1}^N\left[\boldsymbol{F}_0\boldsymbol{l}_{j,0}\boldsymbol{\varepsilon}_j^\top\boldsymbol{F}_0\cdot\mathcal{M}_{R_0,R_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{10.2.j}}^\top\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^e\right)\right)\right]\boldsymbol{O}_{\boldsymbol{F}}^{-1}-\frac{1}{NT}\sum_{j=1}^N\left(\boldsymbol{F}_0\boldsymbol{l}_{j,0}\boldsymbol{\varepsilon}_j^\top\boldsymbol{F}_0\cdot\boldsymbol{\mathcal{U}}_{10.2.j}^e\right)\boldsymbol{O}_{\boldsymbol{F}}^{-1}\\
=&\boldsymbol{\delta}_{10.1}+\cdots+\boldsymbol{\delta}_{10.6}.
\end{aligned}$$

Similar to the proof of $k=12$ in Lemma S.2.3, we have

$$\begin{aligned}
&\frac{\|\boldsymbol{\delta}_{10.1}\|_f^2}{T}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right), \quad \frac{\|\boldsymbol{\delta}_{10.2}\|_f^2}{T}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right),\\
&\frac{\|\boldsymbol{\delta}_{10.3}\|_f^2}{T}=O_p\left(\frac{1}{T^2}\right), \quad \frac{\|\boldsymbol{\delta}_{10.4}\|_f^2}{T}=O_p\left(\frac{1}{NT}\right), \quad \frac{\|\boldsymbol{\delta}_{10.5}\|_f^2}{T}=O_p\left(\frac{1}{T^2}\right), \quad \frac{\|\boldsymbol{\delta}_{10.6}\|_f^2}{T}=O_p\left(\frac{1}{NT}\right).
\end{aligned}$$

It follows that $T^{-1}\|\boldsymbol{\delta}_{10}\|_f^2=O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(m_{NT}^{-4}\right)$. Similar to terms in (S.4.3), we define

$$\begin{aligned}
&\boldsymbol{\mathcal{U}}_{12.3}(\boldsymbol{\pi}_{2ij})=\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4j,0}}{\sqrt{T}}, \quad \boldsymbol{\mathcal{U}}_{12.4}(\boldsymbol{\pi}_{2ij})=\boldsymbol{\mathcal{U}}_{12.3}(\boldsymbol{\pi}_{2ij})\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1},\\
&\boldsymbol{\mathcal{U}}_{12.3.ij}^e=\boldsymbol{\mathcal{U}}_{12.3}\left(\boldsymbol{\pi}_{2ij}^e\right), \quad \boldsymbol{\mathcal{U}}_{12.4.ij}^e=\boldsymbol{\mathcal{U}}_{12.4}\left(\boldsymbol{\pi}_{2ij}^e\right).
\end{aligned}$$

By decomposing $T^{-1}\boldsymbol{X}_i^\top\boldsymbol{\delta}_{12}$:

$$\begin{aligned}
\frac{\boldsymbol{X}_i^\top\boldsymbol{\delta}_{12}}{T}=&\frac{1}{NT}\sum_{j=1}^N\left(\frac{\boldsymbol{X}_i\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{\Phi}}_{4j}-\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4j,0}}{\sqrt{T}}\cdot\boldsymbol{X}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right)\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\\
&+\frac{1}{NT^{3/2}}\sum_{j=1}^N\left[\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4j,0}\cdot\left(\boldsymbol{X}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_j-\boldsymbol{X}_j^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{\varepsilon}_j\right)\boldsymbol{l}_{j,0}^\top\right]\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\\
&+\frac{1}{NT}\sum_{j=1}^N\left[\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.3.ij}}^\top\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^e\right)\right)\cdot\boldsymbol{X}_j^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right]\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\\
&-\frac{1}{NT}\sum_{j=1}^N\left[\mathcal{M}_{p,R_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.4.ij}}^\top\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^e\right)\right)\cdot\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right]\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}
\end{aligned}$$

$$+\frac{1}{NT}\sum_{j=1}^{N}\left(\boldsymbol{\mathcal{U}}^{e}_{12.3.ij}\cdot\boldsymbol{X}_j^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right)\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}-\frac{1}{NT}\sum_{j=1}^{N}\left(\boldsymbol{\mathcal{U}}^{e}_{12.4.ij}\cdot\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right)\cdot\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T},$$

we can show that $\|\Delta_{12}\|_f^2=O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(m_{NT}^{-4}\right)$ by a similar argument for $\boldsymbol{\delta}_{10}$. For $k=13$,

$$\|\Delta_{13}\|_f^2\lesssim\frac{1}{NT^2}\sum_{j=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_j\right\|_2^2\cdot\frac{1}{N}\sum_{j=1}^{N}\left\|\hat{\boldsymbol{V}}_j\right\|_2^2\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f^2}{T}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{T}\right)+O_p\left(\frac{1}{Tm_{NT}^2}\right).$$

For $k=16$,

$$\|\Delta_{16}\|_f^2\lesssim\frac{1}{NT^2}\sum_{j=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_j\right\|_2^2\cdot\frac{1}{NT}\sum_{j=1}^{N}\left\|\hat{\boldsymbol{V}}_j\boldsymbol{\varepsilon}_j\right\|_2^2=O_p\left(\frac{1}{Tm_{NT}^2}\right).$$

For $k=18$, by Lemma S.2.3, we have

$$\|\Delta_{18}\|_f^2\lesssim\frac{1}{NT}\sum_{j=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{\Phi}}_{4j}\right\|_2^2\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\left\|\boldsymbol{X}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_2^2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$$

□

Proof of Lemma S.2.5.

(1) We have

$$\begin{aligned}
&\left\|\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{F}_0\boldsymbol{l}_{i,0}\right\|_2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}}{T}\right\|_2^2\cdot\left\|\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right\|_2\cdot\left\|\boldsymbol{I}_p-\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\boldsymbol{\Theta}_0^\top\right\|_2\right)\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0}{T}\right\|_2^2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right),
\end{aligned}$$

where the last step is due to the consistency rate of $T^{-2}\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_2^2$ established in Lemma S.2.3.

(2) The results follows from

$$\begin{aligned}
&\left\|\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_i\right\|_2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0}{T}\right\|_2\cdot\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i}{T}\right\|_2\right)\\
\le&\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0}{T}\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i}{T}\right\|_2^2\right)^{1/2}\\
=&\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}^2}\right)\right]\cdot O_p\left(\frac{1}{m_{NT}}\right)=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).
\end{aligned}$$

(3) A similar argument as above yields the same consistency rate for this term. □

Proof of Lemma S.2.6. We reshape the vector into a $(p-K_0)\times K_0$ matrix, and decompose it:

$$\mathcal{M}_{p,K_0}\left\{\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_i\right\}$$

$$
\begin{aligned}
=&\frac{1}{N}\sum_{i=1}^{N}\left[\boldsymbol{I}_p-\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\boldsymbol{\Theta}_0^\top\right]\cdot\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_i}{T}\cdot\frac{\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i}{T}\cdot\boldsymbol{\Theta}_0\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\\
=&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\hat{\boldsymbol{\Phi}}_{1i}\\
=&\frac{1}{N}\sum_{i=1}^{N}\left[\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\hat{\boldsymbol{\Phi}}_{1i}-\boldsymbol{S}_{i,0}^\top\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\boldsymbol{\Phi}_{1i,0}\right]\\
&+\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{S}_{i,0}^\top\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\boldsymbol{\Phi}_{1i,0}-\boldsymbol{S}_{i,0}^\top\boldsymbol{\Phi}_{3i,0}\boldsymbol{\Phi}_{3i,0}^\top\boldsymbol{\Phi}_{1i,0}\right)\\
&+\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{S}_{i,0}^\top\boldsymbol{\Phi}_{3i,0}\boldsymbol{\Phi}_{3i,0}^\top\boldsymbol{\Phi}_{1i,0}-\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Phi}_{3i,0}\boldsymbol{\Phi}_{3i,0}^\top\boldsymbol{\Phi}_{1i,0}^e\right)\\
&+\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^{e\top}\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top-\boldsymbol{\Omega}_i)\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i}{T^2}\boldsymbol{\Phi}_{1i,0}^e\\
&+\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{S}_{i,0}^{e\top}\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i}{T}\boldsymbol{\Phi}_{1i,0}^e-\boldsymbol{S}_{i,0}^\top\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i}{T}\boldsymbol{\Phi}_{1i,0}\right)\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^\top\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i}{T}\boldsymbol{\Phi}_{1i,0}\\
=:&\Delta_1+\cdots+\Delta_5+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^\top\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i}{T}\boldsymbol{\Phi}_{1i,0}.
\end{aligned}
\tag{S.4.5}
$$

For Δ_1, we have

$$
\begin{aligned}
\|\Delta_1\|_2=&\left\|\frac{1}{NT}\sum_{i=1}^{N}\left[\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right]^\top\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\hat{\boldsymbol{\Phi}}_{1i}+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^\top\hat{\boldsymbol{\Phi}}_{3i}\hat{\boldsymbol{\Phi}}_{3i}^\top\left(\hat{\boldsymbol{\Phi}}_{1i}-\boldsymbol{\Phi}_{1i,0}\right)\right\|_2\\
\le&\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2\cdot\left(\max_i\left\|\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right\|_2\cdot\max_i\left\|\hat{\boldsymbol{\Phi}}_{1i}\right\|_2+\max_i\|\boldsymbol{S}_{i,0}\|_2\cdot\max_i\left\|\hat{\boldsymbol{\Phi}}_{1i}-\boldsymbol{\Phi}_{1i,0}\right\|_2\right)\\
=&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).
\end{aligned}
$$

For Δ_2, we have

$$
\begin{aligned}
\|\Delta_2\|_2=&\left\|\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^\top\left(\hat{\boldsymbol{\Phi}}_{3i}-\boldsymbol{\Phi}_{3i,0}\right)\hat{\boldsymbol{\Phi}}_{3i}^\top\boldsymbol{\Phi}_{1i,0}+\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{S}_{i,0}^\top\boldsymbol{\Phi}_{3i,0}\left(\hat{\boldsymbol{\Phi}}_{3i}-\boldsymbol{\Phi}_{3i,0}\right)^\top\boldsymbol{\Phi}_{1i,0}\right\|_2\\
\le&\max_i\|\boldsymbol{S}_{i,0}\|_2\cdot\max_i\|\boldsymbol{\Phi}_{1i,0}\|_2\cdot\frac{1}{N}\sum_{i=1}^{N}\left[\left\|\hat{\boldsymbol{\Phi}}_{3i}-\boldsymbol{\Phi}_{3i,0}\right\|_2\cdot\left(\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2+\|\boldsymbol{\Phi}_{3i,0}\|_2\right)\right]\\
\lesssim&\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{3i}-\boldsymbol{\Phi}_{3i,0}\right\|_2^2\right)^{1/2}\cdot\left[\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2\right)^{1/2}+\left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\Phi}_{3i,0}\|_2^2\right)^{1/2}\right]\\
=&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),
\end{aligned}
$$

where

$$
\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{3i}-\boldsymbol{\Phi}_{3i,0}\right\|_2^2=\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right)\boldsymbol{\varepsilon}_i\right\|_2^2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^4}\right).
$$

For Δ_3, since

$$
\boldsymbol{S}_{i,0}-\boldsymbol{S}_{i,0}^e=\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{S}_i}^\top\cdot(\boldsymbol{\pi}_{1i}-\boldsymbol{\pi}_{1i}^e)\right),\quad\boldsymbol{\Phi}_{1i,0}-\boldsymbol{\Phi}_{1i,0}^e=\mathcal{M}_{p,K_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Phi}_{1i}}^\top\cdot(\boldsymbol{\pi}_{1i}-\boldsymbol{\pi}_{1i}^e)\right),
$$

we have

$$\|\Delta_3\|_2 \leq \max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{S}_i}\right\|_2 \cdot \left(\frac{1}{N}\sum_{i=1}^N \|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^2\right)^{1/2} \cdot \left(\frac{1}{N}\sum_{i=1}^N \|\boldsymbol{\Phi}_{3i,0}\|_2^4\right)^{1/2} \cdot \max_i \|\boldsymbol{\Phi}_{1i,0}\|_2$$
$$+ \max_i \left\|\boldsymbol{S}_{i,0}^e\right\|_2 \cdot \left(\frac{1}{N}\sum_{i=1}^N \|\boldsymbol{\Phi}_{3i,0}\|_2^4\right)^{1/2} \cdot \left(\frac{1}{N}\sum_{i=1}^N \|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^2\right)^{1/2} \cdot \max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Phi}_{1i}}\right\|_2 = O_p\left(\frac{1}{T^{3/2}}\right),$$

where $N^{-1}\sum_{i=1}^N \|\boldsymbol{\Phi}_{3i,0}\|_2^4 \lesssim T^{-2} \cdot N^{-1}T^{-2}\sum_{i=1}^N \left(\left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^4 + \|\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\|_2^4\right)$ by Lemma S.4.1 (3e) and (3f), and terms $\max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{S}_i}\right\|_2$, $\max_i \|\boldsymbol{\Phi}_{1i,0}\|_2$, $\max_i \left\|\boldsymbol{S}_{i,0}^e\right\|_2$ and $\max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Phi}_{1i}}\right\|_2$ are all $O_p(1)$. Decompose Δ_4:

$$\begin{aligned}
\Delta_4 =& \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\boldsymbol{\Phi}_{1i,0}^e \\
&- \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\right)^\top \boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\boldsymbol{\Phi}_{1i,0}^e \\
&- \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\right)\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\boldsymbol{\Phi}_{1i,0}^e \\
&- \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^\top\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\right)\boldsymbol{\Phi}_{1i,0}^e \\
&- \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\boldsymbol{\Phi}_{1i,0}^e \\
&+ \frac{1}{NT^2}\sum_{i=1}^N \boldsymbol{S}_{i,0}^{e\top}\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right) + \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\right)^\top \boldsymbol{P}_{\boldsymbol{F}_0}\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{P}_{\boldsymbol{F}_0}\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right) + \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\right)\boldsymbol{\Phi}_{1i,0}^e \\
=:& \Delta_{4.1} + \cdots + \Delta_{4.6}.
\end{aligned}$$

For $\Delta_{4.1}$, we have

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left(\|\Delta_{4.1}\|_f^2\right) \\
=& \frac{1}{N^2T^4}\sum_{i=1}^N \mathbb{E}_{\mathcal{C}}\left\{\operatorname{tr}\left[\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\boldsymbol{\Phi}_{1i,0}^e\boldsymbol{\Phi}_{1i,0}^{e\top}\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\boldsymbol{S}_{i,0}^e\right]\right\} \\
&+ \frac{1}{N^2T^4}\sum_{i\neq j}\operatorname{tr}\left\{\boldsymbol{S}_{i,0}^{e\top}\mathbb{E}_{\mathcal{C}}\left[\boldsymbol{X}_i^\top\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\right]\boldsymbol{\Phi}_{1i,0}^e\boldsymbol{\Phi}_{1j0}^{e\top}\mathbb{E}_{\mathcal{C}}\left[\boldsymbol{X}_j^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top - \boldsymbol{\Omega}_j\right)\boldsymbol{X}_j\right]\boldsymbol{S}_{j,0}^e\right\} \\
=:& \Delta'_{4.1.1} + \Delta'_{4.1.2},
\end{aligned}$$

where we use the cross-sectional independence of $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$ conditional on $\mathcal{C}$ for $\Delta'_{4.1.2}$. For $\Delta'_{4.1.1}$, we have

$$\begin{aligned}
\mathbb{E}\left(\Delta'_{4.1.1}\right) \lesssim& \frac{1}{N^2T^4}\sum_{i=1}^N \mathbb{E}\left(\left\|\boldsymbol{X}_i^\top(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i)\boldsymbol{X}_i\right\|_f^2\right) \\
\lesssim& \frac{1}{NT^2}\cdot\left[\frac{1}{NT^2}\sum_{i=1}^N \mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^4\right) + \frac{1}{NT^2}\sum_{i=1}^N \mathbb{E}\left(\|\boldsymbol{X}_i\|_2^4\right)\right] = O_p\left(\frac{1}{NT^2}\right).
\end{aligned}$$

By Assumptions 6(2) and 6(3), we have

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left[\boldsymbol{X}_i^\top(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^\top - \boldsymbol{\Omega}_i)\boldsymbol{X}_i\right] \\
=& \sum_{s,t=1}^T \mathbb{E}_{\mathcal{C}}\left[\boldsymbol{x}_{is}\boldsymbol{x}_{it}^\top\left(\varepsilon_{is}\varepsilon_{it} - \sigma_{ii,st}\right)\right]
\end{aligned}$$

$$= \sum_{t=1}^{T} \mathbb{E}_{\mathcal{C}}\left[\boldsymbol{x}_{it}\boldsymbol{x}_{it}^{\top} \cdot \mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)\right] + \sum_{s<t} \mathbb{E}_{\mathcal{C}}\left[\boldsymbol{x}_{is}\boldsymbol{x}_{it}^{\top}\varepsilon_{is} \cdot \mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}\left(\varepsilon_{it}\right)\right] + \sum_{t<s} \mathbb{E}_{\mathcal{C}}\left[\left(\boldsymbol{x}_{is}\boldsymbol{x}_{it}^{\top}\varepsilon_{it}\right) \cdot \mathbb{E}_{\mathcal{C}\vee\mathcal{F}_s}\left(\varepsilon_{is}\right)\right] = \mathbf{0}_{p\times p},$$

and it follows that $\Delta'_{4.1.2} = 0$. Therefore, $\Delta_{4.1} = O_p\left(T^{-1}N^{-1/2}\right)$. For $\Delta_{4.2}$, we have

$$\begin{aligned}\|\Delta_{4.2}\|_f \lesssim& \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\left(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)\right)^{\top}\boldsymbol{F}_0\right\|_2^2\right)^{1/2} \cdot \left(\frac{1}{NT^4}\sum_{i=1}^{N}\left\|\boldsymbol{F}_0^{\top}\left(\boldsymbol{\varepsilon}_i\boldsymbol{\varepsilon}_i^{\top} - \boldsymbol{\Omega}_i\right)\boldsymbol{X}_i\right\|_2^2\right)^{1/2} \\ =& O_p\left(\frac{1}{\sqrt{T}}\right) \cdot O_p\left(\frac{1}{T}\right) = O_p\left(\frac{1}{T^{3/2}}\right).\end{aligned}$$

By similar arguments for $\Delta_{4.1}$ and $\Delta_{4.2}$, we have

$$\Delta_{4.3} = O_p\left(\frac{1}{T\sqrt{N}}\right), \quad \Delta_{4.4} = O_p\left(\frac{1}{T^{3/2}}\right), \quad \Delta_{4.5} = O_p\left(\frac{1}{T\sqrt{N}}\right), \quad \Delta_{4.6} = O_p\left(\frac{1}{Tm_{NT}}\right).$$

To conclude, $\Delta_4 = O_p\left(T^{-1}m_{NT}^{-1}\right)$. For Δ_5, we have

$$\begin{aligned}\|\Delta_5\|_2 \leq& \frac{1}{T} \cdot \max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{S}_i}\right\|_2 \cdot \left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^{e}\|_2\right) \cdot \max_i \left\|\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i}{T} \cdot \boldsymbol{\Phi}_{1i,0}^{e}\right\|_2 \\ &+ \frac{1}{T} \cdot \max_i \left\|\boldsymbol{S}_{i,0}^{\top} \cdot \frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i}{T}\right\|_2 \cdot \left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^{e}\|_2\right) \cdot \max_i \left\|\bar{\boldsymbol{\mathcal{D}}}_{\Phi_{1i}}\right\|_2 = O_p\left(\frac{1}{T^{3/2}}\right).\end{aligned}$$

Collecting all the terms above and applying vectorization to (S.4.5), we establish the desired result. □

Proof of Lemma S.2.7. We focus on the expansion of $T^{-1}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0$ in (S.4.2), noting that the second leading term can be interpreted as its weighted average. However, the analysis of $T^{-1}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0$ here requires a more refined treatment than that used in Lemma S.2.3. [3] As pointed out in the proof of Lemma S.2.3, the consistency rate of Δ_k is at least as fast as $T^{-1/2}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_k\right\|_f$ or $T^{-1/2}\|\boldsymbol{\delta}_k\|_f$. Therefore, for certain indices k, it suffices to study the latter two terms. To facilitate the analysis, we divide the terms into two groups:

(1) For $k \in \mathcal{K}_1 = \{1, 3, 4, 5, 6, 9, 10, 11, 13, 14, 19, 20\}$, the corresponding Δ_k terms are negligible.

(2) For $k \in \mathcal{K}_2 = \{2, 7, 8, 12, 15, 16, 17\}$, the Δ_k terms are key contributors to the asymptotic behavior.

(1) For $k = 1$, $T^{-1/2}\|\boldsymbol{\delta}_1\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right)$. For $k = 3$, $T^{-1/2}\|\boldsymbol{\delta}_3\|_f = O_p\left(m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right)$. For $k = 4$, $T^{-1/2}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_4\right\|_f = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right)$. For $k = 5$, $\|\Delta_5\|_f = O_p\left(m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right)$. For $k = 6$, $T^{-1/2}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_6\right\|_f = O_p\left(N^{-1/2}\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(N^{-1/2}m_{NT}^{-2}\right)$. For $k = 9$, we have

$$\frac{\|\boldsymbol{\delta}_9\|_f}{\sqrt{T}} \lesssim \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_2^2\right)^{1/2} \cdot \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{\varepsilon}_i^{\top}\hat{\boldsymbol{F}}\right\|_2^2\right)^{1/2} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

By a similar argument for $k = 7$ in Lemma S.2.3, we can show that

$$\|\Delta_{10}\|_f = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right), \quad \|\Delta_{13}\|_f = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

[3] Although these two lemmas are closely related, they serve distinct purposes. Lemma S.2.3 is an intermediate result for establishing Lemma S.2.4, while Lemma S.2.7 aims to identify the key contributors.

For $k = 11$, we have

$$\frac{\|\boldsymbol{\delta}_{11}\|_f}{\sqrt{T}} \lesssim \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_f^2\right)^{1/2} \cdot \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

By a similar argument for $k = 11$, $T^{-1/2}\|\boldsymbol{\delta}_{14}\|_f = O_p\left(m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(m_{NT}^{-3}\right)$. For $k = 19$, we have

$$\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\delta}_{19}\|_f}{\sqrt{T}} \lesssim \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_f}{\sqrt{T}} \cdot \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_2^2\right)^{1/2} = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

For $k = 20$, we have

$$\frac{\|\boldsymbol{\delta}_{20}\|_f}{\sqrt{T}} \lesssim \frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\right\|_f^2 = O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(\frac{1}{m_{NT}^4}\right).$$

(2) We rewrite $\Delta_{k,i}$ instead of Δ_k for $k \in \mathcal{K}_2$ to derive the exact expression. For $k = 2$, we have

$$\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\left[\boldsymbol{I}_p - \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\right]\Delta_{2,i}\boldsymbol{l}_{i,0}\\
=&\frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^\top\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_j\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]\left(\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right)\\
&+O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right).
\end{aligned}$$

For $k = 7$ and $k = 12$, we have

$$\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\left[\boldsymbol{I}_p - \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\right]\left(\Delta_{7,i}+\Delta_{12,i}\right)\boldsymbol{l}_{i,0}\\
=&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^\top\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{j,0}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left[\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}-\boldsymbol{H}_j\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{\varepsilon}_j+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right).
\end{aligned}$$

For $k = 8, 15, 16, 17$, these terms jointly contribute to $\boldsymbol{\zeta}_{2,NT}^{\dagger}$. Note that

$$\begin{aligned}
&\left(\Delta_{8,i}+\Delta_{15,i}+\Delta_{16,i}+\Delta_{17,i}\right)\boldsymbol{l}_{i,0}+\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}_0^{-1}\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{U}_{j,0}\boldsymbol{\Omega}_j\boldsymbol{U}_{j,0}^\top\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
=&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{V}}_j\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{V}}_j^\top\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{V}}_j\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{V}}_j^\top\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{U}}_j\boldsymbol{\Omega}_j\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]-\boldsymbol{\Xi}_0^{-1}\cdot\ell\left(\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{U}_{j,0}\boldsymbol{\Omega}_j\boldsymbol{U}_{j,0}^\top\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right)\right\}\boldsymbol{l}_{i,0}\\
=:&\Delta_{8,i}'+\Delta_{15,i}'+\Delta_{16,i}'+\Delta_{17,i}'+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{T}\right)+O_p\left(\frac{1}{Tm_{NT}}\right).
\end{aligned}$$

From the equality

$$
\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\left(\Delta_{8,i}+\Delta_{15,i}+\Delta_{16,i}+\Delta_{17,i}\right)\boldsymbol{l}_{i,0}-\frac{\boldsymbol{\zeta}_{2,NT}^{\dagger}}{T}\\
=&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\left(\Delta'_{8,i}+\Delta'_{15,i}+\Delta'_{16,i}+\Delta'_{17,i}\right)+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{T}\right)+O_p\left(\frac{1}{Tm_{NT}}\right),
\end{aligned}
$$

where for each $i=1,\ldots,N$, $\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)$ is bounded, it suffices to show that $\left(\Delta'_{8,i},\Delta'_{15,i},\Delta'_{16,i},\Delta'_{17,i}\right)$ are negligible. The proof uses the same technique as that in proof of Lemma S.2.6. Decompose $\Delta'_{8,i}$:

$$
\begin{aligned}
\Delta'_{8,i}=&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\mathcal{M}_{T,R_0}\left\{\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\bar{\boldsymbol{H}}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\right]\right\}\boldsymbol{l}_{i,0}\\
=:&\Delta'_{8.1,i}+\Delta'_{8.2,i}.
\end{aligned}
$$

Let $\boldsymbol{\mathcal{Q}}=\left(\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}/T\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}$. Further decompose $\Delta'_{8.1,i}$:

$$
\begin{aligned}
\Delta'_{8.1,i}=&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\left(\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\\
=&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}+\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}}{T}\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\\
=&-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}-\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\\
&+\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}}{T}\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\\
&+\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}}{T}\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\\
&+\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}}{T}\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\\
=:&\Delta'_{8.1.1,i}+\cdots+\Delta'_{8.1.5,i},
\end{aligned}
$$

where

$$
\begin{aligned}
&\left\|\Delta'_{8.1.1,i}\right\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{T}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right),\quad\left\|\Delta'_{8.1.2,i}\right\|_2=O_p\left(\frac{1}{T\sqrt{N}}\right),\quad\left\|\Delta'_{8.1.3,i}\right\|_2=O_p\left(\frac{1}{T\sqrt{N}}\right),\\
&\left\|\Delta'_{8.1.4,i}\right\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{T}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right),\quad\left\|\Delta'_{8.1.5,i}\right\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).
\end{aligned}
$$

It follows that $\|\Delta'_{8.1.i}\|_2=O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(m_{NT}^{-3}\right)$. For $\Delta'_{8.2,i}$, since

$$
\begin{aligned}
\Delta'_{8.2,i}=&-\frac{1}{N^2T^2}\sum_{l=1}^{N}\left\{\sum_{j=1}^{N}\boldsymbol{l}_{i,0}^\top\otimes\frac{\boldsymbol{X}_i^\top}{\sqrt{T}}\cdot\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{l,0}\otimes\frac{\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\boldsymbol{X}_l\tilde{\boldsymbol{\Theta}}}{\sqrt{T}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_l^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_l\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\tilde{\boldsymbol{\Theta}}\right.\\
&\left.\cdot\boldsymbol{l}_{l,0}^\top\otimes\boldsymbol{X}_l^\top\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\right]\right\},
\end{aligned}
$$

we have

$$
\begin{aligned}
\left\|\Delta'_{8.2,i}\right\|_2 \lesssim & \frac{1}{N}\sum_{l=1}^{N}\left\|\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_l^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{l,0}\right\|_2 \\
\leq & \frac{1}{N}\sum_{l=1}^{N}\left\|\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_l^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{l,0}\right\|_2+\frac{1}{N}\sum_{l=1}^{N}\left(\frac{\|\boldsymbol{X}_l^\top\hat{\boldsymbol{F}}\|_2}{T}\cdot\left\|\frac{1}{NT^2}\sum_{j=1}^{N}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{l,0}\right\|_2\right) \\
= & O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),
\end{aligned}
$$

In the last step, we establish the same consistency rate using the same proof technique for $\left\|\Delta'_{8.1,i}\right\|_2$. The same reasoning applies to $\Delta_{k.2,i}$ for $k=15,16,17$, and therefore, we only establish the consistency rate for $\Delta_{k.1,i}$ in those cases. Decompose $\Delta'_{15,i}$:

$$
\begin{aligned}
\Delta'_{15,i}=&\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{V}}_j\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0} \\
&+\frac{1}{N}\sum_{l=1}^{N}\left\{\boldsymbol{l}_{i,0}^\top\otimes\frac{\boldsymbol{X}_i^\top}{\sqrt{T}}\cdot\boldsymbol{\Xi}^{-1}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{l,0}\otimes\frac{\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\boldsymbol{X}_l\tilde{\boldsymbol{\Theta}}}{\sqrt{T}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_l^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_l\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\tilde{\boldsymbol{\Theta}}\right. \\
&\left.\cdot\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{X}_l^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{V}}_j\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{l,0}\right\} \\
=:&\Delta'_{15.1,i}+\Delta'_{15.2,i}.
\end{aligned}
$$

Further decompose $\Delta'_{15.1,i}$:

$$
\begin{aligned}
\Delta'_{15.1,i}=&\frac{1}{NT^2}\sum_{j=1}^{N}\left[\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{\Phi}}_{4j}-\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\Phi}_{4j,0})}{\sqrt{T}}\cdot\boldsymbol{X}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4j,0}}{\sqrt{T}}\cdot\boldsymbol{X}_j^\top\left(\boldsymbol{P}_{\boldsymbol{F}_0}-\boldsymbol{P}_{\hat{\boldsymbol{F}}}\right)\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\hat{\boldsymbol{F}}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4j,0}}{\sqrt{T}}\cdot\boldsymbol{X}_j^\top\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\left(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right)\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.1.ji}}^\top\cdot\left(\boldsymbol{\pi}_{2ji}-\boldsymbol{\pi}_{2ji}^e\right)\right)\cdot\boldsymbol{X}_j^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\boldsymbol{\mathcal{U}}_{12.1.ji}^e\cdot\boldsymbol{X}_j^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\mathcal{M}_{p,R_0}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.2.ji}}^\top\cdot\left(\boldsymbol{\pi}_{2ji}-\boldsymbol{\pi}_{2ji}^e\right)\right)\cdot\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
&+\frac{1}{NT^2}\sum_{j=1}^{N}\left[\boldsymbol{\mathcal{U}}_{12.2.ji}^e\cdot\boldsymbol{F}_0^\top\left(\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top-\boldsymbol{\Omega}_j\right)\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\boldsymbol{\mathcal{Q}}\boldsymbol{l}_{i,0}\right] \\
=&\Delta'_{15.1.1,i}+\cdots+\Delta'_{15.1.7,i},
\end{aligned}
$$

where $\left(\boldsymbol{\mathcal{U}}_{12.1.ij}^e,\ \boldsymbol{\mathcal{U}}_{12.2.ij}^e,\ \bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.1.ij}},\ \bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\mathcal{U}}_{12.2.ij}}\right)$ are defined in the proof of Lemma S.2.3, and

$$
\left\|\Delta'_{15.1.1,i}\right\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),
$$

$$\left\|\Delta'_{15.1.2,i}\right\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right) + O_p\left(\frac{1}{m_{NT}^3}\right),$$
$$\left\|\Delta'_{15.1.3,i}\right\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{T}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right),$$
$$\left\|\Delta'_{15.1.4,i}\right\|_2 = O_p\left(\frac{1}{T^{3/2}}\right), \quad \left\|\Delta'_{15.1.5,i}\right\|_2 = O_p\left(\frac{1}{T\sqrt{N}}\right),$$
$$\left\|\Delta'_{15.1.6,i}\right\|_2 = O_p\left(\frac{1}{T^{3/2}}\right), \quad \left\|\Delta'_{15.1.7,i}\right\|_2 = O_p\left(\frac{1}{T\sqrt{N}}\right).$$

It follows that $\left\|\Delta'_{15.1,i}\right\|_2 = O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(m_{NT}^{-3}\right)$. For $k = 16, 17$, We can decompose similarly and show that both $\left\|\Delta'_{16.1,i}\right\|_2$ and $\left\|\Delta'_{17.1,i}\right\|_2$ are $O_p\left(m_{NT}^{-2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(m_{NT}^{-3}\right)$. □

Proof of Lemma S.2.8.

(1) Decompose the second-order derivative:

$$\begin{aligned}
\frac{\partial^2 f_{2,NT}(\boldsymbol{\vartheta})}{\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^\top} = \boldsymbol{J}\cdot\Bigg\{&\frac{2}{NT}\sum_{i=1}^N\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\boldsymbol{y}_i\boldsymbol{y}_i^\top\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\\
&-\frac{2}{NT}\sum_{i=1}^N\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{y}_i\boldsymbol{y}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\\
&\qquad\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\\
&-\frac{2}{NT}\sum_{i=1}^N\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{y}_i\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\\
&\qquad\otimes\boldsymbol{y}_i^\top\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\\
&-\frac{2}{NT}\sum_{i=1}^N\boldsymbol{y}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\otimes\left(\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}\right)^{-1}\boldsymbol{\Theta}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\\
&\qquad\otimes\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\boldsymbol{y}_i\Bigg\}\cdot\boldsymbol{J}^\top\\
=:\ &\Delta_1\left(\boldsymbol{\vartheta}\right)+\Delta_2\left(\boldsymbol{\vartheta}\right)+\Delta_3\left(\boldsymbol{\vartheta}\right)+\Delta_4\left(\boldsymbol{\vartheta}\right).
\end{aligned}\tag{S.4.6}$$

Note that

$$\begin{aligned}
\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp[\boldsymbol{I}_T-\boldsymbol{H}_i(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}})]\boldsymbol{y}_i}{T} =& \frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp[\boldsymbol{I}_T-\boldsymbol{H}_i(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}})](\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i)}{T}\\
=&\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\cdot\frac{\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp(\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i)}{T}\\
=&\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right),
\end{aligned}$$

and

$$\begin{aligned}
\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{y}_i =&\boldsymbol{b}_{i,0}+\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{F}_0\boldsymbol{l}_{i,0}}{T}+\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_i}{T}\right)\\
=&\boldsymbol{b}_{i,0}+\hat{\boldsymbol{\Phi}}_{1i}^\top\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{1i}^\top\hat{\boldsymbol{\Phi}}_{3i}.
\end{aligned}$$

For Δ_1,

$$\|\Delta_1\left(\boldsymbol{\vartheta}_0\right)\|_2 \le \frac{2}{N}\sum_{i=1}^N\left(\left\|\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right\|_2\cdot\left\|\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)\right\|_2^2\right)$$

$$
\begin{aligned}
&\lesssim \frac{1}{N}\sum_{i=1}^{N}\left(\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2^2+\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2\right)\\
&=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^2}\right),
\end{aligned}
$$

where the last step follows from

$$
\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2^2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^2}\right),\quad \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2=O_p\left(\frac{1}{m_{NT}^2}\right).
$$

For Δ_2,

$$
\begin{aligned}
&\left\|\Delta_2\left(\boldsymbol{\vartheta}_0\right)-\left(-\boldsymbol{J}\cdot\frac{2}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\cdot\boldsymbol{J}^{\top}\right)\right\|_2\\
\leq&\left\|\frac{2}{N}\sum_{i=1}^{N}\left[\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{y}_i\boldsymbol{y}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}-\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\right]\otimes\boldsymbol{S}_i^{\top}\left(\hat{\boldsymbol{F}}\right)\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)\right\|_2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{\Phi}}_{1i}^{\top}\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)^{\top}\hat{\boldsymbol{\Phi}}_{1i}+\hat{\boldsymbol{\Phi}}_{1i}^{\top}\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)\boldsymbol{b}_{i,0}^{\top}+\boldsymbol{b}_{i,0}\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)^{\top}\hat{\boldsymbol{\Phi}}_{1i}\right\|_2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2^2+\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2+\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2+\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2\right)\\
=&O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

For Δ_3,

$$
\begin{aligned}
\left\|\Delta_3\left(\boldsymbol{\vartheta}_0\right)\right\|_2\leq&\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\boldsymbol{b}_{i,0}+\hat{\boldsymbol{\Phi}}_{1i}^{\top}\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{1i}^{\top}\hat{\boldsymbol{\Phi}}_{3i}\right\|_2\cdot\left\|\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\hat{\boldsymbol{\Phi}}_{1i}\right\|_2\cdot\left\|\boldsymbol{S}_i^{\top}\left(\hat{\boldsymbol{F}}\right)\left(\hat{\boldsymbol{\Phi}}_{2i}\boldsymbol{l}_{i,0}+\hat{\boldsymbol{\Phi}}_{3i}\right)\right\|_2\right)\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2^2+\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2^2+\left\|\hat{\boldsymbol{\Phi}}_{2i}\right\|_2+\left\|\hat{\boldsymbol{\Phi}}_{3i}\right\|_2\right)\\
=&O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

Since $\Delta_3^{\top}=\Delta_4$, $\|\Delta_4(\boldsymbol{\vartheta}_0)\|_2$ has the same rate as $\|\Delta_3\left(\boldsymbol{\vartheta}_0\right)\|_2$. Therefore,

$$
\begin{aligned}
\frac{\partial^2 f_{2,NT}(\boldsymbol{\vartheta}_0)}{\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^{\top}}=&-\boldsymbol{J}\cdot\frac{2}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\cdot\boldsymbol{J}^{\top}\\
&+O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

(2) It is equivalent to showing that for $\boldsymbol{\vartheta}$ satisfying $\|\boldsymbol{\vartheta}-\boldsymbol{\vartheta}_0\|_2=o(1)$,

$$
\left\|\frac{\partial\ell(\partial^2 f_{2,NT}/\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^{\top})}{\partial\boldsymbol{\vartheta}}\left(\boldsymbol{\vartheta}\right)\right\|_2=O_p(1),\tag{S.4.7}
$$

where $\frac{\partial\ell(\partial^2 f_{2,NT}/\partial\boldsymbol{\vartheta}\partial\boldsymbol{\vartheta}^{\top})}{\partial\boldsymbol{\vartheta}}$ is a $(p-K_0)K_0\times(p-K_0)^2K_0^2$ matrix. The third order derivatives are calculated for $\Delta_1,\ldots,\Delta_4$ in (S.4.6) respectively. For Δ_1,

$$
\Delta_1\left(\boldsymbol{\vartheta}\right)=\boldsymbol{J}\cdot\frac{2}{N}\sum_{i=1}^{N}\left(\frac{\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}}{T}\right)^{-1}\otimes\frac{\boldsymbol{X}_i^{\top}}{\sqrt{T}}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{y}_i\boldsymbol{y}_i^{\top}}{T}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{X}_i}{\sqrt{T}}\cdot\boldsymbol{J}^{\top}
$$

$$=:\boldsymbol{J}\cdot\frac{2}{N}\sum_{i=1}^{N}\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)\otimes\Delta_{1.2,i}\left(\boldsymbol{\vartheta}\right)\cdot\boldsymbol{J}^{\top}.$$

Define

$$\Delta'_{1.1.i}\left(\boldsymbol{\vartheta}\right):=\frac{\partial\ell[\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)]}{\partial\boldsymbol{\vartheta}}=-\boldsymbol{J}\cdot\left(\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)\otimes\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\boldsymbol{\Theta}\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)\right)(\boldsymbol{I}_{K_0^2}+\boldsymbol{C}_{K_0,K_0}),$$

and

$$\begin{aligned}\Delta'_{1.2,i}\left(\boldsymbol{\vartheta}\right):=&\frac{\partial\ell[\Delta_{1.2,i}(\boldsymbol{\vartheta})]}{\partial\boldsymbol{\vartheta}}\\
=&-\boldsymbol{J}\cdot\left\{\left(\frac{\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}}{T}\right)^{-1}\frac{\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}}{\sqrt{T}}\frac{\boldsymbol{y}_i\boldsymbol{y}_i^{\top}}{T}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{X}_i}{\sqrt{T}}\otimes\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}}{\sqrt{T}}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{X}_i}{\sqrt{T}}\right.\\
&\left.+\left(\frac{\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}}{T}\right)^{-1}\frac{\boldsymbol{\Theta}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\otimes\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}}{\sqrt{T}}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{y}_i\boldsymbol{y}_i^{\top}}{T}\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta},\hat{\boldsymbol{F}}\right)\right]\frac{\boldsymbol{X}_i}{\sqrt{T}}\right\}\\
&\cdot\left(\boldsymbol{I}_{p^2}+\boldsymbol{C}_{p,p}\right),\end{aligned}$$

where $\boldsymbol{C}_{m,n}$ is a $mn\times mn$ commutation matrix transforming $\ell(\boldsymbol{M})$ into $\ell\left(\boldsymbol{M}^{\top}\right)$ with $\boldsymbol{M}\in\mathbb{R}^{m\times n}$, that is, $\boldsymbol{C}_{m,n}\cdot\ell(\boldsymbol{M})=\ell\left(\boldsymbol{M}^{\top}\right)$. Then,

$$\frac{\partial\ell[\Delta_1\left(\boldsymbol{\vartheta}\right)]}{\partial\boldsymbol{\vartheta}}=\frac{2}{N}\sum_{i=1}^{N}\left\{\Delta'_{1.1,i}\left(\boldsymbol{\vartheta}\right)\otimes\ell^{\top}[\Delta_{1.2,i}\left(\boldsymbol{\vartheta}\right)]+\ell^{\top}[\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)]\otimes\Delta'_{1.2,i}\left(\boldsymbol{\vartheta}\right)\right\}\cdot\boldsymbol{C}_{pK_0,pK_0}\cdot\boldsymbol{J}^{\top}\otimes\boldsymbol{J}^{\top}.$$

For $\boldsymbol{\vartheta}$ satisfying $\|\boldsymbol{\vartheta}-\boldsymbol{\vartheta}_0\|_2=o(1)$, since $\|\Delta_{1.1,i}\left(\boldsymbol{\vartheta}\right)\|_f$, $\|\Delta_{1.2,i}\left(\boldsymbol{\vartheta}\right)\|_f$, $\left\|\Delta'_{1.1,i}\left(\boldsymbol{\vartheta}\right)\right\|_2$ and $\left\|\Delta'_{1.2,i}\left(\boldsymbol{\vartheta}\right)\right\|_2$ are $O_p(1)$, we have

$$\left\|\frac{\partial\ell[\Delta_1(\boldsymbol{\vartheta})]}{\partial\boldsymbol{\vartheta}}\right\|_2=O_p(1).\tag{S.4.8}$$

Since taking differentiation w.r.t. $\boldsymbol{\vartheta}$ does not introduce higher order terms as in the proof of Δ_1,

$$\left\|\frac{\partial\ell[\Delta_2(\boldsymbol{\vartheta})]}{\partial\boldsymbol{\vartheta}}\right\|_2=O_p(1),\quad\left\|\frac{\partial\ell[\Delta_3(\boldsymbol{\vartheta})]}{\partial\boldsymbol{\vartheta}}\right\|_2=O_p(1),\quad\left\|\frac{\partial\ell[\Delta_4(\boldsymbol{\vartheta})]}{\partial\boldsymbol{\vartheta}}\right\|_2=O_p(1)\tag{S.4.9}$$

can be showed similarly. Equation (S.4.7) is proved by (S.4.6), (S.4.8) and (S.4.9). □

Proof of Lemma S.2.9. By (S.2.4), it follows that

$$\begin{aligned}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{M}_i\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\right)^{\top}\boldsymbol{\varepsilon}_i=&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{M}_i\boldsymbol{\mathcal{Q}}^{\top}\left(\boldsymbol{\delta}_1+\cdots+\boldsymbol{\delta}_{20}\right)^{\top}\boldsymbol{\varepsilon}_i\\
=&\left[\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^{\top}(\boldsymbol{\delta}_1+\cdots+\boldsymbol{\delta}_{20})\otimes\boldsymbol{M}_i\right]\ell\left(\boldsymbol{\mathcal{Q}}^{\top}\right)\\
=:&\left(\Delta_1+\cdots+\Delta_{20}\right)\cdot\ell\left(\boldsymbol{\mathcal{Q}}^{\top}\right),\end{aligned}$$

where $\boldsymbol{\mathcal{Q}}=\left(\boldsymbol{F}_0^{\top}\hat{\boldsymbol{F}}/T\right)^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}$. Decompose Δ_1:

$$\Delta_1=\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^{\top}\boldsymbol{X}_j\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{j,0}\boldsymbol{b}_{j,0}^{\top}\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^{\top}\boldsymbol{X}_j^{\top}\hat{\boldsymbol{U}}_j^{\top}\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i$$

$$
\begin{aligned}
&-\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_j\boldsymbol{X}_j\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{j,0}\boldsymbol{b}_{j,0}^\top\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i\\
&-\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}\hat{\boldsymbol{U}}_j\boldsymbol{X}_j\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{j,0}\boldsymbol{b}_{j,0}^\top\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)^\top\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i\\
=:&\Delta_{1.1}+\Delta_{1.2}+\Delta_{1.3}.
\end{aligned}
$$

For $\Delta_{1.1}$,

$$
\|\Delta_{1.1}\|_2\lesssim\frac{\|\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\|_2^2}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}\boldsymbol{x}_{jt}^\top\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\frac{\|\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\|_2^2}{T^2}\right)^{1/2}.
$$

Since

$$
\begin{aligned}
\mathbb{E}_{\mathcal{C}}\left(\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}\boldsymbol{x}_{jt}^\top\otimes\boldsymbol{M}_i\right\|_f^2\right)=&\frac{1}{NT}\sum_{i,k=1}^{N}\sum_{s,t=1}^{T}\left[\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}\varepsilon_{kt}\boldsymbol{x}_{js}^\top\boldsymbol{x}_{jt}\right)\cdot\operatorname{tr}\left(\boldsymbol{M}_i^\top\boldsymbol{M}_k\right)\right]\\
=&\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left[\sigma_{it}^2\cdot\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{jt}\|_2^2\right)\cdot\operatorname{tr}\left(\boldsymbol{M}_i^\top\boldsymbol{M}_i\right)\right]=O_p(1),
\end{aligned}
$$

where the second step follows from $\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}\varepsilon_{kt}\boldsymbol{x}_{js}^\top\boldsymbol{x}_{jt}\right)=0$ for $i\neq k$ or $s\neq t$ and

$$
\mathbb{E}_{\mathcal{C}}\left[\|\boldsymbol{x}_{jt}\|_2^2\cdot\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]=\mathbb{E}_{\mathcal{C}}\left[\|\boldsymbol{x}_{jt}\|_2^2\cdot\mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]=0.
$$

we have $\|\Delta_{1.1}\|_2=O_p\left((NT)^{-1/2}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right)$. The terms $\Delta_{1.2}$ and $\Delta_{1.3}$ have the same rate as $\Delta_{1.1}$. By a similar argument,

$$
\|\Delta_2\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right),\quad\|\Delta_3\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right),\quad\|\Delta_4\|_2=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right).
$$

Decompose Δ_5:

$$
\begin{aligned}
\Delta_5=&\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\boldsymbol{b}_{j,0}^\top(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}})^\top\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i\\
&-\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{V}}_j\boldsymbol{\varepsilon}_j\boldsymbol{b}_{j,0}^\top(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}})^\top\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i\\
&-\frac{1}{N^2T^2}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}\hat{\boldsymbol{U}}_j\boldsymbol{\varepsilon}_j\boldsymbol{b}_{j,0}^\top(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}})^\top\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\otimes\boldsymbol{M}_i\\
=:&\Delta_{5.1}+\Delta_{5.2}+\Delta_{5.3}.
\end{aligned}
$$

For $\Delta_{5.1}$,

$$
\|\Delta_{5.1}\|_2\leq\left\|\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right\|_2\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\cdot\boldsymbol{M}_i\right)\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\frac{\|\boldsymbol{X}_j^\top\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\|_2^2}{T^2}\right)^{1/2}.
$$

Since

$$
\mathbb{E}_{\mathcal{C}}\left(\left\|\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\cdot\boldsymbol{M}_i\right)\right\|_f^2\right)=\frac{1}{N^2T^2}\sum_{i=1}^{N}\left\{\mathbb{E}_{\mathcal{C}}\left[\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\right)^2\right]\cdot\operatorname{tr}\left(\boldsymbol{M}_i^\top\boldsymbol{M}_i\right)\right\}
$$

$$
\begin{aligned}
&\lesssim \frac{1}{N^2T^2}\sum_{i\neq j}\sum_{t=1}^{T}\left[\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^2\right)\cdot\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{jt}^2\right)\right]+\frac{1}{N^2T^2}\mathbb{E}_{\mathcal{C}}\left[\left(\sum_{t=1}^{T}\varepsilon_{jt}^2\right)^2\right]\\
&=O_p\left(\frac{1}{NT}\right)+O_p\left(\frac{1}{N^2}\right)=O_p\left(\frac{1}{Nm_{NT}^2}\right),
\end{aligned}
$$

we have $\|\Delta_{5.1}\|_2 = O_p\left(N^{-1/2}m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)$, and it follows that $\|\Delta_5\|_2 = O_p\left(N^{-1/2}m_{NT}^{-1}\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)$. For Δ_6,

$$
\begin{aligned}
\|\Delta_6\|_2 &\lesssim \frac{1}{\sqrt{NT}}\cdot\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{F}_0\otimes\boldsymbol{M}_i\right\|_2\cdot\left\|\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{l}_{i,0}\boldsymbol{\varepsilon}_i^\top\hat{\boldsymbol{F}}\right\|_2\\
&=\frac{1}{\sqrt{NT}}\cdot O_p(1)\cdot O_p\left(\frac{1}{m_{NT}}\right)=O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right).
\end{aligned}
$$

Decompose Δ_7:

$$
\Delta_7=\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\otimes\boldsymbol{M}_i-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\frac{\boldsymbol{F}_0^\top\hat{\boldsymbol{F}}}{T}\otimes\boldsymbol{M}_i=:\Delta_{7.1}+\Delta_{7.2}.
$$

Since

$$
\|\Delta_{7.2}\|_2\lesssim\frac{1}{NT}\cdot\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{G}\otimes\boldsymbol{M}_i\right\|_2\cdot\left\|\frac{1}{\sqrt{NT}}\sum_{j=1}^{N}\boldsymbol{G}^\top\boldsymbol{\varepsilon}_j\boldsymbol{l}_{j,0}^\top\right\|_2=O_p\left(\frac{1}{NT}\right),
$$

we have

$$
\begin{aligned}
\Delta_7\ell\left(\boldsymbol{\mathcal{Q}}^\top\right)&=\Delta_{7.1}\ell\left(\boldsymbol{\mathcal{Q}}^\top\right)+O_p\left(\frac{1}{NT}\right)\\
&=\frac{1}{N^2}\sum_{i,j=1}^{N}\boldsymbol{M}_i\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{jt}\right)+O_p\left(\frac{1}{NT}\right).
\end{aligned}
$$

For Δ_8,

$$
\begin{aligned}
\|\Delta_8\|_2\lesssim&\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_j\cdot\boldsymbol{M}_i\right)\right\|_f^2\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{j=1}^{N}\left\|\boldsymbol{\varepsilon}_j^\top\hat{\boldsymbol{F}}\right\|_2^2\right)^{1/2}\\
&+\frac{1}{\sqrt{NT}}\cdot\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{G}\otimes\boldsymbol{M}_i\right\|_2\cdot\left\|\frac{1}{NT^2}\sum_{j=1}^{N}\boldsymbol{G}^\top\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^\top\hat{\boldsymbol{F}}\right\|_2\\
=&O_p\left(\frac{1}{\sqrt{N}m_{NT}}\right)\cdot O_p\left(\frac{1}{m_{NT}}\right)+\frac{1}{\sqrt{NT}}\cdot O_p(1)\cdot O_p\left(\frac{1}{m_{NT}^2}\right)\\
=&O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).
\end{aligned}
$$

For Δ_9,

$$
\begin{aligned}
\|\Delta_9\|_2\lesssim&\frac{1}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_j\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{F}_0\|_2}{\sqrt{T}}\cdot\left(\frac{1}{NT^2}\sum_{j=1}^{N}\left\|\boldsymbol{\varepsilon}_j^\top\hat{\boldsymbol{F}}\right\|_2^2\right)^{1/2}\\
=&\frac{1}{\sqrt{NT}}\cdot O_p\left(1\right)\cdot\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right)\right]\cdot O_p\left(\frac{1}{m_{NT}}\right)\\
=&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{NT}m_{NT}^2}\right).
\end{aligned}
$$

For Δ_{15},

$$\|\Delta_{15}\|_2 \lesssim \frac{1}{\sqrt{NT}} \cdot \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \boldsymbol{X}_j \otimes \boldsymbol{M}_i \right\|_2^2 \right)^{1/2} \cdot \left(\frac{1}{NT^4} \sum_{j=1}^{N} \left\| \boldsymbol{X}_j^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{\varepsilon}_j \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{F}} \right\|_2^2 \right)^{1/2}.$$

Since

$$\frac{1}{NT^4} \sum_{j=1}^{N} \left\| \boldsymbol{X}_j^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{\varepsilon}_j \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{F}} \right\|_2^2 \lesssim \frac{1}{NT^4} \sum_{j=1}^{N} \left(\left\| \boldsymbol{X}_j^\top \boldsymbol{\varepsilon}_j \right\|_2^4 + \left\| \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_j \right\|_2^4 + \left\| \left(\hat{\boldsymbol{F}} \boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0 \right)^\top \boldsymbol{\varepsilon}_j \right\|_2^4 \right) = O_p\left(\left\| \tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0 \right\|_2^4 \right) + O_p\left(\frac{1}{m_{NT}^4} \right),$$

it follows that $\|\Delta_{15}\|_2 = O_p\left((NT)^{-1/2} \left\| \tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0 \right\|_2^2 \right) + O_p\left((NT)^{-1/2} m_{NT}^{-2} \right)$. The consistency rate for other terms are showed below:

$$\|\Delta_{10}\|_2 \lesssim \frac{1}{\sqrt{NT}} \cdot \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{F}_0 \otimes \boldsymbol{M}_i \right\|_2 \cdot \frac{1}{NT} \sum_{j=1}^{N} \left\| \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{V}}_j \hat{\boldsymbol{F}} \right\|_2 = O_p\left(\frac{1}{\sqrt{NT} m_{NT}} \right),$$

$$\begin{aligned}
\|\Delta_{11}\|_2 \lesssim & \frac{1}{\sqrt{NT}} \cdot \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \boldsymbol{X}_j \otimes \boldsymbol{M}_i \right\|_2^2 \right)^{1/2} \cdot \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}} \boldsymbol{F}_0\|_f}{\sqrt{T}} \cdot \left(\frac{1}{NT^2} \sum_{j=1}^{N} \left\| \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{V}}_j^\top \hat{\boldsymbol{F}} \right\|_2^2 \right)^{1/2} \\
= & O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{NT} m_{NT}} \right) + O_p\left(\frac{1}{\sqrt{NT} m_{NT}^2} \right),
\end{aligned}$$

$$\|\Delta_{12}\|_2 \lesssim \frac{1}{\sqrt{NT}} \cdot \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \boldsymbol{X}_j \otimes \boldsymbol{M}_i \right\|_2^2 \right)^{1/2} \cdot \left(\frac{1}{NT^2} \sum_{j=1}^{N} \left\| \boldsymbol{X}_j^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{\varepsilon}_j \right\|_2^2 \right)^{1/2} = O_p\left(\frac{1}{\sqrt{NT} m_{NT}} \right),$$

$$\begin{aligned}
\|\Delta_{13}\|_2 \lesssim & \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{NT} \sum_{i=1}^{N} \left(\boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j \cdot \boldsymbol{M}_i \right) \right\|_f^2 \right)^{1/2} \cdot \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{F}_0\|_f}{\sqrt{T}} \\
& + \frac{1}{T\sqrt{N}} \cdot \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{G} \otimes \boldsymbol{M}_i \right\|_2 \cdot \left(\frac{1}{NT} \sum_{j=1}^{N} \left\| \boldsymbol{G}^\top \boldsymbol{\varepsilon}_j \right\|_2^2 \right)^{1/2} \cdot \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{F}_0\|_f}{\sqrt{T}} \\
= & O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N} m_{NT}} \right) + O_p\left(\frac{1}{\sqrt{N} m_{NT}^2} \right),
\end{aligned}$$

$$\begin{aligned}
\|\Delta_{14}\|_2 \lesssim & \frac{1}{\sqrt{NT}} \cdot \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \boldsymbol{X}_j \otimes \boldsymbol{M}_i \right\|_2^2 \right)^{1/2} \cdot \left(\frac{1}{NT^2} \sum_{j=1}^{N} \left\| \boldsymbol{X}_j^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{\varepsilon}_j \right\|_2^2 \right)^{1/2} \cdot \frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp} \boldsymbol{F}_0\|_f}{\sqrt{T}} \\
= & O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{NT} m_{NT}} \right) + O_p\left(\frac{1}{\sqrt{NT} m_{NT}^2} \right),
\end{aligned}$$

$$\begin{aligned}
\|\Delta_{16}\|_2 \lesssim & \left(\frac{1}{N} \sum_{j=1}^{N} \left\| \frac{1}{NT} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_j \cdot \boldsymbol{M}_i \right\|_2^2 \right)^{1/2} \cdot \left(\frac{1}{NT^2} \sum_{j=1}^{N} \left\| \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{V}}_j^\top \hat{\boldsymbol{F}} \right\|_2^2 \right)^{1/2} \\
& + \frac{1}{T\sqrt{N}} \cdot \left\| \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{\varepsilon}_i^\top \boldsymbol{G} \otimes \boldsymbol{M}_i \right\|_2 \cdot \left(\frac{1}{NT} \sum_{j=1}^{N} \left\| \boldsymbol{G}^\top \boldsymbol{\varepsilon}_j \right\|_2^2 \right)^{1/2} \cdot \left(\frac{1}{NT^2} \sum_{j=1}^{N} \left\| \boldsymbol{\varepsilon}_j^\top \hat{\boldsymbol{V}}_j^\top \hat{\boldsymbol{F}} \right\|_2^2 \right)^{1/2}
\end{aligned}$$

$$=O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right)+O_p\left(\frac{1}{T\sqrt{N}m_{NT}}\right)=O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right),$$

$$\begin{aligned}\|\Delta_{17}\|_2 \lesssim & \frac{1}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^N\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_j\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT^4}\sum_{j=1}^N\left\|\boldsymbol{X}_j^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_j\right\|_2^4\right)^{1/2}\\ =&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{\sqrt{NT}}\right)+O_p\left(\frac{1}{\sqrt{NT}m_{NT}^2}\right),\end{aligned}$$

$$\|\Delta_{18}\|_2\lesssim\frac{1}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^N\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_j\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{F}_0\|_f}{\sqrt{T}}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right)+O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right),$$

$$\|\Delta_{19}\|_2\lesssim\frac{1}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^N\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{F}_0\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{F}_0\|_f}{\sqrt{T}}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right)+O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right),$$

$$\|\Delta_{20}\|_2\lesssim\frac{1}{\sqrt{NT}}\cdot\left(\frac{1}{N}\sum_{j=1}^N\left\|\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_j\otimes\boldsymbol{M}_i\right\|_2^2\right)^{1/2}\cdot\frac{\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{F}_0\|_f^2}{T}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{\sqrt{NT}}\right)+O_p\left(\frac{1}{\sqrt{NT}m_{NT}^2}\right).$$

Collecting all terms above yields the desired result. □

Lemma S.4.2. *Suppose Assumptions 2, 5, and 6 hold. Then:*

(1) $\frac{1}{N}\sum_{i=1}^N\left\|\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{T}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}-\frac{\boldsymbol{X}_i^\top\boldsymbol{F}_0}{T}\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right\|_2^2=O_p\left(\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$

(2) $\frac{1}{NT^2}\sum_{i=1}^N\left\|\boldsymbol{X}_i^\top\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right)\boldsymbol{X}_i\right\|_2^2=O_p\left(\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$

(3) $\frac{1}{N}\sum_{i=1}^N\left\|\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right\|_2^2=O_p\left(\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$

(4) $\frac{1}{N}\sum_{i=1}^N\left\|\boldsymbol{\Phi}_{4i}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)-\boldsymbol{\Phi}_{4i,0}\right\|_2^2=O_p\left(\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2\right)+O_p\left(\frac{1}{m_{NT}^4}\right).$

Proof of Lemma S.2.10. Decompose the term in Lemma S.2.10 as following:

$$\begin{aligned}&\frac{1}{NT}\sum_{i=1}^N\left\{\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)-\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{i,0}\otimes\left[\boldsymbol{I}_T-\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\right]\right\}\boldsymbol{\varepsilon}_i\\ =&\frac{1}{NT}\sum_{i=1}^N\boldsymbol{b}_{i,0}\otimes\boldsymbol{Q}_i^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{\varepsilon}_i-\frac{1}{N^2T}\sum_{i,j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\ &-\frac{1}{N^2T}\sum_{i,j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\bar{\boldsymbol{H}}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\ &+\frac{1}{N^2T}\sum_{i,j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\boldsymbol{\varepsilon}_i\\ &+\frac{1}{N^2T}\sum_{i,j=1}^N\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_j^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\bar{\boldsymbol{H}}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_i\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\boldsymbol{\varepsilon}_i\end{aligned}$$

$$=:\Delta_1+\cdots+\Delta_5.$$

(1) Decompose Δ_1:

$$\begin{aligned}
\Delta_1 =& \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_i^\top\left(\hat{\boldsymbol{F}}\right)\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\\
=& \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\left[\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right]^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\varepsilon}_i\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^\top\left[\boldsymbol{X}_i^\top\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}-\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}^\top\boldsymbol{\varepsilon}_i\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\left(\boldsymbol{S}_{i,0}^\top\boldsymbol{\Sigma}_{i\boldsymbol{F}}-\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\right)\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0-\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top\boldsymbol{\varepsilon}_i\\
&+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0-\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top\boldsymbol{\varepsilon}_i\\
=:&\Delta_{1.1}+\cdots+\Delta_{1.5}.
\end{aligned}$$

For $\Delta_{1.1}$,

$$\|\Delta_{1.1}\|_2\lesssim\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2}=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).$$

For $\Delta_{1.3}$,

$$\begin{aligned}
\|\Delta_{1.3}\|_2\lesssim&\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1}-\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{F}}^\top\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2}\\
=&O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).
\end{aligned}$$

For $\Delta_{1.4}$,

$$\begin{aligned}
\|\Delta_{1.4}\|_2\lesssim&\max_i\|\boldsymbol{S}_{i,0}\|_2\cdot\left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\Sigma}_{i\boldsymbol{F}}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\right)^\top\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2}\\
&+\max_i\left(\|\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\|_2\cdot\left\|\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{S}_i}\right\|_2\right)\cdot\left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\pi}_{1i}-\boldsymbol{\pi}_{1i}^{e}\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\right)^\top\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2}\\
=&O_p(1)\cdot O_p\left(\frac{1}{\sqrt{T}}\right)\cdot\left[O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^2}\right)\right]=O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{T}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right).
\end{aligned}$$

By Lemma S.2.9,

$$\Delta_{1.5}=\frac{\varsigma_{1,NT}}{N}+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right),$$

where

$$\begin{aligned}
\varsigma_{1,NT}=&-\frac{1}{N}\sum_{i,j=1}^{N}\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{jt}\right)\\
=&-\frac{1}{T}\sum_{t=1}^{T}\left\{\left[\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\varepsilon_{it}\otimes\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\right]\cdot\left(\frac{1}{\sqrt{N}}\sum_{j=1}^{N}\boldsymbol{l}_{j,0}\varepsilon_{jt}\right)\right\}=O_p(1).
\end{aligned}$$

To conclude,

$$\Delta_1 = \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{Q}_{i,0}^{\top}\boldsymbol{\varepsilon}_i + \frac{\varsigma_{1,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

(2) For Δ_2,

$$\begin{aligned}
\Delta_2 =& -\frac{1}{N^2T}\sum_{i,j=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \boldsymbol{S}_j^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{X}_j^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i \\
=& -\frac{1}{N^2T}\sum_{i,j=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \left[\boldsymbol{S}_j\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{S}_{j,0}\right]^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i - \frac{1}{N^2T}\sum_{i,j=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i \\
&- \frac{1}{N^2T}\sum_{i,j=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \boldsymbol{S}_{j,0}^{\top}\left[\boldsymbol{X}_j^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1} - \boldsymbol{X}_j^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\boldsymbol{\varepsilon}_i \\
&- \frac{1}{N^2T}\sum_{i,j=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \boldsymbol{S}_{j,0}^{\top}\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{\top}\boldsymbol{\varepsilon}_i \\
=:& \Delta_{2.1} + \cdots + \Delta_{2.4},
\end{aligned}$$

where

$$\Delta_{2.1} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right), \quad \Delta_{2.3} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

For $\Delta_{2.5}$,

$$\begin{aligned}
\Delta_{2.5} =& \frac{1}{N^3}\sum_{i,j,k=1}^{N} w_{ij}\boldsymbol{b}_{j,0} \otimes \boldsymbol{S}_{j,0}^{\top}\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{k,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{kt}\right) + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right) \\
=:& \frac{\varsigma_{2,NT}}{N} + \textit{low-order terms.}
\end{aligned}$$

It follows that

$$\Delta_2 = -\frac{1}{N^2T}\sum_{i,j=1}^{N} \boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top} \otimes \boldsymbol{Q}_{j,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp} \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \otimes \boldsymbol{I}_T \cdot \boldsymbol{l}_{i,0} \otimes \boldsymbol{\varepsilon}_i + \frac{\varsigma_{2,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

(3) Let $\boldsymbol{M}_k(\boldsymbol{F}) = N^{-1}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top} \otimes \boldsymbol{Q}_j^{\top}(\boldsymbol{F})\boldsymbol{P}_{\boldsymbol{G}}^{\perp} \cdot \boldsymbol{\Xi}^{-1}(\boldsymbol{\Theta}_0, \boldsymbol{F}) \cdot \boldsymbol{l}_{k,0} \otimes \boldsymbol{X}_k\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_k^{\top}\boldsymbol{P}_{\boldsymbol{F}}^{\perp}\boldsymbol{X}_k\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^{\top}$. For Δ_3,

$$\begin{aligned}
\Delta_3 =& -\frac{1}{N^2T}\sum_{i,k=1}^{N}\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\boldsymbol{X}_k^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\right] \\
=& -\frac{1}{N^2T}\sum_{i,k=1}^{N}\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\boldsymbol{X}_k^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{\varepsilon}_i\right] \\
&- \frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\left[\boldsymbol{X}_k^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1} - \boldsymbol{X}_k^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{F}}^{\top}\boldsymbol{\varepsilon}_i\right\} \\
&- \frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right] \cdot w_{ik}\boldsymbol{\Sigma}_{k\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{\top}\boldsymbol{\varepsilon}_i\right\} \\
&+ \frac{1}{N^2T}\sum_{i,k=1}^{N}\left[\boldsymbol{M}_k\left(\boldsymbol{F}_0\right) \cdot w_{ik}\boldsymbol{\Sigma}_{k\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0\right)^{\top}\boldsymbol{\varepsilon}_i\right] \\
=:& \Delta_{3.1} + \cdots + \Delta_{3.4}.
\end{aligned}$$

It is straightforward to show that

$$\|\Delta_{3.2}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right), \quad \|\Delta_{3.3}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2^2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

Since

$$\begin{aligned}
&\left\|\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right]\cdot w_{ik}\boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\varepsilon}_i\right\}\right\|_2 \\
\lesssim &\left(\frac{1}{N}\sum_{k=1}^{N}\left\|\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{N}\sum_{k=1}^{N}\left\|\frac{1}{NT}\sum_{i=1}^{N}w_{ik}\boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2} \\
= &\left[O_p\left(\left\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\right\|_2\right) + O_p\left(\frac{1}{m_{NT}}\right)\right]\cdot O_p\left(\frac{1}{\sqrt{NT}}\right) = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right) + O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right),
\end{aligned}$$

it follows that

$$\Delta_{3.1} = -\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}_0^{-1}\cdot \bar{\boldsymbol{H}}_0 \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes \boldsymbol{I}_T \cdot \boldsymbol{l}_{i,0}\otimes \boldsymbol{\varepsilon}_i + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{NT}}\right) + O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right).$$

For $\Delta_{3.4}$,

$$\begin{aligned}
\Delta_{3.4} =& \frac{1}{N^3}\sum_{i,k,l=1}^{N}\left[\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\cdot w_{ik}\boldsymbol{\Sigma}_{k\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{l,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{lt}\right)\right] + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right) \\
=& \frac{1}{N}\cdot\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}_0^{-1}\cdot \frac{1}{N^2}\sum_{i,k,l=1}^{N}w_{ik}\boldsymbol{l}_{k,0}\otimes \frac{\boldsymbol{X}_k\boldsymbol{\Phi}_{4k,0}^\top \boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{l,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{lt}\right) \\
&+ \textit{low-order terms} \\
=:& \frac{\varsigma_{3,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).
\end{aligned}$$

To conclude,

$$\Delta_3 = -\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}_0^{-1}\cdot \bar{\boldsymbol{H}}_0 \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes \boldsymbol{I}_T \cdot \boldsymbol{l}_{i,0}\otimes \boldsymbol{\varepsilon}_i + \frac{\varsigma_{3,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

(4) For Δ_4,

$$\begin{aligned}
\Delta_4 =& \frac{1}{N^2T}\sum_{i,j=1}^{N}\left[w_{ij}\boldsymbol{b}_{j,0}\otimes\left(\frac{\hat{\boldsymbol{Q}}_j^\top(\hat{\boldsymbol{F}})\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}) - \boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\right)\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_i\right] \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left(\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i\right) \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left\{w_{ij}\boldsymbol{b}_{j,0}\otimes\frac{\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\cdot\left[\boldsymbol{X}_i^\top\boldsymbol{F}_0\left(\boldsymbol{F}_0^\top\boldsymbol{F}_0\right)^{-1} - \boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^\top\hat{\boldsymbol{F}}^\top\boldsymbol{\varepsilon}_i\right\} \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left\{w_{ij}\boldsymbol{b}_{j,0}\otimes\left[\frac{\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i}{T}\cdot\frac{\boldsymbol{\Phi}_{4i,0}^\top\boldsymbol{F}_0}{\sqrt{T}} - \frac{\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T}\cdot\frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\right]\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top\boldsymbol{\varepsilon}_i\right\} \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left[w_{ij}\boldsymbol{b}_{j,0}\otimes\frac{\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{\sqrt{T}}\cdot\frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top\boldsymbol{\varepsilon}_i\right] \\
=& \Delta_{4.1} + \cdots + \Delta_{4.5},
\end{aligned}$$

where

$$\Delta_{4.1} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right), \quad \Delta_{4.3} = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

For $\Delta_{4.4}$,

$$\begin{aligned}
\|\Delta_{4.4}\|_2 \lesssim & \max_{i,j}\left\|w_{ij}\boldsymbol{b}_{j,0} \otimes \frac{\boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T}\right\|_2 \cdot \frac{1}{N}\sum_{i=1}^N\left(\left\|\frac{(\boldsymbol{\Phi}_{4i,0} - \boldsymbol{\Phi}_{4i,0}^e)^\top \boldsymbol{F}_0}{\sqrt{T}}\right\|_2 \cdot \left\|\frac{(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}})^\top \boldsymbol{\varepsilon}_i}{T}\right\|_2\right) \\
& + \max_{i,j}\left(\|w_{ij}\boldsymbol{b}_{j,0}\|_2 \cdot \|\boldsymbol{S}_{j,0}\|_2 \cdot \frac{\|\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0\|_2}{\sqrt{T}}\right) \cdot \left[\frac{1}{N^2}\sum_{i=1}^N\left(\left\|\frac{\boldsymbol{X}_i^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i))}{T}\right\|_2 \cdot \left\|\frac{(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}})^\top \boldsymbol{\varepsilon}_i}{T}\right\|_2\right)\right. \\
& \left. + \frac{1}{N^2}\sum_{i=1}^N\sum_{j\neq i}\left(\left\|\frac{\boldsymbol{X}_j^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp(\boldsymbol{X}_i - \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i))}{T}\right\|_2 \cdot \left\|\frac{(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}})^\top \boldsymbol{\varepsilon}_i}{T}\right\|_2\right)\right] \\
= & O_p(1) \cdot O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{N}\right) \cdot \left[O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^2}\right)\right] = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).
\end{aligned}$$

For $\Delta_{4.5}$,

$$\begin{aligned}
\Delta_{4.5} = & -\frac{1}{N^3}\sum_{i,j,k=1}^N\left[w_{ij}\boldsymbol{b}_{j,0} \otimes \frac{\boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T} \cdot \frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{k,0}\left(\frac{1}{T}\sum_{t=1}^T \varepsilon_{it}\varepsilon_{kt}\right)\right] \\
& + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right) \\
=: & \frac{\varsigma_{4,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right) + O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).
\end{aligned}$$

To conclude,

$$\Delta_4 = \frac{1}{N^2T}\sum_{i,j=1}^N\left(\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top \otimes \boldsymbol{Q}_{j,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1} \otimes \boldsymbol{I}_T \cdot \boldsymbol{l}_{i,0} \otimes \boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i\right) + \frac{\varsigma_{4,NT}}{N} + O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right) + O_p\left(\frac{1}{m_{NT}^3}\right).$$

(5) For Δ_5,

$$\begin{aligned}
\Delta_5 = & \frac{1}{N^2T}\sum_{i,k=1}^N\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}_0, \hat{\boldsymbol{F}})}{\sqrt{T}}\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{\varepsilon}_i\right] \\
= & \frac{1}{N^2T}\sum_{i,k=1}^N\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i}(\boldsymbol{\Theta}_0, \hat{\boldsymbol{F}}) - \boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{\varepsilon}_i\right] \\
& + \frac{1}{N^2T}\sum_{i,k=1}^N\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{\varepsilon}_i\right] \\
& + \frac{1}{N^2T}\sum_{i,k=1}^N\left\{\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\left[\boldsymbol{X}_i^\top \boldsymbol{F}_0\left(\boldsymbol{F}_0^\top \boldsymbol{F}_0\right)^{-1} - \boldsymbol{X}_i^\top \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\left(\boldsymbol{O}_{\boldsymbol{F}}^\top \hat{\boldsymbol{F}}^\top \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right]\boldsymbol{O}_{\boldsymbol{F}}^\top \hat{\boldsymbol{F}}^\top \boldsymbol{\varepsilon}_i\right\} \\
& + \frac{1}{N^2T}\sum_{i,k=1}^N\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right] \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\Phi}_{4i,0}}{\sqrt{T}}\boldsymbol{X}_i^\top \boldsymbol{F}_0\left(\boldsymbol{F}_0^\top \boldsymbol{F}_0\right)^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top \boldsymbol{\varepsilon}_i\right\} \\
& + \frac{1}{N^2T}\sum_{i,k=1}^N\left\{\boldsymbol{M}_k\left(\boldsymbol{F}_0\right) \cdot w_{ik}\left[\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \boldsymbol{X}_i}{T} \cdot \frac{\boldsymbol{\Phi}_{4i,0}^\top \boldsymbol{F}_0}{\sqrt{T}} - \frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T} \cdot \frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\right]\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top \boldsymbol{\varepsilon}_i\right\} \\
& + \frac{1}{N^2T}\sum_{i,k=1}^N\left[\boldsymbol{M}_k\left(\boldsymbol{F}_0\right) \cdot w_{ik}\frac{\boldsymbol{X}_k^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp \mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T} \cdot \frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)^\top \boldsymbol{\varepsilon}_i\right]
\end{aligned}$$

$$=:\Delta_{5.1}+\cdots+\Delta_{5.6},$$

where

$$\|\Delta_{5.1}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),\quad \|\Delta_{5.3}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),$$
$$\|\Delta_{5.4}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2^2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right),\quad \|\Delta_{5.5}\|_2 = O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}^2}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).$$

For $\Delta_{5.2}$,

$$\begin{aligned}
\Delta_{5.2} =& \frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\bar{\boldsymbol{H}}_0\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i\\
&+\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right]\cdot w_{ik}\frac{\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp\boldsymbol{\Phi}_{4i,0}-\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp\boldsymbol{\Phi}_{4i,0}^e}{\sqrt{T}}\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\}\\
&+\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right]\cdot w_{ik}\frac{\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp\boldsymbol{\Phi}_{4i,0}^e}{\sqrt{T}}\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\}\\
&-\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right]\cdot w_{ik}\frac{\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp(\boldsymbol{\Phi}_{4i,0}\boldsymbol{\Sigma}_{i\boldsymbol{F}}-\boldsymbol{\Phi}_{4i,0}^e\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e)}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right\}\\
&-\frac{1}{N^2T}\sum_{i,k=1}^{N}\left\{\left[\boldsymbol{M}_k\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\right]\cdot w_{ik}\frac{\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp\boldsymbol{\Phi}_{4i,0}^e\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right\}\\
=& \frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\bar{\boldsymbol{H}}_0\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{T}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right).
\end{aligned}$$

For $\Delta_{5.6}$, we have

$$\begin{aligned}
\Delta_{5.6} =& -\frac{1}{N^3}\sum_{i,k,l=1}^{N}\left[\boldsymbol{M}_k\left(\boldsymbol{F}_0\right)\cdot w_{ik}\frac{\boldsymbol{X}_k^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^\perp\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)}{T}\cdot\frac{\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{l,0}\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it}\varepsilon_{lt}\right)\right]\\
&+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right)\\
=:& \frac{\varsigma_{5,NT}}{N}+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{\sqrt{N}m_{NT}}\right)+O_p\left(\frac{1}{\sqrt{N}m_{NT}^2}\right).
\end{aligned}$$

To conclude,

$$\Delta_5=\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^\top\otimes\boldsymbol{Q}_{j,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\bar{\boldsymbol{H}}_0\cdot\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_T\cdot\boldsymbol{l}_{i,0}\otimes\boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i+\frac{\varsigma_{5,NT}}{N}+O_p\left(\frac{\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\|_2}{m_{NT}}\right)+O_p\left(\frac{1}{m_{NT}^3}\right).$$

Note that $\varsigma_{1,NT}+\cdots+\varsigma_{5,NT}=\boldsymbol{\xi}_{NT}^\dagger$ after relabeling the indices as follows:

(1) $\varsigma_{1,NT}$: unchanged;

(2) $\varsigma_{2,NT}$: relabel $j\mapsto k$, $k\mapsto j$;

(3) $\varsigma_{3,NT}$: relabel $j\mapsto l$, $l\mapsto j$;

(4) $\varsigma_{4,NT}$: relabel $j\mapsto l$, $k\mapsto j$;

(5) $\varsigma_{5,NT}$: relabel $j \mapsto l$, $l \mapsto j$.

Collecting the terms Δ_1–Δ_5 above yields the desired result. □

Proof of Lemma S.2.11. Decompose Δ_1:

$$\Delta_1 = \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{\Psi}_{1i}\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i - \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{\Psi}_{2i}\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i =: \Delta_{1.1} + \Delta_{1.2}.$$

Further decompose $\Delta_{1.1}$:

$$\Delta_{1.1} = \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \left[\boldsymbol{\Psi}_{1i}^e + \mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e)\right) + \boldsymbol{\mathcal{R}}_{\boldsymbol{\Psi}_{1i}}\right] \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i =: \Delta_{1.1.1} + \Delta_{1.1.2} + \Delta_{1.1.3}.$$

Here, $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^e$ denotes the Jacobian matrix of $\boldsymbol{\Psi}_1$ evaluated at $\boldsymbol{\pi}_1 = \boldsymbol{\pi}_{1i}^e$. The (q_1, q_2)-th element of $\boldsymbol{\mathcal{R}}_{\boldsymbol{\Psi}_{1i}}$ is

$$r_{1i,q_1q_2} = (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e)^\top \cdot \bar{\boldsymbol{\mathcal{H}}}_{\boldsymbol{\Psi}_{1i},q_1q_2} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e),$$

where

$$\bar{\boldsymbol{\mathcal{H}}}_{\boldsymbol{\Psi}_{1i},q_1q_2} = \left.\frac{\partial^2 \psi_{1,q_1q_2}(\boldsymbol{\pi}_1)}{\partial \boldsymbol{\pi}_1 \partial \boldsymbol{\pi}_1^\top}\right|_{\boldsymbol{\pi}_1 = \alpha_{q_1q_1}\boldsymbol{\pi}_{1i}^e + (1-\alpha_{q_1q_2})\boldsymbol{\pi}_{1i}},$$

and ψ_{1,q_1q_2} is the (q_1, q_2)-th element of $\boldsymbol{\Psi}_1$ with $0 \le \alpha_{q_1q_2} \le 1$, for $q_1 = 1, \ldots, p$, $q_2 = 1, \ldots, p$. For $\Delta_{1.1.2}$,

$$\begin{aligned}
&\mathbb{E}\left(\|\Delta_{1.1.2} - \mathbb{E}_{\mathcal{C}}(\Delta_{1.1.2})\|_f^2\right) \\
\lesssim &\frac{1}{N^2T^2}\sum_{i=1}^{N} \mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{b}_{i,0} \otimes \left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e)\right) \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i - \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e)\right) \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right]\right\}\right\|_f^2\right)\right] \\
\lesssim &\frac{1}{N^2T^2}\sum_{i=1}^{N} \mathbb{E}\left\{\left[\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^4\right) \cdot \mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^4\right)\right]^{1/2}\right\} \\
\lesssim &\frac{1}{N^2T^2}\sum_{i=1}^{N} \left[\mathbb{E}\left(\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^4\right) \cdot \mathbb{E}\left(\left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^4\right)\right]^{1/2} \le \frac{c}{T^2N},
\end{aligned}$$

where $\mathbb{E}\left(\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^4\right) \le \mathbb{E}\left(\|\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{ii}^e\|_2^4\right) + \mathbb{E}\left(\|\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\|_2^4\right) \le cT^{-2}$. For $\Delta_{1.1.3}$,

$$\begin{aligned}
\|\Delta_{1.1.3}\|_2 \lesssim &\max_{i,q_1,q_2} \left\|\bar{\boldsymbol{\mathcal{H}}}_{1i,q_1q_2}\right\|_2 \cdot \frac{1}{NT}\sum_{i=1}^{N}\left(\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^2 \cdot \left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2\right) \\
=&O_p(1) \cdot \left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e\|_2^4\right)^{1/2} \cdot \left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2} \\
=&O_p(1) \cdot O_p\left(\frac{1}{T}\right) \cdot O_p\left(\frac{1}{\sqrt{T}}\right) = O_p\left(\frac{1}{T^{3/2}}\right).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\Delta_{1.1} =& \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{\Psi}_{1i}^e \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
&+ \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^e)\right) \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right] + O_p\left(\frac{1}{Tm_{NT}}\right).
\end{aligned} \tag{S.4.10}$$

In the same way, we can show that

$$\Delta_{1.2} = -\frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{\Psi}_{2i}^{e} \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i$$
$$- \frac{1}{NT}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{2i}}^{e\top} \cdot (\boldsymbol{\pi}_{1i} - \boldsymbol{\pi}_{1i}^{e})\right) \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right] + O_p\left(\frac{1}{Tm_{NT}}\right). \tag{S.4.11}$$

The result follows from (S.4.10), (S.4.11), and $\boldsymbol{\Psi}_{1i}^{e} \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i - \boldsymbol{\Psi}_{2i}^{e} \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i = \boldsymbol{\Psi}_{1i}^{e} \boldsymbol{\mathfrak{X}}_i^\top \boldsymbol{\varepsilon}_i$. □

Proof of Lemma S.2.12. By the formula $(\boldsymbol{I}_{TR_0} - \boldsymbol{\Upsilon}_0)^{-1} = \boldsymbol{I}_{TR_0} + \sum_{d=1}^{\infty} \boldsymbol{\Upsilon}_0^d$, decompose Δ_2:

$$\begin{aligned}
\Delta_2 =& \frac{1}{N^2T}\sum_{i,j=1}^{N}\left(\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Psi}_{3ij}\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right) - \frac{1}{N^2T}\sum_{i,j=1}^{N}\left(\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Psi}_{4ij}\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right) \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \boldsymbol{X}_j^\top \cdot \sum_{d=1}^{D^{trun}+1} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}_0}^{\perp} \boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right] \\
&- \frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \boldsymbol{X}_j^\top \cdot \sum_{d=1}^{D^{trun}+1} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{X}_i\boldsymbol{\Theta}_0\left(\boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}_0}^{\perp} \boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{F}_0(\boldsymbol{F}_0^\top \boldsymbol{F}_0)^{-1}\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right] \\
&+ \frac{1}{N^2T}\sum_{i,j=1}^{N}\left(\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \boldsymbol{X}_j^\top \cdot \sum_{d=D^{trun}+2}^{\infty} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{H}_{i,0}\boldsymbol{\varepsilon}_i\right) =: \Delta_{2.1} + \cdots + \Delta_{2.5}.
\end{aligned}$$

(1) We begin with $\Delta_{2.1}$:

$$\begin{aligned}
\Delta_{2.1} =& \frac{1}{N^2T}\sum_{i,j=1}^{N} \boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \left[\boldsymbol{\Psi}_{3ij}^{\star} + \mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top} \cdot \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^{\star}\right)\right) + \boldsymbol{\mathcal{R}}_{\boldsymbol{\Psi}_{3ij}}\right]\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
=:& \Delta_{2.1.1} + \Delta_{2.1.2} + \Delta_{2.1.3},
\end{aligned}$$

where both $\boldsymbol{\Psi}_{3ij}^{\star}$ and $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star}$ are both evaluated at $\boldsymbol{\pi}_{2ij}^{\star} := \mathbb{E}_{\mathcal{C}\cup j}(\boldsymbol{\pi}_{2ij})$. Here

$$\begin{aligned}
&\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top} \cdot \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^{\star}\right)\right) \\
=& \left[\left(\boldsymbol{\Sigma}_{ji} - \boldsymbol{\Sigma}_{ji}^{\star}\right) - \boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\right)^\top - \boldsymbol{\Sigma}_{j\boldsymbol{G}}\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\left(\boldsymbol{\Sigma}_{i\boldsymbol{G}} - \boldsymbol{\Sigma}_{i\boldsymbol{G}}^{\star}\right)^\top\right]\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^{\star} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top \\
&- \Big\{\left(\boldsymbol{\Sigma}_{ji}^{\star} - \boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star\top}\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^{\star} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star\top}\right)\boldsymbol{\Theta}_0\right]^{-1} \\
&\cdot \boldsymbol{\Theta}_0^\top\left[\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{ii}^{\star}\right) - \left(\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\right)\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star\top} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\right)^\top\right]\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}^{\star} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{\star\top}\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top\Big\}.
\end{aligned}$$

The (q_1, q_2)-th element of $\boldsymbol{\mathcal{R}}_{\boldsymbol{\Psi}_{3ij}}$ is

$$r_{3ij,q_1q_2} = \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^{\star}\right)^\top \cdot \bar{\boldsymbol{\mathcal{H}}}_{\boldsymbol{\Psi}_{3ij},q_1q_2} \cdot \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^{\star}\right),$$

where

$$\bar{\boldsymbol{\mathcal{H}}}_{\boldsymbol{\Psi}_{3ij},q_1q_2} = \left.\frac{\partial^2\psi_{3,q_1q_2}}{\partial\boldsymbol{\pi}_2\partial\boldsymbol{\pi}_2^\top}\right|_{\boldsymbol{\pi}_2=\alpha_{q_1q_2}\boldsymbol{\pi}_{2ij}^{\star}+(1-\alpha_{q_1q_2})\boldsymbol{\pi}_{2ij}},$$

and ψ_{3,q_2q_2} is the (q_1, q_2)-th element of $\boldsymbol{\Psi}_3$ with $0 \le \alpha_{q_1q_2} \le 1$, for $q_1, q_2 = 1, \ldots, p$. For $\Delta_{2.1.1}$, we have

$$\Delta_{2.1.1} = \frac{1}{N^2T}\sum_{i=1}^{N} w_{ii}\boldsymbol{b}_{i,0} \otimes \boldsymbol{S}_{i,0}^\top \boldsymbol{\Psi}_{3ii}\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i$$

$$
\begin{aligned}
&+\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{e}\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\\
&+\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\left(\boldsymbol{\Psi}_{3ij}^{\star}-\boldsymbol{\Psi}_{3ij}^{e}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\\
&+\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1j}}^{\top}\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^{e}\right)\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{\star}\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\\
=:&\Delta_{2.1.1.1}+\cdots+\Delta_{2.1.1.4},
\end{aligned}
$$

where

$$
\Delta_{2.1.1.1}=O_p\left(\frac{1}{N\sqrt{T}}\right),\quad \Delta_{2.1.1.3}=O_p\left(\frac{1}{N\sqrt{T}}\right),\quad \Delta_{2.1.1.4}=O_p\left(\frac{1}{T\sqrt{N}}\right).
$$

Therefore,

$$
\Delta_{2.1.1}=\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{e}\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i+O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right).
$$

For $\Delta_{2.1.2}$, we have

$$
\begin{aligned}
&\left\|\Delta_{2.1.2}-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\right\|_2\\
\lesssim&\frac{1}{N^2T}\sum_{j=1}^{N}\left\|\sum_{i\neq j}\tilde{\boldsymbol{l}}_{i,0}\otimes\left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\right\}\right\|_2\\
&\cdot\max_j\left\|\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\right\|_2=O_p\left(\frac{1}{T\sqrt{N}}\right)\cdot O_p(1)=O_p\left(\frac{1}{T\sqrt{N}}\right).
\end{aligned}
$$

Furthermore,

$$
\begin{aligned}
&\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\\
=&\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\left\{\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\right.\\
&\left.+\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]-\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\right\}\\
&+\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1j}}^{\top}\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^{e}\right)\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\\
=&\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e\top}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]+O_p\left(\frac{1}{Tm_{NT}}\right).
\end{aligned}
$$

where the second term

$$
\begin{aligned}
&\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}w_{ij}\boldsymbol{b}_{j,0}\otimes\boldsymbol{\Psi}_{1j}^{e}\left\{\mathbb{E}_{\mathcal{C}\cup j}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]-\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\cdot\left(\boldsymbol{\pi}_{2ij}-\boldsymbol{\pi}_{2ij}^{e}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\right\}\\
=&\frac{1}{N^2T}\sum_{j=1}^{N}\sum_{i\neq j}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{i,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\\
&\cdot\left\{\left[\left(\boldsymbol{\Sigma}_{ji}^{e}-\boldsymbol{\Sigma}_{ji}^{\star}\right)-\left(\boldsymbol{\Sigma}_{j\boldsymbol{F}}-\boldsymbol{\Sigma}_{j\boldsymbol{F}}^{e}\right)\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\left(\boldsymbol{\Sigma}_{i\boldsymbol{F}}-2\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\right)-\left(\boldsymbol{\Sigma}_{j\boldsymbol{G}}-\boldsymbol{\Sigma}_{j\boldsymbol{G}}^{e}\right)\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\left(\boldsymbol{\Sigma}_{i\boldsymbol{G}}-2\boldsymbol{\Sigma}_{i\boldsymbol{G}}^{e}\right)\right]\right.
\end{aligned}
$$

$$
\begin{aligned}
&\cdot \boldsymbol{\Theta}_0 \left[\boldsymbol{\Theta}_0^\top \left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1} \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right) \boldsymbol{\Theta}_0\right]^{-1} \boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
&+ \left[\left(\boldsymbol{\Sigma}_{ji}^e - \boldsymbol{\Sigma}_{ji}^\star\right) - \left(\boldsymbol{\Sigma}_{j\boldsymbol{F}}^e - \boldsymbol{\Sigma}_{j\boldsymbol{F}}\right) \boldsymbol{\Sigma}_{\boldsymbol{F}} \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right] \boldsymbol{\Theta}_0 \left[\boldsymbol{\Theta}_0^\top \left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1} \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right) \boldsymbol{\Theta}_0\right]^{-1} \\
&\cdot \boldsymbol{\Theta}_0^\top \left[\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{ii}^e\right) - \left(\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\right) \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1} \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1} \left(\boldsymbol{\Sigma}_{i\boldsymbol{F}} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\right)^\top\right] \boldsymbol{\Theta}_0 \left[\boldsymbol{\Theta}_0^\top \left(\boldsymbol{\Sigma}_{ii}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}^e \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1} \boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e\top}\right) \boldsymbol{\Theta}_0\right]^{-1} \boldsymbol{\Theta}_0^\top \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \Bigg\} \\
&= O_p\left(\frac{1}{\sqrt{NT} m_{NT}}\right),
\end{aligned}
$$

and the last term is $O_p\left(T^{-3/2}\right)$. For $\Delta_{2.1.3}$, we have

$$
\begin{aligned}
&\left\|\Delta_{2.1.3}\right\|_2 \\
\lesssim & \max_{i,j,q_1,q_2} \left(\left\|\boldsymbol{S}_{j,0}\right\|_2 \cdot \left\|\bar{\boldsymbol{\mathcal{H}}}_{\boldsymbol{\Psi}_{3ij}, q_1 q_2}\right\|_2\right) \cdot \frac{1}{N^2 T} \sum_{i,j=1}^N \left(\left\|\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^\star\right\|_2^2 \cdot \left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2\right) \\
\lesssim & O_p(1) \cdot \left(\frac{1}{N^2} \sum_{i=1}^N \sum_{j \neq i} \left\|\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^\star\right\|_2^4\right)^{1/2} \cdot \left(\frac{1}{NT^2} \sum_{i=1}^N \left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2} \\
= & O_p(1) \cdot O_p\left(\frac{1}{T}\right) \cdot O_p\left(\frac{1}{\sqrt{T}}\right) = O_p\left(\frac{1}{T^{3/2}}\right).
\end{aligned}
$$

To conclude,

$$
\begin{aligned}
\Delta_{2.1} = & \frac{1}{N^2 T} \sum_{j=1}^N \sum_{i \neq j} \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Psi}_{3ij}^e \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
& + \frac{1}{N^2 T} \sum_{j=1}^N \sum_{i \neq j} \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \mathbb{E}_{\mathcal{C}} \left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \cdot \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^e\right)\right) \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right] + O_p\left(\frac{1}{\sqrt{T} m_{NT}^2}\right).
\end{aligned}
$$

(2) By the same approach as that in the proof of $\Delta_{2.1}$, we can show that

$$
\begin{aligned}
\Delta_{2.2} = & -\frac{1}{N^2 T} \sum_{j=1}^N \sum_{i \neq j} \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Psi}_{4ij}^e \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i \\
& - \frac{1}{N^2 T} \sum_{j=1}^N \sum_{i \neq j} \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \mathbb{E}_{\mathcal{C}} \left[\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{4ij}}^{e\top} \cdot \left(\boldsymbol{\pi}_{2ij} - \boldsymbol{\pi}_{2ij}^e\right)\right) \boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right] + O_p\left(\frac{1}{\sqrt{T} m_{NT}^2}\right).
\end{aligned}
$$

(3) For $\Delta_{2.3}$, we have

$$
\begin{aligned}
\Delta_{2.3} = & \frac{1}{N^2 T} \sum_{i,j=1}^N \Bigg\{\boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{S}_{j,0}^\top \boldsymbol{X}_j^\top \cdot \sum_{d=0}^{D^{trun}} \boldsymbol{\Upsilon}_0^d \cdot \frac{1}{N} \sum_{k=1}^N \tilde{\boldsymbol{l}}_{k,0} \tilde{\boldsymbol{l}}_{k,0}^\top \otimes \boldsymbol{P}_{(\boldsymbol{F}_0, \boldsymbol{G})}^\perp \boldsymbol{X}_k \boldsymbol{\Theta}_0 \left(\boldsymbol{\Theta}_0^\top \boldsymbol{X}_k^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{X}_k \boldsymbol{\Theta}_0\right)^{-1} \boldsymbol{\Theta}_0^\top \\
& \left[\boldsymbol{\Psi}_{3ik}^\star + \mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{\star\top} \cdot \left(\boldsymbol{\pi}_{2ik} - \boldsymbol{\pi}_{2ik}^\star\right)\right) + \boldsymbol{\mathcal{R}}_{\boldsymbol{\Psi}_{3ik}}\right] \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\Bigg\} =: \Delta_{2.3.1} + \Delta_{2.3.2} + \Delta_{2.3.3}.
\end{aligned}
$$

For $\Delta_{2.3.1}$,

$$
\begin{aligned}
\Delta_{2.3.1} = & \frac{1}{N^2 T^{3/2}} \sum_{i,j=1}^N \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^\top\right) \cdot \sum_{d=1}^{D^{trun}+1} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Phi}_{4i,0}^e \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
& + \frac{1}{N^2 T^{3/2}} \sum_{i,j=1}^N \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \boldsymbol{\Psi}_{1j}^e \left(\boldsymbol{X}_j - \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j\right)\right)^\top \cdot \sum_{d=1}^{D^{trun}+1} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Phi}_{4i,0}^e \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i \\
& + \frac{1}{N^2 T^{3/2}} \sum_{i,j=1}^N \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^\top \otimes \mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1j}}^\top \cdot \left(\boldsymbol{\pi}_{1j} - \boldsymbol{\pi}_{1j}^e\right)\right) \boldsymbol{X}_j^\top \cdot \sum_{d=1}^{D^{trun}+1} \boldsymbol{\Upsilon}_0^d \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\Phi}_{4i,0}^e \boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i
\end{aligned}
$$

$$=:\Delta_{2.3.1.1}+\Delta_{2.3.1.2}+\Delta_{2.3.1.3},$$

where it is straightforward to show that

$$\Delta_{2.3.1.2}=O_p\left(\frac{1}{N\sqrt{T}}\right),\quad \Delta_{2.3.1.3}=O_p\left(\frac{1}{T\sqrt{N}}\right).$$

Let

$$\boldsymbol{\mathcal{V}}^\top=\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^\top\right),\quad \boldsymbol{\Upsilon}_{0,-k}=\frac{1}{N}\sum_{i\neq k}\tilde{\boldsymbol{l}}_{i,0}\tilde{\boldsymbol{l}}_{i,0}^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{H}_{i,0}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}.$$

Then, given $2\leq D_0\leq D^{trun}+1$,

$$\begin{aligned}
&\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0^{D_0}-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{D_0}\right)\right]\\
=&\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0^{D_0-1}-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{D_0-1}\right)\right]\boldsymbol{\Upsilon}_0+\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{D_0-1}\right)\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\right]\\
&+\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{D_0-1}-\boldsymbol{\Upsilon}_{0,-k}^{D_0-1}\right)\mathbb{E}_{\mathcal{C}}\left(\tilde{\boldsymbol{H}}_{k0}\right)+\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left[\left(\boldsymbol{\Upsilon}_{0,-k}^{D_0-1}-\boldsymbol{\Upsilon}_0^{D_0-1}\right)\tilde{\boldsymbol{H}}_{k0}\right]\\
=&\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\right]\boldsymbol{\Upsilon}_0^{D_0-1}+\sum_{d=1}^{D_0-1}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_0)\right]\boldsymbol{\Upsilon}_0^{D_0-1-d}\\
&+\sum_{d=1}^{D_0-1}\left\{\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}-\boldsymbol{\Upsilon}_{0,-k}^{d}\right)\mathbb{E}_{\mathcal{C}}\left(\tilde{\boldsymbol{H}}_{k0}\right)\boldsymbol{\Upsilon}_0^{D_0-1-d}\right\}+\sum_{d=1}^{D_0-1}\left\{\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left[\left(\boldsymbol{\Upsilon}_{0,-k}^{d}-\boldsymbol{\Upsilon}_0^{d}\right)\tilde{\boldsymbol{H}}_{k0}\right]\boldsymbol{\Upsilon}_0^{D_0-1-d}\right\}\\
=:&\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\right]\boldsymbol{\Upsilon}_0^{D_0-1}+\sum_{l=1}^{3}\sum_{d=1}^{D_0-1}\boldsymbol{\mathcal{W}}_l^{(D_0-1,d)}\\
=:&\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\right]\boldsymbol{\Upsilon}_0^{D_0-1}+\sum_{l=1}^{3}\boldsymbol{\mathcal{W}}_l^{(D_0)},
\end{aligned}$$

where

$$\sum_{d=1}^{D^{trun}+1}\boldsymbol{\mathcal{V}}^\top\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0\right)\right]\boldsymbol{\Upsilon}_0^{d-1}=O_p\left(\frac{1}{\sqrt{N}}\right)\cdot\sum_{d=1}^{D^{trun}+1}\boldsymbol{\Upsilon}_0^{d-1}=O_p\left(\frac{1}{\sqrt{N}}\right).$$

Since

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{d=1}^{d'-1}\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_0)\right]\boldsymbol{\Upsilon}_0^{d'-1-d}\right\|_f\right)\\
\leq&\sum_{d=1}^{d'-1}\left\{\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\mathcal{V}}^\top\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\left[\boldsymbol{\Upsilon}_0-\mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_0)\right]\right\|_f^2\right)\right]^{1/2}\cdot\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\Upsilon}_0^{d'-1-d}\right\|_2^2\right)\right]^{1/2}\right\}\\
\lesssim&\sum_{d=1}^{d'-1}\left(\frac{a_2^d}{\sqrt{N}}\cdot a_2^{d'-1-d}\right)\leq\frac{c_1d'a_2^{d'}}{\sqrt{N}},
\end{aligned}$$

we have

$$\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{d'=2}^{D^{trun}+1}\boldsymbol{\mathcal{W}}_1^{(d')}\right\|_f\right)\leq\sum_{d'=2}^{D^{trun}+1}\sum_{d=1}^{d'-1}\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\mathcal{W}}_1^{(d'-1,d)}\right\|_2\right)\lesssim\sum_{d'=2}^{D^{trun}+1}\frac{d'a_2^{d'}}{\sqrt{N}}=O\left(\frac{1}{\sqrt{N}}\right).$$

Since $\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\Upsilon}_0^{d'}-\boldsymbol{\Upsilon}_{0,-k}^{d'}\right\|_2\right)\leq N^{-1}\sum_{d=1}^{d'}\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\Upsilon}_0^{d}\tilde{\boldsymbol{H}}_{k0}\boldsymbol{\Upsilon}_{0,-k}^{d'-d-1}\right\|_2\right)\lesssim N^{-1}d'a_2^{d'}$, it is straightforward to show that

$$\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{d'=2}^{D^{trun}+1}\boldsymbol{\mathcal{W}}_2^{(d')}\right\|_f\right)\leq\sum_{d'=2}^{D^{trun}+1}\sum_{d=1}^{d'-1}\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{\mathcal{W}}_2^{(d'-1,d)}\right\|_f\right)\lesssim\sum_{d'=2}^{D^{trun}+1}\sum_{d=1}^{d'-1}\left(\frac{da_2^d}{N}\cdot a_2^{d'-1-d}\right)=O_p\left(\frac{1}{N}\right),$$

and similarly, $\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{d'=2}^{D^{trun}+1}\boldsymbol{\mathcal{W}}_3^{(d')}\right\|_f\right)=O_p\left(N^{-1}\right)$. Therefore,

$$\Delta_{2.3.1}=\frac{1}{N^2T^{3/2}}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i+O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right).$$

For $\Delta_{2.3.2}$,

$$\begin{aligned}\Delta_{2.3.2}=&\frac{1}{N^3T^{3/2}}\sum_{i,j,k=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\\
&\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\Big\{\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i-\mathbb{E}_{\mathcal{C}\cup k}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\\
&+\mathbb{E}_{\mathcal{C}\cup k}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]-\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\cdot\left(\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{e}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\Big\}\\
&+\frac{1}{N^3T^{3/2}}\sum_{i,j,k=1}^{N}\Big\{\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\left(\boldsymbol{\Phi}_{4k,0}-\boldsymbol{\Phi}_{4k,0}^{e}\right)\\
&+\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\left[\boldsymbol{\Upsilon}_0^d-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\right]\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\\
&+\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\left[\boldsymbol{X}_j-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j\right)\right]^{\top}\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^d\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\\
&+\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1j}}^{\top}\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^{e}\right)\right)\boldsymbol{X}_j^{\top}\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^d\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\Big\}\\
&\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{\star\top}\cdot\left(\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{\star}\right)\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\\
=:&\Delta_{2.3.2.1}+\cdots+\Delta_{2.3.2.7},\end{aligned}$$

where

$$\Delta_{2.3.2.2}=O_p\left(\frac{1}{T\sqrt{N}}\right),\quad\Delta_{2.3.2.3}=O_p\left(\frac{1}{T\sqrt{N}}\right),\quad\Delta_{2.3.2.4}=O_p\left(\frac{1}{T^{3/2}}\right),$$
$$\Delta_{2.3.2.5}=O_p\left(\frac{1}{T\sqrt{N}}\right),\quad\Delta_{2.3.2.6}=O_p\left(\frac{1}{T\sqrt{N}}\right),\quad\Delta_{2.3.2.7}=O_p\left(\frac{1}{T^{3/2}}\right).$$

Therefore,

$$\begin{aligned}\Delta_{2.3.2}=&\frac{1}{N^3T^{3/2}}\sum_{i,j,k=1}^{N}\Big\{\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\\
&\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\Big\}+O_p\left(\frac{1}{Tm_{NT}}\right).\end{aligned}$$

For $\Delta_{2.3.3}$, we have

$$\|\Delta_{2.3.3}\|_2\lesssim O_p(1)\cdot\left(\frac{1}{N^2}\sum_{i=1}^{N}\sum_{k\neq i}\|\boldsymbol{\pi}_{2ik}-\boldsymbol{\pi}_{2ik}^{\star}\|_2^4\right)^{1/2}\cdot\left(\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right\|_2^2\right)^{1/2}=O_p\left(\frac{1}{T^{3/2}}\right).$$

To conclude,

$$\Delta_{2.3}=\frac{1}{N^2T^{3/2}}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]$$

$$+\frac{1}{N^3T^{3/2}}\sum_{i,j,k=1}^{N}\Bigg\{\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}$$

$$\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right]\Bigg\}+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right).$$

(4) By the same approach as that in the proof of $\Delta_{2.3}$, we can show that

$$\Delta_{2.4}=-\frac{1}{N^2T^{3/2}}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^{e}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]$$

$$-\frac{1}{N^3T^{3/2}}\sum_{i,j,k=1}^{N}\Bigg\{\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j^{\top}\right)\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}$$

$$\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,R_0}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{4ik}}^{e\top}\boldsymbol{\pi}_{2ik}\right)\boldsymbol{F}_0^{\top}\boldsymbol{\varepsilon}_i\right]\Bigg\}+O_p\left(\frac{1}{\sqrt{T}m_{NT}^2}\right).$$

(5) The last term is negligible, since $\left\|\sum_{d=D^{trun}+2}^{\infty}\boldsymbol{\Upsilon}_0^d\right\|_2=O_p\left(a_1^{D^{trun}}\right)$ with $a_1<1$, and $D^{trun}/\log N\to\infty$. □

Proof of Lemma S.2.13. Decompose

$$\begin{aligned}\Delta_3=&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\sum_{d=1}^{D^{trun}+1}\boldsymbol{\Upsilon}_0^d\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\sum_{d=D^{trun}+2}^{\infty}\boldsymbol{\Upsilon}_0^d\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\=:&\,\Delta_{3.1}+\Delta_{3.2}+\Delta_{3.3}.\end{aligned}$$

For $\Delta_{3.1}$,

$$\begin{aligned}\Delta_{3.1}=&-\boldsymbol{\mathcal{V}}^{\top}\left(\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\varepsilon}_i\right)\\&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{1j}^{e}\left[\boldsymbol{X}_j-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_j\right)\right]^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\varepsilon}_i\\&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\bar{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1j}}^{\top}\cdot\left(\boldsymbol{\pi}_{1j}-\boldsymbol{\pi}_{1j}^{e}\right)\right)\boldsymbol{X}_j^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\varepsilon}_i\\=:&\,\Delta_{3.1.1}+\Delta_{3.1.2}+\Delta_{3.1.3},\end{aligned}$$

where

$$\Delta_{3.1.2}=O_p\left(\frac{1}{N\sqrt{T}}\right),\quad\Delta_{3.1.3}=O_p\left(\frac{1}{T\sqrt{N}}\right).$$

For $\Delta_{3.2}$,

$$\begin{aligned}\Delta_{3.2}=&-\left[\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\left(\boldsymbol{\Psi}_{1j}\boldsymbol{X}_j^{\top}-\boldsymbol{\Psi}_{1j}^{e}\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_j^{\top})\right)\right]\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^d\cdot\boldsymbol{\Upsilon}_0\cdot\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\varepsilon}_i\\&-\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=0}^{D^{trun}}\left[\boldsymbol{\Upsilon}_0^d-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\right]\cdot\boldsymbol{\Upsilon}_0\cdot\frac{1}{N\sqrt{T}}\sum_{i=1}^{N}\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\varepsilon}_i\end{aligned}$$

$$
\begin{aligned}
&-\boldsymbol{\mathcal{V}}^\top \cdot \sum_{d=0}^{D^{trun}} \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right) \cdot \frac{1}{N\sqrt{T}} \sum_{k=1}^{N} \tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\left(\boldsymbol{\Phi}_{4k,0}-\boldsymbol{\Phi}_{4k,0}^e\right)\boldsymbol{X}_k^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp} \cdot \frac{1}{N\sqrt{T}} \sum_{i=1}^{N} \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\varepsilon}_i \\
&-\boldsymbol{\mathcal{V}}^\top \cdot \sum_{d=0}^{D^{trun}} \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right) \cdot \frac{1}{N\sqrt{T}} \sum_{k=1}^{N} \tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^e\left(\boldsymbol{X}_k-\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_k)\right)^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp} \cdot \frac{1}{N\sqrt{T}} \sum_{i=1}^{N} \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\varepsilon}_i \\
&-\boldsymbol{\mathcal{V}}^\top \cdot \sum_{d=0}^{D^{trun}} \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right) \cdot \frac{1}{N\sqrt{T}} \sum_{k=1}^{N} \tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^e\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_k^\top)\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp} \cdot \frac{1}{N\sqrt{T}} \sum_{i=1}^{N} \tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{\varepsilon}_i \\
&=:\Delta_{3.2.1}+\cdots+\Delta_{3.2.5},
\end{aligned}
$$

where

$$
\Delta_{3.2.1}=O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right), \quad \Delta_{3.2.2}=O_p\left(\frac{1}{N\sqrt{T}}\right), \quad \Delta_{3.2.3}=O_p\left(\frac{1}{T\sqrt{N}}\right), \quad \Delta_{3.2.4}=O_p\left(\frac{1}{N\sqrt{T}}\right).
$$

Therefore,

$$
\Delta_{3.2}=-\boldsymbol{\mathcal{V}}^\top \cdot \sum_{d=0}^{D^{trun}} \mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right) \cdot \frac{1}{N^2T} \sum_{i,k=1}^{N} \tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top\tilde{\boldsymbol{l}}_{i,0} \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^e\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_k^\top)\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\varepsilon}_i+O_p\left(\frac{1}{\sqrt{NT}m_{NT}}\right).
$$

The last term $\Delta_{3.3}$ is negligible as $\Delta_{2.5}$ in the proof of Lemma S.2.13. □

Proof of Lemma S.2.14. It suffices to verify the following statements:

(1) For any $\boldsymbol{e}\in\mathbb{R}^{(p-K_0)K_0}$ with $\|\boldsymbol{e}\|_2=1$, we have $\dfrac{(NT)^{-1/2}\sum_{i=1}^N \boldsymbol{e}^\top\boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i}{[(NT)^{-1}\sum_{i=1}^N\sum_{t=1}^T(\varepsilon_{it}^2\cdot\boldsymbol{e}^\top\boldsymbol{J}_{\boldsymbol{\mathfrak{z}}_{it,0}}\boldsymbol{\mathfrak{z}}_{it,0}^\top\boldsymbol{J}^\top\boldsymbol{e})]^{1/2}} \xrightarrow{\mathbb{D}} \mathcal{N}(0,1)$, where $\boldsymbol{\mathfrak{z}}_{it,0}^\top$ is the t-th row of $\boldsymbol{\mathfrak{Z}}_{i,0}$.

(2) $\dfrac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\varepsilon_{it}^2\left(\boldsymbol{\mathfrak{z}}_{it,0}\boldsymbol{\mathfrak{z}}_{it,0}^\top-\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^\top\right)=o_p(1)$.

(3) $\dfrac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^\top=o_p(1)$.

Under Assumption 7, statements (1), (2), and (3) together imply that for any $\boldsymbol{e}\in\mathbb{R}^{(p-K_0)K_0}$ with $\|\boldsymbol{e}\|_2=1$,

$$
\frac{1}{\sqrt{NT}}\sum_{i=1}^N\boldsymbol{e}^\top\boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i \xrightarrow{\mathbb{D}} \mathcal{N}\left(0,\boldsymbol{e}^\top\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\boldsymbol{e}\right),
$$

The result then follows by the Cramer-Wold device.

(1) We use the same idea of proving Lemma B.2 in Moon and Weidner (2017) to show statement (1). Define $\varrho_{it}=\boldsymbol{e}^\top\boldsymbol{J}_{\boldsymbol{\mathfrak{z}}_{it,0}}\varepsilon_{it}$, and $\varrho_m=\varrho_{it}$ with $m=T(i-1)+t\in\{1,\ldots,M\}$ and $M=NT$. From the decomposition $(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0)^{-1}=\boldsymbol{I}_{TR_0}+(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0)^{-1}\boldsymbol{\Upsilon}_0$, we have

$$
\begin{aligned}
\boldsymbol{\mathfrak{z}}_{it,0}=&\left[\boldsymbol{b}_{i,0}\otimes\boldsymbol{\Psi}_{1i}^e+\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}_{1j}^e\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^e+\boldsymbol{\mathcal{V}}^\top\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_0^d)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^e\right]\boldsymbol{\mathfrak{x}}_{it} \\
&-\frac{1}{N}\sum_{j=1}^N\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{1j}^e\mathbb{E}_{\mathcal{C}}(\boldsymbol{x}_{jt}) \\
&-\boldsymbol{\mathcal{V}}^\top\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\frac{1}{N}\sum_{k=1}^N\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^\top\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^e\mathbb{E}_{\mathcal{C}}(\boldsymbol{x}_{kt})
\end{aligned}
$$

$$+\boldsymbol{\mathcal{V}}^{\top}\cdot\left[\boldsymbol{I}_{TR_0}+\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^d\right)\cdot\frac{1}{N\sqrt{T}}\sum_{k=1}^{N}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_k^{\top}\right)\right]$$
$$\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\sqrt{T}\left[\boldsymbol{G}\left(\boldsymbol{G}^{\top}\boldsymbol{G}\right)^{-1}\boldsymbol{g}_t+\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\boldsymbol{f}_{t,0}\right],$$

where $\boldsymbol{\mathfrak{x}}_{it}^{\top}$ is the t-th row of $\boldsymbol{\mathfrak{X}}_i$. We can show that:

(1.i) The sequence $\{\varrho_m\}_{m=1}^{M}$ is a martingale difference sequence under the filtration $\mathcal{C}\vee\sigma(\{\varrho_n:n<m\})$.

(1.ii) $\mathbb{E}\left(\varrho_{it}^4\right)$ is uniformly bounded since

$$\begin{aligned}
&\mathbb{E}\left(\varrho_{it}^4\right)\\
=&\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^4\right)\right]\\
\leq&\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{\mathfrak{z}}_{it,0}\|_2^4\cdot\varepsilon_{it}^4\right)\right]\\
\lesssim&\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\|_2^4\cdot\varepsilon_{it}^4\right)\right]+\mathbb{E}\left[\|\boldsymbol{f}_{t,0}\|_2^4\cdot\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^4\right)\right]+\mathbb{E}\left[\|\boldsymbol{g}_t\|_2^4\cdot\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^4\right)\right]+\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^4\right)\right],
\end{aligned}$$

and $\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^8\right)$, $\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\|_2^8\right)$, and $\mathbb{E}\left(\|\boldsymbol{f}_{t,0}\|_2^4\right)$ are uniformly bounded.

(1.iii) From

$$\begin{aligned}
\mathbb{E}_{\mathcal{C}}\left\{\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varrho_{it}^2-\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^2\right)\right)\right]^2\right\}=&\frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{s,t=1}^{T}\mathbb{E}_{\mathcal{C}}\left[\left(\varrho_{is}^2-\mathbb{E}_{\mathcal{C}}\left(\varrho_{is}^2\right)\right)\left(\varrho_{it}^2-\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^2\right)\right)\right]\\
\leq&\frac{1}{N^2T}\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbb{E}_{\mathcal{C}}\left[\left(\varrho_{it}^2-\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^2\right)\right)^2\right]\\
\leq&\frac{1}{N^2T}\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^4\right)=O_p\left(\frac{1}{N}\right),
\end{aligned}$$

we have $(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varrho_{it}^2-\mathbb{E}_{\mathcal{C}}\left(\varrho_{it}^2\right)\right)=o_p(1)$.

By Corollary 5.26 in White (2001), statements (1.i)–(1.iii) together lead to

$$\frac{(NT)^{-1/2}\sum_{i=1}^{N}\sum_{t=1}^{T}\varrho_{it}}{[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\varrho_{it}^2]^{1/2}}\xrightarrow{\mathbb{D}}\mathcal{N}(0,1).$$

(2) Since

$$\begin{aligned}
&\left\|\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\left(\boldsymbol{\mathfrak{z}}_{it,0}\boldsymbol{\mathfrak{z}}_{it,0}^{\top}-\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^{\top}\right)\right\|_2\\
\leq&\left\|\frac{1}{2NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\left[(\boldsymbol{\mathfrak{z}}_{it,0}+\boldsymbol{z}_{it,0})(\boldsymbol{\mathfrak{z}}_{it,0}-\boldsymbol{z}_{it,0})^{\top}+(\boldsymbol{\mathfrak{z}}_{it,0}-\boldsymbol{z}_{it,0})(\boldsymbol{\mathfrak{z}}_{it,0}+\boldsymbol{z}_{it,0})^{\top}\right]\right\|_2\\
\leq&\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^4\cdot\left\|\boldsymbol{\mathfrak{z}}_{it,0}+\boldsymbol{z}_{it,0}\right\|_2^2\right)\right]^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{\mathfrak{z}}_{it,0}-\boldsymbol{z}_{it,0}\right\|_2^2\right)^{1/2},
\end{aligned}$$

it suffices to show that (2.i) $(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^4\cdot\left\|\boldsymbol{\mathfrak{z}}_{it,0}+\boldsymbol{z}_{it,0}\right\|_2^2\right)=O_p(1)$; (2.ii) $(NT)^{-1}\sum_{i=1}^{N}\left\|\boldsymbol{\mathfrak{Z}}_{i,0}-\boldsymbol{Z}_{i,0}\right\|_f^2=o_p(1)$.

(2.i) The result follows from

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^{4}\cdot\left\|\boldsymbol{\mathfrak{z}}_{it,0}+\boldsymbol{z}_{it,0}\right\|_{2}^{2}\right)\lesssim\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^{8}\right)^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\left\|\boldsymbol{z}_{it,0}\right\|_{2}^{4}+\left\|\boldsymbol{\mathfrak{z}}_{it,0}\right\|_{2}^{4}\right)\right]^{1/2},$$

and

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^{8}=O_p(1),\quad \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{z}_{it}\right\|_{2}^{4}=O_p(1),\quad \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{\mathfrak{z}}_{it}\right\|_{2}^{4}=O_p(1).$$

(2.ii) Decompose $(NT)^{-1}\sum_{i=1}^{N}\left\|\boldsymbol{Z}_{i,0}-\boldsymbol{\mathfrak{Z}}_{i,0}\right\|_{f}^{2}$:

$$\begin{aligned}
&\frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{Z}_{i,0}-\boldsymbol{\mathfrak{Z}}_{i,0}\right\|_{f}^{2}\\
\lesssim&\Bigg(\max_{i}\left\|\boldsymbol{b}_{i,0}\otimes\boldsymbol{\Psi}_{1i}^{e}+\frac{1}{N}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{e}+\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}(\boldsymbol{\Upsilon}_{0}^{d})\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\right\|_{2}^{2}\\
&\cdot\frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{\mathfrak{X}}_{i}-\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_{i}\right\|_{f}^{2}\Bigg)\\
&+\frac{1}{T}\max_{i}\left\|\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_{i}\right\|_{f}^{2}\cdot\Bigg(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{b}_{i,0}\otimes(\boldsymbol{\Psi}_{1i}-\boldsymbol{\Psi}_{1i}^{e})\right\|_{f}^{2}\\
&+\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_{j}^{\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}-\frac{1}{N}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Psi}_{3ij}^{e}\right\|_{f}^{2}\\
&+\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_{j}^{\top}\cdot\sum_{d=1}^{D^{trun}+1}\boldsymbol{\Upsilon}_{0}^{d}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}-\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=1}^{D^{trun}+1}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_{0}^{d}\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}^{e}\right\|_{f}^{2}\\
&+\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_{j}^{\top}\cdot\sum_{d=D^{trun}+2}^{\infty}\boldsymbol{\Upsilon}_{0}^{d}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{\Phi}_{4i,0}\right\|_{f}^{2}\Bigg)\\
&+\max_{i}\left\|\tilde{\boldsymbol{l}}_{i,0}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\right\|_{2}\cdot\Bigg(\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\boldsymbol{X}_{j}^{\top}-\boldsymbol{\mathcal{V}}^{\top}\right\|_{f}^{2}\\
&+\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_{j}^{\top}\cdot\sum_{d=1}^{D^{trun}+1}\boldsymbol{\Upsilon}_{0}^{d}-\boldsymbol{\mathcal{V}}^{\top}\cdot\sum_{d=0}^{D^{trun}}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_{0}^{d}\right)\cdot\frac{1}{N\sqrt{T}}\sum_{k=1}^{N}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_{k}^{\top}\right)\right\|_{f}^{2}\\
&+\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}\boldsymbol{X}_{j}^{\top}\cdot\sum_{d=D^{trun}+2}^{\infty}\boldsymbol{\Upsilon}_{0}^{d}\right\|_{f}^{2}\Bigg)=:\Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.1}+\cdots+\Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.8}.
\end{aligned}$$

By the approaches in the proof of Lemmas S.2.11–S.2.13, we can show that

$$\Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.1}=O_p\left(\frac{1}{T}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.2}=O_p\left(\frac{1}{T}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.3}=O_p\left(\frac{1}{m_{NT}^{2}}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.4}=O_p\left(\frac{1}{m_{NT}^{2}}\right),$$
$$\Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.5}=O_p\left(\frac{1}{N}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.6}=O_p\left(\frac{1}{m_{NT}^{2}}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.7}=O_p\left(\frac{1}{m_{NT}^{2}}\right),\quad \Delta_{\boldsymbol{Z}\boldsymbol{\mathfrak{Z}}.8}=O_p\left(\frac{1}{N}\right).$$

(3) We first decompose $\{\boldsymbol{Z}_{i,0}\}_{i=1}^{N}$:

$$\boldsymbol{Z}_{i,0}=\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{Q}_{i,0}-\frac{1}{N}\sum_{j=1}^{N}w_{ij}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}-\frac{1}{N}\sum_{j=1}^{N}\tilde{\boldsymbol{l}}_{i,0}^{\top}\otimes\boldsymbol{I}_{T}\cdot\boldsymbol{\Upsilon}_{0}\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_{0}\right)^{-1}\cdot\tilde{\boldsymbol{l}}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}$$

$$+\frac{1}{N}\sum_{j=1}^{N}\tilde{\boldsymbol{l}}_{i,0}^{\top}\otimes\boldsymbol{H}_{i,0}\cdot\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\cdot\tilde{\boldsymbol{l}}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}=:\boldsymbol{Z}_{i.1,0}+\boldsymbol{Z}_{i.2,0}+\boldsymbol{Z}_{i.3,0}+\boldsymbol{Z}_{i.4,0}.$$

Then,

$$\Delta_Z=\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^{\top}=\sum_{l_1,l_2=1}^{4}\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\boldsymbol{z}_{it.l_1,0}\boldsymbol{z}_{it.l_2,0}^{\top}\right]=:\sum_{l_1,l_2=1}^{4}\Delta_{\boldsymbol{Z}.l_1l_2}.$$

For $(l_1,l_2)=(1,1)$, decompose

$$\begin{aligned}\Delta_{\boldsymbol{Z}.11}=&\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{S}_{i,0}^{\top}\left[\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]\boldsymbol{S}_{i,0}\\&+\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{S}_{i,0}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\left[\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^{\top}\boldsymbol{X}_i\boldsymbol{S}_{i,0}\\&-\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{S}_{i,0}^{\top}\left[\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^{\top}\boldsymbol{X}_i\boldsymbol{S}_{i,0}\\&-\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{S}_{i,0}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{F}_0\left(\boldsymbol{F}_0^{\top}\boldsymbol{F}_0\right)^{-1}\left[\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{x}_{it}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right]\boldsymbol{S}_{i,0}.\end{aligned}$$

The result that $\Delta_{\boldsymbol{Z}.11}=O_p(T^{-1/2})$ follows from the uniform boundedness of $\left\{\left(\boldsymbol{b}_{i,0},\boldsymbol{S}_{i,0},T^{-1}\boldsymbol{X}_i^{\top}\boldsymbol{F}_0\right)\right\}_{i=1}^{N}$ w.p.a. 1, together with the fact that the following three terms are bounded by c_1T^{-1}:

$$\begin{aligned}\mathbb{E}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_f^2\right)=&\frac{1}{T}\cdot\mathbb{E}\left\{\frac{1}{T}\sum_{t=1}^{T}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^2\cdot\|\boldsymbol{x}_{it}\|_2^4\right]\right\}\\&+\frac{2}{T}\cdot\mathbb{E}\left[\sum_{t=2}^{T}\sum_{s=1}^{t-1}\mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\cdot\|\boldsymbol{x}_{it}^{\top}\boldsymbol{x}_{is}\|_2^2\cdot\left(\varepsilon_{is}^2-\sigma_{is}^2\right)\right]\\=&\frac{1}{T}\cdot\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^2\cdot\|\boldsymbol{x}_{it}\|_2^4\right]\\\leq&\frac{1}{T}\cdot\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left(\left\{\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^4\right]\cdot\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\|_2^8\right)\right\}^{1/2}\right)=O\left(\frac{1}{T}\right),\\\mathbb{E}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_f^2\right)\leq&\frac{1}{T}\cdot\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left\{\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^2\right]\cdot\|\boldsymbol{f}_{t,0}\|_2^4\right\}=O\left(\frac{1}{T}\right),\\\mathbb{E}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_f^2\right)\leq&\frac{1}{T}\cdot\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left(\left\{\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^4\right]\cdot\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\|_2^4\right)\right\}^{1/2}\cdot\|\boldsymbol{f}_{t,0}\|_2^2\right)=O\left(\frac{1}{T}\right).\end{aligned}$$

Similar structures can be identified in $\Delta_{\boldsymbol{Z}.l_1l_2}$, for other index combinations (l_1,l_2) with $1\leq l_1,l_2\leq 4$:

$$\begin{aligned}&\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{jt}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{g}_t^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\\&\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{jt}\boldsymbol{x}_{jt}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{jt}\boldsymbol{x}_{kt}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{jt}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{jt}\boldsymbol{g}_t^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\\&\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{f}_{t,0}^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{g}_t\boldsymbol{g}_t^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\quad\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{g}_t^{\top}\left(\varepsilon_{it}^2-\sigma_{it}^2\right),\end{aligned}$$

for $i,j,k=1,\ldots,N$, $i\neq j\neq k$. The same rate $\Delta_{\boldsymbol{Z}.l_1l_2}=O_p(T^{-1/2})$ therefore holds immediately for these cases. To conclude, $\Delta_{\boldsymbol{Z}}=O_p(T^{-1/2})$. This completes the proof. □

Proof of Lemma S.2.15.

(1) Decompose $\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}\left(\boldsymbol{F}_0\right)$:

$$
\begin{aligned}
&\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}\left(\boldsymbol{F}_0\right)\\
=&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\left[\boldsymbol{Q}_i^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}^{\top}\boldsymbol{Q}_{i,0}\right]\\
&-\frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_j^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_{j,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{i,0}\right]\\
=:&\Delta_1+\Delta_2.
\end{aligned}
$$

For Δ_1,

$$
\Delta_1=\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\frac{\boldsymbol{Q}_i^{\top}(\hat{\boldsymbol{F}})[\boldsymbol{Q}_i(\hat{\boldsymbol{F}})-\boldsymbol{Q}_{i,0}]}{T}+\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\frac{[\boldsymbol{Q}_i(\hat{\boldsymbol{F}})-\boldsymbol{Q}_{i,0}]^{\top}\boldsymbol{Q}_{i,0}}{T}=:\Delta_{1.1}+\Delta_{1.2},
$$

where

$$
\begin{aligned}
\|\Delta_{1.1}\|_2\leq&\bar{\boldsymbol{b}}^2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right),\\
\|\Delta_{1.2}\|_2\leq&\bar{\boldsymbol{b}}^2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_{i,0}\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

For Δ_2,

$$
\begin{aligned}
\Delta_2=&\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_j^{\top}\left(\hat{\boldsymbol{F}}\right)\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\left[\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)-\boldsymbol{\Xi}_0^{-1}\right]\cdot\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)\\
&+\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\left[\boldsymbol{Q}_j\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{j,0}\right]^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)\\
&+\frac{1}{N^2T}\sum_{i,j=1}^{N}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\otimes\boldsymbol{Q}_{j,0}^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\boldsymbol{l}_{i,0}\boldsymbol{b}_{i,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left[\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}\right]=:\Delta_{2.1}+\Delta_{2.2}+\Delta_{2.3},
\end{aligned}
$$

where

$$
\begin{aligned}
\|\Delta_{2.1}\|_2\leq&\bar{\boldsymbol{b}}^2\bar{\boldsymbol{l}}^2\cdot\frac{1}{T}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)\right\|_2^2\cdot\left\|\boldsymbol{\Xi}^{-1}\left(\boldsymbol{\Theta}_0,\hat{\boldsymbol{F}}\right)-\boldsymbol{\Xi}_0^{-1}\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right),\\
\|\Delta_{2.2}\|_2\leq&\bar{\boldsymbol{b}}^2\bar{\boldsymbol{l}}^2\cdot\left\|\boldsymbol{\Xi}_0^{-1}\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right),\\
\|\Delta_{2.3}\|_2\leq&\bar{\boldsymbol{b}}^2\bar{\boldsymbol{l}}^2\cdot\left\|\boldsymbol{\Xi}_0^{-1}\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_{i,0}\right\|_2\cdot\frac{1}{\sqrt{T}}\max_i\left\|\boldsymbol{Q}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{Q}_{i,0}\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

To conclude,

$$
\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}\left(\boldsymbol{F}_0\right)=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}}\right)=o_p(1).
$$

(2)-(3) In the proof of Proposition S.2.1, we show that

$$
\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^{\top}\right]^{-1}=O_p(1),\quad \boldsymbol{D}^{+}(\hat{\boldsymbol{F}})=\boldsymbol{J}^{\top}\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^{\top}\right]^{-1}\boldsymbol{J}=O_p(1).
$$

Corollary S.2.1, Corollary S.2.2, and Lemma S.2.14 together lead to the following equation:

$$\begin{aligned}\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0 =& \boldsymbol{J}\cdot\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\cdot\frac{1}{NT}\sum_{i=1}^N\boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i+\frac{\boldsymbol{\xi}_{NT}}{N}+\frac{\boldsymbol{\zeta}_{NT}+\boldsymbol{\eta}_{NT}}{T}\\ &+o_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\frac{1}{m_{NT}^3}\right)+o_p\left(\frac{1}{\sqrt{NT}}\right).\end{aligned}\tag{S.4.12}$$

The term $(NT)^{-1/2}\sum_{i=1}^N\boldsymbol{J}\boldsymbol{\mathfrak{Z}}_{i,0}^\top\boldsymbol{\varepsilon}_i$ is $O_p(1)$ by Lemma S.2.14. Furthermore,

$$\boldsymbol{\zeta}_{NT}=\boldsymbol{J}\cdot\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\left(\boldsymbol{\zeta}_{1,NT}^\dagger+\boldsymbol{\zeta}_{2,NT}^\dagger\right),\quad \boldsymbol{\xi}_{NT}=\boldsymbol{J}\cdot\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\xi}_{NT}^\dagger,\quad \boldsymbol{\eta}_{NT}=\boldsymbol{J}\cdot\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{\eta}_{NT}^\dagger$$

are all $O_p(1)$ by (S.2.7), (S.2.8), (S.2.11), and (S.2.15) respectively. To conclude, $\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0=O_p\left(m_{NT}^{-2}\right)$. Therefore, $\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}\left(\boldsymbol{F}_0\right)=O_p\left(m_{NT}^{-1}\right)$, and

$$\begin{aligned}\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)=&\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}-\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\boldsymbol{F}_0\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}\\ =&\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\cdot\boldsymbol{J}^\top\right]^{-1}\cdot\boldsymbol{J}\left[\boldsymbol{D}\left(\boldsymbol{F}_0\right)-\boldsymbol{D}\left(\hat{\boldsymbol{F}}\right)\right]\boldsymbol{J}^\top\cdot\left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\boldsymbol{F}_0\right)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}^\top=O_p\left(\frac{1}{m_{NT}}\right).\end{aligned}$$

Both $\sqrt{T/N}O_p(m_{NT}^{-1})$ and $\sqrt{N/T}O_p(m_{NT}^{-1})$ are $o_p(1)$ since $\sqrt{T}/N\to 0$ and $\sqrt{N}/T\to 0$ respectively.

(4) Since

$$\sqrt{\frac{T}{N}}\left[\boldsymbol{\xi}_{NT}-\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\boldsymbol{\xi}_{NT}^\dagger\right]=\sqrt{\frac{T}{N}}\left[\boldsymbol{D}^+\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\right]\cdot\boldsymbol{\xi}_{NT}^\dagger=o_p(1)$$

by statement (2), it suffices to study the difference

$$\begin{aligned}&\sqrt{\frac{T}{N}}\cdot\left[\mathbb{B}^{\boldsymbol{\xi}}-\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\boldsymbol{\xi}_{NT}^\dagger\right]\\ =&\sqrt{\frac{T}{N}}\cdot\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\left\{\frac{1}{N}\sum_{i=1}^N\boldsymbol{b}_{i,0}\otimes\left[\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\cdot\sum_{j=1}^N\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)-\boldsymbol{S}_{i,0}^\top\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right]\right.\\ &\left.-\frac{1}{N^2}\sum_{i,k=1}^N w_{ik}\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^\top\boldsymbol{\Sigma}_{k\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\cdot\left[\sum_{j=1}^N\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)-\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right]\right\}\\ &+\sqrt{\frac{T}{N}}\cdot\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\frac{1}{N\sqrt{T}}\sum_{l=1}^N\boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^\top\otimes\boldsymbol{Q}_{l,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\\ &\left\{\frac{1}{N}\sum_{i=1}^N\boldsymbol{l}_{i,0}\otimes\left[\frac{\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)\boldsymbol{\Phi}_{4i,0}^{e\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\cdot\sum_{j=1}^N\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)-\frac{\boldsymbol{X}_i\boldsymbol{\Phi}_{4i,0}^\top\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\cdot\sum_{j=1}^N\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right]\right.\\ &\left.-\frac{1}{N^2}\sum_{i,k=1}^N w_{ik}\boldsymbol{l}_{k,0}\otimes\frac{\boldsymbol{X}_k\boldsymbol{\Phi}_{4k,0}^\top\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\cdot\left[\sum_{j=1}^N\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)-\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right]\right\}\\ =&\sqrt{\frac{T}{N}}\cdot\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\left(\Delta_1+\Delta_2\right)+\sqrt{\frac{T}{N}}\cdot\boldsymbol{D}^+\left(\boldsymbol{F}_0\right)\cdot\frac{1}{N\sqrt{T}}\sum_{l=1}^N\boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^\top\otimes\boldsymbol{Q}_{l,0}^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\boldsymbol{\Xi}_0^{-1}\cdot\left(\Delta_3+\Delta_4\right).\end{aligned}$$

Decompose Δ_1:

$$\Delta_1=\frac{1}{N}\sum_{i,j=1}^N\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{j,0}\left(\frac{1}{T}\sum_{t=1}^T\varepsilon_{it}\varepsilon_{jt}\right)-\frac{1}{N}\sum_{i=1}^N\boldsymbol{b}_{i,0}\otimes\boldsymbol{S}_{i,0}^\top\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2$$

$$
\begin{aligned}
&= \frac{1}{N}\sum_{i,j=1}^{N} \boldsymbol{b}_{i,0} \otimes \boldsymbol{M}_i^e \boldsymbol{l}_{j,0} \left[\frac{1}{T}\sum_{t=1}^{T}\left(\varepsilon_{it}\varepsilon_{jt} - \sigma_{ij,tt}\right)\right] + \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \left(\boldsymbol{M}_i^e - \boldsymbol{M}_i\right) \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2 \\
&=: \Delta_{1.1} + \Delta_{1.2},
\end{aligned}
$$

where $\boldsymbol{M}_i^e = \boldsymbol{S}_{i,0}^{e\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}$, $\boldsymbol{M}_i = \boldsymbol{S}_{i,0}^{\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}$, $\sigma_{ii,tt} = \mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^2\right)$ and $\sigma_{ij,tt} = 0$ for $i \neq j$. For $\Delta_{1.1}$,

$$
\begin{aligned}
\mathbb{E}_{\mathcal{C}}\left(\|\Delta_{1.1}\|_2^2\right) &= \frac{1}{N^2T^2}\sum_{i,j,k,l=1}^{N}\sum_{s,t=1}^{T} \boldsymbol{b}_{i,0}^{\top}\boldsymbol{b}_{k,0}\boldsymbol{l}_{j,0}^{\top}\boldsymbol{M}_i^{e\top}\boldsymbol{M}_k^e\boldsymbol{l}_{l,0}\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}\varepsilon_{jt} - \sigma_{ij,tt}\right)\left(\varepsilon_{ks}\varepsilon_{ls} - \sigma_{kl,ss}\right)\right] \\
&= \frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{j\neq i}\sum_{t=1}^{T} \boldsymbol{b}_{i,0}^{\top}\boldsymbol{b}_{i,0}\boldsymbol{l}_{j,0}^{\top}\boldsymbol{M}_i^{e\top}\boldsymbol{M}_i^e\boldsymbol{l}_{j,0}\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}\varepsilon_{jt} - \sigma_{ij,tt}\right)^2\right] \\
&\quad + \frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{j\neq i}\sum_{t=1}^{T} \boldsymbol{b}_{i,0}^{\top}\boldsymbol{b}_{j,0}\boldsymbol{l}_{j,0}^{\top}\boldsymbol{M}_i^{e\top}\boldsymbol{M}_j^e\boldsymbol{l}_{i,0}\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}\varepsilon_{jt} - \sigma_{ij,tt}\right)^2\right] \\
&\quad + \frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{t=1}^{T} \boldsymbol{b}_{i,0}^{\top}\boldsymbol{b}_{i,0}\boldsymbol{l}_{i,0}^{\top}\boldsymbol{M}_i^{e\top}\boldsymbol{M}_i^e\boldsymbol{l}_{i,0}\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)^2\right] = O_p\left(\frac{1}{T}\right),
\end{aligned}
$$

where $\mathbb{E}_{\mathcal{C}}\left[\left(\varepsilon_{it}\varepsilon_{jt} - \sigma_{ij,tt}\right)\left(\varepsilon_{ks}\varepsilon_{ls} - \sigma_{kl,ss}\right)\right] \neq 0$ only when $s = t$ and (a) $i = j = k = l$ or (b) $(i = k) \neq (j = l)$ or (c) $(i = l) \neq (j = k)$. For $\Delta_{1.2}$,

$$
\begin{aligned}
\Delta_{1.2} &= \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{b}_{i,0} \otimes \left[\left(\boldsymbol{S}_{i,0}^e - \boldsymbol{S}_{i,0}\right)^{\top}\boldsymbol{\Sigma}_{i\boldsymbol{F}} + \boldsymbol{S}_{i,0}^{e\top}\left(\boldsymbol{\Sigma}_{i\boldsymbol{F}}^e - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\right)\right]\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2 \\
&= O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right) = O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}
$$

Therefore, we have

$$
\sqrt{\frac{T}{N}} \cdot \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right) \cdot \Delta_1 = \sqrt{\frac{T}{N}} \cdot O_p(1) \cdot O_p\left(\frac{1}{m_{NT}}\right) = o_p(1).
$$

The other counterparts Δ_2-Δ_4 can be analyzed similarly.

(5)–(6) Since $\left(\boldsymbol{\zeta}_{1,NT}^{\dagger}, \boldsymbol{\zeta}_{2,NT}^{\dagger}, \boldsymbol{\eta}_{NT}^{\dagger}\right)$ are all $O_p(1)$, and

$$
\begin{aligned}
\sqrt{\frac{N}{T}}\left(\boldsymbol{\zeta}_{NT} - \mathbb{B}^{\boldsymbol{\zeta}}\right) &= \sqrt{\frac{N}{T}}\left[\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right)\right] \cdot \left(\boldsymbol{\zeta}_{1,NT}^{\dagger} + \boldsymbol{\zeta}_{2,NT}^{\dagger}\right), \\
\sqrt{\frac{N}{T}}\left(\boldsymbol{\eta}_{NT} - \mathbb{B}^{\boldsymbol{\eta}}\right) &= \sqrt{\frac{N}{T}}\left[\boldsymbol{D}^{+}\left(\hat{\boldsymbol{F}}\right) - \boldsymbol{D}^{+}\left(\boldsymbol{F}_0\right)\right] \cdot \boldsymbol{\eta}_{NT}^{\dagger},
\end{aligned}
$$

statements (5)–(6) follow from (3). □

S.4.2 Proofs of Lemmas in Section S.2.4

Lemma S.4.3. *Suppose Assumptions 2, 5–7, and 9 hold. If $T = O(N)$, then:*

(1) $\max_t \left\|\hat{\boldsymbol{f}}_t\right\|_2 = O_p(1)$.

(2a) $\dfrac{\|\hat{\boldsymbol{L}}^{\top} - \boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^{\top}\|_f^2}{N} = O_p\left(\dfrac{1}{N}\right)$.

(2b) $\left(\dfrac{\hat{\boldsymbol{L}}^{\top}\hat{\boldsymbol{L}}}{N}\right)^{-1} - \boldsymbol{O}_{\boldsymbol{F}}^{-\top}\left(\dfrac{\boldsymbol{L}_0^{\top}\boldsymbol{L}_0}{N}\right)^{-1}\boldsymbol{O}_{\boldsymbol{F}}^{-1} = O_p\left(\dfrac{1}{\sqrt{N}}\right)$.

(2c) $\max_i \left\|\hat{\boldsymbol{l}}_i\right\|_2 = O_p(1)$.

(3a) $\dfrac{\|\tilde{\boldsymbol{B}} - \boldsymbol{B}_0\|_f^2}{N} = O_p\left(\dfrac{1}{N}\right)$.

(3b) $\max_i \left\|\tilde{\boldsymbol{b}}_i\right\|_2 = O_p(1)$.

(4a) $\dfrac{1}{NT}\sum_{i=1}^N \|\hat{\boldsymbol{\varepsilon}}_i - \boldsymbol{\varepsilon}_i\|_2^2 = O_p\left(\dfrac{1}{N}\right)$.

(4b) $\max_{i,t} |\hat{\varepsilon}_{it} - \varepsilon_{it}| = O_p(1)$.

(5a) $\max_i \dfrac{\|\hat{\boldsymbol{Q}}_i - \boldsymbol{Q}_{i,0}\|_f}{\sqrt{T}} = O_p\left(\dfrac{1}{\sqrt{N}}\right)$.

(5b) $\max_i \left\|\hat{\boldsymbol{S}}_i - \boldsymbol{S}_{i,0}\right\|_f = O_p\left(\dfrac{1}{\sqrt{N}}\right)$.

(5c) $\max_i \left\|\hat{\boldsymbol{V}}_i - \boldsymbol{V}_{i,0}\right\|_f = O_p\left(\dfrac{1}{\sqrt{N}}\right)$.

(5d) $\left\|\boldsymbol{O}_{\boldsymbol{F}}^\top \otimes \boldsymbol{I}_T \cdot \hat{\boldsymbol{\Xi}}^{-1} \cdot \boldsymbol{O}_{\boldsymbol{F}} \otimes \boldsymbol{I}_T - \boldsymbol{\Xi}_0^{-1}\right\|_2 = O_p\left(\dfrac{1}{\sqrt{N}}\right)$.

Proof of Lemma S.2.16. We use the identity mentioned in the Supplement to Bai (2009) in proof: for $n \geq 1$,

$$\prod_{i=1}^n a_i' - \prod_{i=1}^n a_i = \sum_{i^*=1}^n \left[(a_{i^*}' - a_{i^*}) \cdot \prod_{i_1=1}^{i^*-1} a_{i_1} \cdot \prod_{i_2=i^*+1}^n a_{i_2}'\right], \tag{S.4.13}$$

where we define $\prod_{i_1=1}^0 a_{i_1} = 1$ and $\prod_{i_2=n+1}^n a_{i_2}' = 1$. For example, when $n = 3$, the identity is $a_1'a_2'a_3' - a_1a_2a_3 = (a_1' - a_1)a_2'a_3' + a_1(a_2' - a_2)a_3' + a_1a_2(a_3' - a_3)$.

(1) Since the denominator $\tilde{\boldsymbol{D}}_0^+$ in $\tilde{\mathbb{B}}^{\boldsymbol{\xi}}$ satisfies $\tilde{\boldsymbol{D}}_0^+ - \boldsymbol{D}^+(\boldsymbol{F}_0) = O_p\left(N^{-1/2}\right)$ (see the proof of Proposition 3.1), it suffices to show the difference between numerators of $\tilde{\mathbb{B}}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\xi}}$,

$$\begin{aligned}
&\frac{1}{N}\sum_{i=1}^N \left(\boldsymbol{b}_{i,0} \otimes \boldsymbol{S}_{i,0}^\top \frac{\boldsymbol{X}_i^\top \boldsymbol{F}_0}{T} \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2 - \tilde{\boldsymbol{b}}_i \otimes \hat{\boldsymbol{S}}_i^\top \frac{\boldsymbol{X}_i^\top \hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2\right)\\
&-\frac{1}{N^2}\sum_{i,k=1}^N \left(w_{ik}\boldsymbol{b}_{k,0} \otimes \boldsymbol{S}_{k,0}^\top \frac{\boldsymbol{X}_k^\top \boldsymbol{F}_0}{T} \boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2 - \hat{w}_{ik}\tilde{\boldsymbol{b}}_k \otimes \hat{\boldsymbol{S}}_k^\top \frac{\boldsymbol{X}_k^\top \hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2\right)\\
&+\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^N \boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^\top \otimes \boldsymbol{Q}_{l,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}_0^{-1} \cdot \frac{1}{N}\sum_{i=1}^N \boldsymbol{l}_{i,0} \otimes \frac{\boldsymbol{V}_{i,0}^\top \boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right.\\
&\left.-\frac{1}{N\sqrt{T}}\sum_{l=1}^N \tilde{\boldsymbol{b}}_l\hat{\boldsymbol{l}}_l^\top \otimes \hat{\boldsymbol{Q}}_l^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \hat{\boldsymbol{\Xi}}^{-1} \cdot \frac{1}{N}\sum_{i=1}^N \hat{\boldsymbol{l}}_i \otimes \frac{\hat{\boldsymbol{V}}_i^\top \hat{\boldsymbol{F}}}{\sqrt{T}}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2\right)\\
&-\left(\frac{1}{N\sqrt{T}}\sum_{l=1}^N \boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^\top \otimes \boldsymbol{Q}_{l,0}^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \boldsymbol{\Xi}_0^{-1} \cdot \frac{1}{N^2}\sum_{i,k=1}^N w_{ik}\boldsymbol{l}_{k,0} \otimes \frac{\boldsymbol{V}_{k0}^\top \boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right.\\
&\left.-\frac{1}{N\sqrt{T}}\sum_{l=1}^N \tilde{\boldsymbol{b}}_l\hat{\boldsymbol{l}}_l^\top \otimes \hat{\boldsymbol{Q}}_l^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \cdot \hat{\boldsymbol{\Xi}}^{-1} \cdot \frac{1}{N^2}\sum_{i,k=1}^N \hat{w}_{ik}\hat{\boldsymbol{l}}_k \otimes \frac{\hat{\boldsymbol{V}}_k^\top \hat{\boldsymbol{F}}}{\sqrt{T}}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2\right)\\
&=:\Delta_{\boldsymbol{\xi}.1} + \Delta_{\boldsymbol{\xi}.2} + \Delta_{\boldsymbol{\xi}.3} + \Delta_{\boldsymbol{\xi}.4} = o_p(1).
\end{aligned}$$

We study the differences of five parts in $\Delta_{\boldsymbol{\xi}.1}$: $\boldsymbol{b}_{i,0}$, $\boldsymbol{S}_{i,0}$, $\boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}$, $\boldsymbol{l}_{i,0}$ and $\bar{\sigma}_i^2$. Decompose $\Delta_{\boldsymbol{\xi}.1}$:

$$\begin{aligned}\Delta_{\boldsymbol{\xi}.1} =& \frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{b}_{i,0}-\tilde{\boldsymbol{b}}_i\right)\otimes\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2+\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{b}}_i\otimes\left(\boldsymbol{S}_{i,0}-\hat{\boldsymbol{S}}_i\right)^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\\
&+\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\left[\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}-\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{T}\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\boldsymbol{O}_{\boldsymbol{F}}\right]\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\\
&+\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\left(\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}-\hat{\boldsymbol{l}}_i\right)\bar{\sigma}_i^2+\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\left(\bar{\sigma}_i^2-\hat{\bar{\sigma}}_i^2\right)\\
=:&\Delta_{\boldsymbol{\xi}.1.1}+\cdots+\Delta_{\boldsymbol{\xi}.1.5},\end{aligned}$$

where

$$\begin{aligned}
&\|\Delta_{\boldsymbol{\xi}.1.1}\|_2\le\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{b}_{i,0}-\tilde{\boldsymbol{b}}_i\right\|_2^2\right)^{1/2}\cdot\max_i\left\|\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right\|_2=O_p\left(\frac{1}{\sqrt{N}}\right),\\
&\|\Delta_{\boldsymbol{\xi}.1.2}\|_2\le\max_i\left\|\tilde{\boldsymbol{b}}_i\right\|_2\cdot\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{S}_{i,0}-\hat{\boldsymbol{S}}_i\right\|_2\cdot\max_i\left\|\frac{\boldsymbol{X}_i^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right\|_2=O_p\left(\frac{1}{\sqrt{N}}\right),\\
&\|\Delta_{\boldsymbol{\xi}.1.3}\|_2\le\max_i\left\|\tilde{\boldsymbol{b}}_i\otimes\frac{\hat{\boldsymbol{S}}_i^{\top}\boldsymbol{X}_i}{\sqrt{T}}\right\|_2\cdot\left\|\frac{\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}-\frac{\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{\sqrt{T}}\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\boldsymbol{O}_{\boldsymbol{F}}\right\|_2\cdot\max_i\left\|\boldsymbol{l}_{i,0}\bar{\sigma}_i^2\right\|_2=O_p\left(\frac{1}{\sqrt{N}}\right),\\
&\|\Delta_{\boldsymbol{\xi}.1.4}\|_2\le\max_i\left\|\bar{\sigma}_i^2\tilde{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\right\|_2\cdot\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}-\hat{\boldsymbol{l}}_i\right\|_2^2\right)^{1/2}=O_p\left(\frac{1}{\sqrt{N}}\right),\\
&\|\Delta_{\boldsymbol{\xi}.1.5}\|_2\le\max_i\left\|\tilde{\boldsymbol{b}}_i\otimes\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\hat{\boldsymbol{F}}}{T}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\right\|_2\cdot\frac{1}{N}\sum_{i=1}^{N}\left|\bar{\sigma}_i^2-\hat{\bar{\sigma}}_i^2\right|=O_p\left(\frac{1}{\sqrt{N}}\right).
\end{aligned}$$

Note that for $\Delta_{\boldsymbol{\xi}.1.5}$,

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^{N}\left|\bar{\sigma}_i^2-\hat{\bar{\sigma}}_i^2\right|\le&\frac{1}{N}\sum_{i=1}^{N}\left|\frac{1}{T}\sum_{t=1}^{T}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right|+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left|\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right|\\
=&O_p\left(\frac{1}{\sqrt{T}}\right)+O_p\left(\frac{1}{\sqrt{N}}\right)=O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where

$$\begin{aligned}\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left|\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right|\le&\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}+\varepsilon_{it}\right)^2\right]^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}\\
\le&\left[\frac{2}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2+\frac{8}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\right]^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}\\
=&O_p(1)\cdot O_p\left(\frac{1}{\sqrt{N}}\right)=O_p\left(\frac{1}{\sqrt{N}}\right).\end{aligned}$$

This establishes that $\Delta_{\boldsymbol{\xi}.1}=O_p(N^{-1/2})$. For $\Delta_{\boldsymbol{\xi}.2}$, as $(\boldsymbol{b}_{k,0},\ \boldsymbol{S}_{k,0},\ \boldsymbol{F}_0\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1},\ \boldsymbol{l}_{i,0})$ are analyzed in $\Delta_{\boldsymbol{\xi}.1}$, we only show that

$$\frac{1}{N^2}\sum_{i,k=1}^{N}\left(w_{ik}-\hat{w}_{ik}\right)\boldsymbol{b}_{k,0}\otimes\boldsymbol{S}_{k,0}^{\top}\frac{\boldsymbol{X}_k^{\top}\boldsymbol{F}_0}{T}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2=o_p(1).$$

From the decomposition

$$w_{ik}-\hat{w}_{ik}=\boldsymbol{l}_{i,0}^{\top}\left(\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}-\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\boldsymbol{O}_{\boldsymbol{F}}\right)\boldsymbol{l}_{k,0}+\left(\boldsymbol{l}_{i,0}^{\top}\boldsymbol{O}_{\boldsymbol{F}}^{\top}-\hat{\boldsymbol{l}}_i^{\top}\right)\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{k,0}+\hat{\boldsymbol{l}}_i^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\left(\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{k,0}-\hat{\boldsymbol{l}}_k\right),$$

the above statement is true since

$$\frac{1}{N^2}\sum_{i,k=1}^{N}(w_{ik}-\hat{w}_{ik})^2 \lesssim \frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{l}_{i,0}\|_2^2\cdot\left\|\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}-\boldsymbol{O}_{\boldsymbol{F}}^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\boldsymbol{O}_{\boldsymbol{F}}\right\|_2^2\cdot\frac{1}{N}\sum_{k=1}^{N}\|\boldsymbol{l}_{k,0}\|_2^2$$
$$+\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{l}}_i-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right\|_2^2\cdot\left\|\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\right\|_2^2\cdot\frac{1}{N}\sum_{k=1}^{N}\|\boldsymbol{l}_{k,0}\|_2^2$$
$$+\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{l}}_i\right\|_2^2\cdot\left\|\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\right\|_2^2\cdot\frac{1}{N}\sum_{k=1}^{N}\left\|\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{k,0}-\hat{\boldsymbol{l}}_k\right\|_2^2=O_p\left(\frac{1}{N}\right).$$

We rewrite $\Delta_{\boldsymbol{\xi}.3}$ as

$$\Delta_{\boldsymbol{\xi}.3}=-\frac{1}{N\sqrt{T}}\sum_{l=1}^{N}\tilde{\boldsymbol{b}}_l\hat{\boldsymbol{l}}_l^{\top}\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\otimes\hat{\boldsymbol{Q}}_l^{\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^{\top}\otimes\boldsymbol{I}_T\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\boldsymbol{O}_{\boldsymbol{F}}\otimes\boldsymbol{I}_T\right)\cdot\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{O}_{\boldsymbol{F}}^{-1}\hat{\boldsymbol{l}}_i\otimes\frac{\hat{\boldsymbol{V}}_i^{\top}\hat{\boldsymbol{F}}}{\sqrt{T}}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\hat{\boldsymbol{l}}_i\hat{\bar{\sigma}}_i^2$$
$$+\frac{1}{N\sqrt{T}}\sum_{l=1}^{N}\boldsymbol{b}_{l,0}\boldsymbol{l}_{l,0}^{\top}\otimes\boldsymbol{Q}_{l,0}^{\top}\cdot\boldsymbol{\Xi}_0^{-1}\cdot\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{l}_{i,0}\otimes\frac{\boldsymbol{V}_{i,0}^{\top}\boldsymbol{F}_0}{\sqrt{T}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{\boldsymbol{L}}^{-1}\boldsymbol{l}_{i,0}\bar{\sigma}_i^2,$$

It suffices to analyze the new components $\left(\boldsymbol{Q}_{i,0},\ \boldsymbol{V}_{i,0},\ \boldsymbol{\Xi}_0^{-1}\right)$ in $\Delta_{\boldsymbol{\xi}.3}$. We have

$$\frac{1}{NT}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{Q}}_i-\boldsymbol{Q}_{i,0}\right\|_f^2=O_p\left(\frac{1}{N}\right),\quad \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{V}}_i-\boldsymbol{V}_{i,0}\right\|_f^2=O_p\left(\frac{1}{N}\right),$$
$$\left\|\boldsymbol{O}_{\boldsymbol{F}}^{\top}\otimes\boldsymbol{I}_T\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\boldsymbol{O}_{\boldsymbol{F}}\otimes\boldsymbol{I}_T-\boldsymbol{\Xi}_0^{-1}\right\|_2=O_p\left(\frac{1}{\sqrt{N}}\right).$$

All components in $\Delta_{\boldsymbol{\xi}.4}$ are analyzed before. It follows that $\Delta_{\boldsymbol{\xi}.4}=O_p\left(N^{-1/2}\right)$. Therefore, the difference between numerators of $\tilde{\mathbb{B}}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\xi}}$ are

$$\Delta_{\boldsymbol{\xi}.1}+\cdots+\Delta_{\boldsymbol{\xi}.4}=O_p\left(\frac{1}{\sqrt{N}}\right).$$

This completes the proof of statement (1).

(2) Since

$$\tilde{\boldsymbol{D}}_0^{+}-\boldsymbol{D}^{+}(\boldsymbol{F}_0)=O_p\left(\frac{1}{\sqrt{N}}\right),\quad \tilde{\mathbb{B}}^{\boldsymbol{\zeta}}-\mathbb{B}^{\boldsymbol{\zeta}}=\left[\tilde{\mathbb{B}}^{\boldsymbol{\zeta}}-\tilde{\boldsymbol{D}}_0^{+}\cdot\left(\boldsymbol{\zeta}_{1,NT}^{\dagger}+\boldsymbol{\zeta}_{2,NT}^{\dagger}\right)\right]+\left[\tilde{\boldsymbol{D}}_0^{+}-\boldsymbol{D}^{+}(\boldsymbol{F}_0)\right]\cdot\left(\boldsymbol{\zeta}_{1,NT}^{\dagger}+\boldsymbol{\zeta}_{2,NT}^{\dagger}\right),$$

it suffices to show the difference between $\boldsymbol{\zeta}_{1,NT}^{\dagger}+\boldsymbol{\zeta}_{2,NT}^{\dagger}$ and the numerator of $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}}$ is $o_p(1)$. The difference between $\boldsymbol{\zeta}_{1,NT}^{\dagger}$ and the numerator of $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}_1}$ is

$$\frac{1}{N}\sum_{i=1}^{N}\left[\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\hat{\boldsymbol{\Omega}}_i\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}-\boldsymbol{S}_{i,0}^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{\Omega}_i\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i}{T}\boldsymbol{\Theta}_0\left(\frac{\boldsymbol{\Theta}_0^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right].$$

Since differences in other parts are discussed before, we only analyze terms involving $\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i$, that is,

$$\left\|\frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{S}}_i^{\top}\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}(\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i)\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i}{T}\tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^{\top}\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\hat{\boldsymbol{F}}}^{\perp}\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\right\|_2$$
$$\lesssim\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\boldsymbol{X}_i^{\top}(\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i)\boldsymbol{X}_i\right\|_2}{T}+\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\hat{\boldsymbol{F}}^{\top}(\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i)\hat{\boldsymbol{F}}\right\|_2}{T}$$

$$
\begin{aligned}
&+\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\boldsymbol{X}_i^\top(\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i)\hat{\boldsymbol{F}}\right\|_2}{T}+\frac{1}{N}\sum_{i=1}^{N}\frac{\left\|\hat{\boldsymbol{F}}^\top(\hat{\boldsymbol{\Omega}}_i-\boldsymbol{\Omega}_i)\boldsymbol{X}_i\right\|_2}{T}\\
=:&\Delta_{\boldsymbol{\Omega}.1}+\Delta_{\boldsymbol{\Omega}.2}+\Delta_{\boldsymbol{\Omega}.3}+\Delta_{\boldsymbol{\Omega}.4}.
\end{aligned}
\tag{S.4.14}
$$

For $\Delta_{\boldsymbol{\Omega}.1}$,

$$
\Delta_{\boldsymbol{\Omega}.1}\le\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_2+\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right)\right\|_2=O_p\left(\frac{1}{\sqrt{N}}\right),
$$

where

$$
\begin{aligned}
\mathbb{E}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_2\right)\le&\mathbb{E}\left\{\frac{1}{N}\sum_{i=1}^{N}\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_f^2\right)\right]^{1/2}\right\}\\
=&\frac{1}{\sqrt{T}}\cdot\mathbb{E}\left(\frac{1}{N}\sum_{i=1}^{N}\left\{\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{\mathcal{C}}\left[\|\boldsymbol{x}_{it}\|_2^4\cdot\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^2\right]\right\}^{1/2}\right)\\
=&O\left(\frac{1}{\sqrt{T}}\right),
\end{aligned}
$$

and

$$
\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right)\right\|_2\\
\le&\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\|\boldsymbol{x}_{it}\|_2^2\cdot|\hat{\varepsilon}_{it}+\varepsilon_{it}|\cdot|\hat{\varepsilon}_{it}-\varepsilon_{it}|\right)\\
\le&\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}+\varepsilon_{it}\right)^4\right]^{1/4}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\|\boldsymbol{x}_{it}\|_2^8\right)^{1/4}\\
=&O_p\left(\frac{1}{\sqrt{N}}\right)\cdot O_p(1)\cdot O_p(1)=O_p\left(\frac{1}{\sqrt{N}}\right).
\end{aligned}
$$

Note that

$$
\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}+\varepsilon_{it}\right)^4=\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left[\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)+2\varepsilon_{it}\right]^4\le 8\max_{i,t}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^4+\frac{128}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^4=O_p(1).
$$

For $\Delta_{\boldsymbol{\Omega}.2}$,

$$
\begin{aligned}
\Delta_{\boldsymbol{\Omega}.2}\le&\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{f}}_t\hat{\boldsymbol{f}}_t^\top\left(\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right)\right\|_2+\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\left(\hat{\boldsymbol{f}}_t-\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right)\hat{\boldsymbol{f}}_t^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_2\\
&+\left\|\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2\cdot\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\left(\hat{\boldsymbol{f}}_t-\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right)^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_2+\left\|\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2^2\cdot\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{f}_{t,0}\boldsymbol{f}_{t,0}^\top\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\right\|_2\\
=:&\Delta_{\boldsymbol{\Omega}.2.1}+\Delta_{\boldsymbol{\Omega}.2.2}+\Delta_{\boldsymbol{\Omega}.2.3}+\Delta_{\boldsymbol{\Omega}.2.4}=O_p\left(\frac{1}{\sqrt{N}}\right),
\end{aligned}
$$

where

$$
\begin{aligned}
\Delta_{\boldsymbol{\Omega}.2.1}&\le\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2^2\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}+\varepsilon_{it}\right)^2\right]^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}=O_p\left(\frac{1}{\sqrt{N}}\right),\\
\Delta_{\boldsymbol{\Omega}.2.2}&\le\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2\cdot\left(\frac{1}{T}\sum_{t=1}^{T}\left\|\hat{\boldsymbol{f}}_t-\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right\|_2^2\right)^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^2-\sigma_{it}^2\right)^2\right]^{1/2}=O_p\left(\frac{1}{\sqrt{N}}\right),
\end{aligned}
$$

$$\Delta_{\boldsymbol{\Omega}.2.3} \lesssim \left(\frac{1}{T}\sum_{t=1}^{T}\left\|\hat{\boldsymbol{f}}_t - \boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right\|_2^2\right)^{1/2} \cdot \left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)^2\right]^{1/2} = O_p\left(\frac{1}{\sqrt{N}}\right), \quad \Delta_{\boldsymbol{\Omega}.2.4} = O_p\left(\frac{1}{\sqrt{T}}\right).$$

For $\Delta_{\boldsymbol{\Omega}.3}$,

$$\begin{aligned}\Delta_{\boldsymbol{\Omega}.3} \leq & \frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\hat{\boldsymbol{f}}_t^\top\left(\hat{\varepsilon}_{it}^2 - \varepsilon_{it}^2\right)\right\|_2 + \frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\left(\hat{\boldsymbol{f}}_t - \boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right)^\top\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)\right\|_2 \\ & + \left\|\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2 \cdot \frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_{it}\boldsymbol{f}_{t,0}^\top\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)\right\|_2 \\ =: & \Delta_{\boldsymbol{\Omega}.3.1} + \Delta_{\boldsymbol{\Omega}.3.2} + \Delta_{\boldsymbol{\Omega}.3.3} = O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where

$$\begin{aligned}\Delta_{\boldsymbol{\Omega}.3.1} \leq & \max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2 \cdot \left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{x}_{it}\right\|_2^4\right)^{1/4} \cdot \left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it} + \varepsilon_{it}\right)^4\right]^{1/4} \cdot \left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it} - \varepsilon_{it}\right)^2\right]^{1/2} \\ = & O_p(1)\cdot O_p(1)\cdot O_p(1)\cdot O_p\left(\frac{1}{\sqrt{N}}\right) = O_p\left(\frac{1}{\sqrt{N}}\right), \\ \Delta_{\boldsymbol{\Omega}.3.2} \leq & \left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{x}_{it}\right\|_2^4\right)^{1/4} \cdot \left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\hat{\boldsymbol{f}}_t - \boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{f}_{t,0}\right\|_2^2\right)^{1/2} \cdot \left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)^4\right]^{1/4} \\ = & O_p(1)\cdot O_p\left(\frac{1}{\sqrt{N}}\right)\cdot O_p(1) = O_p\left(\frac{1}{\sqrt{N}}\right), \\ \Delta_{\boldsymbol{\Omega}.3.3} = & O_p\left(\frac{1}{\sqrt{T}}\right).\end{aligned}$$

In the same way, we can show that $\Delta_{\boldsymbol{\Omega}.4} = O_p(N^{-1/2})$. This establishes that $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}_1} - \mathbb{B}^{\boldsymbol{\zeta}_1} = O_p(N^{-1/2})$. For the difference between $\boldsymbol{\zeta}_{2,NT}^\dagger$ and the numerator of $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}_2}$, it suffices to study

$$\begin{aligned}& \frac{1}{N^2T}\sum_{i,j=1}^{N}\left\{\tilde{\boldsymbol{b}}_i\hat{\boldsymbol{l}}_i^\top \otimes \hat{\boldsymbol{Q}}_i^\top \cdot \hat{\boldsymbol{\Xi}}^{-1}\cdot\ell\left[\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\hat{\boldsymbol{U}}_j\left(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j\right)\hat{\boldsymbol{U}}_j^\top\hat{\boldsymbol{F}}\right]\right\} \\ = & -\frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\tilde{\boldsymbol{b}}_i\hat{\boldsymbol{l}}_i^\top \otimes \hat{\boldsymbol{Q}}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\boldsymbol{I}_{R_0}\otimes\hat{\boldsymbol{V}}_j\left(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j\right)\left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_j\right)^\top\hat{\boldsymbol{F}}\cdot\ell(\boldsymbol{I}_{R_0})\right] \\ & + \frac{1}{N^3T}\sum_{i,j,k=1}^{N}\left[\tilde{\boldsymbol{b}}_i\hat{\boldsymbol{l}}_i^\top \otimes \hat{\boldsymbol{Q}}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\hat{\boldsymbol{l}}_k\hat{\boldsymbol{l}}_k^\top\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{P}_{\boldsymbol{G}}^\perp\hat{\boldsymbol{H}}_k\boldsymbol{P}_{\boldsymbol{G}}^\perp\left(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j\right)\left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_j\right)^\top\hat{\boldsymbol{F}}\cdot\ell(\boldsymbol{I}_{R_0})\right] \\ & + \frac{1}{N^2T}\sum_{i,j=1}^{N}\left[\tilde{\boldsymbol{b}}_i\hat{\boldsymbol{l}}_i^\top\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\otimes\hat{\boldsymbol{Q}}_i^\top\boldsymbol{P}_{\boldsymbol{G}}^\perp\left(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j\right)\left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_j\right)^\top\hat{\boldsymbol{F}}\cdot\ell(\boldsymbol{I}_{R_0})\right].\end{aligned}$$

In each term, we can identify similar structures as in (S.4.14) by expanding $\left(\hat{\boldsymbol{V}}_j,\ \hat{\boldsymbol{H}}_k,\ \hat{\boldsymbol{Q}}_i\right)$. Specifically, they take the form of various bilinear expressions involving $\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j$:

$$\begin{aligned}& \frac{\hat{\boldsymbol{F}}^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\hat{\boldsymbol{F}}}{T}, \quad \frac{\boldsymbol{X}_j^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\hat{\boldsymbol{F}}}{T}, \quad \frac{\hat{\boldsymbol{F}}^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\boldsymbol{X}_j}{T}, \quad \frac{\boldsymbol{X}_j^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\boldsymbol{X}_j}{T}, \\ & \frac{\boldsymbol{X}_k^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\hat{\boldsymbol{F}}}{T}, \quad \frac{\boldsymbol{X}_k^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\boldsymbol{X}_j}{T}, \quad \frac{\boldsymbol{G}^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\hat{\boldsymbol{F}}}{T}, \quad \frac{\boldsymbol{G}^\top(\hat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j)\boldsymbol{X}_j}{T}.\end{aligned}$$

By a similar argument for $\mathbb{B}^{\boldsymbol{\zeta}_1}$, we obtain $\tilde{\mathbb{B}}^{\boldsymbol{\zeta}_2} - \mathbb{B}^{\boldsymbol{\zeta}_2} = O_p(N^{-1/2})$. This completes the proof of statement (2).

(3) It suffices to show that the difference between the numerators of $\tilde{\mathbb{B}}^{\boldsymbol{\eta}}$ and $\mathbb{B}^{\boldsymbol{\eta}}$ is $o_p(1)$. Let

$$\boldsymbol{\varpi}_{1is} = \left[\ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^\top\right)\right]^\top, \quad \tilde{\boldsymbol{\varpi}}_{1is} = \left[\ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}\right)\right]^\top,$$

$$\boldsymbol{\varpi}_{2ijs} = \left[\ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{js}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{g}_s^\top\right),\ \ell^\top\left(\boldsymbol{x}_{js}\boldsymbol{f}_{s,0}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{js}\boldsymbol{g}_s^\top\right)\right]^\top,$$

$$\tilde{\boldsymbol{\varpi}}_{2ijs} = \left[\ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{js}^\top\right),\ \ell^\top\left(\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}\right),\ \ell^\top\left(\boldsymbol{x}_{is}\boldsymbol{g}_s^\top\right),\ \ell^\top\left(\boldsymbol{x}_{js}\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}\right),\ \ell^\top\left(\boldsymbol{x}_{js}\boldsymbol{g}_s^\top\right)\right]^\top.$$

By replacing $(\boldsymbol{F}_0,\ \boldsymbol{\Theta}_0)$ in $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}$ and $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}$ with $\left(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}},\ \tilde{\boldsymbol{\Theta}}\right)$, we obtain $\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}$ and $\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}$. Note that $\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^\top\hat{\boldsymbol{\varpi}}_{1is} = \tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\tilde{\boldsymbol{\varpi}}_{1is}$ and $\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^\top\hat{\boldsymbol{\varpi}}_{2ijs} = \tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^\top\tilde{\boldsymbol{\varpi}}_{2ijs}$. We only study the first, third, and fifth differences between the numerators of $\tilde{\mathbb{B}}^{\boldsymbol{\eta}}$ and $\mathbb{B}^{\boldsymbol{\eta}}$, since the second, fourth, and sixth differences can be proved in the same way.

(3.i) We begin with the first part of the difference:

$$\begin{aligned}
\Delta_1 =& \frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{b}}_i\otimes\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^\top\hat{\boldsymbol{\varpi}}_{1is}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it} - \frac{1}{N}\sum_{i=1}^N\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\pi}_{1i}\right)\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right]\\
=& \frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\left(\sum_{s=t+1}^{(t+m)\wedge T} - \sum_{s=1}^T\right)\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\\
&+\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{b}_{i,0}\otimes\left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it} - \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\right\}\\
&+\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\left[\tilde{\boldsymbol{b}}_i\otimes\mathcal{M}_{p,p}\left(\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^\top\tilde{\boldsymbol{\varpi}}_{1is}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it} - \boldsymbol{b}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\\
=:& \Delta_{1.1} + \Delta_{1.2} + \Delta_{1.3}.
\end{aligned}$$

Since $\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right] = \boldsymbol{0}_p$ for $s \le t$, we have

$$\begin{aligned}
\|\Delta_{1.1}\|_2 =& \left\|\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\sum_{s=t+m+1}^T\boldsymbol{b}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\right\|_2\\
\lesssim& \frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T\sum_{s=t+m+1}^T\left\{\alpha_i\left(|s-t|\right)^{1-1/4-1/4}\cdot\left[\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{\varpi}_{1is}\|_2^4\right)\right]^{1/4}\cdot\left[\mathbb{E}_{\mathcal{C}}\left(\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^4\right)\right]^{1/4}\right\}\\
\lesssim& \frac{1}{T}\sum_{t=1}^T\sum_{s=t+m+1}^T(s-t)^{-\gamma/2} = O(m^{-\gamma/2+1}) = o_p(1).
\end{aligned}$$

For $\Delta_{1.2}$,

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left(\|\Delta_{1.2}\|_2^2\right)\\
\le& \frac{1}{N^2T^2}\sum_{i=1}^N\mathbb{E}_{\mathcal{C}}\left(\left\|\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{b}_{i,0}\otimes\left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it} - \mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\right\}\right\|_2^2\right)\\
=& O_p\left(\frac{m^2}{N}\right) = o_p(1).
\end{aligned}$$

To show $\Delta_{1.3} = o_p(1)$, we first specify $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^\top\boldsymbol{\varpi}_{1is}$:

$$\begin{aligned}
\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^\top\boldsymbol{\varpi}_{1is}\right) =& -\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top - \boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^\top\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{f}_{s,0}\boldsymbol{x}_{is}^\top\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top\\
&+\left\{\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top\right.\\
&\left.\cdot\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top - \boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^\top\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{f}_{s,0}\boldsymbol{x}_{is}^\top\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii} - \boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top\right\}.
\end{aligned}$$

Decompose $\Delta_{1.3}$:

$$
\begin{aligned}
\Delta_{1.3} =& \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\left(\tilde{\boldsymbol{b}}_i-\boldsymbol{b}_{i,0}\right)\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\varepsilon_{it}\\
&+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{b}}_i\otimes\mathcal{M}_{p,p}\left[\left(\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}-\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e}\right)^{\top}\boldsymbol{\varpi}_{1is}\right]\boldsymbol{x}_{it}\varepsilon_{it}\\
&+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{b}}_i\otimes\mathcal{M}_{p,p}\left(\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^{\top}\boldsymbol{\varpi}_{1is}\right)\boldsymbol{x}_{it}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)\\
&+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{b}}_i\otimes\mathcal{M}_{p,p}\left[\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}^{\top}\left(\tilde{\boldsymbol{\varpi}}_{1is}-\boldsymbol{\varpi}_{1is}\right)\right]\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\\
=&\Delta_{1.3.1}+\cdots+\Delta_{1.3.4}.
\end{aligned}
$$

For $\Delta_{1.3.1}$,

$$
\begin{aligned}
\|\Delta_{1.3.1}\|_2 \le& \left(\frac{m}{N}\sum_{i=1}^{N}\left\|\tilde{\boldsymbol{b}}_i-\boldsymbol{b}_{i,0}\right\|_2^2\right)^{1/2}\cdot\max_i\left\|\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e}\right\|_2\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\left(\|\boldsymbol{\varpi}_{1is}\|_2^2\cdot\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^2\right)\right]^{1/2}\\
=&O_p\left(\sqrt{\frac{m}{N}}\right)\cdot O_p(1)\cdot O_p(\sqrt{m})=O_p\left(\frac{m}{\sqrt{N}}\right)=o_p(1),
\end{aligned}
$$

where

$$
\begin{aligned}
&\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\left(\|\boldsymbol{\varpi}_{1is}\|_2^2\cdot\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^2\right)\\
\le&\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\|\boldsymbol{\varpi}_{1is}\|_2^4\right)^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^4\right)^{1/2}\\
\lesssim&\left[\frac{m}{NT}\sum_{i=1}^{N}\sum_{s=1}^{T}\left(\left\|\boldsymbol{x}_{is}\boldsymbol{x}_{is}^{\top}\right\|_2^4+\left\|\boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^{\top}\right\|_2^4\right)\right]^{1/2}\cdot\left(\frac{m}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^4\right)^{1/2}\\
=&O_p(\sqrt{m})\cdot O_p(\sqrt{m})=O_p(m).
\end{aligned}
$$

For $\Delta_{1.3.2}$,

$$
\begin{aligned}
\|\Delta_{1.3.2}\|_2 \lesssim& \max_i\left\|\tilde{\boldsymbol{b}}_i\right\|_2\cdot\left[\frac{m}{N}\sum_{i=1}^{N}\left(\left\|\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}-\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}\right\|_2^2+\left\|\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}-\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{1i}}^{e}\right\|_2^2\right)\right]^{1/2}\\
&\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\left(\|\boldsymbol{\varpi}_{1is}\|_2^2\cdot\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^2\right)\right]^{1/2}\\
=&O_p(1)\cdot O_p\left(\sqrt{\frac{m}{N}}\right)\cdot O_p(\sqrt{m})=O_p\left(\frac{m}{\sqrt{N}}\right)=o_p(1).
\end{aligned}
$$

For $\Delta_{1.3.3}$,

$$
\begin{aligned}
\|\Delta_{1.3.3}\|_2 \lesssim& \max_i\left\|\tilde{\boldsymbol{b}}_i\right\|_2\cdot\max_i\left\|\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}\right\|_2\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\left(\|\boldsymbol{\varpi}_{1is}\|_2^2\cdot\|\boldsymbol{x}_{it}\|_2^2\right)\right]^{1/2}\cdot\left[\frac{m}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}\\
=&O_p(1)\cdot O_p(1)\cdot O_p(\sqrt{m})\cdot O_p\left(\sqrt{\frac{m}{N}}\right)=O_p\left(\frac{m}{\sqrt{N}}\right)=o_p(1).
\end{aligned}
$$

For $\Delta_{1.3.4}$,

$$\begin{aligned}\|\Delta_{1.3.4}\|_2 \lesssim & \max_i \left\|\tilde{\boldsymbol{b}}_i\right\|_2 \cdot \max_i \left\|\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{1i}}\right\|_2 \cdot \left(\frac{m}{NT}\sum_{i=1}^N\sum_{s=1}^T \|\tilde{\boldsymbol{\varpi}}_{1is}-\boldsymbol{\varpi}_{1is}\|_2^2\right)^{1/2} \cdot \left(\frac{m}{NT}\sum_{i=1}^N\sum_{t=1}^T \|\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\|_2^2\right)^{1/2} \\ =& O_p(1)\cdot O_p(1)\cdot O_p\left(\frac{\sqrt{m}}{N^{3/8}}\right)\cdot O_p(\sqrt{m}) = O_p\left(\frac{m}{N^{3/8}}\right) = o_p(1),\end{aligned}$$

where

$$\begin{aligned}\frac{1}{NT}\sum_{i=1}^N\sum_{s=1}^T \|\tilde{\boldsymbol{\varpi}}_{1is}-\boldsymbol{\varpi}_{1is}\|_2^2 \leq & \frac{1}{NT}\sum_{i=1}^N\sum_{s=1}^T \left\|\boldsymbol{x}_{is}\left(\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{f}_{s,0}^\top\right)\right\|_2^2 \\ \leq & \left(\frac{1}{NT}\sum_{i=1}^N\sum_{s=1}^T \|\boldsymbol{x}_{is}\|_2^8\right)^{1/4}\cdot\left(\frac{1}{T}\sum_{s=1}^T \left\|\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{f}_{s,0}^\top\right\|_2^{8/3}\right)^{3/4} \\ \lesssim & O_p(1)\cdot\left(\frac{1}{T}\sum_{s=1}^T \left\|\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{f}_{s,0}^\top\right\|_2^2\right)^{3/4}\cdot \max_s \left\|\hat{\boldsymbol{f}}_s^\top\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{f}_{s,0}^\top\right\|_2^{1/2} \\ = & O_p(1)\cdot O_p\left(\frac{1}{N^{3/4}}\right)\cdot O_p(1) = O_p\left(\frac{1}{N^{3/4}}\right),\end{aligned}$$

and

$$\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T \|\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\|_2^2 \leq \max_{i,t}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\cdot\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T \|\boldsymbol{x}_{it}\|_2^2 + \frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T \|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^2 = O_p(1).$$

This establishes that $\Delta_1 = o_p(1)$.

(3.ii) Decompose the third part of the difference:

$$\begin{aligned}\Delta_3 =& \frac{1}{N^2T}\sum_{i,j=1}^N\sum_{t=1}^T\left(\sum_{s=t+1}^{(t+m)\wedge T}-\sum_{s=1}^T\right)\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}_{1j}^e\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\varpi}_{2ijs}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right] \\ &+\frac{1}{N^2T}\sum_{i,j=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{\Psi}_{1j}^e\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\varpi}_{2ijs}\right)\boldsymbol{x}_{it}\varepsilon_{it}-\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\varpi}_{2ijs}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\right\} \\ &+\frac{1}{N^2T}\sum_{i,j=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\left(\boldsymbol{\Psi}_{1j}-\boldsymbol{\Psi}_{1j}^e\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\varpi}_{2ijs}\right)\boldsymbol{x}_{it}\varepsilon_{it} \\ &+\left[\frac{1}{N^2T}\sum_{i,j=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{b}}_j\hat{\boldsymbol{l}}_j^\top\otimes\hat{\boldsymbol{S}}_j^\top\cdot\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{I}_p\cdot\hat{\boldsymbol{l}}_i\otimes\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^\top\hat{\boldsymbol{\varpi}}_{2ijs}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\right. \\ &\left.-\frac{1}{N^2T}\sum_{i,j=1}^N\sum_{t=1}^T\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^\top\otimes\boldsymbol{S}_{j,0}^\top\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top}\boldsymbol{\varpi}_{2ijs}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right] \\ =:&\Delta_{3.1}+\cdots+\Delta_{3.4}.\end{aligned}$$

We first specify $\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^\top\boldsymbol{\varpi}_{2ijs}$:

$$\begin{aligned}\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^\top\boldsymbol{\varpi}_{2ijs}\right) =& \left(\boldsymbol{x}_{js}\boldsymbol{x}_{is}^\top-\boldsymbol{x}_{js}\boldsymbol{f}_{s,0}^\top\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top-\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{f}_{s,0}\boldsymbol{x}_{is}^\top-\boldsymbol{x}_{js}\boldsymbol{g}_s^\top\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{G}}^\top-\boldsymbol{\Sigma}_{j\boldsymbol{G}}\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\boldsymbol{g}_s\boldsymbol{x}_{is}^\top\right)\boldsymbol{\Theta}_0 \\ &\cdot\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top \\ &-\left(\boldsymbol{\Sigma}_{ji}-\boldsymbol{\Sigma}_{j\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top-\boldsymbol{\Sigma}_{j\boldsymbol{G}}\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{G}}^\top\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1} \\ &\cdot\boldsymbol{\Theta}_0^\top\left(\boldsymbol{x}_{is}\boldsymbol{x}_{is}^\top-\boldsymbol{x}_{is}\boldsymbol{f}_{s,0}^\top\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{f}_{s,0}\boldsymbol{x}_{is}^\top\right)\boldsymbol{\Theta}_0\left[\boldsymbol{\Theta}_0^\top\left(\boldsymbol{\Sigma}_{ii}-\boldsymbol{\Sigma}_{i\boldsymbol{F}}\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\boldsymbol{\Sigma}_{i\boldsymbol{F}}^\top\right)\boldsymbol{\Theta}_0\right]^{-1}\boldsymbol{\Theta}_0^\top.\end{aligned}$$

For $\Delta_{3.1}$,

$$\begin{aligned}
\|\Delta_{3.1}\|_2 =& \left\| \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+m+1}^{T} \boldsymbol{b}_{j,0} \tilde{\boldsymbol{l}}_{j,0}^{\top} \otimes \boldsymbol{\Psi}_{1j}^{e} \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \mathbb{E}_{\mathcal{C}} \left[\mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \varepsilon_{it} \right] \right\|_2 \\
\lesssim& \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+m+1}^{T} \left\| \mathbb{E}_{\mathcal{C}} \left[\mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \varepsilon_{it} \right] \right\|_2 \\
\lesssim& \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+m+1}^{T} \left\{ \alpha_i \left(|s-t| \right)^{1-1/4-1/4} \cdot \left[\mathbb{E}_{\mathcal{C}} \left(\|\boldsymbol{\varpi}_{2ijs}\|_2^4 \right) \right]^{1/4} \cdot \left[\mathbb{E}_{\mathcal{C}} \left(\|\boldsymbol{x}_{it}\varepsilon_{it}\|_2^4 \right) \right]^{1/4} \right\} \\
\lesssim& \frac{1}{T} \sum_{t=1}^{T} \sum_{s=t+m+1}^{T} (s-t)^{-\gamma/2} = O\left(m^{-\gamma/2+1} \right) = o_p(1).
\end{aligned}$$

It is straightforward to show that

$$\Delta_{3.2} = O_p\left(\frac{m}{\sqrt{N}} \right) = o_p(1), \quad \Delta_{3.3} = O_p\left(\frac{m}{\sqrt{T}} \right) = o_p(1).$$

Further decompose $\Delta_{3.4}$:

$$\begin{aligned}
\Delta_{3.4} =& \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} w_{ij} \left(\tilde{\boldsymbol{b}}_j - \boldsymbol{b}_{j,0} \right) \otimes \boldsymbol{S}_{j,0}^{\top} \mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \varepsilon_{it} \\
&+ \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \left(\hat{w}_{ij} - w_{ij} \right) \tilde{\boldsymbol{b}}_j \otimes \boldsymbol{S}_{j,0}^{\top} \mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \varepsilon_{it} \\
&+ \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{w}_{ij} \tilde{\boldsymbol{b}}_j \otimes \left(\hat{\boldsymbol{S}}_j - \boldsymbol{S}_{j,0} \right)^{\top} \mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \varepsilon_{it} \\
&+ \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{w}_{ij} \tilde{\boldsymbol{b}}_j \otimes \hat{\boldsymbol{S}}_j^{\top} \mathcal{M}_{p,p} \left[\left(\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}} - \boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ij}}^{e} \right)^{\top} \boldsymbol{\varpi}_{2ijs} \right] \boldsymbol{x}_{it} \varepsilon_{it} \\
&+ \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{w}_{ij} \tilde{\boldsymbol{b}}_j \otimes \hat{\boldsymbol{S}}_j^{\top} \mathcal{M}_{p,p} \left(\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^{\top} \boldsymbol{\varpi}_{2ijs} \right) \boldsymbol{x}_{it} \left(\hat{\varepsilon}_{it} - \varepsilon_{it} \right) \\
&+ \frac{1}{N^2T} \sum_{i,j=1}^{N} \sum_{t=1}^{T} \sum_{s=t+1}^{(t+m)\wedge T} \hat{w}_{ij} \tilde{\boldsymbol{b}}_j \otimes \hat{\boldsymbol{S}}_j^{\top} \mathcal{M}_{p,p} \left[\tilde{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ij}}^{\top} \left(\tilde{\boldsymbol{\varpi}}_{2ijs} - \boldsymbol{\varpi}_{2ijs} \right) \right] \boldsymbol{x}_{it} \hat{\varepsilon}_{it} \\
=:& \Delta_{3.4.1} + \cdots + \Delta_{3.4.6},
\end{aligned}$$

where

$$\begin{aligned}
\Delta_{3.4.1} =& O_p\left(\frac{m}{\sqrt{N}} \right), \quad \Delta_{3.4.2} = O_p\left(\frac{m}{\sqrt{N}} \right), \quad \Delta_{3.4.3} = O_p\left(\frac{m}{\sqrt{N}} \right), \\
\Delta_{3.4.4} =& O_p\left(\frac{m}{\sqrt{N}} \right), \quad \Delta_{3.4.5} = O_p\left(\frac{m}{\sqrt{N}} \right), \quad \Delta_{3.4.6} = O_p\left(\frac{m}{N^{3/8}} \right).
\end{aligned}$$

This establishes that $\Delta_3 = o_p(1)$.

(3.iii) Decompose the fifth part of the difference:

$$\begin{aligned}
\Delta_5 =& \boldsymbol{\mathcal{V}}^{\top} \cdot \sum_{d=0}^{D^{trun}} \mathbb{E}_{\mathcal{C}} \left(\boldsymbol{\Upsilon}_0^d \right) \cdot \frac{1}{N^2T} \sum_{i,k=1}^{N} \sum_{t=1}^{T} \\
& \left\{ \left(\sum_{s=t+1}^{(t+m)\wedge T} - \sum_{s=1}^{T} \right) \tilde{\boldsymbol{l}}_{k,0} \tilde{\boldsymbol{l}}_{k,0}^{\top} \otimes \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp} \boldsymbol{\Phi}_{4k,0}^{e} \cdot \tilde{\boldsymbol{l}}_{i,0} \otimes \mathbb{E}_{\mathcal{C}} \left[\mathcal{M}_{p,p} \left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top} \boldsymbol{\varpi}_{2iks} \right) \boldsymbol{x}_{it} \varepsilon_{it} \right] \right.
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}^{e}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\left\{\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}-\mathbb{E}_{\mathcal{C}}\left[\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\right\}\\
&+\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\left(\boldsymbol{\Phi}_{4k,0}-\boldsymbol{\Phi}_{4k,0}^{e}\right)\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\Bigg\}\\
&+\Bigg\{\boldsymbol{\nu}^{\top}\cdot\sum_{d=0}^{D^{trun}}\left[\boldsymbol{\Upsilon}_0^{d}-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{\Upsilon}_0^{d}\right)\right]+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{\Psi}_{1j}^{e}\left[\boldsymbol{X}_j-\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_j)\right]^{\top}\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^{d}\\
&+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\left(\boldsymbol{S}_{j,0}-\boldsymbol{S}_{j,0}^{e}\right)^{\top}\boldsymbol{X}_j^{\top}\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^{d}\Bigg\}\\
&\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\\
&+\Bigg[\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\tilde{\boldsymbol{b}}_j\hat{\boldsymbol{l}}_j^{\top}\otimes\hat{\boldsymbol{S}}_j^{\top}\boldsymbol{X}_j^{\top}\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\hat{\boldsymbol{l}}_k\hat{\boldsymbol{l}}_k^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1}\otimes\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\hat{\boldsymbol{\Phi}}_{4k}\cdot\hat{\boldsymbol{l}}_i\otimes\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ik}}^{\top}\hat{\boldsymbol{\varpi}}_{2iks}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\\
&-\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\cdot\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^{d}\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\\
&\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\Bigg]=:\Delta_{5.1}+\cdots+\Delta_{5.7}.
\end{aligned}
$$

Similar to the proof of $\Delta_{3.1}$, we can show that $\Delta_{5.1}=O\left(m^{-\gamma/2+1}\right)=o(1)$. Similar to the proof of $\Delta_{2.3.1}$ in Lemma S.2.12, we can show that $\Delta_{5.4}=O_p\left(m/\sqrt{N}\right)=o_p(1)$. It is straightforward to show that

$$
\Delta_{5.2}=O_p\left(\frac{m}{\sqrt{N}}\right),\quad \Delta_{5.3}=O_p\left(\frac{m}{\sqrt{T}}\right),\quad \Delta_{5.5}=O_p\left(\frac{m}{\sqrt{N}}\right),\quad \Delta_{5.6}=O_p\left(\frac{m}{\sqrt{T}}\right).
$$

Further decompose $\Delta_{5.7}$:

$$
\begin{aligned}
\Delta_{5.7}=&\Bigg\{\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\cdot\left[\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}-\sum_{d=0}^{D^{trun}}\boldsymbol{\Upsilon}_0^{d}\right]\\
&+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\left(\tilde{\boldsymbol{b}}_j\hat{\boldsymbol{l}}_j^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\otimes\hat{\boldsymbol{S}}_j^{\top}\boldsymbol{X}_j^{\top}-\boldsymbol{b}_{j,0}\tilde{\boldsymbol{l}}_{j,0}^{\top}\otimes\boldsymbol{S}_{j,0}^{\top}\boldsymbol{X}_j^{\top}\right)\cdot\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\Bigg\}\\
&\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\tilde{\boldsymbol{l}}_{k,0}\tilde{\boldsymbol{l}}_{k,0}^{\top}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\cdot\tilde{\boldsymbol{l}}_{i,0}\otimes\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\\
&+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\tilde{\boldsymbol{b}}_j\hat{\boldsymbol{l}}_j^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\otimes\hat{\boldsymbol{S}}_j^{\top}\boldsymbol{X}_j^{\top}\cdot\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\\
&\left[\hat{w}_{ik}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\hat{\boldsymbol{l}}_k\otimes\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\hat{\boldsymbol{\Phi}}_{4k}\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ik}}^{\top}\hat{\boldsymbol{\varpi}}_{2iks}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it}-w_{ik}\tilde{\boldsymbol{l}}_{k,0}\otimes\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{G})}^{\perp}\boldsymbol{\Phi}_{4k,0}\mathcal{M}_{p,p}\left(\boldsymbol{\mathcal{D}}_{\boldsymbol{\Psi}_{3ik}}^{e\top}\boldsymbol{\varpi}_{2iks}\right)\boldsymbol{x}_{it}\varepsilon_{it}\right]\\
&+\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\tilde{\boldsymbol{b}}_j\hat{\boldsymbol{l}}_j^{\top}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\otimes\hat{\boldsymbol{S}}_j^{\top}\boldsymbol{X}_j^{\top}\cdot\left[\left(\boldsymbol{I}_{TR_0}-\hat{\boldsymbol{\Upsilon}}\right)^{-1}-\left(\boldsymbol{I}_{TR_0}-\boldsymbol{\Upsilon}_0\right)^{-1}\right]\\
&\cdot\frac{1}{N^2T}\sum_{i,k=1}^{N}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\hat{w}_{ik}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\hat{\boldsymbol{l}}_k\otimes\boldsymbol{P}_{(\hat{\boldsymbol{F}},\boldsymbol{G})}^{\perp}\hat{\boldsymbol{\Phi}}_{4k}\mathcal{M}_{p,p}\left(\hat{\boldsymbol{\mathcal{D}}}_{\boldsymbol{\Psi}_{3ik}}^{\top}\hat{\boldsymbol{\varpi}}_{2iks}\right)\boldsymbol{x}_{it}\hat{\varepsilon}_{it}\\
=:&\Delta_{5.7.1}+\cdots+\Delta_{5.7.4},
\end{aligned}
$$

where $\hat{\boldsymbol{\Upsilon}} = N^{-1}\sum_{i=1}^{N}\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\hat{\boldsymbol{l}}_i\hat{\boldsymbol{l}}_i^\top\hat{\boldsymbol{\Sigma}}_{\boldsymbol{L}}^{-1/2}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{H}}_i\boldsymbol{P}_{\boldsymbol{G}}^{\perp}$, and

$$\Delta_{5.7.1} = O_p\left(ma_1^{D^{trun}}\right) = o_p(1), \quad \Delta_{5.7.2} = O_p\left(\frac{m}{\sqrt{N}}\right), \quad \Delta_{5.7.3} = O_p\left(\frac{m}{N^{3/8}}\right), \quad \Delta_{5.7.4} = O_p\left(\frac{m}{\sqrt{N}}\right).$$

This establishes that $\Delta_5 = o_p(1)$. □

Proof of Proposition 3.1. Let $\boldsymbol{D}_{\boldsymbol{Z}}^* = (NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_{it}^2\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^\top$. Since $\hat{\boldsymbol{A}}^\top\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{D}}_{\boldsymbol{Z}}\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{A}} = \tilde{\boldsymbol{A}}^\top\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{A}}$, we have

$$\begin{aligned}\hat{\boldsymbol{A}}^\top\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{D}}_{\boldsymbol{Z}}\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{A}} - \boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0 =& \left[\tilde{\boldsymbol{A}}^\top\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{A}} - \boldsymbol{A}_0^\top\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{A}_0\right]\\ &+\left[\boldsymbol{A}_0^\top\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{A}_0 - \boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0\right]\\ =&\tilde{\boldsymbol{A}}^\top\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{A}} - \boldsymbol{A}_0^\top\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{A}_0 + o_p(1).\end{aligned}$$

The last step follows from

$$\begin{aligned}\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{D}^+(\boldsymbol{F}_0) - \boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+ =& \boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}(\boldsymbol{F}_0)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{J}^\top\left[\boldsymbol{J}\cdot\boldsymbol{D}(\boldsymbol{F}_0)\cdot\boldsymbol{J}^\top\right]^{-1}\boldsymbol{J}\\ &-\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top\left(\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top\right)^{-1}\boldsymbol{J} = o_p(1),\end{aligned}$$

since $\boldsymbol{D}(\boldsymbol{F}_0) - \boldsymbol{D}_0 = o_p(1)$ and $\boldsymbol{D}_{\boldsymbol{Z}}^* - \boldsymbol{D}_{\boldsymbol{Z}} = o_p(1)$. Therefore, it suffices to show

$$\tilde{\boldsymbol{A}}^\top\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}\tilde{\boldsymbol{D}}_0^+\tilde{\boldsymbol{A}} - \boldsymbol{A}_0^\top\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{D}_{\boldsymbol{Z}}^*\boldsymbol{D}^+(\boldsymbol{F}_0)\boldsymbol{A}_0 = o_p(1).$$

It is straightforward to show $\tilde{\boldsymbol{A}} - \boldsymbol{A}_0 = o_p(1)$. It remains to show $\tilde{\boldsymbol{D}}_0 - \boldsymbol{D}(\boldsymbol{F}_0) = o_p(1)$ and $\tilde{\boldsymbol{D}}_{\boldsymbol{Z}} - \boldsymbol{D}_{\boldsymbol{Z}}^* = o_p(1)$.

(1) We begin with

$$\begin{aligned}\left\|\tilde{\boldsymbol{D}}_0 - \boldsymbol{D}(\boldsymbol{F}_0)\right\|_2 =& \frac{1}{NT}\left\|\sum_{i=1}^{N}\left(\tilde{\boldsymbol{Z}}_i^\top\tilde{\boldsymbol{Z}}_i - \boldsymbol{Z}_{i,0}^\top\boldsymbol{Z}_{i,0}\right)\right\|_2\\ \le& \frac{1}{NT}\sum_{i=1}^{N}\left\|\tilde{\boldsymbol{Z}}_i - \boldsymbol{Z}_{i,0}\right\|_2^2 + \frac{2}{NT}\sum_{i=1}^{N}\left\|\left(\tilde{\boldsymbol{Z}}_i - \boldsymbol{Z}_{i,0}\right)^\top\boldsymbol{Z}_{i,0}\right\|_2\\ =:&\Delta_{D_0.1} + \Delta_{D_0.2}.\end{aligned}$$

For $\Delta_{D_0.1}$, we have

$$\begin{aligned}\Delta_{D_0.1} \lesssim& \frac{1}{NT}\sum_{i=1}^{N}\left\|\tilde{\boldsymbol{b}}_i^\top\otimes\hat{\boldsymbol{Q}}_i - \boldsymbol{b}_{i,0}^\top\otimes\boldsymbol{Q}_{i,0}\right\|_2^2\\ &+\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{l}_{i,0}^\top\otimes\boldsymbol{H}_{i,0}^{\perp}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\otimes\boldsymbol{I}_T\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\boldsymbol{O}_{\boldsymbol{F}}\otimes\boldsymbol{I}_T - \boldsymbol{\Xi}_0^{-1}\right)\cdot\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}\right\|_2^2\\ &+\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{l}_{i,0}^\top\boldsymbol{O}_{\boldsymbol{F}}^\top\otimes\boldsymbol{H}_{i,0}^{\perp}\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\left(\hat{\boldsymbol{l}}_j\tilde{\boldsymbol{b}}_j^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j - \boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}\right)\right\|_2^2\\ &+\frac{1}{N}\sum_{i=1}^{N}\left\|\left(\hat{\boldsymbol{l}}_i^\top\otimes\hat{\boldsymbol{H}}_i^{\perp} - \boldsymbol{l}_{i,0}^\top\boldsymbol{O}_{\boldsymbol{F}}^\top\otimes\boldsymbol{H}_{i,0}^{\perp}\right)\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\hat{\boldsymbol{l}}_j\tilde{\boldsymbol{b}}_j^\top\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j\right\|_2^2\\ =:&\Delta_{\boldsymbol{Z}.1} + \Delta_{\boldsymbol{Z}.2} + \Delta_{\boldsymbol{Z}.3} + \Delta_{\boldsymbol{Z}.4}.\end{aligned}$$

For $\Delta_{\boldsymbol{Z}.1}$,

$$\begin{aligned}\Delta_{\boldsymbol{Z}.1} \lesssim & \frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{b}_{i,0}^{\top}\otimes\left(\hat{\boldsymbol{Q}}_i-\boldsymbol{Q}_{i,0}\right)\right\|_2^2+\frac{1}{NT}\sum_{i=1}^{N}\left\|\left(\tilde{\boldsymbol{b}}_i-\boldsymbol{b}_{i,0}\right)^{\top}\otimes\hat{\boldsymbol{Q}}_i\right\|_2^2\\ \leq & \frac{\bar{b}^2}{NT}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{Q}}_i-\boldsymbol{Q}_{i,0}\right\|_2^2+\frac{\left\|\tilde{\boldsymbol{B}}-\boldsymbol{B}_0\right\|_f^2}{N}\cdot\left(\frac{1}{T}\max_i\left\|\hat{\boldsymbol{Q}}_i\right\|_2^2\right)=O_p\left(\frac{1}{N}\right).\end{aligned}$$

For $\Delta_{\boldsymbol{Z}.2}$,

$$\Delta_{\boldsymbol{Z}.2}\leq\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{H}_{i,0}^{\perp}\right\|_2^2\cdot\left\|\boldsymbol{O}_{\boldsymbol{F}}^{\top}\otimes\boldsymbol{I}_T\cdot\hat{\boldsymbol{\Xi}}^{-1}\cdot\boldsymbol{O}_{\boldsymbol{F}}\otimes\boldsymbol{I}_T-\boldsymbol{\Xi}_0^{-1}\right\|_2^2\cdot\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}\right\|_2^2=O_p\left(\frac{1}{N}\right).$$

For $\Delta_{\boldsymbol{Z}.3}$, we have

$$\begin{aligned}\Delta_{\boldsymbol{Z}.3}\lesssim & \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{l}_{i,0}^{\top}\otimes\boldsymbol{H}_{i,0}^{\perp}\right\|_2^2\cdot\left\|\hat{\boldsymbol{\Xi}}^{-1}\right\|_2^2\cdot\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\left(\hat{\boldsymbol{l}}_j\tilde{\boldsymbol{b}}_j^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{Q}_{j,0}\right)\right\|_2^2\\ \lesssim & \left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\hat{\boldsymbol{Q}}_j-\boldsymbol{Q}_{j,0}\right)\right\|_2^2+\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\left(\tilde{\boldsymbol{b}}_j-\boldsymbol{b}_{j,0}\right)^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j\right\|_2^2\\ & +\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\left(\hat{\boldsymbol{l}}_j-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\right)\tilde{\boldsymbol{b}}_j^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j\right\|_2^2\\ =: & \Delta_{\boldsymbol{Z}.3.1}+\Delta_{\boldsymbol{Z}.3.2}+\Delta_{\boldsymbol{Z}.3.3},\end{aligned}$$

where

$$\begin{aligned}\Delta_{\boldsymbol{Z}.3.1}\leq & \frac{1}{N}\sum_{j=1}^{N}\left\|\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\boldsymbol{b}_{j,0}^{\top}\right\|_2^2\cdot\left(\frac{1}{NT}\sum_{j=1}^{N}\left\|\hat{\boldsymbol{Q}}_j-\boldsymbol{Q}_{j,0}\right\|_2^2\right)=O_p(1)\cdot O_p\left(\frac{1}{N}\right)=O_p\left(\frac{1}{N}\right),\\ \Delta_{\boldsymbol{Z}.3.2}\leq & \frac{1}{NT}\sum_{j=1}^{N}\left(\left\|\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\right\|_2^2\cdot\left\|\hat{\boldsymbol{Q}}_j\right\|_2^2\right)\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\tilde{\boldsymbol{b}}_j-\boldsymbol{b}_{j,0}\right\|_2^2\right)=O_p(1)\cdot O_p\left(\frac{1}{N}\right)=O_p\left(\frac{1}{N}\right),\\ \Delta_{\boldsymbol{Z}.3.3}\leq & \frac{1}{NT}\sum_{j=1}^{N}\left(\left\|\tilde{\boldsymbol{b}}_j\right\|_2^2\cdot\left\|\hat{\boldsymbol{Q}}_j\right\|_2^2\right)\cdot\left(\frac{1}{N}\sum_{j=1}^{N}\left\|\hat{\boldsymbol{l}}_j-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{j,0}\right\|_2^2\right)=O_p(1)\cdot O_p\left(\frac{1}{N}\right)=O_p\left(\frac{1}{N}\right).\end{aligned}$$

For $\Delta_{\boldsymbol{Z}.4}$, we have

$$\begin{aligned}\Delta_{\boldsymbol{Z}.4}\leq & \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{l}}_i^{\top}\otimes\hat{\boldsymbol{H}}_i^{\perp}-\boldsymbol{l}_{i,0}^{\top}\boldsymbol{O}_{\boldsymbol{F}}^{\top}\otimes\boldsymbol{H}_{i,0}^{\perp}\right\|_2^2\cdot\left\|\hat{\boldsymbol{\Xi}}^{-1}\right\|_2^2\cdot\left\|\frac{1}{N\sqrt{T}}\sum_{j=1}^{N}\hat{\boldsymbol{l}}_j\tilde{\boldsymbol{b}}_j^{\top}\otimes\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{Q}}_j\right\|_2^2\\ \lesssim & \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{l}_{i,0}^{\top}\boldsymbol{O}_{\boldsymbol{F}}^{\top}\otimes\left(\hat{\boldsymbol{H}}_i-\boldsymbol{H}_{i,0}\right)\right\|_2^2+\frac{1}{N}\sum_{i=1}^{N}\left\|\left(\hat{\boldsymbol{l}}_i-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right)^{\top}\otimes\hat{\boldsymbol{H}}_i^{\perp}\right\|_2^2\\ \lesssim & \max_i\left\|\boldsymbol{H}_{i,0}-\hat{\boldsymbol{H}}_i\right\|_2^2+\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{l}}_i-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right\|_2^2=O_p\left(\frac{1}{N}\right).\end{aligned}$$

Collecting all terms above, it follows that $\Delta_{D_0.1}=O_p\left(N^{-1}\right)$. For $\Delta_{D_0.2}$, we have

$$\Delta_{D_0.2}\lesssim\left(\frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{Z}_{i,0}\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\left\|\tilde{\boldsymbol{Z}}_i-\boldsymbol{Z}_{i,0}\right\|_2^2\right)^{1/2}=O_p(1)\cdot\Delta_{D_0.1}^{1/2}=O_p\left(\frac{1}{\sqrt{N}}\right).$$

To conclude, $\tilde{\boldsymbol{D}}_0-\boldsymbol{D}\left(\boldsymbol{F}_0\right)=O_p\left(N^{-1/2}\right)$.

(2) Decompose $\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}-\boldsymbol{D}_{\boldsymbol{Z}}^{*}$:

$$\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}-\boldsymbol{D}_{\boldsymbol{Z}}^{*}=\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left[\left(\varepsilon_{it}^2-\sigma_{it}^2\right)\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^{\top}+\varepsilon_{it}^2\left(\tilde{\boldsymbol{z}}_{it}\tilde{\boldsymbol{z}}_{it}^{\top}-\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^{\top}\right)+\left(\hat{\varepsilon}_{it}^2-\varepsilon_{it}^2\right)\tilde{\boldsymbol{z}}_{it}\tilde{\boldsymbol{z}}_{it}^{\top}\right]$$

$$=:\Delta_{\boldsymbol{D_Z}.1}+\Delta_{\boldsymbol{D_Z}.2}+\Delta_{\boldsymbol{D_Z}.3}.$$

We showed $\Delta_{\boldsymbol{D_Z}.1}=O_p\left(T^{-1/2}\right)$ in statement (3) of the proof for Lemma S.2.14. Decompose $\Delta_{\boldsymbol{D_Z}.2}$:

$$\begin{aligned}\Delta_{\boldsymbol{D_Z}.2}=&\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\left(\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right)\boldsymbol{z}_{it,0}^{\top}+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\boldsymbol{z}_{it,0}\left(\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right)^{\top}\\&+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^2\left(\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right)\left(\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right)^{\top}=:\Delta_{\boldsymbol{D_Z}.2.1}+\Delta_{\boldsymbol{D_Z}.2.2}+\Delta_{\boldsymbol{D_Z}.2.3}.\end{aligned}$$

For $\Delta_{\boldsymbol{D_Z}.2.1}$,

$$\begin{aligned}\left\|\Delta_{\boldsymbol{D_Z}.2.1}\right\|_2\leq&\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^4\cdot\left\|\boldsymbol{z}_{it,0}\right\|_2^2\right)^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right\|_2^2\right)^{1/2}\\\leq&\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it}^8\right)^{1/4}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{z}_{it,0}\right\|_2^4\right)^{1/4}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\left\|\tilde{\boldsymbol{Z}}_i-\boldsymbol{Z}_{i,0}\right\|_2^2\right)^{1/2}\\=&O_p(1)\cdot O_p(1)\cdot O_p\left(\frac{1}{\sqrt{N}}\right)=O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{z}_{it,0}\right\|_2^4\lesssim\frac{1}{T}\sum_{t=1}^{T}\left\|\boldsymbol{f}_{t,0}\right\|_2^4+\frac{1}{T}\sum_{t=1}^{T}\left\|\boldsymbol{g}_t\right\|_2^4+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{x}_{it}\right\|_2^4=O_p(1).$$

In the same way, we can show that $\left\|\Delta_{\boldsymbol{D_Z}.2.2}\right\|_2=O_p(N^{-1/2})$. For $\Delta_{\boldsymbol{D_Z}.2.3}$,

$$\begin{aligned}\left\|\Delta_{\boldsymbol{D_Z}.2.3}\right\|_2\leq&\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\tilde{\boldsymbol{z}}_{it}-\boldsymbol{z}_{it,0}\right\|_2^2\cdot\max_{i,t}\varepsilon_{it}^2\\=&O_p\left(\frac{1}{N}\right)\cdot O_p\left((NT)^{1/4}\right)=O_p\left(\frac{1}{\sqrt{N}}\right).\end{aligned}$$

To conclude, $\Delta_{\boldsymbol{D_Z}.2}=O_p(N^{-1/2})$. For $\Delta_{\boldsymbol{D_Z}.3}$,

$$\begin{aligned}\left\|\Delta_{\boldsymbol{D_Z}.3}\right\|_f=&\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}-\varepsilon_{it}\right)^2\right]^{1/2}\cdot\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{it}+\varepsilon_{it}\right)^4\right]^{1/4}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\tilde{\boldsymbol{z}}_{it}\right\|_2^8\right)^{1/4}\\=&O_p\left(\frac{1}{\sqrt{N}}\right)\cdot O_p(1)\cdot O_p(1)=O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\tilde{\boldsymbol{z}}_{it}\right\|_2^8\lesssim\frac{1}{T}\sum_{t=1}^{T}\left\|\hat{\boldsymbol{f}}_t\right\|_2^8+\frac{1}{T}\sum_{t=1}^{T}\left\|\boldsymbol{g}_t\right\|_2^8+\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left\|\boldsymbol{x}_{it}\right\|_2^8=O_p(1).$$

To conclude,

$$\tilde{\boldsymbol{D}}_{\boldsymbol{Z}}-\boldsymbol{D}_{\boldsymbol{Z}}^{*}=\Delta_{\boldsymbol{D_Z}.1}+\Delta_{\boldsymbol{D_Z}.2}+\Delta_{\boldsymbol{D_Z}.3}=O_p\left(\frac{1}{\sqrt{N}}\right).$$

□

S.4.3 Proofs of Lemmas in Section S.2.5

The estimators $\left(\hat{\boldsymbol{\Theta}}^{(K)}_{(K,R)},\ \hat{\boldsymbol{B}}^{(K)}_{(K,R)},\ \hat{\boldsymbol{F}}^{(R)}_{(K,R)},\ \hat{\boldsymbol{L}}^{(R)}_{(K,R)}\right)$ are

$$\begin{aligned}\left(\hat{\boldsymbol{\Theta}}^{(K)}_{(K,R)},\hat{\boldsymbol{B}}^{(K)}_{(K,R)},\hat{\boldsymbol{F}}^{(R)}_{(K,R)},\hat{\boldsymbol{L}}^{(R)}_{(K,R)}\right) &= \underset{(\boldsymbol{\Theta}^{(K)},\boldsymbol{B}^{(K)},\boldsymbol{F}^{(R)},\boldsymbol{L}^{(R)}):\boldsymbol{F}^{(R)}\perp\boldsymbol{G}}{\arg\min} f_{K,R}\left(\boldsymbol{\Theta}^{(K)},\boldsymbol{B}^{(K)},\boldsymbol{F}^{(R)},\boldsymbol{L}^{(R)}\right)\\ &= \underset{(\boldsymbol{\Theta}^{(K)},\boldsymbol{B}^{(K)},\boldsymbol{F}^{(R)},\boldsymbol{L}^{(R)}):\boldsymbol{F}^{(R)}\perp\boldsymbol{G}}{\arg\min} \frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{y}_i-\boldsymbol{X}_i\boldsymbol{\Theta}^{(K)}\boldsymbol{b}_i^{(K)}-\boldsymbol{F}^{(R)}\boldsymbol{l}_i^{(R)}\right\|_2^2.\end{aligned}$$

We keep the subscript above because $\left(\hat{\boldsymbol{\Theta}}^{(K)},\ \hat{\boldsymbol{B}}^{(K)}\right)$ are usually different for different pairs $(K,\ R_1)$ and $(K,\ R_2)$, and also $\left(\hat{\boldsymbol{F}}^{(R)},\ \hat{\boldsymbol{L}}^{(R)}\right)$ are usually different for different pairs $(K_1,\ R)$ and $(K_2,\ R)$. To simplify the notation, we omit the subscript $(K,\ R)$ hereafter.

Proof of Lemma S.2.17. For a pair $(K,\ R)$ where $K_0 \le K \le K_{max}$ and $R_0 \le R \le R_{max}$,

$$f^*_{K,R} \le f^*_{K_0,R_0} \le f_{K_0,R_0}\left(\boldsymbol{\Theta}_0,\boldsymbol{B}_0,\boldsymbol{F}_0,\boldsymbol{L}_0\right) = \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i.$$

On the other hand,

$$\begin{aligned}f^*_{K,R} \ge& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{y}_i-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\hat{\boldsymbol{b}}_i^{(K)}-\hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)}\right)^\top \boldsymbol{P}^\perp_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\left(\boldsymbol{y}_i-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\hat{\boldsymbol{b}}_i^{(K)}-\hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)}\right)\\ =&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i\\ =&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i-\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i,\end{aligned}$$

where $\boldsymbol{P}^\perp_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\boldsymbol{F}^{(R)})} = \boldsymbol{I}_T - \boldsymbol{P}_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}$, and $\boldsymbol{P}_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}$ is the projection matrix of the $T\times(p+R_0+R)$ matrix $\left(\boldsymbol{X}_i,\quad \boldsymbol{F}_0,\quad \boldsymbol{F}^{(R)}\right)$. It suffices to show that

$$\begin{aligned}&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{X}_i,\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i\\ \le&\sup_{\boldsymbol{F}^{(R_{max})}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{X}_i^h,\boldsymbol{X}_i^l,\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})})}\boldsymbol{\varepsilon}_i\\ \le&\sup_{\boldsymbol{F}^{(R_{max})}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{\varepsilon}_i\\ &+\sup_{\boldsymbol{F}^{(R_{max})}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{\varepsilon}_i\\ =&O_p\left(\frac{1}{m_{NT}^2}\right).\end{aligned}$$

To ease notation, let $R' = R_0 + R_{max} + p_2$. By Assumption 10(2), it follows that

$$\sup_{\boldsymbol{F}^{(R_{max})}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{\varepsilon}_i \le \sup_{\boldsymbol{F}^{(R')}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top P_{\boldsymbol{F}^{(R')}}\boldsymbol{\varepsilon}_i \le \frac{R'\|\boldsymbol{\mathcal{E}}\|_2^2}{NT} = O_p\left(\frac{1}{m_{NT}^2}\right).$$

By Assumption 10(3),

$$\sup_{\boldsymbol{F}^{(R_{max})}}\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})}\boldsymbol{\varepsilon}_i$$

$$
\begin{aligned}
&\lesssim \sup_{\boldsymbol{F}^{(R_{max})}} \frac{1}{NT^2} \sum_{i\in\mathcal{I}_{\boldsymbol{F}^{(R_{max})}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})} \boldsymbol{X}_i^h \boldsymbol{X}_i^{h\top} \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R_{max})},\boldsymbol{G})} \boldsymbol{\varepsilon}_i + \sup_{\boldsymbol{F}^{(R_{max})}} \frac{1}{NT} \sum_{i\in\mathcal{I}^c_{\boldsymbol{F}^{(R_{max})}}} \boldsymbol{\varepsilon}_i^\top \boldsymbol{\varepsilon}_i \\
&=: \Delta_1 + \Delta_2,
\end{aligned}
$$

where

$$
\begin{aligned}
\Delta_1 \lesssim & \frac{1}{NT^2} \sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top \boldsymbol{X}_i^h \boldsymbol{X}_i^{h\top} \boldsymbol{\varepsilon}_i + \sup_{\boldsymbol{F}^{(R')}} \frac{1}{NT} \sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top \frac{\boldsymbol{F}^{(R')}}{\sqrt{T}} \frac{\boldsymbol{F}^{(R')\top} \boldsymbol{X}_i^h}{T} \frac{\boldsymbol{X}_i^{h\top} \boldsymbol{F}^{(R')}}{T} \frac{\boldsymbol{F}^{(R')\top}}{\sqrt{T}} \boldsymbol{\varepsilon}_i \\
\lesssim & \frac{1}{T} \cdot \frac{1}{N} \sum_{i=1}^N \frac{\boldsymbol{\varepsilon}_i^\top \boldsymbol{X}_i^h \boldsymbol{X}_i^{h\top} \boldsymbol{\varepsilon}_i}{T} + \frac{R' \|\boldsymbol{\mathcal{E}}\|_2^2}{NT} \\
= & O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{m_{NT}^2}\right) = O_p\left(\frac{1}{m_{NT}^2}\right),
\end{aligned}
$$

and

$$
\Delta_2 \le \frac{\phi \|\boldsymbol{\mathcal{E}}\|_2^2}{NT} = O_p\left(\frac{1}{m_{NT}^2}\right).
$$

□

Proof of Lemma S.2.18. For a pair (K, R) where $K < K_0$,

$$
\begin{aligned}
f^*_{K,R} \ge & \frac{1}{NT} \sum_{i=1}^N \left(\boldsymbol{y}_i - \boldsymbol{X}_i \hat{\boldsymbol{\Theta}}^{(K)} \hat{\boldsymbol{b}}_i^{(K)} - \hat{\boldsymbol{F}}^{(R)} \hat{\boldsymbol{l}}_i^{(R)}\right)^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \left(\boldsymbol{y}_i - \boldsymbol{X}_i \hat{\boldsymbol{\Theta}}^{(K)} \hat{\boldsymbol{b}}_i^{(K)} - \hat{\boldsymbol{F}}^{(R)} \hat{\boldsymbol{l}}_i^{(R)}\right) \\
= & \frac{1}{NT} \sum_{i=1}^N \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \hat{\boldsymbol{b}}_i^{(K)}\right) + \boldsymbol{\varepsilon}_i\right]^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \hat{\boldsymbol{b}}_i^{(K)}\right) + \boldsymbol{\varepsilon}_i\right] \\
\ge & \frac{1}{NT} \sum_{i=1}^N \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) + \boldsymbol{\varepsilon}_i\right]^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) + \boldsymbol{\varepsilon}_i\right] \\
= & \frac{1}{NT} \sum_{i=1}^N \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right)^\top \boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) \\
& + \frac{1}{NT} \sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \boldsymbol{\varepsilon}_i + \frac{2}{NT} \sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) \\
=: & \Delta_1 + \Delta_2 + \Delta_3.
\end{aligned}
$$

In the third step, $\tilde{\boldsymbol{B}}^{(K)}$ is obtained by solving

$$
\tilde{\boldsymbol{B}}^{(K)} = \arg\min_{\boldsymbol{B}^{(K)}} \frac{1}{NT} \sum_{i=1}^N \left\| \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \boldsymbol{b}_i^{(K)}\right) + \boldsymbol{\varepsilon}_i\right] \right\|_2^2.
$$

(1) For Δ_1, we have

$$
\begin{aligned}
\Delta_1 \gtrsim & \frac{1}{N} \sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}} \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right)^\top \frac{\boldsymbol{X}_i^\top \boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})} \boldsymbol{X}_i}{T} \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) \\
\gtrsim & \frac{1}{N} \sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}} \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right)^\top \left(\boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)} \tilde{\boldsymbol{b}}_i^{(K)}\right) \\
\ge & \operatorname{tr}\left[\boldsymbol{\Theta}_0^\top \left(\boldsymbol{I}_p - \boldsymbol{P}_{\hat{\boldsymbol{\Theta}}^{(K)}}\right) \boldsymbol{\Theta}_0 \cdot \frac{1}{N} \sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}} \boldsymbol{b}_{i,0} \boldsymbol{b}_{i,0}^\top\right] \ge \tau_K.
\end{aligned}
$$

In the third step, $\boldsymbol{b}_i^{(K)} = \left(\hat{\boldsymbol{\Theta}}^{(K)\top}\hat{\boldsymbol{\Theta}}^{(K)}\right)^{-}\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}$ minimizes $N^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\left\|\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} - \hat{\boldsymbol{\Theta}}^{(K)}\boldsymbol{b}_i^{(K)}\right\|_2^2$. In the last step, $N^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^\top$ is positive definite by Assumption 2(3), and $\boldsymbol{\Theta}_0^\top\left(\boldsymbol{I}_p - \boldsymbol{P}_{\hat{\boldsymbol{\Theta}}^{(K)}}\right)\boldsymbol{\Theta}_0$ is positive semi-definite for $K < K_0$.

(2) For Δ_2,

$$\left|\Delta_2 - \frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\right| = \frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i \le \frac{(R_0+R)\|\boldsymbol{\mathcal{E}}\|_2^2}{NT} = O_p\left(\frac{1}{m_{NT}^2}\right).$$

(3) Since

$$\tilde{\boldsymbol{b}}_i^{(K)} = \left(\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\right)^{+}\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\left(\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} + \boldsymbol{\varepsilon}_i\right),$$

decompose Δ_3:

$$\begin{aligned}
\Delta_3 =& \frac{2}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} \\
&- \frac{2}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\left(\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\right)^{+}\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i \\
&- \frac{2}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\left(\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\right)^{+}\hat{\boldsymbol{\Theta}}^{(K)\top}\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0} \\
=:& \Delta_{3.1} + \Delta_{3.2} + \Delta_{3.3}.
\end{aligned}$$

For $\Delta_{3.1}$, we have

$$\Delta_{3.1} \lesssim \left|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right| + \sup_{\boldsymbol{F}^{(R)}}\left|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right| = O_p\left(\frac{1}{m_{NT}}\right),$$

where

$$\left|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right| \lesssim \frac{1}{NT}\sum_{i=1}^N \left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2 = O_p\left(\frac{1}{\sqrt{T}}\right),$$

and

$$\begin{aligned}
\sup_{\boldsymbol{F}^{(R)}}\left|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right| \le& \sup_{\boldsymbol{F}^{(R)}}\frac{1}{NT}\sum_{i=1}^N\left(\left\|\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}\boldsymbol{\varepsilon}_i\right\|_2\cdot\left\|\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right\|_2\right) \\
\le& \left(\sup_{\boldsymbol{F}^{(R)}}\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R)})}\boldsymbol{\varepsilon}_i\right)^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^N\left\|\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\right\|_2^2\right)^{1/2} \\
\le& O_p\left(\frac{1}{m_{NT}}\right)\cdot O_p(1) = O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}$$

For $\Delta_{3.2}$, we have

$$\begin{aligned}
\Delta_{3.2} \lesssim& \frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}} \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\boldsymbol{X}_i^\top\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i + \frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i \\
\lesssim& \frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}} \boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{X}_i\boldsymbol{X}_i^\top\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i + \frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{X}_i\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i + \frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i
\end{aligned}$$

$$
\begin{aligned}
&\lesssim \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top \boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i + \frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^2 + \frac{\phi\|\boldsymbol{\mathcal{E}}\|_2^2}{NT}\\
&= O_p\left(\frac{1}{m_{NT}^2}\right) + O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{m_{NT}^2}\right) = O_p\left(\frac{1}{m_{NT}^2}\right).
\end{aligned}
$$

For $\Delta_{3.3}$, we have

$$
\Delta_{3.3} \lesssim |\Delta_{3.2}|^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\|\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}\|_2^2\right)^{1/2} = O_p\left(\frac{1}{m_{NT}}\right)\cdot O_p(1) = O_p\left(\frac{1}{m_{NT}}\right),
$$

where the first step follows from the inequality $\sum_{i=1}^{N}\boldsymbol{a}_i^\top\boldsymbol{b}_i \le \left(\sum_{i=1}^{N}\|\boldsymbol{a}_i\|_2^2\right)^{1/2}\cdot\left(\sum_{i=1}^{N}\|\boldsymbol{b}_i\|_2^2\right)^{1/2}$. Collecting the terms above leads to the result. □

Proof of Lemma S.2.19. For a pair (K,R) where $R<R_0$, we have

$$
\begin{aligned}
f_{K,R}^{\star} \ge& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\hat{\boldsymbol{b}}_i^{(K)} - \hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)}\right)^\top \boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}^{(K)}\hat{\boldsymbol{b}}_i^{(K)} - \hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)}\right)\\
=& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)} + \boldsymbol{\varepsilon}_i\right)^\top \boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\hat{\boldsymbol{l}}_i^{(R)} + \boldsymbol{\varepsilon}_i\right)\\
\ge& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)} + \boldsymbol{\varepsilon}_i\right)^\top \boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)} + \boldsymbol{\varepsilon}_i\right)\\
=& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)^\top \boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right) + \frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\boldsymbol{\varepsilon}_i\\
&+ \frac{2}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)\\
=:& \Delta_1 + \Delta_2 + \Delta_3.
\end{aligned}
$$

In the third step, $\tilde{\boldsymbol{L}}^{(R)}$ is obtained by solving

$$
\tilde{\boldsymbol{L}}^{(R)} = \arg\min_{\boldsymbol{L}^{(R)}} \frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{P}_{\boldsymbol{X}_i}^{\perp}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\boldsymbol{l}_i^{(R)} + \boldsymbol{\varepsilon}_i\right)\right\|_2^2.
$$

(1) For Δ_1, we have

$$
\begin{aligned}
\Delta_1 =& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)^\top\left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp} - \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right]\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)\\
=& \frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)^\top\left[\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\right]\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)\\
\gtrsim& \frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)^\top\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)\\
\ge& \operatorname{tr}\left[\frac{\boldsymbol{F}_0^\top(\boldsymbol{I}_T - \boldsymbol{P}_{\hat{\boldsymbol{F}}^{(R)}})\boldsymbol{F}_0}{T}\cdot\frac{1}{N}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top\right] \ge \tau_R.
\end{aligned}
$$

The first step follows from $\boldsymbol{P}_{\boldsymbol{X}_i} = \boldsymbol{P}_{(\boldsymbol{X}_i^h,\boldsymbol{G})} = \boldsymbol{P}_{\boldsymbol{G}} + \boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}$. In the fourth step, $\boldsymbol{l}_i^{(R)} = \left(\hat{\boldsymbol{F}}^{(R)\top}\hat{\boldsymbol{F}}^{(R)}\right)^+\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{F}_0\boldsymbol{l}_{i,0}$ minimizes $(NT)^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\left\|\boldsymbol{F}_0\boldsymbol{l}_{i,0} - \hat{\boldsymbol{F}}^{(R)}\boldsymbol{l}_i^{(R)}\right\|_2^2$. In the last step, $N^{-1}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top$

is positive definite by Assumption 2(2), and $T^{-1}\boldsymbol{F}_0^\top\left(\boldsymbol{I}_T-\boldsymbol{P}_{\hat{\boldsymbol{F}}^{(R)}}\right)\boldsymbol{F}_0$ is positive semi-definite for $R<R_0$ by Assumption 2(1). The third step is verified by the following detailed analysis. Let

$$\boldsymbol{e}_i=\frac{\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}}{\|\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\|_2},\quad i=1,\ldots,N.$$

For $i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}$, we have

$$\begin{aligned}
&\boldsymbol{e}_i^\top\left[\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right]\boldsymbol{e}_i\\
=&1-\lambda_{max}\left[\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)^{-1/2}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}_{\boldsymbol{e}_i}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)^{-1/2}\right]\\
=&\lambda_{min}\left[\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h}{T}\right)^{-1/2}\cdot\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{e}_i})\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h}{T}\cdot\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h}{T}\right)^{-1/2}\right]\\
\geq&\lambda_{min}\left[\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h}{T}\right)^{-1}\right]\cdot\lambda_{min}\left(\frac{\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp(\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{e}_i})\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h}{T}\right)\\
\geq&\frac{\lambda_{min}\left(T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{X}_i^h\right)}{\lambda_{max}\left(T^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)}\\
\geq&\frac{C_7}{C_5},
\end{aligned}$$

where the third step is due to $\lambda_{min}\left(\boldsymbol{A}_1^{1/2}\boldsymbol{A}_2\boldsymbol{A}_1^{1/2}\right)\geq\lambda_{min}(\boldsymbol{A}_1)\cdot\lambda_{min}(\boldsymbol{A}_2)$, the fourth step is due to $\boldsymbol{P}_{\boldsymbol{e}_i}\leq\boldsymbol{P}_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}$ and $\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{P}_{\boldsymbol{G}}^\perp=\boldsymbol{P}^\perp_{(\boldsymbol{F}_0,\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}$, and the last inequality is due to Assumptions 5(1) and 10(2). Therefore:

$$\begin{aligned}
&\frac{1}{NT}\sum_{i=1}^N\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)^\top\left[\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)^+\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right]\left(\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right)\\
\geq&\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\left\{\boldsymbol{e}_i^\top\left[\boldsymbol{I}_T-\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\boldsymbol{X}_i^h\right)^{-1}\boldsymbol{X}_i^{h\top}\boldsymbol{P}_{\boldsymbol{G}}^\perp\right]\boldsymbol{e}_i\cdot\left\|\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right\|_2^2\right\}\\
\geq&\frac{C_7}{C_5}\cdot\frac{1}{NT}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\left\|\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\hat{\boldsymbol{F}}^{(R)}\tilde{\boldsymbol{l}}_i^{(R)}\right\|_2^2.
\end{aligned}$$

(2) For Δ_2, it is straightforward to show that

$$\left|\Delta_2-\frac{1}{NT}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\right|=\frac{1}{NT}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}\boldsymbol{\varepsilon}_i=O_p\left(\frac{1}{T}\right).$$

(3) Since

$$\tilde{\boldsymbol{l}}_i^{(R)}=\left(\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\hat{\boldsymbol{F}}^{(R)}\right)^+\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp(\boldsymbol{F}_0\boldsymbol{l}_{i,0}+\boldsymbol{\varepsilon}_i),$$

decompose Δ_3:

$$\begin{aligned}
\Delta_3=&\frac{2}{NT}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\frac{2}{NT}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\hat{\boldsymbol{F}}^{(R)}\left(\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\hat{\boldsymbol{F}}^{(R)}\right)^+\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\boldsymbol{\varepsilon}_i\\
&-\frac{2}{NT}\sum_{i=1}^N\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\hat{\boldsymbol{F}}^{(R)}\left(\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\hat{\boldsymbol{F}}^{(R)}\right)^+\hat{\boldsymbol{F}}^{(R)\top}\boldsymbol{P}_{\boldsymbol{X}_i}^\perp\boldsymbol{F}_0\boldsymbol{l}_{i,0}
\end{aligned}$$

$$=:\Delta_{3.1}+\Delta_{3.2}+\Delta_{3.3}.$$

For $\Delta_{3.1}$, we have

$$\Delta_{3.1}\lesssim\left|\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{F}_0\boldsymbol{l}_{i,0}\right|+\left|\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{\boldsymbol{X}_i}\boldsymbol{F}_0\boldsymbol{l}_{i,0}\right|=o_p(1).$$

For $\Delta_{3.2}$, we have

$$\begin{aligned}\Delta_{3.2}\lesssim&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\boldsymbol{X}_i,\hat{\boldsymbol{F}}^{(R)})}\boldsymbol{\varepsilon}_i\\ \leq&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{\varepsilon}_i+\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{X}_i^h\left(\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{X}_i^h\right)^{+}\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{\varepsilon}_i\\ \lesssim&\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{\varepsilon}_i+\frac{1}{NT^2}\sum_{i\in\mathcal{I}_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{P}^{\perp}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{X}_i^h\boldsymbol{X}_i^{h\top}\boldsymbol{P}^{\perp}_{(\hat{\boldsymbol{F}}^{(R)},\boldsymbol{G})}\boldsymbol{\varepsilon}_i+\frac{1}{NT}\sum_{i\in\mathcal{I}^c_{\hat{\boldsymbol{F}}^{(R)}}}\boldsymbol{\varepsilon}_i^\top\boldsymbol{\varepsilon}_i\\ =&O_p\left(\frac{1}{m_{NT}^2}\right).\end{aligned}$$

For $\Delta_{3.3}$, we have

$$\Delta_{3.3}\lesssim|\Delta_{3.2}|^{1/2}\cdot\left(\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{l}_{i,0}^\top\boldsymbol{F}_0^\top\boldsymbol{F}_0\boldsymbol{l}_{i,0}\right)^{1/2}=O_p\left(\frac{1}{m_{NT}}\right)\cdot O_p(1)=O_p\left(\frac{1}{m_{NT}}\right).$$

Collecting the terms above leads to the result. □

S.4.4 Proofs of Auxiliary Lemmas in Section S.4

Proof of Lemma S.4.1.

(1) We have

$$\begin{aligned}\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f^2=&2R_0-\operatorname{tr}\left(\boldsymbol{P}_{\boldsymbol{F}_0}\boldsymbol{P}_{\hat{\boldsymbol{F}}}\right)-\operatorname{tr}\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\right)\\ =&2\operatorname{tr}\left(\boldsymbol{P}_{\boldsymbol{F}_0}\right)-2\operatorname{tr}\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\right)\\ =&2\operatorname{tr}\left(\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\right)\\ =&2\operatorname{tr}\left(\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\right).\end{aligned}$$

Since

$$\operatorname{tr}\left(\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\right)=\operatorname{tr}\left[\frac{\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{F}_0}{\sqrt{T}}\cdot\left(\frac{\boldsymbol{F}_0^\top\boldsymbol{F}_0}{T}\right)^{-1}\cdot\frac{\boldsymbol{F}_0^\top\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}}{\sqrt{T}}\right]\leq\lambda_{max}\left(\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right)\cdot\frac{\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{F}_0\|_f^2}{T},$$

and

$$\operatorname{tr}\left(\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{P}_{\boldsymbol{F}_0}\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\right)\geq\lambda_{min}\left(\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right)\cdot\frac{\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{F}_0\|_f^2}{T},$$

it follows that

$$\frac{\sqrt{2}}{\lambda_{max}^{1/2}(\boldsymbol{\Sigma}_{\boldsymbol{F}})}\cdot\frac{\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{F}_0\|_f}{\sqrt{T}}\leq\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f\leq\frac{\sqrt{2}}{\lambda_{min}^{1/2}(\boldsymbol{\Sigma}_{\boldsymbol{F}})}\cdot\frac{\|\boldsymbol{P}^{\perp}_{\hat{\boldsymbol{F}}}\boldsymbol{F}_0\|_f}{\sqrt{T}}.$$

(2a) The following two inequalities are used in the proof:

$$\left\|\boldsymbol{A}_1^{-1}\right\|_2 \leq \frac{\left\|\boldsymbol{A}_2^{-1}\right\|_2}{1-\|\boldsymbol{A}_1-\boldsymbol{A}_2\|_2\cdot\|\boldsymbol{A}_2^{-1}\|_2} \tag{S.4.15}$$

for matrix $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ satisfying $\|\boldsymbol{A}_1-\boldsymbol{A}_2\|_2\cdot\|\boldsymbol{A}_2^{-1}\|_2<1$, and

$$\left\|\boldsymbol{A}_1^{-1}-\boldsymbol{A}_2^{-1}\right\|_2 \leq \left\|\boldsymbol{A}_1^{-1}\right\|_2\cdot\|\boldsymbol{A}_1-\boldsymbol{A}_2\|_2\cdot\left\|\boldsymbol{A}_2^{-1}\right\|_2. \tag{S.4.16}$$

By the inequality (S.4.15), we obtain

$$\begin{aligned}
&\left\|\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\right\|_2\\
\leq&\frac{\left\|\left(T^{-1}\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}\right)^{-1}\right\|_2}{1-\left\|T^{-1}\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp-\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\right)\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}\right\|_2\cdot\left\|\left(T^{-1}\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}\right)^{-1}\right\|_2}\\
\leq&\frac{\lambda_{min}^{-1}\left(\tilde{\boldsymbol{\Theta}}^\top\tilde{\boldsymbol{\Theta}}\right)\cdot\lambda_{min}^{-1}\left(T^{-1}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\right)}{1-\lambda_{max}\left(\tilde{\boldsymbol{\Theta}}^\top\tilde{\boldsymbol{\Theta}}\right)\cdot\lambda_{min}^{-1}\left(\tilde{\boldsymbol{\Theta}}^\top\tilde{\boldsymbol{\Theta}}\right)\cdot\lambda_{max}\left(T^{-1}\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\cdot\lambda_{min}^{-1}\left(T^{-1}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\right)\cdot\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_2}.
\end{aligned}$$

Since $\left\|\tilde{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0\right\|_2=o_p(1)$, $\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_2=o_p(1)$, $\lambda_{max}\left(T^{-1}\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)$ and $\lambda_{min}\left(T^{-1}\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\right)$ are uniformly upper bounded by Assumption 5(1),

$$\max_i\left\|\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\right\|\leq c_1$$

for some $c_1>0$. Since

$$\max_i\left\|\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}-\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right\|_2\leq c_2\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2+\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f\right),$$

by the inequality (S.4.16), we obtain

$$\begin{aligned}
&\max_i\left\|\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}-\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right\|_2\\
\leq&\max_i\left\|\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}-\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right\|_2\cdot\max_i\left\|\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\right\|_2\cdot\max_i\left\|\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right\|_2\\
\leq&c_3\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2+\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f\right).
\end{aligned}$$

The result follows from the decomposition

$$\begin{aligned}
\hat{\boldsymbol{W}}_i-\boldsymbol{W}_{i,0}=&\frac{\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{\sqrt{T}}\cdot\left[\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}-\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\right]\cdot\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top}{\sqrt{T}}\\
&+\frac{\boldsymbol{X}_i(\tilde{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0)}{\sqrt{T}}\cdot\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\cdot\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top}{\sqrt{T}}\\
&+\frac{\boldsymbol{X}_i\boldsymbol{\Theta}_0}{\sqrt{T}}\cdot\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\cdot\frac{(\tilde{\boldsymbol{\Theta}}-\boldsymbol{\Theta}_0)^\top\boldsymbol{X}_i^\top}{\sqrt{T}}.
\end{aligned}$$

(2b)–(2f) A similar argument for (2a) yields the desired result.

(2g) Using the same approach, we can first show that

$$\left\|\boldsymbol{\Xi}\left(\tilde{\boldsymbol{\Theta}},\hat{\boldsymbol{F}}\right)-\boldsymbol{\Xi}_0\right\|_2=O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2\right)+O_p\left(\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f\right).$$

A consecutive use of (S.4.15) and (S.4.16) leads to the desired result.

(3a) Since

$$\mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_i-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\right\|_2^4\right)=\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_i-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\right\|_2^4\right)\right],$$

it suffices to show that $\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_i-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\right\|_2^4\right)$ is of order $O(T^2)$. By a conditional version of Theorem 2.2 in Rio et al. (2017), we have

$$\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_i-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top\boldsymbol{X}_i\right)\right\|_2^4\right)\lesssim\left[\sum_{t=1}^{T}M_{2,\alpha,T}(Q_{1it})\right]^2+\sum_{t=1}^{T}M_{4,\alpha,T}(Q_{1it}),$$

where $M_{q,\alpha,T}(Q_{1it})=\int_0^1[\alpha^{-1}(u)\wedge T]^{q-1}Q_{1it}^q(u)du$, and $Q_{1it}(u)$ denotes the conditional u-th quantile of $\|\boldsymbol{x}_{it}\|_2^2$ given $\mathcal{C}$. To ensure that the above bound is $O(T^2)$, by (C.16) in Rio et al. (2017), it suffices to assume

$$\mathbb{P}\left(\|\boldsymbol{x}_{it}\|_2^2>a\middle|\mathcal{C}\right)\le\left(\frac{c_1}{a}\right)^r,\quad\alpha_i=O\left(i^{-2r/(r-4)}\right).$$

Let $r=4+\epsilon/2$. The first condition follows from Markov's inequality together with the existence of a finite $2r$-th moment of $\|\boldsymbol{x}_{it}\|_2$, while the second condition is ensured by Assumption 6(5).

(3b) Similarly,

$$\begin{aligned}&\mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{F}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)^\top\boldsymbol{F}_0\right\|_2^4\right)\\=&\mathbb{E}\left[\mathbb{E}_{\mathcal{C}}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{F}_0-\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i\right)^\top\boldsymbol{F}_0\right\|_2^4\right)\right]\\\lesssim&\mathbb{E}\left\{\left[\sum_{t=1}^{T}\left(M_{2,\alpha,T}(Q_{2it})\cdot\|\boldsymbol{f}_{t,0}\|_2^2\right)\right]^2\right\}+\sum_{t=1}^{T}\mathbb{E}\left(M_{4,\alpha,T}(Q_{2it})\cdot\|\boldsymbol{f}_{t,0}\|_2^4\right),\end{aligned}$$

where $Q_{2it}(u)$ denotes the conditional u-th quantile of $\|\boldsymbol{x}_{it}\|_2$ given $\mathcal{C}$. To ensure that $M_{2,\alpha,T}(Q_{2it})\le c_1$ and $M_{4,\alpha,T}(Q_{2it})\le c_2T$, it suffices to assume

$$\mathbb{P}\left(\|\boldsymbol{x}_{it}\|_2>a\middle|\mathcal{C}\right)\le\left(\frac{c_1}{a}\right)^r,\quad\alpha_i=O\left(i^{-2r/(r-4)}\right),$$

which holds by taking $r=8+\epsilon$.

(3c) The result follows by the same argument as in (3b).

(3d) Note that $\mathbb{E}_{\mathcal{C}\cup j}(\boldsymbol{X}_i)=\mathbb{E}_{\mathcal{C}}(\boldsymbol{X}_i)$ for $i\ne j$, since cross-sectional units i and j are independent conditional on $\mathcal{C}$, where $\mathcal{C}\cup j$ denotes the σ-field generated by $\mathcal{C}$ and $\boldsymbol{X}_j$. Applying the same moment inequality to

$$\mathbb{E}_{\mathcal{C}\cup j}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{X}_j-\mathbb{E}_{\mathcal{C}\cup j}(\boldsymbol{X}_i)^\top\boldsymbol{X}_j\right\|_2^4\right)$$

yields the desired bound. The remaining steps are identical to those in (3b).

(3e) Since $\mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}(\boldsymbol{x}_{it}\varepsilon_{it}) = \boldsymbol{x}_{it}\mathbb{E}_{\mathcal{C}\vee\mathcal{F}_t}(\varepsilon_{it}) = \mathbf{0}_p$, $\{\sum_{s=1}^t \boldsymbol{x}_{is}\varepsilon_{is}\}_{t\geq 1}$ is a discrete-time martingale with respect to the filtration $\{\mathcal{C}\vee\mathcal{F}_t\}_{t\geq 1}$. By BDG inequality (Davis and Song, 2011),

$$\begin{aligned}\mathbb{E}\left(\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^4\right) \lesssim& \sum_{j=1}^p \mathbb{E}\left[\left(\sum_{t=1}^T x_{it,j}^2\varepsilon_{it}^2\right)^2\right]\\ \lesssim& T\sum_{j=1}^p \mathbb{E}\left(\sum_{t=1}^T x_{it,j}^4\varepsilon_{it}^4\right)\\ \lesssim& T\sum_{j=1}^p\sum_{t=1}^T\left[\mathbb{E}\left(x_{it,j}^8\right)\cdot\mathbb{E}\left(\varepsilon_{it}^8\right)\right]^{1/2} \leq c_1T^2.\end{aligned}$$

(3f) By BDG inequality,

$$\begin{aligned}\mathbb{E}\left(\left\|\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\right\|_2^4\right) \leq& \sum_{r=1}^{R_0} \mathbb{E}\left[\left(\sum_{t=1}^T f_{tr,0}^2\varepsilon_{it}^2\right)^2\right]\\ \leq& \sum_{r=1}^{R_0} \mathbb{E}\left(\sum_{s,t=1}^T f_{sr,0}^2 f_{tr,0}^2\varepsilon_{is}^2\varepsilon_{it}^2\right)\\ \lesssim& \sum_{r=1}^{R_0} \mathbb{E}\left[\sum_{s,t=1}^T f_{sr,0}^2 f_{tr,0}^2\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}^2\varepsilon_{it}^2\right)\right]\\ \lesssim& \sum_{r=1}^{R_0} \mathbb{E}\left(\sum_{s,t=1}^T f_{sr,0}^2 f_{tr,0}^2\right) \leq c_1T^2,\end{aligned}$$

where the fourth step follows from $\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}^2\varepsilon_{it}^2\right) \leq \left[\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{is}^4\right)\cdot\mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^4\right)\right]^{1/2}$, and the last step follows from $\mathbb{E}\left(f_{sr,0}^2 f_{tr,0}^2\right) \leq \left[\mathbb{E}\left(f_{sr,0}^4\right)\cdot\mathbb{E}\left(f_{tr,0}^4\right)\right]^{1/2}$. □

Proof of Lemma S.4.2. The proof relies on Lemma S.2.4.

(1) The result follows from

$$\begin{aligned}&\frac{1}{N}\sum_{i=1}^N\left\|\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{T}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1} - \frac{\boldsymbol{X}_i^\top\boldsymbol{F}_0}{T}\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right\|_2^2\\ \leq& \frac{2}{N}\sum_{i=1}^N\frac{\|\boldsymbol{X}_i^\top(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)\|_2^2}{T^2}\cdot\left\|\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\right\|_2^2 + \frac{2}{N}\sum_{i=1}^N\frac{\|\boldsymbol{X}_i^\top\boldsymbol{F}_0\|_2^2}{T^2}\cdot\left\|(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}})^{-1}-\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right\|_2^2\\ =& O_p\left(\left\|\tilde{\boldsymbol{\vartheta}}-\boldsymbol{\vartheta}_0\right\|_2^2\right) + O_p\left(\frac{1}{m_{NT}^4}\right),\end{aligned}$$

where the last step is due to Lemma S.2.4 (1) and (4).

(2) The result follows from

$$\begin{aligned}&\frac{1}{NT^2}\sum_{i=1}^N\left\|\boldsymbol{X}_i^\top\left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}-\boldsymbol{P}_{\boldsymbol{F}_0}\right)\boldsymbol{X}_i\right\|_2^2\\ \lesssim& \frac{1}{N}\sum_{i=1}^N\left\|\left[\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{T}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1} - \frac{\boldsymbol{X}_i^\top\boldsymbol{F}_0}{T}\cdot\boldsymbol{\Sigma}_{\boldsymbol{F}}^{-1}\right]\cdot\frac{\boldsymbol{F}_0^\top\boldsymbol{X}_i}{T}\right\|_2^2\\ &+\frac{1}{N}\sum_{i=1}^N\left\|\frac{\boldsymbol{X}_i^\top\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}}{T}\cdot\left(\boldsymbol{O}_{\boldsymbol{F}}^\top\boldsymbol{O}_{\boldsymbol{F}}\right)^{-1}\cdot\frac{(\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0)^\top\boldsymbol{X}_i}{T}\right\|_2^2,\end{aligned}$$

and the consistency rate established in (1).

(3) The result follows from

$$
\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{S}_i\left(\hat{\boldsymbol{F}}\right)-\boldsymbol{S}_{i,0}\right\|_2^2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left\|\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i-\left(\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0\right)^{-1}\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\right\|_2^2\\
\lesssim&\frac{2}{N}\sum_{i=1}^{N}\left\|\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\cdot\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top(\boldsymbol{P}_{\boldsymbol{F}_0}-\boldsymbol{P}_{\hat{\boldsymbol{F}}})\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\cdot\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\cdot\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i}{T}\right\|_2^2\\
&+\frac{2}{N}\sum_{i=1}^{N}\left\|\left(\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\boldsymbol{F}_0}^\perp\boldsymbol{X}_i\boldsymbol{\Theta}_0}{T}\right)^{-1}\cdot\frac{\boldsymbol{\Theta}_0^\top\boldsymbol{X}_i^\top(\boldsymbol{P}_{\boldsymbol{F}_0}-\boldsymbol{P}_{\hat{\boldsymbol{F}}})\boldsymbol{X}_i}{T}\right\|_2^2,
\end{aligned}
$$

and the consistency rate established in (2).

(4) By a similar argument for (3), we can show the result in (4). □

Proof of Lemma S.4.3.

(1) We have

$$
\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2\le\max_t\left\|\boldsymbol{f}_{t,0}\right\|_2\cdot\left\|\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2+\max_t\left\|\hat{\boldsymbol{f}}_t^\top-\boldsymbol{f}_{t,0}^\top\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2.
$$

The result follows from the boundedness of $\|\boldsymbol{f}_{t,0}\|_2$ by Assumption 9, and

$$
\max_t\left\|\hat{\boldsymbol{f}}_t^\top-\boldsymbol{f}_{t,0}^\top\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2\le\sqrt{T}\cdot\frac{\|\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}-\boldsymbol{F}_0\|_f}{\sqrt{T}}\cdot\left\|\boldsymbol{O}_{\boldsymbol{F}}^{-1}\right\|_2=O_p(1).
$$

(2a) We have

$$
\begin{aligned}
\hat{\boldsymbol{l}}_i-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}=&\frac{1}{T}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{y}_i-\boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\\
=&\frac{1}{T}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i\right)\boldsymbol{X}_i\boldsymbol{\Theta}_0\boldsymbol{b}_{i,0}+\left[\frac{1}{T}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i\right)\boldsymbol{F}_0\boldsymbol{l}_{i,0}-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right]+\frac{1}{T}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i\right)\boldsymbol{\varepsilon}_i\\
=:&\Delta_{\boldsymbol{L}.1i}+\Delta_{\boldsymbol{L}.2i}+\Delta_{\boldsymbol{L}.3i}.
\end{aligned}
\tag{S.4.17}
$$

In the second step, we replace $\left\{\hat{\boldsymbol{b}}_i\right\}_{i=1}^N$ with (S.2.3). It follows that

$$
\begin{aligned}
\frac{\|\hat{\boldsymbol{L}}^\top-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^\top\|_f^2}{N}=&\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{l}}_i-\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right\|_2^2\\
\lesssim&\frac{1}{N}\sum_{i=1}^{N}\left(\|\Delta_{\boldsymbol{L}.1i}\|_2^2+\|\Delta_{\boldsymbol{L}.2i}\|_2^2+\|\Delta_{\boldsymbol{L}.3i}\|_2^2\right)\\
=:&\Delta_{\boldsymbol{L}.1}+\Delta_{\boldsymbol{L}.2}+\Delta_{\boldsymbol{L}.3}.
\end{aligned}
$$

For $\Delta_{\boldsymbol{L}.1}$, since $\left(\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i\right)\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}=\boldsymbol{0}_{T\times K_0}$, we have

$$
\frac{1}{N}\sum_{i=1}^{N}\|\Delta_{\boldsymbol{L}.1i}\|_2^2=\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{F}}^\top\left(\boldsymbol{I}_T-\hat{\boldsymbol{V}}_i\right)\boldsymbol{X}_i\left(\boldsymbol{\Theta}_0-\tilde{\boldsymbol{\Theta}}\right)\boldsymbol{b}_{i,0}\right\|_2^2
$$

$$\leq \left\|\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right\|_2^2 \cdot \frac{\bar{\boldsymbol{b}}^2}{NT} \sum_{i=1}^N \left\|\left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_i\right) \boldsymbol{X}_i\right\|_2^2 = O_p\left(\frac{1}{N^2}\right).$$

For $\Delta_{\boldsymbol{L}.2}$, since $\hat{\boldsymbol{V}}_i \boldsymbol{F}_0 = \hat{\boldsymbol{V}}_i \left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)$ and $T^{-1}\hat{\boldsymbol{F}}^\top \boldsymbol{F}_0 - \boldsymbol{O}_{\boldsymbol{F}} = T^{-1}\hat{\boldsymbol{F}}^\top \left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)$, we have

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^N \|\Delta_{\boldsymbol{L}.2i}\|_2^2 =& \frac{1}{NT^2}\sum_{i=1}^N \left\|\hat{\boldsymbol{F}}^\top \left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_i\right)\left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right)\boldsymbol{l}_{i,0}\right\|_2^2 \\
\leq& \frac{\|\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\|_f^2}{T} \cdot \frac{\bar{\boldsymbol{l}}^2}{N}\sum_{i=1}^N \left\|\boldsymbol{I}_T - \hat{\boldsymbol{V}}_i\right\|_2^2 = O_p\left(\frac{1}{N}\right).
\end{aligned}$$

Further decompose $\Delta_{\boldsymbol{L}.3}$:

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^N \|\Delta_{\boldsymbol{L}.3i}\|_2^2 \lesssim& \frac{1}{N}\sum_{i=1}^N \frac{\|(\hat{\boldsymbol{F}} - \boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1})^\top \boldsymbol{\varepsilon}_i\|_2^2}{T^2} + \frac{1}{NT}\sum_{i=1}^N \frac{\|\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\|_2^2}{T} \\
&+ \frac{1}{N}\sum_{i=1}^N \left(\frac{\|\hat{\boldsymbol{F}}^\top \boldsymbol{X}_i\|_2^2}{T^2} \cdot \left\|\tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \tilde{\boldsymbol{\Theta}}}{T}\right)^{-1} \tilde{\boldsymbol{\Theta}}^\top\right\|_2^2 \cdot \frac{\|\boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{\varepsilon}_i\|_2^2}{T^2}\right) \\
=:& \Delta_{\boldsymbol{L}.3.1} + \Delta_{\boldsymbol{L}.3.2} + \Delta_{\boldsymbol{L}.3.3},
\end{aligned}$$

where

$$\begin{aligned}
\Delta_{\boldsymbol{L}.3.1} \leq& \left\|\frac{1}{NT}\sum_{i=1}^N \boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_i^\top\right\|_2 \cdot \frac{\|\hat{\boldsymbol{F}} - \boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\|_f^2}{T} = O_p\left(\frac{1}{N^2}\right), \\
\Delta_{\boldsymbol{L}.3.2} \lesssim& \frac{1}{T} \cdot \frac{1}{NT}\sum_{i=1}^N \left\|\boldsymbol{F}_0^\top \boldsymbol{\varepsilon}_i\right\|_2^2 = O_p\left(\frac{1}{T}\right), \\
\Delta_{\boldsymbol{L}.3.3} \lesssim& \frac{1}{T} \cdot \frac{1}{NT}\sum_{i=1}^N \left\|\boldsymbol{X}_i^\top \boldsymbol{\varepsilon}_i\right\|_2^2 + \frac{1}{N}\sum_{i=1}^N \left(\frac{\|\boldsymbol{X}_i^\top \hat{\boldsymbol{F}}\|_2^2}{T^2} \cdot \frac{\|(\hat{\boldsymbol{F}} - \boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1} + \boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1})^\top \boldsymbol{\varepsilon}_i\|_2^2}{T^2}\right) \\
=& O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{N^2} + \frac{1}{T}\right) = O_p\left(\frac{1}{N}\right).
\end{aligned}$$

Collecting all terms above leads to statement (2a).

(2b) The result follows from

$$\begin{aligned}
&\left(\frac{\hat{\boldsymbol{L}}^\top \hat{\boldsymbol{L}}}{N}\right)^{-1} - \boldsymbol{O}_{\boldsymbol{F}}^{-\top}\left(\frac{\boldsymbol{L}_0^\top \boldsymbol{L}_0}{N}\right)^{-1}\boldsymbol{O}_{\boldsymbol{F}}^{-1} \\
=& \left(\frac{\hat{\boldsymbol{L}}^\top \hat{\boldsymbol{L}}}{N}\right)^{-1} \cdot \left(\frac{\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^\top \boldsymbol{L}_0 \boldsymbol{O}_{\boldsymbol{F}}^\top - \hat{\boldsymbol{L}}^\top \hat{\boldsymbol{L}}}{N}\right) \cdot \left(\frac{\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^\top \boldsymbol{L}_0 \boldsymbol{O}_{\boldsymbol{F}}^\top}{N}\right)^{-1} \\
=& O_p(1) \cdot \left[-\frac{(\hat{\boldsymbol{L}} - \boldsymbol{L}_0\boldsymbol{O}_{\boldsymbol{F}}^\top)^\top \boldsymbol{L}_0\boldsymbol{O}_{\boldsymbol{F}}^\top}{N} - \frac{\boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^\top(\hat{\boldsymbol{L}} - \boldsymbol{L}_0\boldsymbol{O}_{\boldsymbol{F}}^\top)}{N} - \frac{(\hat{\boldsymbol{L}} - \boldsymbol{L}_0\boldsymbol{O}_{\boldsymbol{F}}^\top)^\top(\hat{\boldsymbol{L}} - \boldsymbol{L}_0\boldsymbol{O}_{\boldsymbol{F}}^\top)}{N}\right] \cdot O_p(1) \\
=& O_p(1) \cdot \left[O_p\left(\frac{1}{\sqrt{N}}\right) + O_p\left(\frac{1}{\sqrt{N}}\right) + O_p\left(\frac{1}{N}\right)\right] \cdot O_p(1) = O_p\left(\frac{1}{\sqrt{N}}\right).
\end{aligned}$$

(2c) The result follows from the inequality:

$$\begin{aligned}
\max_i \left\|\hat{\boldsymbol{l}}_i\right\|_2 \leq& \max_i \|\boldsymbol{l}_{i,0}\|_2 \cdot \|\boldsymbol{O}_{\boldsymbol{F}}\|_2 + \max_i \left\|\hat{\boldsymbol{l}}_i - \boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{l}_{i,0}\right\|_2 \\
\leq& \bar{\boldsymbol{l}} \cdot \|\boldsymbol{O}_{\boldsymbol{F}}\|_2 + \sqrt{N} \cdot \frac{\|\hat{\boldsymbol{L}}^\top - \boldsymbol{O}_{\boldsymbol{F}}\boldsymbol{L}_0^\top\|_2}{\sqrt{N}} = O_p(1).
\end{aligned}$$

(3a) We have

$$\begin{aligned}\tilde{\boldsymbol{b}}_i - \boldsymbol{b}_{i,0} =& \left(\tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \tilde{\boldsymbol{\Theta}}\right)^{-1} \tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{y}_i - \boldsymbol{b}_{i,0} \\ =& \left(\tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \tilde{\boldsymbol{\Theta}}\right)^{-1} \tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \left[\boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right) \boldsymbol{b}_{i,0} + (\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}) \boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i\right] \\ =:& \Delta_{\boldsymbol{B}.1i} + \Delta_{\boldsymbol{B}.2i} + \Delta_{\boldsymbol{B}.3i}.\end{aligned}$$

By a similar argument as in the proof of (2a), it is straightforward to show that

$$\begin{aligned}\frac{\|\tilde{\boldsymbol{B}} - \boldsymbol{B}_0\|_f^2}{N} =& \frac{1}{N}\sum_{i=1}^N \left\|\tilde{\boldsymbol{b}}_i - \boldsymbol{b}_{i,0}\right\|_2^2 \\ \lesssim& \frac{1}{N}\sum_{i=1}^N \left(\|\Delta_{\boldsymbol{B}.1i}\|_2^2 + \|\Delta_{\boldsymbol{B}.2i}\|_2^2 + \|\Delta_{\boldsymbol{B}.3i}\|_2^2\right) \\ =:& \Delta_{\boldsymbol{B}.1} + \Delta_{\boldsymbol{B}.2} + \Delta_{\boldsymbol{B}.3} = O_p\left(\frac{1}{N}\right),\end{aligned}$$

where

$$\Delta_{\boldsymbol{B}.1} = O_p\left(\frac{1}{N^2}\right), \quad \Delta_{\boldsymbol{B}.2} = O_p\left(\frac{1}{N}\right), \quad \Delta_{\boldsymbol{B}.3} = O_p\left(\frac{1}{N}\right).$$

(3b) The result follows from the inequality:

$$\begin{aligned}\max_i \left\|\tilde{\boldsymbol{b}}_i\right\|_2 \le& \max_i \|\boldsymbol{b}_{i,0}\|_2 + \max_i \left\|\tilde{\boldsymbol{b}}_i - \boldsymbol{b}_{i,0}\right\|_2 \\ \le& \bar{\boldsymbol{b}} + \sqrt{N} \cdot \frac{\|\tilde{\boldsymbol{B}} - \boldsymbol{B}_0\|_2}{\sqrt{N}} = O_p(1).\end{aligned}$$

(4a) We have

$$\begin{aligned}\hat{\boldsymbol{\varepsilon}}_i - \boldsymbol{\varepsilon}_i =& \boldsymbol{y}_i - \boldsymbol{X}_i \tilde{\boldsymbol{\Theta}} \tilde{\boldsymbol{b}}_i - \hat{\boldsymbol{F}} \hat{\boldsymbol{l}}_i - \boldsymbol{\varepsilon}_i \\ =& \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \left(\boldsymbol{I}_T - \hat{\boldsymbol{V}}_i\right) \boldsymbol{y}_i - \boldsymbol{\varepsilon}_i \\ =& \left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp - \hat{\boldsymbol{H}}_i\right) \boldsymbol{X}_i \left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right) \boldsymbol{b}_{i,0} + \left(\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp - \hat{\boldsymbol{H}}_i\right) \left(\boldsymbol{F}_0 - \hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}}\right) \boldsymbol{l}_{i,0} - \left(\boldsymbol{P}_{\hat{\boldsymbol{F}}} + \hat{\boldsymbol{H}}_i\right) \boldsymbol{\varepsilon}_i \\ =:& \Delta_{\boldsymbol{\varepsilon}.1i} + \Delta_{\boldsymbol{\varepsilon}.2i} + \Delta_{\boldsymbol{\varepsilon}.3i}.\end{aligned}$$

It is straightforward to show that

$$\frac{1}{NT}\sum_{i=1}^N \|\hat{\boldsymbol{\varepsilon}}_i - \boldsymbol{\varepsilon}_i\|_2^2 \lesssim \frac{1}{NT}\sum_{i=1}^N \left(\|\Delta_{\boldsymbol{\varepsilon}.1i}\|_2^2 + \|\Delta_{\boldsymbol{\varepsilon}.2i}\|_2^2 + \|\Delta_{\boldsymbol{\varepsilon}.3i}\|_2^2\right) =: \Delta_{\boldsymbol{\varepsilon}.1} + \Delta_{\boldsymbol{\varepsilon}.2} + \Delta_{\boldsymbol{\varepsilon}.3} = O_p\left(\frac{1}{N}\right),$$

where

$$\Delta_{\boldsymbol{\varepsilon}.1} = O_p\left(\frac{1}{N^2}\right), \quad \Delta_{\boldsymbol{\varepsilon}.2} = O_p\left(\frac{1}{N}\right), \quad \Delta_{\boldsymbol{\varepsilon}.3} = O_p\left(\frac{1}{N}\right).$$

(4b) We have

$$\hat{\varepsilon}_{it} - \varepsilon_{it} = \left(\boldsymbol{x}_{it}^\top - \hat{\boldsymbol{f}}_t^\top \cdot \frac{\hat{\boldsymbol{F}}^\top \boldsymbol{X}_i}{T}\right) \cdot \left[\boldsymbol{I}_p - \tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \tilde{\boldsymbol{\Theta}}}{T}\right)^{-1} \frac{\tilde{\boldsymbol{\Theta}}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i}{T}\right] \cdot \left(\boldsymbol{\Theta}_0 - \tilde{\boldsymbol{\Theta}}\right) \boldsymbol{b}_{i,0}$$

$$
\begin{aligned}
&+\left(\boldsymbol{f}_{t,0}^\top-\hat{\boldsymbol{f}}_t^\top\boldsymbol{O}_{\boldsymbol{F}}\right)\boldsymbol{l}_{i,0}-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\hat{\boldsymbol{F}}^\top(\boldsymbol{F}_0-\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}})\boldsymbol{l}_{i,0}}{T}\\
&-\left(\boldsymbol{x}_{it}^\top-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\hat{\boldsymbol{F}}^\top\boldsymbol{X}_i}{T}\right)\cdot\tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp(\boldsymbol{F}_0-\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}})\boldsymbol{l}_{i,0}}{T}\\
&-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i}{T}-\hat{\boldsymbol{f}}_t^\top\cdot\frac{(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1})^\top\boldsymbol{\varepsilon}_i}{T}\\
&-\left(\boldsymbol{x}_{it}^\top-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\hat{\boldsymbol{F}}^\top\boldsymbol{X}_i}{T}\right)\cdot\tilde{\boldsymbol{\Theta}}\left(\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{X}_i\tilde{\boldsymbol{\Theta}}}{T}\right)^{-1}\frac{\tilde{\boldsymbol{\Theta}}^\top\boldsymbol{X}_i^\top\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp\boldsymbol{\varepsilon}_i}{T}\\
=:&\Delta_{\varepsilon.1it}+\cdots+\Delta_{\varepsilon.7it}.
\end{aligned}
$$

Since $\max_t\left\|\boldsymbol{f}_{t,0}^\top-\hat{\boldsymbol{f}}_t^\top\boldsymbol{O}_{\boldsymbol{F}}\right\|_2=O_p(1)$ and $\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2=O_p(1)$, we have

$$
\max_{i,t}|\Delta_{\varepsilon.2it}|=O_p(1),\quad \max_{i,t}|\Delta_{\varepsilon.3it}|=O_p\left(\frac{1}{\sqrt{N}}\right).
$$

Since

$$
\max_{i,t}\left\|\boldsymbol{x}_{it}^\top-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\hat{\boldsymbol{F}}^\top\boldsymbol{X}_i}{T}\right\|_2\lesssim\max_{i,t}\|\boldsymbol{x}_{it}\|_2+\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2=O_p\left((NT)^{1/8}\right),
$$

we have

$$
\begin{aligned}
\max_{i,t}|\Delta_{\varepsilon.1.it}|&=O_p\left((NT)^{1/8}\right)\cdot O_p\left(\frac{1}{N}\right)=O_p\left(\frac{1}{N^{3/4}}\right),\\
\max_{i,t}|\Delta_{\varepsilon.4.it}|&=O_p\left((NT)^{1/8}\right)\cdot O_p\left(\frac{1}{\sqrt{N}}\right)=O_p\left(\frac{1}{N^{1/4}}\right).
\end{aligned}
$$

By Lemma S.4.1 (4b), $T^{-1/2}\max_i\|\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\|_2=O_p\left(N^{1/8}\right)$, and it follows that

$$
\max_{i,t}|\Delta_{\varepsilon.5it}|\lesssim\frac{1}{\sqrt{T}}\cdot\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2\cdot\max_i\frac{\|\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\|_2}{\sqrt{T}}=O_p\left(\frac{1}{N^{1/4}}\right).
$$

For $\Delta_{\varepsilon.6it}$,

$$
\begin{aligned}
\max_{i,t}|\Delta_{\varepsilon.6it}|\leq&\max_t\left\|\hat{\boldsymbol{f}}_t\right\|_2\cdot\frac{\|\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1}\|_2}{\sqrt{T}}\cdot\max_i\frac{\|\boldsymbol{\varepsilon}_i\|_2}{\sqrt{T}}\\
=&O_p(1)\cdot O_p\left(\frac{1}{\sqrt{N}}\right)\cdot O_p(N^{1/8})=O_p\left(\frac{1}{N^{1/4}}\right).
\end{aligned}
$$

By Lemma S.4.1 (4a), $\max_i T^{-1/2}\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2=O_p(N^{1/4})$, and it follows that

$$
\begin{aligned}
\max_{i,t}|\Delta_{\varepsilon.7it}|\leq&\max_{i,t}\left\|\boldsymbol{x}_{it}^\top-\hat{\boldsymbol{f}}_t^\top\cdot\frac{\hat{\boldsymbol{F}}^\top\boldsymbol{X}_i}{T}\right\|_2\cdot\max_i\left(\frac{\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\|_2}{T}+\frac{\|\boldsymbol{O}_{\boldsymbol{F}}^{-\top}\boldsymbol{F}_0^\top\boldsymbol{\varepsilon}_i\|_2}{T}+\frac{\|(\hat{\boldsymbol{F}}-\boldsymbol{F}_0\boldsymbol{O}_{\boldsymbol{F}}^{-1})\boldsymbol{\varepsilon}_i\|_2}{T}\right)\\
=&O_p\left((NT)^{1/8}\right)\cdot\left[O_p\left(\frac{1}{N^{1/4}}\right)+O_p\left(\frac{1}{N^{1/4}}\right)+O_p\left(\frac{1}{N^{3/8}}\right)\right]=O_p(1).
\end{aligned}
$$

Collecting all terms above leads to the desired result. □

S.5 Other Simulation Results

This section collects supplementary results that complement the main text. It includes verification examples related to Section 2, additional simulation results for the model selection criteria, results on bias correction, and supplementary singular value decomposition (SVD) analyses for the real data applications.

S.5.1 Verification Examples related to Section 2

Example 2.1 is straightforward, which shows that $\boldsymbol{F}_0$ may not be unique without the orthogonality restrictions.

The need for rank-one decomposition. We consider the model with $(p, K_0, R_0) = (3, 2, 2)$:

$$\boldsymbol{y}_i = \boldsymbol{X}_i \boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} + \boldsymbol{F}_0 \boldsymbol{l}_{i,0} + \sigma \boldsymbol{\varepsilon}_i. \tag{S.5.1}$$

The first two predictors $x_{it,1}$ and $x_{it,2}$ are independently generated from $\mathcal{N}(0, 1)$. The third predictor is low-rank and admits the decomposition $\boldsymbol{X}_{,3} = \boldsymbol{g}_{\cdot 1}\boldsymbol{\gamma}_{\cdot 1}^{\top} + \boldsymbol{g}_{\cdot 2}\boldsymbol{\gamma}_{\cdot 2}^{\top}$ with $\mathrm{Rank}(\boldsymbol{X}_{,3}) = 2$, and $\boldsymbol{F}_0 = \left(\boldsymbol{g}_{\cdot 2} + \boldsymbol{g}_{\cdot 3}, \quad \boldsymbol{g}_{\cdot 4}\right)$, where $\boldsymbol{g}_{\cdot j_1}^{\top}\boldsymbol{g}_{\cdot j_2} = 0$ for $j_1 \neq j_2$. All elements of $\boldsymbol{B}_0$, $\boldsymbol{L}_0$, and $\left(\boldsymbol{\gamma}_{\cdot 1}, \quad \boldsymbol{\gamma}_{\cdot 2}\right)$ are independently generated from $\mathrm{Unif}[0.5, 1.5]$, and

$$\boldsymbol{\Theta}_0 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 \end{pmatrix}.$$

The error ε_{it} is independently generated from $\mathcal{N}(0, 1)$.

Suppose we put the orthogonality restriction $\boldsymbol{F}_0 \perp \left(\boldsymbol{g}_{\cdot 1}, \quad \boldsymbol{g}_{\cdot 2}\right)$ on the original model (S.5.1). Under this restriction, this model can no longer be equivalently represented in the form of (2.2). However, after decomposing the low-rank predictor as

$$\boldsymbol{X}_{,3} = \boldsymbol{X}_{,3}^{\star} + \boldsymbol{X}_{,4}^{\star}, \quad \text{with } \boldsymbol{X}_{,3}^{\star} = \boldsymbol{g}_{\cdot 1}\boldsymbol{\gamma}_{\cdot 1}^{\top} \text{ and } \boldsymbol{X}_{,4}^{\star} = \boldsymbol{g}_{\cdot 2}\boldsymbol{\gamma}_{\cdot 2}^{\top},$$

the model can be equivalently rewritten as

$$\boldsymbol{y}_i = \boldsymbol{X}_i^{\star} \boldsymbol{\Theta}_0^{\star} \boldsymbol{b}_{i,0}^{\star} + \boldsymbol{F}_0^{\star} \boldsymbol{l}_{i,0} + \sigma \boldsymbol{\varepsilon}_i, \quad \text{s.t. } \boldsymbol{F}_0^{\star} \perp \left(\boldsymbol{g}_{\cdot 1}, \quad \boldsymbol{g}_{\cdot 2}\right), \tag{S.5.2}$$

where $(p^{\star}, K_0^{\star}, R_0^{\star}) = (4, 3, 2)$ and

$$\boldsymbol{x}^{\star} = \left(x_{it,1}, x_{it,2}, x_{it,3}^{\star}, x_{it,4}^{\star}\right)^{\top}, \quad \boldsymbol{\Theta}_0^{\star} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{b}_{i,0}^{\star} = \begin{pmatrix} \boldsymbol{b}_{i,0} \\ b_{i20} + \frac{l_{1i0}}{w_{2i}} \end{pmatrix}, \quad \boldsymbol{F}_0^{\star} = \left(\boldsymbol{g}_{\cdot 3}, \quad \boldsymbol{g}_{\cdot 4}\right).$$

Table S.1 reports the mean squared error

$$MSE = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(y_{it} - \hat{y}_{it}\right)^2$$

under different configurations. The first row corresponds to the noise-free case ($\sigma = 0$) and the noisy case ($\sigma = 0.2$). In the second row, "BD" and "AD" denote estimation before and after rank-one decomposition, respectively, while (K, R) specifies the numbers of latent factors used in estimation.

Comparing the second and fourth columns, we observe that under the orthogonality restriction, the information contained in observed predictors and latent factor structures cannot be fully recovered by the model without rank-one decomposition. Even when over-specifying the model by setting $K = p = 3$ and $R = 10$, the fitted model still fails to capture the underlying structure, as shown in the third column. When noise is present, the uniformly superior performance of "AD: (3, 2)" relative to the two "BD" specifications further confirms the necessity of rank-one decomposition under orthogonality.

Table S.1: Simulation results of MSE to verify the need for rank-one decomposition under orthogonality restriction.

(N, T)	$\sigma = 0$			$\sigma = 0.2$		
	BD: (2, 2)	BD: (3, 10)	AD: (3, 2)	BD: (2, 2)	BD: (3, 10)	AD: (3, 2)
(30, 30)	0.5149	0.2754	0	0.5630	0.2928	0.0310
(50, 50)	0.5353	0.3913	0	0.5717	0.4089	0.0345
(70, 70)	0.5322	0.4273	0	0.5717	0.4527	0.0361
(100, 100)	0.5371	0.4618	0	0.5683	0.4831	0.0372
(150, 150)	0.5406	0.4884	0	0.5771	0.5170	0.0381
(200, 200)	0.5378	0.4981	0	0.5774	0.5329	0.0386

Loss of parsimony under orthogonality restriction. We consider the model with $(p, K_0, R_0) = (3, 1, 2)$:

$$\boldsymbol{y}_i = \boldsymbol{X}_i \boldsymbol{\theta}_0 b_{i,0} + \boldsymbol{F}_0 \boldsymbol{l}_{i,0} + \sigma \boldsymbol{\varepsilon}_i. \tag{S.5.3}$$

Here $\boldsymbol{X}_{\cdot j} = \boldsymbol{g}_{\cdot j} \boldsymbol{\gamma}_{\cdot j}^{\top}$ are low-rank predictors for $j = 1, 2, 3$, $\boldsymbol{F}_0 = \left(\boldsymbol{g}_{\cdot 3} + \boldsymbol{g}_{\cdot 4}, \quad \boldsymbol{g}_{\cdot 5}\right)$, where $\boldsymbol{g}_{\cdot j_1}^{\top} \boldsymbol{g}_{\cdot j_2} = 0$ for $j_1 \neq j_2$. All elements in $\{(b_{i,0}, \boldsymbol{l}_{i,0})\}_{i=1}^{N}$ and $\left(\boldsymbol{\gamma}_{\cdot 1}, \quad \boldsymbol{\gamma}_{\cdot 2}, \quad \boldsymbol{\gamma}_{\cdot 3}\right)$ are independently generated from $\mathrm{Unif}[0.5, 1.5]$, and $\boldsymbol{\theta}_0 = (1, 1, 1)^{\top}$. The error ε_{it} is independently generated from $\mathcal{N}(0, 1)$. If we put the restriction $\boldsymbol{F}_0 \perp \left(\boldsymbol{g}_{\cdot 1}, \quad \boldsymbol{g}_{\cdot 2}, \quad \boldsymbol{g}_{\cdot 3}\right)$ on the original model (S.5.3), this model cannot be written in the form of (2.2) if we keep $K_0 = 1$. However, we can rewrite (S.5.3) as

$$\boldsymbol{y}_i = \boldsymbol{X}_i \boldsymbol{\Theta}_0^{\star} \boldsymbol{b}_{i,0}^{\star} + \boldsymbol{F}_0^{\star} \boldsymbol{l}_{i,0} + \sigma \boldsymbol{\varepsilon}_i, \quad \text{s.t. } \boldsymbol{F}_0^{\star} \perp \left(\boldsymbol{g}_{\cdot 1}, \quad \boldsymbol{g}_{\cdot 2}, \quad \boldsymbol{g}_{\cdot 3}\right), \tag{S.5.4}$$

where $(p, K_0^{\star}, R_0^{\star}) = (3, 2, 2)$, and

$$\boldsymbol{\Theta}_0^{\star} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \boldsymbol{b}_{i,0}^{\star} = \begin{pmatrix} b_{i,0} \\ \frac{l_{1i,0}}{w_{3i}} \end{pmatrix}, \quad \boldsymbol{F}_0^{\star} = \left(\boldsymbol{g}_{\cdot 4}, \quad \boldsymbol{g}_{\cdot 5}\right).$$

Table S.2 reports the MSE under different choices of (K, R) in both the noise-free case ($\sigma = 0$) and the noisy case ($\sigma = 0.2$). Under the orthogonality restriction, fitting the model with the true dimension $K = 1$ results in non-negligible errors, even when the number of latent factors is substantially over-specified (e.g., $(1, 10)$). By contrast, allowing an additional linear combination of predictors and fitting the model with $(K, R) = (2, 2)$ achieves exact recovery when $\sigma = 0$ and uniformly smaller errors when noise is present. This demonstrates that imposing orthogonality forces an increase in the effective dimension of the reduced-rank structure, leading to a loss of parsimony relative to the original data-generating process.

Table S.2: Simulation results of MSE to verify the loss of parsimony under orthogonality restriction.

(N, T)	$\sigma = 0$			$\sigma = 0.2$		
	$(1, 2)$	$(1, 10)$	$(2, 2)$	$(1, 2)$	$(1, 10)$	$(2, 2)$
$(30, 30)$	0.0537	0.0396	0	0.0885	0.0664	0.0324
$(50, 50)$	0.0537	0.0453	0	0.0915	0.0748	0.0353
$(70, 70)$	0.0541	0.0475	0	0.0918	0.0785	0.0367
$(100, 100)$	0.0538	0.0495	0	0.0931	0.0831	0.0377
$(150, 150)$	0.0548	0.0519	0	0.0934	0.0863	0.0385
$(200, 200)$	0.0551	0.0533	0	0.0941	0.0886	0.0388

S.5.2 Simulation Results for Bias Correction

We consider the model

$$y_{it} = (x_{it,1}\theta_{1,0} + x_{it,2}\theta_{2,0} + \rho_y y_{i,t-1})\, b_{i,0} + l_{i,0} f_{t,0} + \varepsilon_{it},$$

where $x_{it,1}$ are independently generated from $\mathcal{N}(0, 1)$, $x_{it,2}$ is a cross-sectionally invariant predictor, and the last predictor is the one-period lag of the response. Both $b_{i,0}$ and $l_{i,0}$ represent heterogeneous loadings and are generated independently from $\text{Unif}[0.5, 1.5]$. In this model, $\boldsymbol{\theta}_0 = (\rho_y, \theta_{2,0}, \theta_{3,0})^\top$, $p = 3$, and $K_0 = R_0 = 1$.

We evaluate estimation accuracy using the projection matrix error $\left\|\boldsymbol{P}_{\hat{\boldsymbol{\theta}}} - \boldsymbol{P}_{\boldsymbol{\theta}_0}\right\|_f$. To assess the effect of bias correction, we compare the uncorrected error with its bias-corrected counterpart $\left\|\boldsymbol{P}_{\hat{\boldsymbol{\theta}}} - \boldsymbol{P}_{\boldsymbol{\theta}_0} - \hat{\mathbb{B}}\right\|_f$, where $\hat{\mathbb{B}}$ denotes the estimated bias term for the projection matrix estimator. Table S.3 reports the results under three error settings. Across all scenarios, bias correction leads to a systematic, albeit moderate, reduction in projection estimation error, and remains robust under heavy-tailed and heteroskedastic disturbances.

S.5.3 Simulation Results in Section 4.1

This section presents comprehensive simulation results extending the analysis in Section 4.1. While the main text focuses on PC_{p1} and IC_{p1} for conciseness, here we provide complete results for all MSC variants (PC_{p2}–PC_{p3}, IC_{p2}–IC_{p3}) across all error designs (E1-E3) and different (K_0, R_0) configurations. Note that

Table S.3: Projection matrix estimation error with and without bias correction under different error settings. Here `Proj` denotes the uncorrected projection error, and `Proj.BC` denotes the bias-corrected counterpart.

(N, T)	E1: i.i.d. normal		E2: heavy-tailed		E3: heteroskedastic	
	Proj	Proj.BC	Proj	Proj.BC	Proj	Proj.BC
(30, 30)	0.1072	0.1044	0.1157	0.1111	0.1067	0.1049
(50, 50)	0.0592	0.0552	0.0611	0.0580	0.0603	0.0562
(70, 70)	0.0356	0.0338	0.0437	0.0406	0.0363	0.0345
(100, 100)	0.0267	0.0240	0.0295	0.0272	0.0271	0.0246
(150, 150)	0.0179	0.0162	0.0187	0.0170	0.0178	0.0160
(200, 200)	0.0129	0.0117	0.0160	0.0145	0.0129	0.0117

when $(K_0, R_0) = (0, 0)$, the results are identical across all error designs (E1–E3), since the null model is correctly identified in all cases and no difference between projections needs to be computed. Therefore, for this configuration, only the E1 case is reported.

S.5.3.1 Error Type: I.I.D. Normal

Table S.4: Estimation efficiency and selection frequencies under error design E1 (i.i.d. normal) with true $K_0 = 2$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	0.2124	0.6365	PCp1	0	34	**66**	0	0	0	10	**90**	0	0
			PCp2	0	38	**62**	0	0	0	12	**88**	0	0
			PCp3	0	31	**69**	0	0	0	5	**95**	0	0
			ICp1	0	43	**57**	0	0	4	24	**72**	0	0
			ICp2	0	46	**54**	0	0	8	27	**65**	0	0
			ICp3	0	32	**68**	0	0	2	12	**86**	0	0
(50, 200)	0.1342	0.4048	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	1	**99**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	2	**98**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(100, 200)	0.0921	0.2863	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 200)	0.0617	0.2019	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 20)	0.2461	0.2270	PCp1	0	4	**96**	0	0	0	29	**71**	0	0
			PCp2	0	6	**94**	0	0	0	35	**65**	0	0
			PCp3	0	1	**99**	0	0	0	20	**80**	0	0
			ICp1	0	7	**93**	0	0	4	33	**63**	0	0
			ICp2	0	9	**91**	0	0	7	32	**61**	0	0
			ICp3	0	5	**95**	0	0	3	25	**72**	0	0
(200, 50)	0.1351	0.2100	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 100)	0.0921	0.2070	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0

Table S.5: Estimation efficiency and selection frequencies under error design E1 (i.i.d. normal) with true $K_0 = 0$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	**K=0**	K=1	K=2	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	–	0.6333	PCp1	**100**	0	0	0	0	0	9	**91**	0	0
			PCp2	**100**	0	0	0	0	0	9	**91**	0	0
			PCp3	**100**	0	0	0	0	0	5	**95**	0	0
			ICp1	**100**	0	0	0	0	1	18	**81**	0	0
			ICp2	**100**	0	0	0	0	3	19	**78**	0	0
			ICp3	**100**	0	0	0	0	1	11	**88**	0	0
(50, 200)	–	0.4027	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(100, 200)	–	0.2854	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 200)	–	0.2010	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 20)	–	0.2083	PCp1	**100**	0	0	0	0	0	18	**82**	0	0
			PCp2	**100**	0	0	0	0	0	19	**81**	0	0
			PCp3	**100**	0	0	0	0	0	15	**85**	0	0
			ICp1	**100**	0	0	0	0	5	33	**62**	0	0
			ICp2	**100**	0	0	0	0	6	33	**61**	0	0
			ICp3	**100**	0	0	0	0	2	30	**68**	0	0
(200, 50)	–	0.2059	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 100)	–	0.2055	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0

Table S.6: Estimation efficiency and selection frequencies under error design E1 (i.i.d. normal) with true $K_0 = 2$ and $R_0 = 0$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	**R=0**	R=1	R=2	R=3	R=4
(20, 200)	0.1459	–	PCp1	0	1	**99**	0	0	**100**	0	0	0	0
			PCp2	0	1	**99**	0	0	**100**	0	0	0	0
			PCp3	0	1	**99**	0	0	**100**	0	0	0	0
			ICp1	0	3	**97**	0	0	**100**	0	0	0	0
			ICp2	0	4	**96**	0	0	**100**	0	0	0	0
			ICp3	0	2	**98**	0	0	**100**	0	0	0	0
(50, 200)	0.0885	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(100, 200)	0.0630	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 200)	0.0445	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 20)	0.1563	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 50)	0.0892	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 100)	0.0612	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0

Table S.7: Estimation efficiency and selection frequencies under error design E1 (i.i.d. normal) with true $K_0 = 0$ and $R_0 = 0$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	**K=0**	K=1	K=2	K=3	K=4	**R=0**	R=1	R=2	R=3	R=4
(20, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(50, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(100, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(200, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(200, 20)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(200, 50)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0
(200, 100)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
			PCp2	**100**	0	0	0	0	**100**	0	0	0	0
			PCp3	**100**	0	0	0	0	**100**	0	0	0	0
			ICp1	**100**	0	0	0	0	**100**	0	0	0	0
			ICp2	**100**	0	0	0	0	**100**	0	0	0	0
			ICp3	**100**	0	0	0	0	**100**	0	0	0	0

S.5.3.2 Error Type: Heavy-tailed

Table S.8: Estimation efficiency and selection frequencies under error design E2 (heavy-tailed) with true $K_0 = 2$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	0.2126	0.6328	PCp1	0	42	**58**	0	0	0	6	**94**	0	0
			PCp2	0	42	**58**	0	0	0	7	**93**	0	0
			PCp3	0	40	**60**	0	0	0	2	**98**	0	0
			ICp1	0	47	**53**	0	0	6	17	**77**	0	0
			ICp2	0	**55**	45	0	0	9	23	**68**	0	0
			ICp3	0	45	**55**	0	0	2	10	**88**	0	0
(50, 200)	0.1219	0.4037	PCp1	0	1	**99**	0	0	0	0	**100**	0	0
			PCp2	0	3	**97**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	2	**98**	0	0	0	0	**100**	0	0
			ICp2	0	3	**97**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(100, 200)	0.0888	0.2847	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 200)	0.0632	0.2016	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 20)	0.2508	0.2399	PCp1	0	6	**94**	0	0	2	31	**67**	0	0
			PCp2	0	7	**93**	0	0	2	35	**63**	0	0
			PCp3	0	2	**98**	0	0	1	25	**74**	0	0
			ICp1	0	14	**86**	0	0	12	34	**54**	0	0
			ICp2	0	19	**81**	0	0	15	36	**49**	0	0
			ICp3	0	7	**93**	0	0	6	29	**65**	0	0
(200, 50)	0.1325	0.2120	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 100)	0.0865	0.2022	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0

Table S.9: Estimation efficiency and selection frequencies under error design E2 (heavy-tailed) with true $K_0 = 0$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	**K=0**	K=1	K=2	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	–	0.6281	PCp1	**100**	0	0	0	0	0	3	**97**	0	0
			PCp2	**100**	0	0	0	0	0	7	**93**	0	0
			PCp3	**100**	0	0	0	0	0	2	**98**	0	0
			ICp1	**100**	0	0	0	0	1	10	**89**	0	0
			ICp2	**100**	0	0	0	0	4	11	**85**	0	0
			ICp3	**100**	0	0	0	0	0	7	**93**	0	0
(50, 200)	–	0.4019	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(100, 200)	–	0.2834	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 200)	–	0.2008	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 20)	–	0.2230	PCp1	**100**	0	0	0	0	1	19	**80**	0	0
			PCp2	**100**	0	0	0	0	1	23	**76**	0	0
			PCp3	**100**	0	0	0	0	0	16	**84**	0	0
			ICp1	**100**	0	0	0	0	10	35	**55**	0	0
			ICp2	**100**	0	0	0	0	14	37	**49**	0	0
			ICp3	**100**	0	0	0	0	6	32	**62**	0	0
(200, 50)	–	0.2069	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 100)	–	0.2004	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0

Table S.10: Estimation efficiency and selection frequencies under error design E2 (heavy-tailed) with true $K_0 = 2$ and $R_0 = 0$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	**R=0**	R=1	R=2	R=3	R=4
(20, 200)	0.1433	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	1	**99**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	3	**97**	0	0	**100**	0	0	0	0
			ICp2	0	6	**94**	0	0	**100**	0	0	0	0
			ICp3	0	1	**99**	0	0	**100**	0	0	0	0
(50, 200)	0.0891	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(100, 200)	0.0625	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 200)	0.0442	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 20)	0.1483	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 50)	0.0864	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 100)	0.0611	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0

S.5.3.3 Error Type: Heteroskedastic

Table S.11: Estimation efficiency and selection frequencies under error design E3 (heteroskedastic) with true $K_0 = 2$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	0.2125	0.6366	PCp1	0	34	**66**	0	0	0	10	**90**	0	0
			PCp2	0	37	**63**	0	0	0	13	**87**	0	0
			PCp3	0	31	**69**	0	0	0	6	**94**	0	0
			ICp1	0	41	**59**	0	0	5	23	**72**	0	0
			ICp2	0	46	**54**	0	0	8	27	**65**	0	0
			ICp3	0	32	**68**	0	0	3	13	**84**	0	0
(50, 200)	0.1333	0.4047	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	1	**99**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	1	**99**	0	0	0	0	**100**	0	0
			ICp2	0	2	**98**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(100, 200)	0.0923	0.2867	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 200)	0.0619	0.2015	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 20)	0.2508	0.2262	PCp1	0	3	**97**	0	0	0	32	**68**	0	0
			PCp2	0	5	**95**	0	0	0	36	**64**	0	0
			PCp3	0	0	**100**	0	0	0	20	**80**	0	0
			ICp1	0	8	**92**	0	0	4	33	**63**	0	0
			ICp2	0	10	**90**	0	0	6	33	**61**	0	0
			ICp3	0	5	**95**	0	0	3	25	**72**	0	0
(200, 50)	0.1362	0.2105	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0
(200, 100)	0.0920	0.2072	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
			PCp2	0	0	**100**	0	0	0	0	**100**	0	0
			PCp3	0	0	**100**	0	0	0	0	**100**	0	0
			ICp1	0	0	**100**	0	0	0	0	**100**	0	0
			ICp2	0	0	**100**	0	0	0	0	**100**	0	0
			ICp3	0	0	**100**	0	0	0	0	**100**	0	0

Table S.12: Estimation efficiency and selection frequencies under error design E3 (heteroskedastic) with true $K_0 = 0$ and $R_0 = 2$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	**K=0**	K=1	K=2	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
(20, 200)	–	0.6334	PCp1	**100**	0	0	0	0	0	9	**91**	0	0
			PCp2	**100**	0	0	0	0	0	9	**91**	0	0
			PCp3	**100**	0	0	0	0	0	5	**95**	0	0
			ICp1	**100**	0	0	0	0	1	17	**82**	0	0
			ICp2	**100**	0	0	0	0	3	19	**78**	0	0
			ICp3	**100**	0	0	0	0	1	12	**87**	0	0
(50, 200)	–	0.4024	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(100, 200)	–	0.2859	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 200)	–	0.2007	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 20)	–	0.2084	PCp1	**100**	0	0	0	0	0	17	**83**	0	0
			PCp2	**100**	0	0	0	0	0	19	**81**	0	0
			PCp3	**100**	0	0	0	0	0	13	**87**	0	0
			ICp1	**100**	0	0	0	0	5	33	**62**	0	0
			ICp2	**100**	0	0	0	0	6	33	**61**	0	0
			ICp3	**100**	0	0	0	0	2	30	**68**	0	0
(200, 50)	–	0.2065	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0
(200, 100)	–	0.2057	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
			PCp2	**100**	0	0	0	0	0	0	**100**	0	0
			PCp3	**100**	0	0	0	0	0	0	**100**	0	0
			ICp1	**100**	0	0	0	0	0	0	**100**	0	0
			ICp2	**100**	0	0	0	0	0	0	**100**	0	0
			ICp3	**100**	0	0	0	0	0	0	**100**	0	0

Table S.13: Estimation efficiency and selection frequencies under error design E3 (heteroskedastic) with true $K_0 = 2$ and $R_0 = 0$.

(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	**R=0**	R=1	R=2	R=3	R=4
(20, 200)	0.1459	–	PCp1	0	1	**99**	0	0	**100**	0	0	0	0
			PCp2	0	1	**99**	0	0	**100**	0	0	0	0
			PCp3	0	1	**99**	0	0	**100**	0	0	0	0
			ICp1	0	3	**97**	0	0	**100**	0	0	0	0
			ICp2	0	4	**96**	0	0	**100**	0	0	0	0
			ICp3	0	2	**98**	0	0	**100**	0	0	0	0
(50, 200)	0.0886	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(100, 200)	0.0637	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 200)	0.0443	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 20)	0.1555	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 50)	0.0895	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0
(200, 100)	0.0607	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
			PCp2	0	0	**100**	0	0	**100**	0	0	0	0
			PCp3	0	0	**100**	0	0	**100**	0	0	0	0
			ICp1	0	0	**100**	0	0	**100**	0	0	0	0
			ICp2	0	0	**100**	0	0	**100**	0	0	0	0
			ICp3	0	0	**100**	0	0	**100**	0	0	0	0

S.5.4 Additional Results for Real Datasets

This section reports additional results of explanatory power and singular value decomposition of $\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{B}}^\top$ under alternative (K, R) specifications selected by different criteria.

As discussed in Section 4.2, combining reduced-rank predictors with latent factors can achieve higher explanatory power than using all 11 predictors directly; see Tables S.14–S.15. Across different choices of (K, R), we find that the dominant directions are highly stable. In particular, the leading directions consistently capture persistence in real GDP per capita through lags of log(`rgdpe/pop`), together with capital intensity and human capital as reflected in log(`rnna/emp`) and log(`hc`), respectively. Additional directions associated with larger values of K typically have smaller singular values and more diffuse economic interpretations. Overall, these results support using parsimonious specifications for substantive interpretation, while larger (K, R) primarily serve to absorb residual variation.

Table S.14: Explanatory power and singular value decomposition under $(K, R) = (6, 6)$ for `PWT`.

Predictors	$\boldsymbol{\theta}_1$	$\boldsymbol{\theta}_2$	$\boldsymbol{\theta}_3$	$\boldsymbol{\theta}_4$	$\boldsymbol{\theta}_5$	$\boldsymbol{\theta}_6$	only $\boldsymbol{X}$
log(`rgdpe/pop`)$_{t-1}$	-0.560	**-0.797**	0.079	0.151	0.048	0.025	–
log(`rgdpe/pop`)$_{t-2}$	0.167	-0.012	0.019	0.093	0.364	0.342	–
log(`rnna/emp`)$_t$	**-0.729**	0.488	0.021	-0.331	-0.159	0.073	–
log(`rnna/emp`)$_{t-1}$	0.313	-0.305	-0.031	**-0.767**	-0.148	0.043	–
log(`hc`)	-0.089	-0.091	**-0.926**	-0.107	-0.042	0.180	–
log(`emp/pop`)	-0.140	0.007	0.165	-0.410	**0.717**	0.145	–
log(`pl_i/pl_c`)$_t$	0.008	-0.007	-0.044	-0.035	0.201	-0.180	–
log(`pl_i/pl_c`)$_{t-1}$	-0.010	-0.012	-0.006	0.051	-0.119	-0.096	–
`csh_i`	-0.015	-0.008	-0.021	0.006	-0.026	-0.222	–
`csh_g`	0.029	0.075	0.013	0.215	-0.021	**0.785**	–
`year`	0.020	-0.135	0.325	-0.201	-0.496	0.340	–
Cumulative R^2	0.9766	0.9913	0.9953	0.9973	0.9976	0.9979	–
Latent factors	$\boldsymbol{f}_{\cdot 1}$	$\boldsymbol{f}_{\cdot 2}$	$\boldsymbol{f}_{\cdot 3}$	$\boldsymbol{f}_{\cdot 4}$	$\boldsymbol{f}_{\cdot 5}$	$\boldsymbol{f}_{\cdot 6}$	–
Cumulative R^2	0.9983	0.9986	0.9988	0.9990	0.9992	0.9994	0.9984

Table S.15: Explanatory power and singular value decomposition under $(K, R) = (4, 3)$ for PWT.

Predictors	$\boldsymbol{\theta}_1$	$\boldsymbol{\theta}_2$	$\boldsymbol{\theta}_3$	$\boldsymbol{\theta}_4$	$\boldsymbol{f}_{\cdot 1}$	$\boldsymbol{f}_{\cdot 2}$	$\boldsymbol{f}_{\cdot 3}$	only $\boldsymbol{X}$
$\log(\texttt{rgdpe/pop})_{t-1}$	**-0.958**	-0.116	0.154	0.102	–	–	–	–
$\log(\texttt{rgdpe/pop})_{t-2}$	0.182	-0.002	-0.058	0.163	–	–	–	–
$\log(\texttt{rnna/emp})_{t}$	-0.190	**0.730**	-0.514	-0.061	–	–	–	–
$\log(\texttt{rnna/emp})_{t-1}$	-0.008	-0.477	-0.272	-0.339	–	–	–	–
$\log(\texttt{hc})$	-0.076	-0.445	**-0.555**	-0.088	–	–	–	–
$\log(\texttt{emp/pop})$	-0.070	0.109	-0.362	-0.426	–	–	–	–
$\log(\texttt{pl_i/pl_c})_{t}$	-0.002	-0.009	-0.072	0.066	–	–	–	–
$\log(\texttt{pl_i/pl_c})_{t-1}$	-0.016	0.012	0.088	-0.020	–	–	–	–
csh_i	-0.022	-0.026	-0.037	0.122	–	–	–	–
csh_g	0.018	-0.002	0.000	0.252	–	–	–	–
year	-0.020	0.123	0.426	**-0.756**	–	–	–	–
Cumulative R^2	0.9918	0.9949	0.9967	0.9974	0.9978	0.9985	0.9988	0.9984

S.6 Nuclear Norm Regularization Estimation

The Nuclear Norm Regularization (NNR) estimation is motivated by the framework of Chernozhukov et al. (2019). We begin by introducing necessary notation and definitions. We then study the optimization problem (S.6.1), propose the estimation procedure, and establish its convergence properties. Moreover, we prove the consistency of the NNR estimator, present the model selection approach, and show that it consistently estimates (K_0, R_0) under suitable assumptions. In Section S.6.4, we discuss the tuning parameter choices in (S.6.1). Proofs of theorems are in Section S.6.5.

S.6.1 Notation and Definitions

Let

$$\boldsymbol{M}^{(1)} = \boldsymbol{\Theta}\boldsymbol{B}^\top = \left(\boldsymbol{\Theta}\boldsymbol{b}_1, \quad \ldots, \quad \boldsymbol{\Theta}\boldsymbol{b}_N\right), \quad \boldsymbol{M}^{(2)} = \boldsymbol{F}\boldsymbol{L}^\top = \begin{pmatrix} \boldsymbol{f}_1^\top \boldsymbol{l}_1 & \ldots & \boldsymbol{f}_1^\top \boldsymbol{l}_N \\ \vdots & \ddots & \vdots \\ \boldsymbol{f}_T^\top \boldsymbol{l}_1 & \ldots & \boldsymbol{f}_T^\top \boldsymbol{l}_N \end{pmatrix},$$

$$\boldsymbol{Y} = \begin{pmatrix} y_{11} & \ldots & y_{N1} \\ \vdots & \ddots & \vdots \\ y_{1T} & \ldots & y_{NT} \end{pmatrix}, \quad \boldsymbol{Z} = \left((\boldsymbol{X}_{,1} \odot \boldsymbol{\mathcal{E}})^\top \mathbf{1}_T, \quad \ldots, \quad (\boldsymbol{X}_{,p} \odot \boldsymbol{\mathcal{E}})^\top \mathbf{1}_T\right),$$

where $\odot$ denotes the Kronecker product. The following definitions from Ma et al. (2011) are also needed later in our analysis of the optimization problem (S.6.1).

Definition 1 (Nonnegative Vector Shrinkage Operator)**.** *Assume $\boldsymbol{x} \in \mathbb{R}_+^n$. For any $\nu > 0$, the nonnegative vector shrinkage operator $s_\nu(\cdot)$ is defined as*

$$s_\nu(\boldsymbol{x}) = \bar{\boldsymbol{x}},$$

where the i–th element of $\bar{\boldsymbol{x}}$ is $\bar{x}_i = \max(x_i - v, 0)$.

Definition 2 (Matrix Shrinkage Operator)**.** *Assume* $\boldsymbol{X} \in \mathbb{R}^{m\times n}$ *and the singular value decomposition (SVD) of* $\boldsymbol{X}$ *is given by* $\boldsymbol{X} = \boldsymbol{U}\,\mathrm{Diag}\left(\boldsymbol{\sigma}\right)\boldsymbol{V}^{\top}$, $\boldsymbol{U} \in \mathbb{R}^{m\times r}$, $\boldsymbol{\sigma} \in \mathbb{R}_{+}^{r}$, $\boldsymbol{V} \in \mathbb{R}^{n\times r}$. *For any* $\nu > 0$, *the matrix shrinkage operator* $S_{\nu}\left(\cdot\right)$ *is defined as*

$$S_{\nu}\left(\boldsymbol{X}\right) = \boldsymbol{U}\,\mathrm{Diag}\left(\bar{\boldsymbol{\sigma}}\right)\boldsymbol{V}^{\top},$$

where $\bar{\boldsymbol{\sigma}} = s_{\nu}\left(\boldsymbol{\sigma}\right)$.

S.6.2 Algorithm and Convergence

We consider the optimization problem:

$$\begin{aligned}\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right) &= \underset{(\boldsymbol{M}^{(1)},\boldsymbol{M}^{(2)}):\boldsymbol{M}^{(2)}\perp\boldsymbol{G}}{\arg\min} \sum_{i=1}^{N}\left\|\boldsymbol{y}_i - \boldsymbol{X}_i\boldsymbol{m}_i^{(1)} - \boldsymbol{m}_i^{(2)}\right\|_2^2 + \nu_1\left\|\boldsymbol{M}^{(1)}\right\|_n + \nu_2\left\|\boldsymbol{M}^{(2)}\right\|_n \\ &= \underset{(\boldsymbol{M}^{(1)},\boldsymbol{M}^{(2)}):\boldsymbol{M}^{(2)}\perp\boldsymbol{G}}{\arg\min} \left\|\boldsymbol{Y} - \sum_{j=1}^{p}\boldsymbol{X}_{,j}\odot\boldsymbol{1}_T\boldsymbol{m}_{\cdot j}^{(1)\top} - \boldsymbol{M}^{(2)}\right\|_f^2 + \nu_1\left\|\boldsymbol{M}^{(1)}\right\|_n + \nu_2\left\|\boldsymbol{M}^{(2)}\right\|_n, \quad \text{(S.6.1)}\end{aligned}$$

where $\|\cdot\|_n$ denotes the nuclear norm of a matrix, and $\nu_1, \nu_2 > 0$ are tuning parameters. We solve this problem via alternating minimization, using singular value thresholding at each step.

Let $\left(\boldsymbol{M}^{(1)k},\ \boldsymbol{M}^{(2)k}\right)$ denote the solution after the k–th iteration. In the $(k+1)$–th iteration, given $\boldsymbol{M}^{(2)k}$, we first minimize (S.6.1) with respect to $\boldsymbol{M}^{(1)}$. The exact minimizer $\tilde{\boldsymbol{M}}^{(1)k+1}$ satisfies the following KKT condition (Ma et al., 2011): for any $\tau > 0$,

$$\tilde{\boldsymbol{M}}^{(1)k+1} = S_{\tau\nu_1/2}\left(\tilde{\boldsymbol{M}}^{(1)k+1} - \tau\boldsymbol{A}\left(\tilde{\boldsymbol{M}}^{(1)k+1}, \boldsymbol{M}^{(2)k}\right)\right),$$

where

$$\boldsymbol{A}\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(2)}\right) = \left(\boldsymbol{X}_1^{\top}\left(\boldsymbol{X}_1\boldsymbol{m}_1^{(1)} + \boldsymbol{m}_1^{(2)} - \boldsymbol{y}_1\right), \quad \ldots, \quad \boldsymbol{X}_N^{\top}\left(\boldsymbol{X}_N\boldsymbol{m}_N^{(1)} + \boldsymbol{m}_N^{(2)} - \boldsymbol{y}_N\right)\right).$$

Following Chernozhukov et al. (2019), the KKT condition motivates the following gradient descent procedure for computing the exact minimizer $\tilde{\boldsymbol{M}}^{(1)k+1}$:

$$\tilde{\boldsymbol{M}}^{(1)k+1,r+1} = S_{\tau\nu_1/2}\left(\tilde{\boldsymbol{M}}^{(1)k+1,r} - \tau\boldsymbol{A}\left(\tilde{\boldsymbol{M}}^{(1)k+1,r}, \boldsymbol{M}^{(2)k}\right)\right), \quad r = 0, 1, \ldots,$$

initialized at $\tilde{\boldsymbol{M}}^{(1)k+1,0} = \boldsymbol{M}^{(1)k}$. To reduce computation, we update $\boldsymbol{M}^{(1)k}$ using the one-step estimator $\tilde{\boldsymbol{M}}^{(1)k+1,1}$, and define $\boldsymbol{M}^{(1)k+1} := \tilde{\boldsymbol{M}}^{(1)k+1,1}$. This yields the simplified update:

$$\boldsymbol{M}^{(1)k+1} = S_{\tau\nu_1/2}\left(\boldsymbol{M}^{(1)k} - \tau\boldsymbol{A}^{k}\right), \quad \text{(S.6.2)}$$

where

$$\boldsymbol{A}^{k} := \boldsymbol{A}\left(\boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right) = \left(\boldsymbol{X}_1^{\top}\left(\boldsymbol{X}_1\boldsymbol{m}_1^{(1)k} + \boldsymbol{m}_1^{(2)k} - \boldsymbol{y}_1\right), \quad \ldots, \quad \boldsymbol{X}_N^{\top}\left(\boldsymbol{X}_N\boldsymbol{m}_N^{(1)k} + \boldsymbol{m}_N^{(2)k} - \boldsymbol{y}_N\right)\right). \quad \text{(S.6.3)}$$

We later show this one-step update suffices for global convergence of the algorithm when the step size τ is properly chosen. Given $\boldsymbol{M}^{(1)k+1}$, we next minimize (S.6.1) with respect to $\boldsymbol{M}^{(2)}$. For any $\boldsymbol{M}^{(2)}$, the constraint $\boldsymbol{M}^{(2)} \perp \boldsymbol{G}$ yields the decomposition

$$\left\| \boldsymbol{Y} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T \boldsymbol{m}_{\cdot j}^{(1)\top} - \boldsymbol{M}^{(2)} \right\|_f^2 = \left\| \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \left(\boldsymbol{Y} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T \boldsymbol{m}_{\cdot j}^{(1)\top} - \boldsymbol{M}^{(2)} \right) \right\|_f^2 + \left\| \boldsymbol{P}_{\boldsymbol{G}} \left(\boldsymbol{Y} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T \boldsymbol{m}_{\cdot j}^{(1)\top} \right) \right\|_f^2,$$

where the second term on the RHS does not depend on $\boldsymbol{M}^{(2)}$. Therefore, the update reduces to

$$\boldsymbol{M}^{(2)k+1} = \underset{\boldsymbol{M}^{(2)}:\boldsymbol{M}^{(2)} \perp \boldsymbol{G}}{\arg\min} \left\| \boldsymbol{P}_{\boldsymbol{G}}^{\perp} \left(\boldsymbol{Y} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T \boldsymbol{m}_{\cdot j}^{(1)k+1\top} \right) - \boldsymbol{M}^{(2)} \right\|_f^2 + \nu_2 \left\| \boldsymbol{M}^{(2)} \right\|_n.$$

The corresponding unconstrained problem has the closed-form solution

$$\boldsymbol{M}_u^{(2)k+1} = S_{\nu_2/2} \left[\boldsymbol{P}_{\boldsymbol{G}}^{\perp} \left(\boldsymbol{Y} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T \boldsymbol{m}_{\cdot j}^{(1)k+1\top} \right) \right], \tag{S.6.4}$$

which automatically lies in the orthogonal complement of $\boldsymbol{G}$. Hence $\boldsymbol{M}^{(2)k+1} = \boldsymbol{M}_u^{(2)k+1}$. We alternate between updating $\boldsymbol{M}^{(1)}$ in (S.6.2) and $\boldsymbol{M}^{(2)}$ in (S.6.4) until convergence.

Proposition S.6.1 (Convergence of NNR Iteration)**.** *Let* $\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right)$ *be a global minimum of* (S.6.1)*, and define*

$$g\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(2)}\right) = \sum_{i=1}^{N} \left\| \boldsymbol{y}_i - \boldsymbol{X}_i \boldsymbol{m}_i^{(1)} - \boldsymbol{m}_i^{(2)} \right\|_2^2 + \nu_1 \left\| \boldsymbol{M}^{(1)} \right\|_n + \nu_2 \left\| \boldsymbol{M}^{(2)} \right\|_n.$$

Given $\tau \in \left(0, 1/\max_i \lambda_{max}\left(\boldsymbol{X}_i^\top \boldsymbol{X}_i\right)\right)$*, for any initial* $\left(\boldsymbol{M}^{(1)0}, \boldsymbol{M}^{(2)0}\right)$*, we have:*

$$g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k+1}\right) \le g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k}\right) \le g\left(\boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right), \quad k \ge 0.$$

Furthermore,

$$g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k+1}\right) - g\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right) \le \frac{1}{k\tau} \left\| \boldsymbol{M}^{(1)1} - \hat{\boldsymbol{M}}^{(1)} \right\|_f^2, \quad k \ge 1. \tag{S.6.5}$$

Following Chernozhukov et al. (2019), Proposition S.6.1 shows that, starting from any initialization, the objective value $g\left(\boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right)$ decreases monotonically and converges to the global minimum at the rate $O\left(k^{-1}\right)$. An important feature of the convergence bound is that it depends only on $\boldsymbol{M}^{(1)1} - \hat{\boldsymbol{M}}^{(1)}$, and not on $\boldsymbol{M}^{(2)1} - \hat{\boldsymbol{M}}^{(2)}$, since $\boldsymbol{M}^{(2)1}$ has a closed-form expression which is fully determined by $\boldsymbol{M}^{(1)1}$; see the update above. Moreover, the step size τ must not exceed $1/\max_i \lambda_{max}\left(\boldsymbol{X}_i^\top \boldsymbol{X}_i\right)$.

S.6.3 Singular Value Thresholding Approach for Model Selection

We select (K, R) by the following singular value thresholding (SVT) approach:

$$\hat{K}_{NNR} = \sum_{j=1}^{p} 1\left\{\sqrt{T}\psi_j\left(\hat{\boldsymbol{M}}^{(1)}\right) \geq c_1\left(\frac{\nu_1}{\sqrt{T}}\right)^{1/2-\delta_K}\left(\sqrt{T}\left\|\hat{\boldsymbol{M}}^{(1)}\right\|_2\right)^{1/2+\delta_K}\right\},$$

$$\hat{R}_{NNR} = \sum_{j=1}^{m_{NT}^2} 1\left\{\psi_j\left(\hat{\boldsymbol{M}}^{(2)}\right) \geq c_2\nu_2^{1/2-\delta_R}\left\|\hat{\boldsymbol{M}}^{(2)}\right\|_2^{1/2+\delta_R}\right\}, \tag{S.6.6}$$

for some $\delta_K, \delta_R \in (0, 1/2)$, where $\psi_j(\cdot)$ is the j–th largest singular value, and c_1 and c_2 are positive constants. Before stating the assumptions required to guarantee selection consistency, we introduce some additional notation.

Suppose that $\boldsymbol{M}_0^{(1)} = \boldsymbol{\Theta}_0\boldsymbol{B}_0^\top$ and $\boldsymbol{M}_0^{(2)} = \boldsymbol{F}_0\boldsymbol{L}_0^\top$ admit the following SVD:

$$\boldsymbol{M}_0^{(1)} = \boldsymbol{U}_1\boldsymbol{\Psi}_1\boldsymbol{V}_1^\top, \quad \boldsymbol{U}_1 \in \mathbb{R}^{p\times p},\ \boldsymbol{\Psi}_1 \in \mathbb{R}^{p\times N},\ \boldsymbol{V}_1 \in \mathbb{R}^{N\times N},$$

$$\boldsymbol{M}_0^{(2)} = \boldsymbol{U}_2\boldsymbol{\Psi}_2\boldsymbol{V}_2^\top, \quad \boldsymbol{U}_2 \in \mathbb{R}^{T\times T},\ \boldsymbol{\Psi}_2 \in \mathbb{R}^{T\times N},\ \boldsymbol{V}_2 \in \mathbb{R}^{N\times N}.$$

Further decompose

$$\boldsymbol{U}_1 = \left(\boldsymbol{U}_{1r}, \quad \boldsymbol{U}_{1c}\right), \quad \boldsymbol{V}_1 = \left(\boldsymbol{V}_{1r}, \quad \boldsymbol{V}_{1c}\right),$$

where $(\boldsymbol{U}_{1r}, \boldsymbol{V}_{1r})$ are singular vectors corresponding to the non-zero singular values, and $(\boldsymbol{U}_{1c}, \boldsymbol{V}_{1c})$ are singular vectors corresponding to the zero singular values. For a matrix $\Delta_1 \in \mathbb{R}^{p\times N}$, let

$$\boldsymbol{\mathcal{P}}_1^\perp(\Delta_1) = \boldsymbol{U}_{1c}\boldsymbol{U}_{1c}^\top\Delta_1\boldsymbol{V}_{1c}\boldsymbol{V}_{1c}^\top, \quad \boldsymbol{\mathcal{P}}_1(\Delta_1) = \Delta_1 - \boldsymbol{\mathcal{P}}_1^\perp(\Delta_1).$$

Similarly, we can define

$$\boldsymbol{U}_2 = \left(\boldsymbol{U}_{2r}, \quad \boldsymbol{U}_{2c}\right), \quad \boldsymbol{V}_2 = \left(\boldsymbol{V}_{2r}, \quad \boldsymbol{V}_{2c}\right),$$

$$\boldsymbol{\mathcal{P}}_2^\perp(\Delta_2) = \boldsymbol{U}_{2c}\boldsymbol{U}_{2c}^\top\Delta_2\boldsymbol{V}_{2c}\boldsymbol{V}_{2c}^\top, \quad \boldsymbol{\mathcal{P}}_2(\Delta_2) = \Delta_2 - \boldsymbol{\mathcal{P}}_2^\perp(\Delta_2),$$

where $\Delta_2 \in \mathbb{R}^{T\times N}$. Here $\boldsymbol{\mathcal{P}}_1(\cdot)$ and $\boldsymbol{\mathcal{P}}_2(\cdot)$ are considered as the projection matrices onto the reduced-rank spaces of $\boldsymbol{M}_0^{(1)}$ and low-rank spaces of $\boldsymbol{M}_0^{(2)}$ respectively, and $\boldsymbol{\mathcal{P}}_1^\perp(\cdot)$ and $\boldsymbol{\mathcal{P}}_2^\perp(\cdot)$ are projections onto their orthogonal spaces. To establish the asymptotic result, we make the following assumption.

Assumption S.1 (Restricted Strong Convexity)**.** *Define the restricted low-rank set as, for some $c > 0$,*

$$\mathcal{C}(c) = \left\{(\Delta_1, \Delta_2) : \sqrt{T}\left\|\boldsymbol{\mathcal{P}}_1^\perp(\Delta_1)\right\|_n + \left\|\boldsymbol{\mathcal{P}}_2^\perp(\Delta_2)\right\|_n \leq c\sqrt{T}\left\|\boldsymbol{\mathcal{P}}_1(\Delta_1)\right\|_n + c\left\|\boldsymbol{\mathcal{P}}_2(\Delta_2)\right\|_n\right\}. \tag{S.6.7}$$

For any $c > 0$, there is a constant $\kappa_c > 0$ such that for all $(\Delta_1, \Delta_2) \in \mathcal{C}(c)$, the following inequality holds:

$$\left\|\sum_{j=1}^{p}\boldsymbol{X}_{\cdot j} \odot \mathbf{1}_T\left(\boldsymbol{m}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right\|_f^2 \geq \kappa_c T\left\|\Delta_1\right\|_f^2 + \kappa_c\left\|\Delta_2\right\|_f^2. \tag{S.6.8}$$

Analogous to Assumption 3.1 in Chernozhukov et al. (2019), Assumption S.1 imposes the so-called invertibility condition (S.6.8) on (Δ_1, Δ_2), when restricted to the cone or the low-rank set defined by (S.6.7). Since $\Delta_1 \in \mathbb{R}^{p\times N}$ and $\Delta_2 \in \mathbb{R}^{T\times N}$, we rescale Δ_1 by a factor $\sqrt{T}$ to ensure that Δ_1 and Δ_2 are on the same scale.

Theorem S.6.1. *Suppose Assumptions 2 and S.1 hold. If* $2\|\boldsymbol{Z}\|_2 \le (1-c)\nu_1$, $2\|\boldsymbol{\mathcal{E}}\|_2 \le (1-c)\nu_2$, *and* $T^{-1/2}\nu_1 \asymp \nu_2 = O_p(M_{NT})$, *then:*

(1) $\dfrac{\|\hat{\boldsymbol{M}}^{(1)} - \boldsymbol{M}_0^{(1)}\|_f^2}{N} = O_p\left(\dfrac{1}{m_{NT}^2}\right), \quad \dfrac{\|\hat{\boldsymbol{M}}^{(2)} - \boldsymbol{M}_0^{(2)}\|_f^2}{NT} = O_p\left(\dfrac{1}{m_{NT}^2}\right).$

(2) $\lim\limits_{(N,T)\to\infty} \mathbb{P}\left[\left(\hat{K}_{NNR}, \hat{R}_{NNR}\right) = (K_0, R_0)\right] = 1.$

S.6.4 Choice of Tuning Parameters

In Theorem S.6.1, we require that $\|\boldsymbol{\mathcal{E}}\|_2 = O_p(M_{NT})$, and $\|\boldsymbol{X}_{,j} \odot \boldsymbol{\mathcal{E}}\|_2 = O_p(M_{NT})$ for $j = 1, \ldots, p$, since

$$\|\boldsymbol{Z}\|_2 \le \sum_{j=1}^{p} \left\|(\boldsymbol{X}_{,j} \odot \boldsymbol{\mathcal{E}})^\top \mathbf{1}_T\right\|_2 \le \sqrt{T} \sum_{j=1}^{p} \|\boldsymbol{X}_{,j} \odot \boldsymbol{\mathcal{E}}\|_2 = O_p\left(\sqrt{T} M_{NT}\right).$$

The condition on $\boldsymbol{\mathcal{E}}$ is identical to that in Assumption 6(1). Similar arguments and examples, such as those in Latala (2005) and Moon and Weidner (2017) apply to the term $\boldsymbol{X}_{,j} \odot \boldsymbol{\mathcal{E}}$. An additional example of $\|\boldsymbol{X}_{,j} \odot \boldsymbol{\mathcal{E}}\|_2 = O_p(M_{NT})$ is provided in Chernozhukov et al. (2019); see Assumption 3.4 and Lemma C.1 therein. Accordingly, in these examples, the tuning parameters can be set to satisfy $T^{-1/2}\nu_1 \asymp \nu_2 = O_p(M_{NT})$.

In the discussion above, we argued that such tuning parameters generally exist in theory. We now outline the approach to choose the tuning parameters in our simulation studies, following the approach proposed in Chernozhukov et al. (2019). Suppose that $\{\varepsilon_{it}\}$ are independent across (i,t) and $\varepsilon_{it} \sim \mathcal{N}\left(0, \sigma_i^2\right)$. To approximate the distribution of relevant stochastic terms, we generate a series of independent copies of $\boldsymbol{\mathcal{E}}$, denoted by $\tilde{\boldsymbol{\mathcal{E}}}$, where $\{\tilde{\varepsilon}_{it}\}$ are independent across (i,t) and $\tilde{\varepsilon}_{it} \sim \mathcal{N}(0, \sigma_i^2)$. Let $Q(U;q)$ denote the q–th quantile of a random variable U. We then choose the tuning parameters as

$$\nu_1 = 2(1+c_1)\cdot Q\left(\left\|\tilde{\boldsymbol{Z}}\right\|_2; 1-\delta_{NT}\right), \quad \nu_2 = 2(1+c_1)\cdot Q\left(\left\|\tilde{\boldsymbol{\mathcal{E}}}\right\|_2; 1-\delta_{NT}\right),$$

where $\tilde{\boldsymbol{Z}} = \left(\left(\boldsymbol{X}_{,1} \odot \tilde{\boldsymbol{\mathcal{E}}}\right)^\top \mathbf{1}_T, \quad \ldots, \quad \left(\boldsymbol{X}_{,p} \odot \tilde{\boldsymbol{\mathcal{E}}}\right)^\top \mathbf{1}_T\right)$, and $\delta_{NT} = o_p(1)$. Under this choice, the following inequalities hold with probability at least $1-\delta_{NT}$:

$$2\|\boldsymbol{Z}\|_2 \le \left(1 - \frac{c_1}{1+c_1}\right)\nu_1, \quad 2\|\boldsymbol{\mathcal{E}}\|_2 \le \left(1 - \frac{c_1}{1+c_1}\right)\nu_2.$$

Following the practical implementation in Chernozhukov et al. (2019), we can use an initial consistent estimator of σ_i^2 to compute the empirical quantiles.

S.6.5 Proofs of Theorems in Section S.6

The proof of Lemmas S.6.1–S.6.3 are omitted, as similar arguments can be found in Chernozhukov et al. (2019).

Lemma S.6.1. *The following results hold:*

(1) $\boldsymbol{M}^{(1)k+1} = \underset{\boldsymbol{M}^{(1)}}{\arg\min}\, h\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right) + \nu_1 \left\|\boldsymbol{M}^{(1)}\right\|_n$, *where* $h\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right) := \tau^{-1}\left\|\boldsymbol{M}^{(1)k} - \boldsymbol{M}^{(1)}\right\|_f^2 - 2\,\mathrm{tr}\left[\left(\boldsymbol{M}^{(1)k} - \boldsymbol{M}^{(1)}\right)^\top \boldsymbol{A}^k\right]$ *with* $\boldsymbol{A}^k$ *is defined in* (S.6.3).

(2) For any $\tau \in \left(0, 1/\max_i \lambda_{max}\left(\boldsymbol{X}_i^\top \boldsymbol{X}_i\right)\right)$,

$$g\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(2)k}\right) \le h\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right) + \nu_1 \left\|\boldsymbol{M}^{(1)}\right\|_n + \nu_2 \left\|\boldsymbol{M}^{(2)k}\right\|_n + \left\|\boldsymbol{Y} - \boldsymbol{M}^{(2)k} - \sum_{j=1}^{p} \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \boldsymbol{m}_{\cdot j}^{(1)k\top}\right\|_f^2,$$

where $g\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(2)}\right)$ *is defined in Proposition S.6.1.*

(3) $g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k+1}\right) \le g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k}\right) \le g\left(\boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right)$.

Lemma S.6.2. *For any* $\tau \in \left(0, 1/\max_i \lambda_{max}\left(\boldsymbol{X}_i^\top \boldsymbol{X}_i\right)\right)$, *any* $\left(\boldsymbol{M}^{(1)k},\ \boldsymbol{M}^{(2)k}\right)$ *and any* $k \ge 1$, *we have*

$$g\left(\boldsymbol{M}^{(1)}, \boldsymbol{M}^{(2)}\right) - g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k+1}\right) \ge \frac{1}{\tau}\left(\left\|\boldsymbol{M}^{(1)k+1} - \boldsymbol{M}^{(1)}\right\|_f^2 - \left\|\boldsymbol{M}^{(1)k} - \boldsymbol{M}^{(1)}\right\|_f^2\right).$$

Proof of Proposition S.6.1. The inequality (S.6.5) follows from

$$\begin{aligned}
k\left[g\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right) - g\left(\boldsymbol{M}^{(1)k+1}, \boldsymbol{M}^{(2)k+1}\right)\right] &\ge \sum_{l=1}^{k}\left[g\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right) - g\left(\boldsymbol{M}^{(1)l+1}, \boldsymbol{M}^{(2)l+1}\right)\right] \\
&\ge \frac{1}{\tau}\sum_{l=1}^{k}\left(\left\|\boldsymbol{M}^{(1)l+1} - \hat{\boldsymbol{M}}^{(1)}\right\|_f^2 - \left\|\boldsymbol{M}^{(1)l} - \hat{\boldsymbol{M}}^{(1)}\right\|_f^2\right) \\
&= \frac{1}{\tau}\left(\left\|\boldsymbol{M}^{(1)k+1} - \hat{\boldsymbol{M}}^{(1)}\right\|_f^2 - \left\|\boldsymbol{M}^{(1)1} - \hat{\boldsymbol{M}}^{(1)}\right\|_f^2\right) \\
&\ge -\frac{1}{\tau}\left\|\boldsymbol{M}^{(1)1} - \hat{\boldsymbol{M}}^{(1)}\right\|_f^2,
\end{aligned}$$

where the first step is due to the monotonic decreasing property of the sequence $g\left(\boldsymbol{M}^{(1)k}, \boldsymbol{M}^{(2)k}\right)$ as established in Lemma S.6.1 (3), and the second step is due to Lemma S.6.2. □

Lemma S.6.3. *For any matrix* $\Delta \in \mathbb{R}^{p\times N}$, *the following results hold:*

(1) $\left\|\boldsymbol{M}_0^{(1)} + \boldsymbol{\mathcal{P}}_1^\perp(\Delta)\right\|_n = \left\|\boldsymbol{M}_0^{(1)}\right\|_n + \left\|\boldsymbol{\mathcal{P}}_1^\perp(\Delta)\right\|_n$.

(2) $\|\Delta\|_f^2 = \left\|\boldsymbol{\mathcal{P}}_1^\perp(\Delta)\right\|_f^2 + \left\|\boldsymbol{\mathcal{P}}_1(\Delta)\right\|_f^2$.

(3) $\mathrm{Rank}\left(\boldsymbol{\mathcal{P}}_1(\Delta)\right) \le 2K_0$.

(4) $\|\Delta\|_n^2 \le \|\Delta\|_f^2 \cdot \mathrm{Rank}(\Delta)$.

(5) For any matrix Δ_1, Δ_2, $|\mathrm{tr}(\Delta_1\Delta_2)| \le \|\Delta_1\|_n \cdot \|\Delta_2\|_2$.

Similar results hold if we replace $\boldsymbol{M}^{(1)}$ with $\boldsymbol{M}^{(2)}$, $\boldsymbol{\mathcal{P}}_1$ with $\boldsymbol{\mathcal{P}}_2$, and K_0 with R_0.

Proof of Theorem S.6.1.

(1) Let

$$\Delta_1 := \hat{\boldsymbol{M}}^{(1)} - \boldsymbol{M}_0^{(1)} = \left(\hat{\boldsymbol{m}}_{\cdot 1}^{(1)} - \boldsymbol{m}_{\cdot 1,0}^{(1)}, \quad \ldots, \quad \hat{\boldsymbol{m}}_{\cdot p}^{(1)} - \boldsymbol{m}_{\cdot p,0}^{(1)}\right)^\top,$$
$$\Delta_2 := \hat{\boldsymbol{M}}^{(2)} - \boldsymbol{M}_0^{(2)} = \left(\hat{\boldsymbol{m}}_1^{(2)} - \boldsymbol{m}_{1,0}^{(2)}, \quad \ldots, \quad \hat{\boldsymbol{m}}_N^{(2)} - \boldsymbol{m}_{N,0}^{(2)}\right).$$

Then,

$$\begin{aligned}
&\left\|\boldsymbol{Y} - \sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \hat{\boldsymbol{m}}_{\cdot j}^{(1)\top} - \hat{\boldsymbol{M}}^{(2)}\right\|_f^2 \\
=&\left\|\boldsymbol{Y} - \sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\boldsymbol{m}_{\cdot j,0}^{(1)} + \hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top - \left(\boldsymbol{M}_0^{(2)} + \Delta_2\right)\right\|_f^2 \\
=&\|\boldsymbol{\mathcal{E}}\|_f^2 - 2\,\mathrm{tr}\left\{\boldsymbol{\mathcal{E}}^\top\left[\sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right]\right\} + \left\|\sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right\|_f^2.
\end{aligned}$$

Since

$$\mathrm{tr}\left\{\sum_{j=1}^p \boldsymbol{\mathcal{E}}^\top\left[\boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top\right]\right\} = \mathrm{tr}\left[\sum_{j=1}^p \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)\boldsymbol{1}_T^\top\left(\boldsymbol{X}_{\cdot j} \odot \boldsymbol{\mathcal{E}}\right)\right] = \mathrm{tr}\left(\Delta_1 \boldsymbol{Z}\right),$$

we have

$$\begin{aligned}
\left|2\,\mathrm{tr}\left\{\boldsymbol{\mathcal{E}}^\top\left[\sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right]\right\}\right| \le& 2\|\boldsymbol{Z}\|_2 \cdot \|\Delta_1\|_n + 2\|\boldsymbol{\mathcal{E}}\|_2 \cdot \|\Delta_2\|_n \\
\le& (1-c)\,\nu_1\|\Delta_1\|_n + (1-c)\,\nu_2\|\Delta_2\|_n.
\end{aligned}$$

Since $\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right)$ is the minimizer of (S.6.1), we have

$$\begin{aligned}
0 \ge& g\left(\hat{\boldsymbol{M}}^{(1)}, \hat{\boldsymbol{M}}^{(2)}\right) - g\left(\boldsymbol{M}_0^{(1)}, \boldsymbol{M}_0^{(2)}\right) \\
=&\left(\left\|\boldsymbol{Y} - \sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \hat{\boldsymbol{m}}_{\cdot j}^{(1)\top} - \hat{\boldsymbol{M}}^{(2)}\right\|_f^2 - \|\boldsymbol{\mathcal{E}}\|_f^2\right) + \nu_1\left(\|\hat{\boldsymbol{M}}^{(1)}\|_n - \left\|\boldsymbol{M}_0^{(1)}\right\|_n\right) + \nu_2\left(\|\hat{\boldsymbol{M}}^{(2)}\|_n - \left\|\boldsymbol{M}_0^{(2)}\right\|_n\right) \\
\ge&\left\|\sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right\|_f^2 + \nu_1\left[\|\hat{\boldsymbol{M}}^{(1)}\|_n - \left\|\boldsymbol{M}_0^{(1)}\right\|_n - (1-c)\|\Delta_1\|_n\right] \\
&+ \nu_2\left[\|\hat{\boldsymbol{M}}^{(2)}\|_n - \left\|\boldsymbol{M}_0^{(2)}\right\|_n - (1-c)\|\Delta_2\|_n\right] \\
\ge&\left\|\sum_{j=1}^p \boldsymbol{X}_{\cdot j} \odot \boldsymbol{1}_T \left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^\top + \Delta_2\right\|_f^2 + \nu_1\left[c\left\|\boldsymbol{\mathcal{P}}_1^\perp(\Delta_1)\right\|_n - (2-c)\|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n\right] \\
&+ \nu_2\left[c\left\|\boldsymbol{\mathcal{P}}_2^\perp(\Delta_2)\right\|_n - (2-c)\|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_n\right].
\end{aligned} \tag{S.6.9}$$

Here in the last step,

$$\|\Delta_1\|_n = \left\|\boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1) + \boldsymbol{\mathcal{P}}_1(\Delta_1)\right\|_n \le \left\|\boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1)\right\|_n + \|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n,$$
$$\|\Delta_2\|_n = \left\|\boldsymbol{\mathcal{P}}_2^{\perp}(\Delta_2) + \boldsymbol{\mathcal{P}}_2(\Delta_2)\right\|_n \le \left\|\boldsymbol{\mathcal{P}}_2^{\perp}(\Delta_2)\right\|_n + \|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_n,$$

and by Lemma S.6.3 (1),

$$\begin{aligned}
\left\|\hat{\boldsymbol{M}}^{(1)}\right\|_n &= \left\|\boldsymbol{M}_0^{(1)} + \boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1) + \boldsymbol{\mathcal{P}}_1(\Delta_1)\right\|_n \\
&\ge \left\|\boldsymbol{M}_0^{(1)} + \boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1)\right\|_n - \|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n \\
&= \left\|\boldsymbol{M}_0^{(1)}\right\|_n + \left\|\boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1)\right\|_n - \|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n, \\
\left\|\hat{\boldsymbol{M}}^{(2)}\right\|_n &\ge \left\|\boldsymbol{M}_0^{(2)}\right\|_n + \left\|\boldsymbol{\mathcal{P}}_2^{\perp}(\Delta_2)\right\|_n - \|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_n.
\end{aligned}$$

From inequality (S.6.9), we have

$$\sqrt{T}\left\|\boldsymbol{\mathcal{P}}_1^{\perp}(\Delta_1)\right\|_n + \frac{\sqrt{T}\nu_2}{\nu_1}\left\|\boldsymbol{\mathcal{P}}_2^{\perp}(\Delta_2)\right\|_n \le \frac{2-c}{c}\left(\sqrt{T}\|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n + \frac{\sqrt{T}\nu_2}{\nu_1}\|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_n\right)$$

Since $\nu_1 \asymp \sqrt{T}\nu_2$, $(\Delta_1, \Delta_2) \in \mathcal{C}(c_1)$ for some $c_1 > 0$. By Assumption S.1, we obtain

$$\begin{aligned}
T\|\Delta_1\|_f^2 + \|\Delta_2\|_f^2 &\lesssim \left\|\sum_{j=1}^{p} \boldsymbol{X}_{\cdot,j} \odot \mathbf{1}_T\left(\hat{\boldsymbol{m}}_{\cdot j}^{(1)} - \boldsymbol{m}_{\cdot j,0}^{(1)}\right)^{\top} + \Delta_2\right\|_f^2 \\
&\lesssim \frac{\nu_1}{\sqrt{T}} \cdot \sqrt{T}\|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_n + \nu_2\|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_n \\
&\lesssim \frac{\nu_1}{\sqrt{T}} \cdot \sqrt{T}\|\boldsymbol{\mathcal{P}}_1(\Delta_1)\|_f \cdot \sqrt{2K_0} + \nu_2\|\boldsymbol{\mathcal{P}}_2(\Delta_2)\|_f \cdot \sqrt{2R_0} \\
&\lesssim \frac{\nu_1}{\sqrt{T}} \cdot \sqrt{T}\|\Delta_1\|_f + \nu_2\|\Delta_2\|_f \\
&\lesssim \max\left(\frac{\nu_1}{\sqrt{T}}, \nu_2\right) \cdot \sqrt{T\|\Delta_1\|_f^2 + \|\Delta_2\|_f^2},
\end{aligned}$$

where the second step is due to inequality (S.6.9), the third step is due to Lemma S.6.3 (3) and (4), and the fourth step is due to Lemma S.6.3 (2). It follows that

$$T\|\Delta_1\|_f^2 + \|\Delta_2\|_f^2 \le c_2\left(\frac{\nu_1^2}{T} + \nu_2^2\right).$$

This completes the proof of (1).

(2) Let $\lambda_j^{(1)}$ be the j–th largest eigenvalues of $\boldsymbol{\Theta}_0\boldsymbol{\Sigma}_{\boldsymbol{B}}\boldsymbol{\Theta}_0^{\top}$, for $j = 1, \ldots, p$. By Assumption 2(3), we have $\lambda_1^{(1)} \ge \ldots \ge \lambda_{K_0}^{(1)} > \lambda_{K_0+1}^{(1)} = \ldots = \lambda_p^{(1)} = 0$. By Weyl's theorem, we have

$$\begin{aligned}
\left|\frac{1}{N}\psi_j^2\left(\hat{\boldsymbol{M}}^{(1)}\right) - \lambda_j^{(1)}\right| &\le \frac{\|\hat{\boldsymbol{M}}^{(1)}\hat{\boldsymbol{M}}^{(1)\top} - \boldsymbol{M}_0^{(1)}\boldsymbol{M}_0^{(1)\top}\|_2}{N} \\
&\le \frac{\|\hat{\boldsymbol{M}}^{(1)} - \boldsymbol{M}_0^{(1)}\|_2^2}{N} + \frac{2\|\hat{\boldsymbol{M}}^{(1)} - \boldsymbol{M}_0^{(1)}\|_2}{\sqrt{N}} \cdot \frac{\|\boldsymbol{M}_0^{(1)}\|_2}{\sqrt{N}} = O_p\left(\frac{1}{m_{NT}}\right).
\end{aligned}$$

Since $\lambda_j^{(1)} > 0$ for $1 \le j \le K_0$ and $\lambda_j^{(1)} = 0$ for $K_0 < j \le p$, we have

$$\min_{1\le j\le K_0} \psi_j\left(\hat{\boldsymbol{M}}^{(1)}\right) \ge \frac{\sqrt{\lambda_{K_0}^{(1)} N}}{2}, \quad \max_{K_0<j\le p} \psi_j\left(\hat{\boldsymbol{M}}^{(1)}\right) = O_p\left(\frac{\sqrt{N}}{m_{NT}^{1/2}}\right).$$

On the other hand, $\left(T^{-1/2}\nu_1\right)^{1/2-\delta_K}\left(\sqrt{T}\left\|\hat{\boldsymbol{M}}^{(1)}\right\|_2\right)^{1/2+\delta_K} \asymp M_{NT}^{1/2-\delta_K}\left(\sqrt{NT}\right)^{1/2+\delta_K}$. Therefore, for a small $\delta_K \in (0, 1/2)$,

$$\frac{\min_{j\le K_0}\sqrt{T}\psi_j(\hat{\boldsymbol{M}}^{(1)})}{(T^{-1/2}\nu_1)^{1/2-\delta_K}(\sqrt{T}\|\hat{\boldsymbol{M}}^{(1)}\|_2)^{1/2+\delta_K}} = O_p\left(\left(\frac{\sqrt{NT}}{M_{NT}}\right)^{1/2-\delta_K}\right) = O_p\left(m_{NT}^{1/2-\delta_K}\right) \to \infty,$$

$$\frac{\max_{j>K_0}\sqrt{T}\psi_j(\hat{\boldsymbol{M}}^{(1)})}{(T^{-1/2}\nu_1)^{1/2-\delta_K}(\sqrt{T}\|\hat{\boldsymbol{M}}^{(1)}\|_2)^{1/2+\delta_K}} = O_p\left(\left(\frac{\sqrt{NT}}{M_{NT}}\right)^{1/2-\delta_K}\cdot\frac{1}{m_{NT}^{1/2}}\right) = O_p\left(\frac{1}{m_{NT}^{\delta_K}}\right) \to 0.$$

This establishes that $\lim_{(N,T)\to\infty}\mathbb{P}\left(\hat{K}_{NNR} = K_0\right) = 1$. The other part $\lim_{(N,T)\to\infty}\mathbb{P}\left(\hat{R}_{NNR} = R_0\right) = 1$ can be shown similarly; see also the proof of Proposition 3.1 in Chernozhukov et al. (2019). □

References

Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica*, 77(4):1229–1279.

Cai, T., Li, H., and Ma, R. (2021). Optimal structured principal subspace estimation: Metric entropy and minimax rates. *Journal of Machine Learning Research*, 22(46):1–45.

Chernozhukov, V., Hansen, C., Liao, Y., and Zhu, Y. (2019). Inference for heterogeneous effects using low-rank estimation of factor slopes. Technical report, Cornell University Library, arXiv.org.

Davis, B. and Song, R. (2011). Integral inequalities for convex functions of operators on martingales. In *Selected Works of Donald L. Burkholder*, pages 181–198. Springer.

Latala, R. (2005). Some estimates of norms of random matrices. *Proceedings of the American Mathematical Society*, 133(5):1273–1282.

Ma, S., Goldfarb, D., and Chen, L. (2011). Fixed point and bregman iterative methods for matrix rank minimization. *Mathematical Programming*, 128(1-2):321–353.

Moon, H. R. and Weidner, M. (2017). Dynamic linear panel regression models with interactive fixed effects. *Econometric Theory*, 33(1):158–195.

Rio, E. et al. (2017). *Asymptotic theory of weakly dependent random processes*, volume 80. Springer.

Tropp, J. A. et al. (2015). An introduction to matrix concentration inequalities. *Foundations and Trends® in Machine Learning*, 8(1-2):1–230.

Vershynin, R. (2020). High-dimensional probability. *University of California, Irvine*, 10:11.

White, H. (2001). *Asymptotic theory for econometricians*. Academic Press, San Diego, rev. edition.

Zhang, F. (2006). *The Schur complement and its applications*, volume 4. Springer Science & Business Media.